สยามชำนาญกิจ
SIAM CHAMNAN KIT

Software design
Software architecture
Microservices



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Module 1 : Design

Cloud Native Application
Evolution of architecture
Microservice architecture
How to decompose app to Microservice
Communication between service
Data consistency
Workshop



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Module 2 : Develop + Testing

Recap Microservice
Properties of Microservice
Microservice 1.0 - 4.0 (4.5)
How to develop Microservice ?
How to test Microservice ?
12-factors app
Observable services



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Module 3 : Deploy

How to deploy Microservice ?
Continuous Integration and Delivery
Practices of Continuous Integration
Deployment strategies
Working with containerization (Docker)
Workshop



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“ Don’t use Microservice ”

Start with your **problems** ?



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Design

Develop
Test

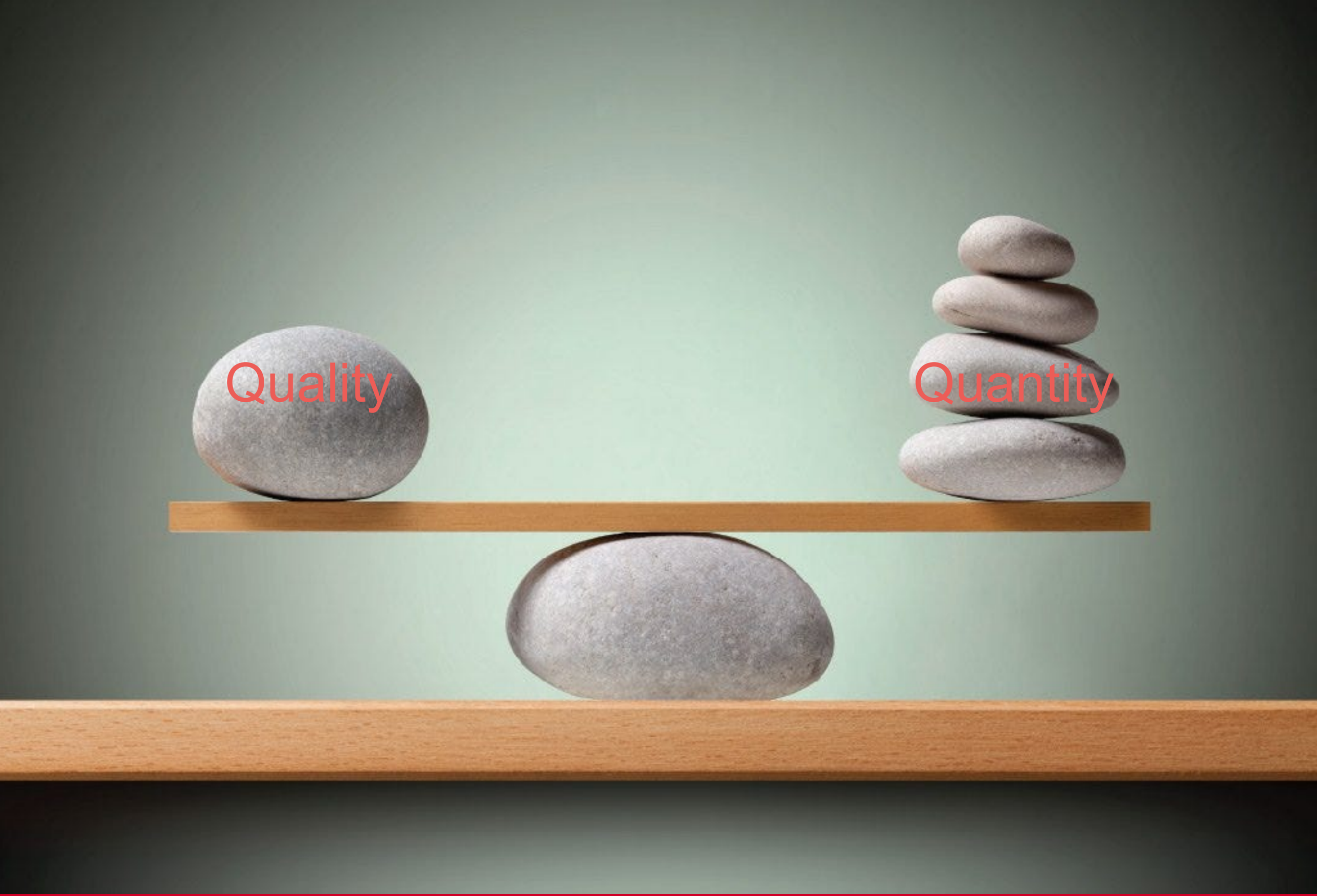
Deploy

Observability

Continuous Integration/Delivery



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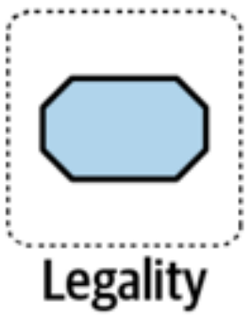


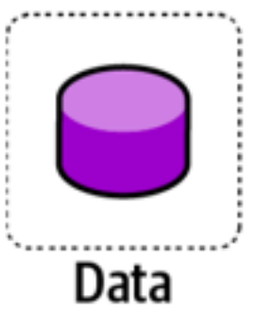


Software Architecture

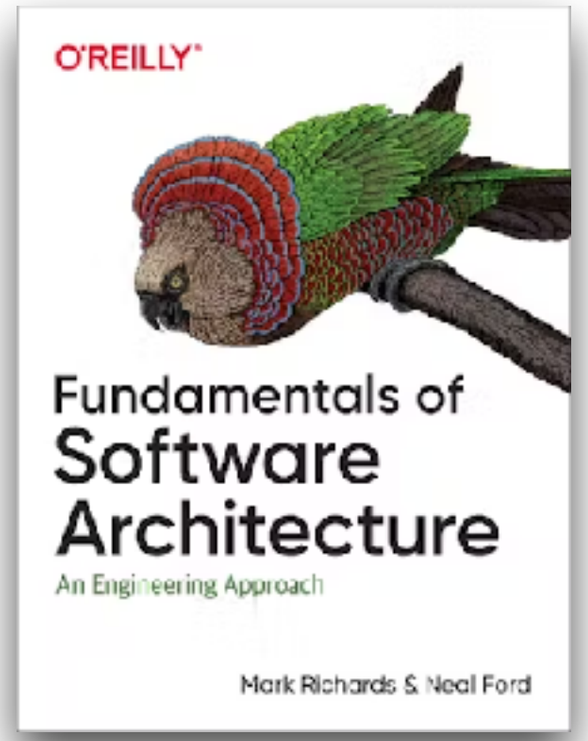


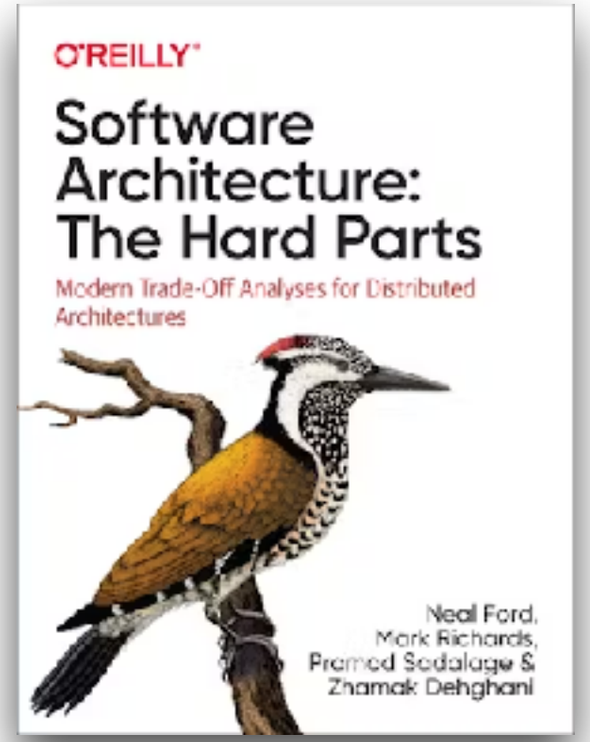


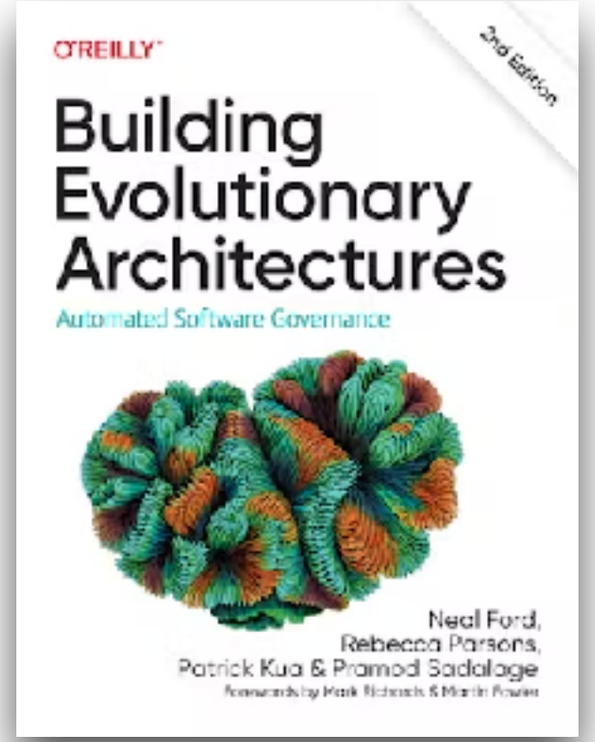


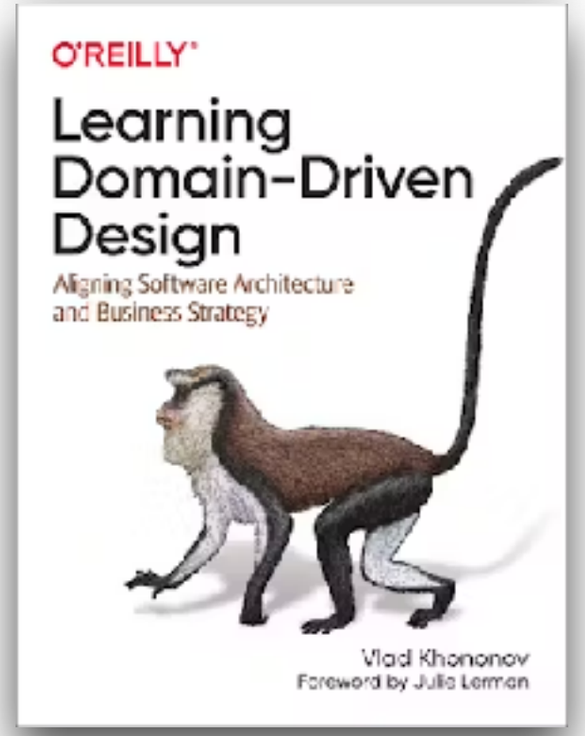


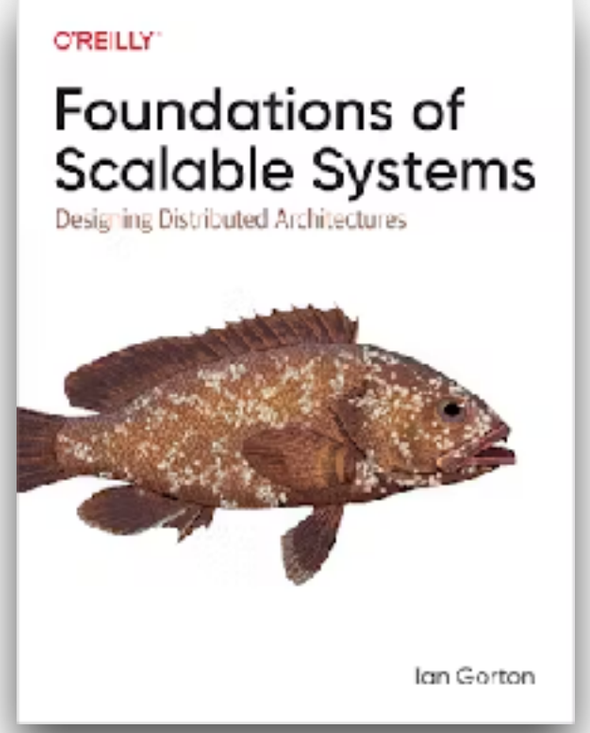
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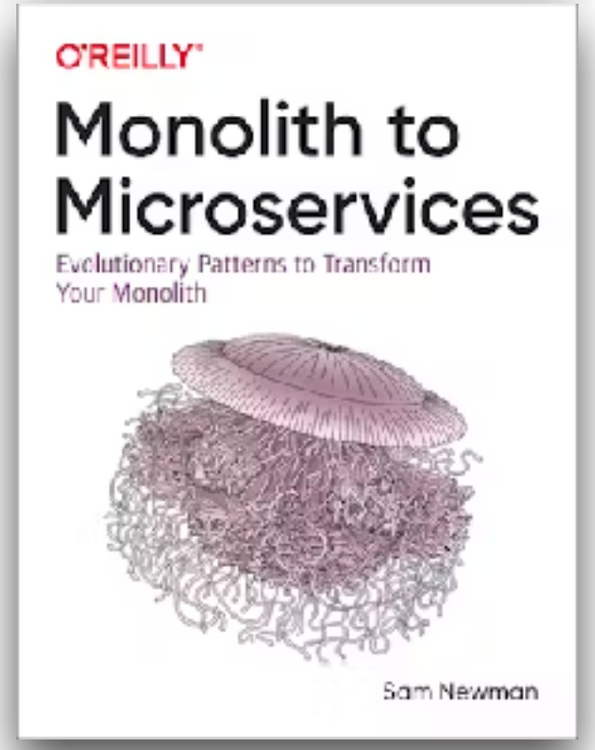












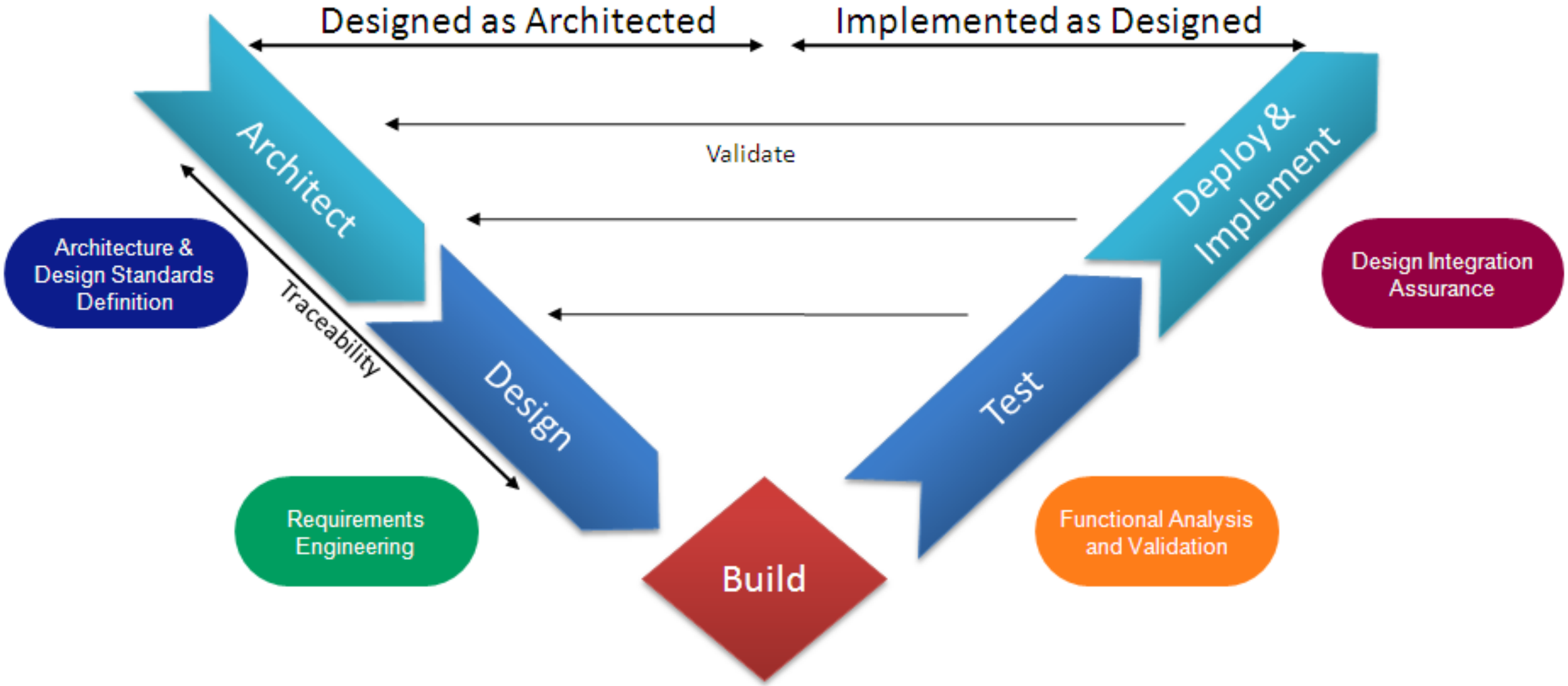
https://www.amazon.com/Fundamentals-Software-Architecture-Comprehensive-Characteristics/dp/1492043451

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Traditional Process



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Architectural Thinking ?

Architecture view	Developer view

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Architectural Thinking ?

Different between architecture and developer/design

Collaborate with development team

Wide breadth of **technical** knowledge (more solutions)

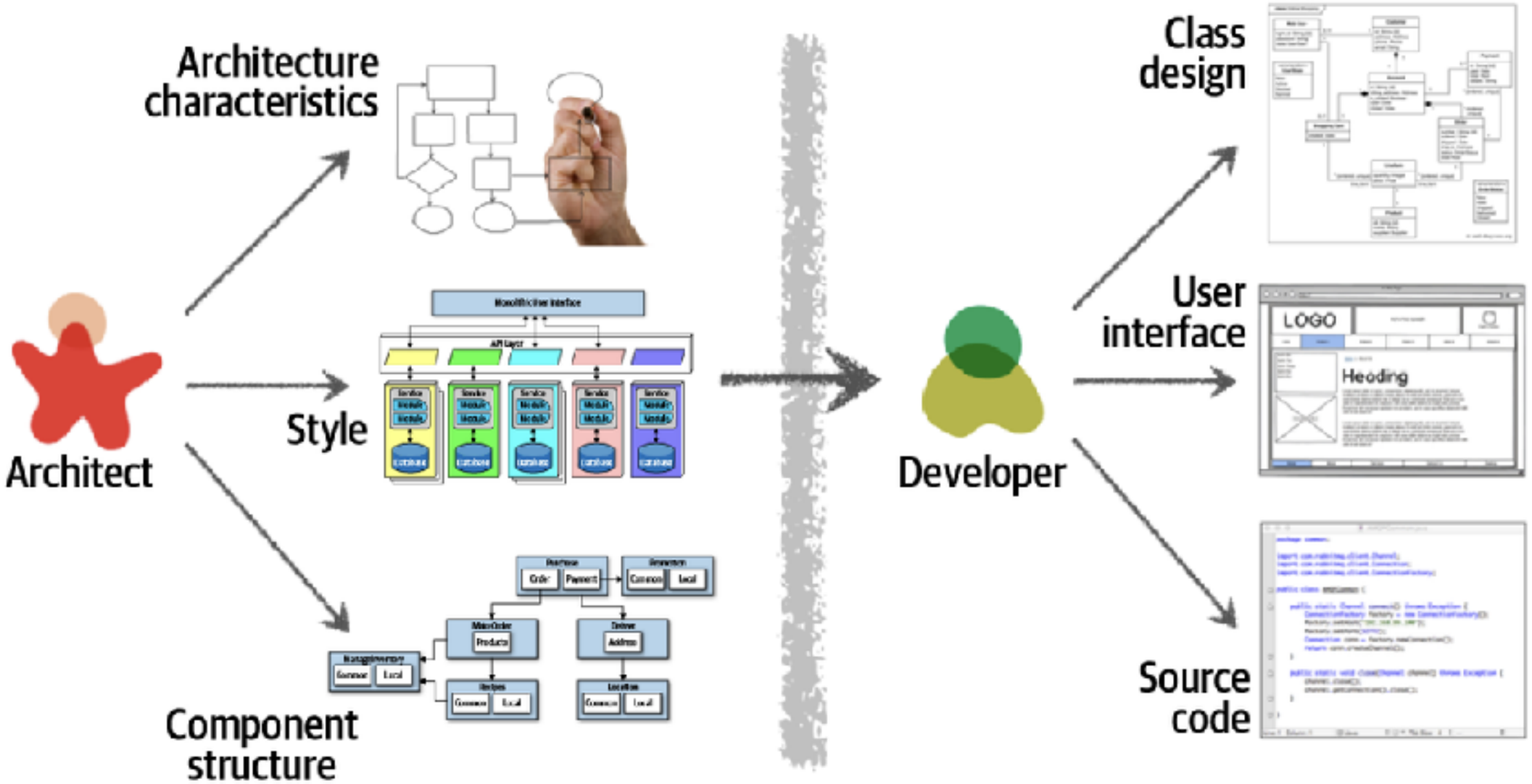
Understand, analyze and **trade-offs** (solutions and technical)

Understand the importance of **business** drivers



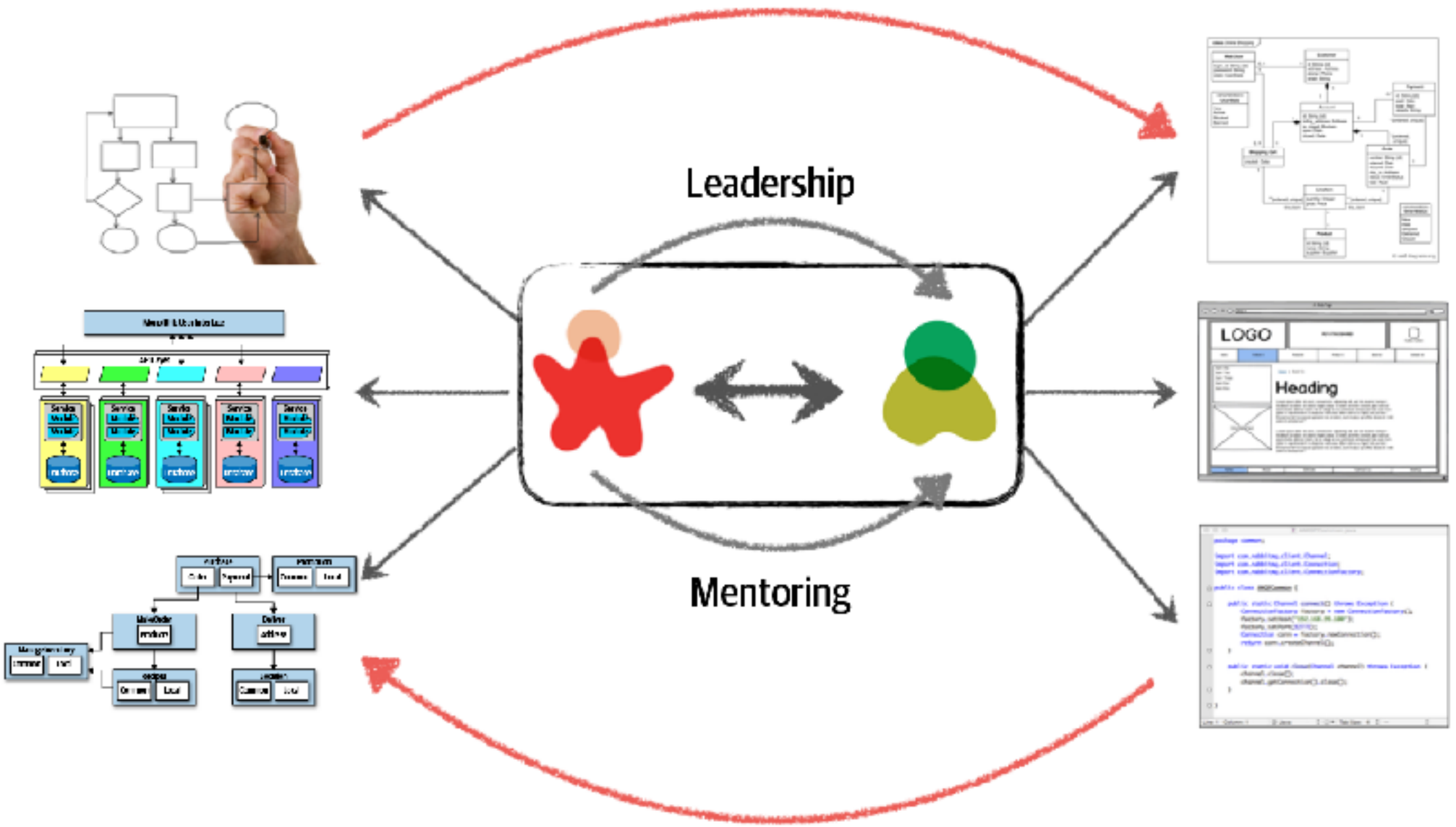
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Architecture vs Developer ?



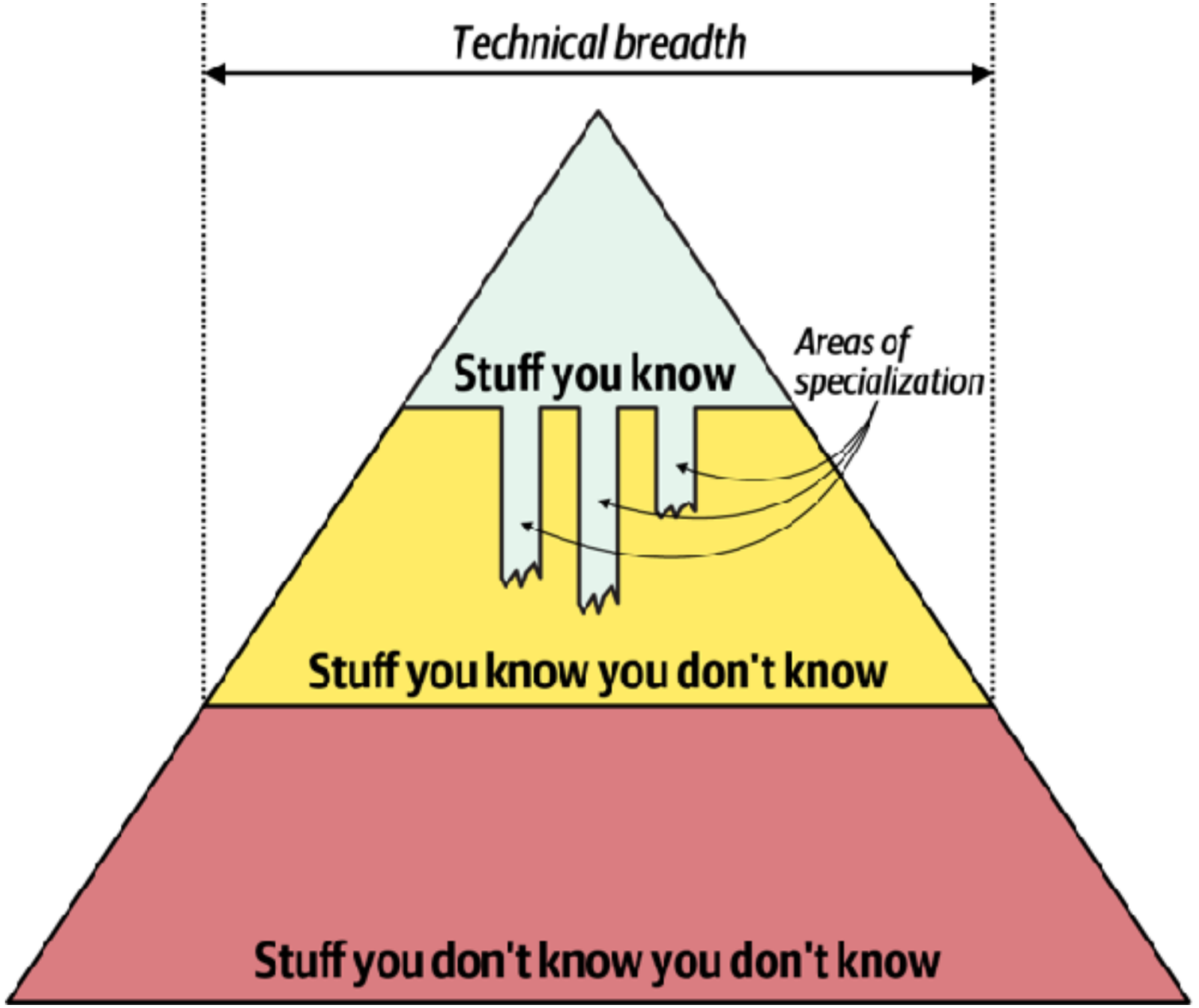
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Improve Collaboration

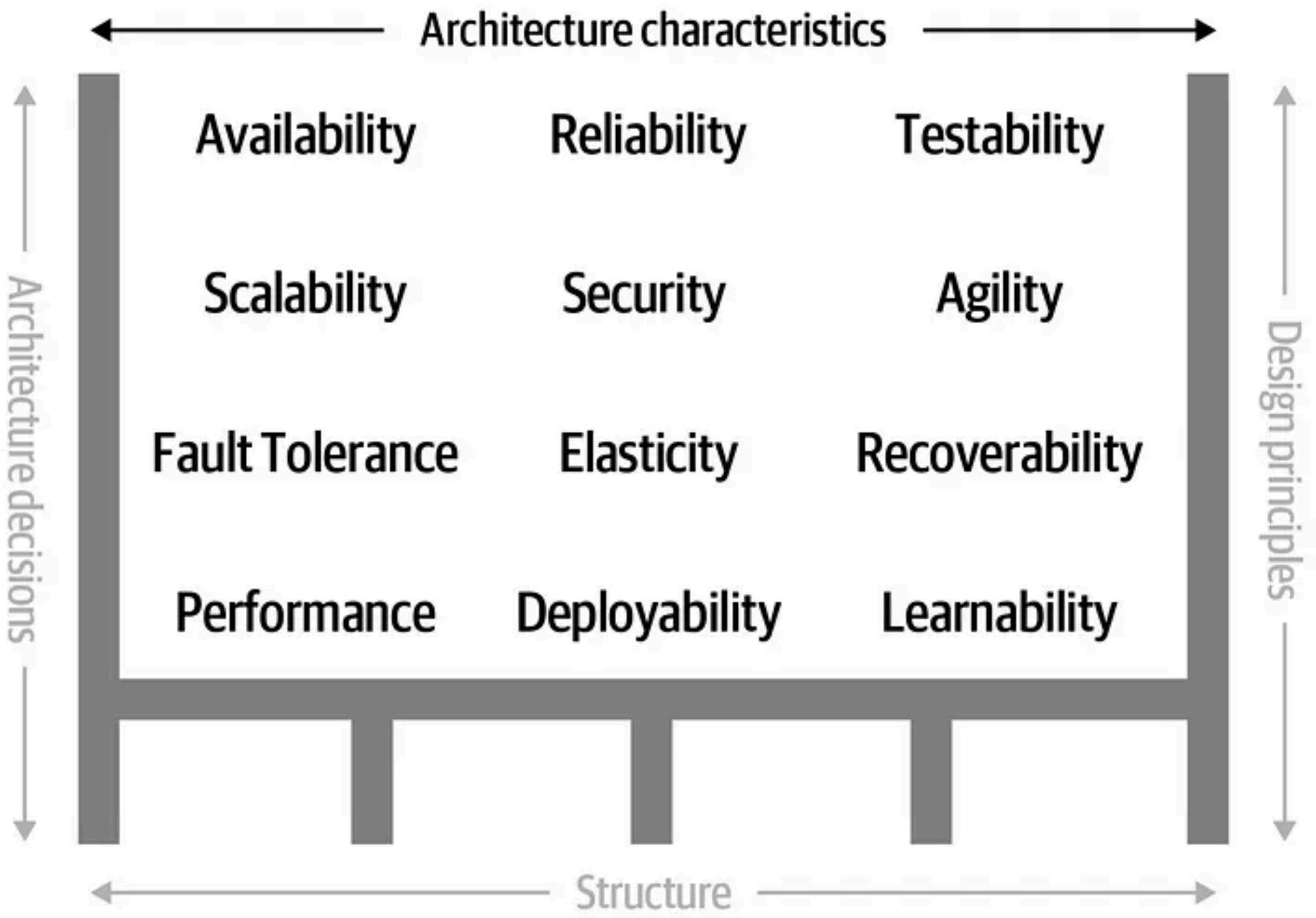


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Always Improve your knowledge

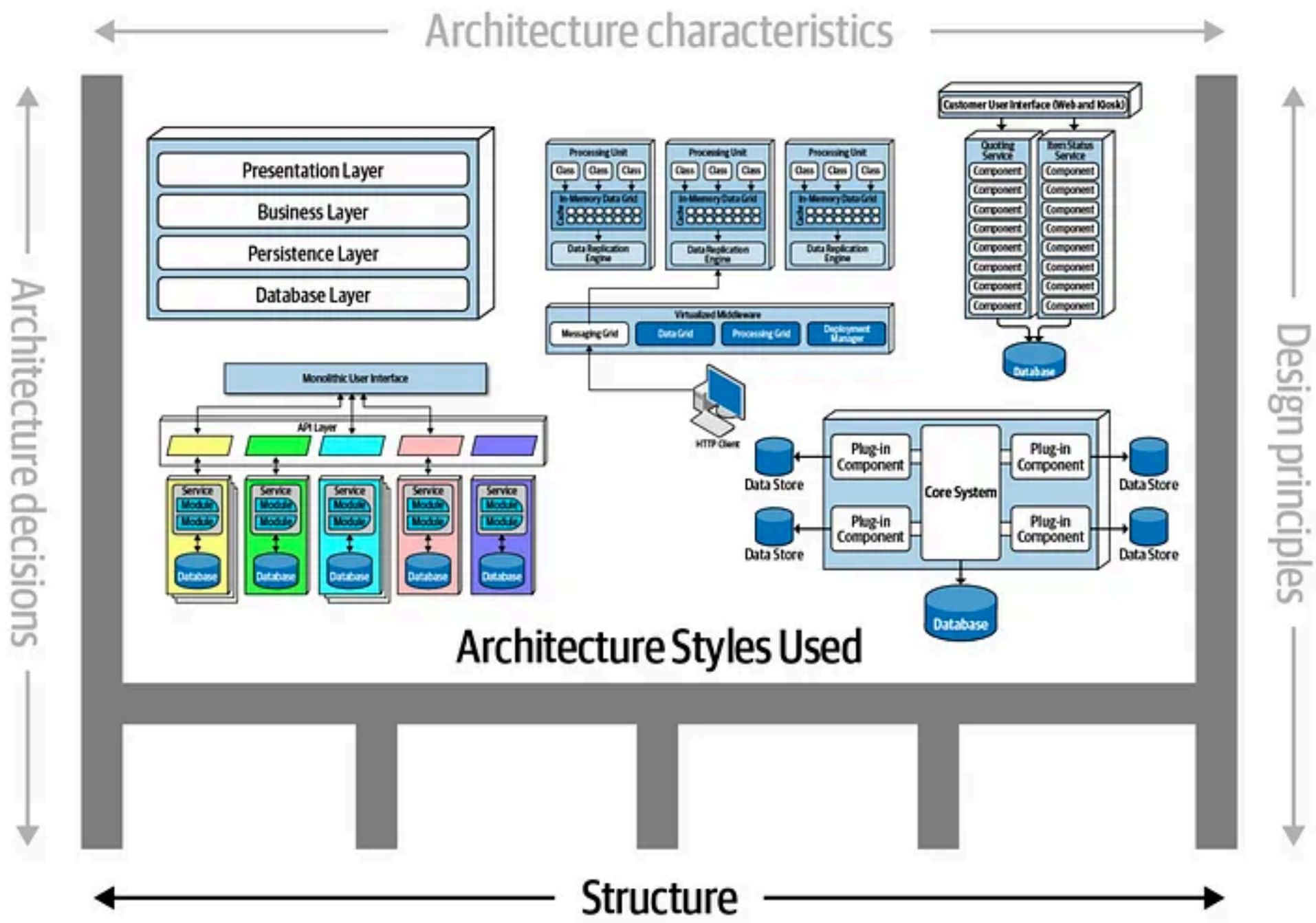


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Module 1 : Design

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Design Decision ?

Reusability

Maintainability

Changeability

Functional and
Non-functional requirements

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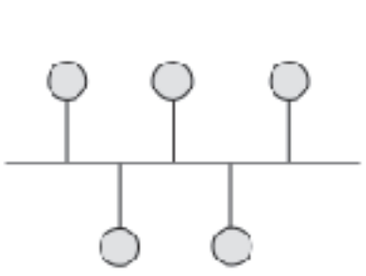
Architecture Smells

Increase technical debt
Degrade the quality of system
Affect when/where/how much change code !!
Increase your effort

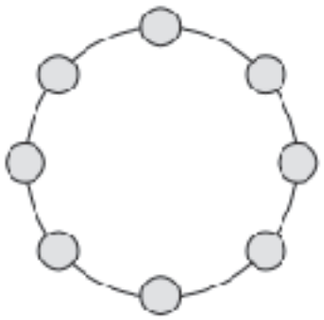


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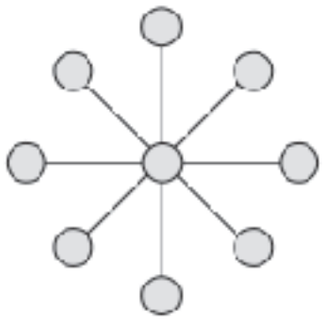
Cyclic Dependencies ?



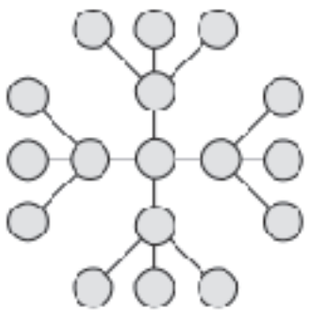
Bus



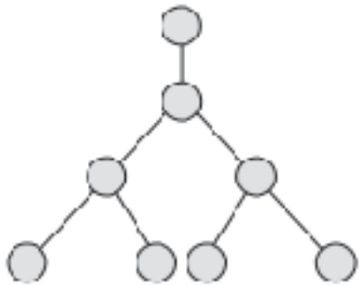
Ring



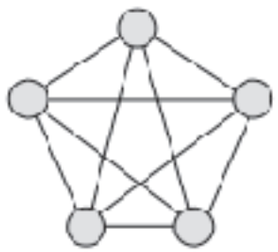
Star



Extended Star

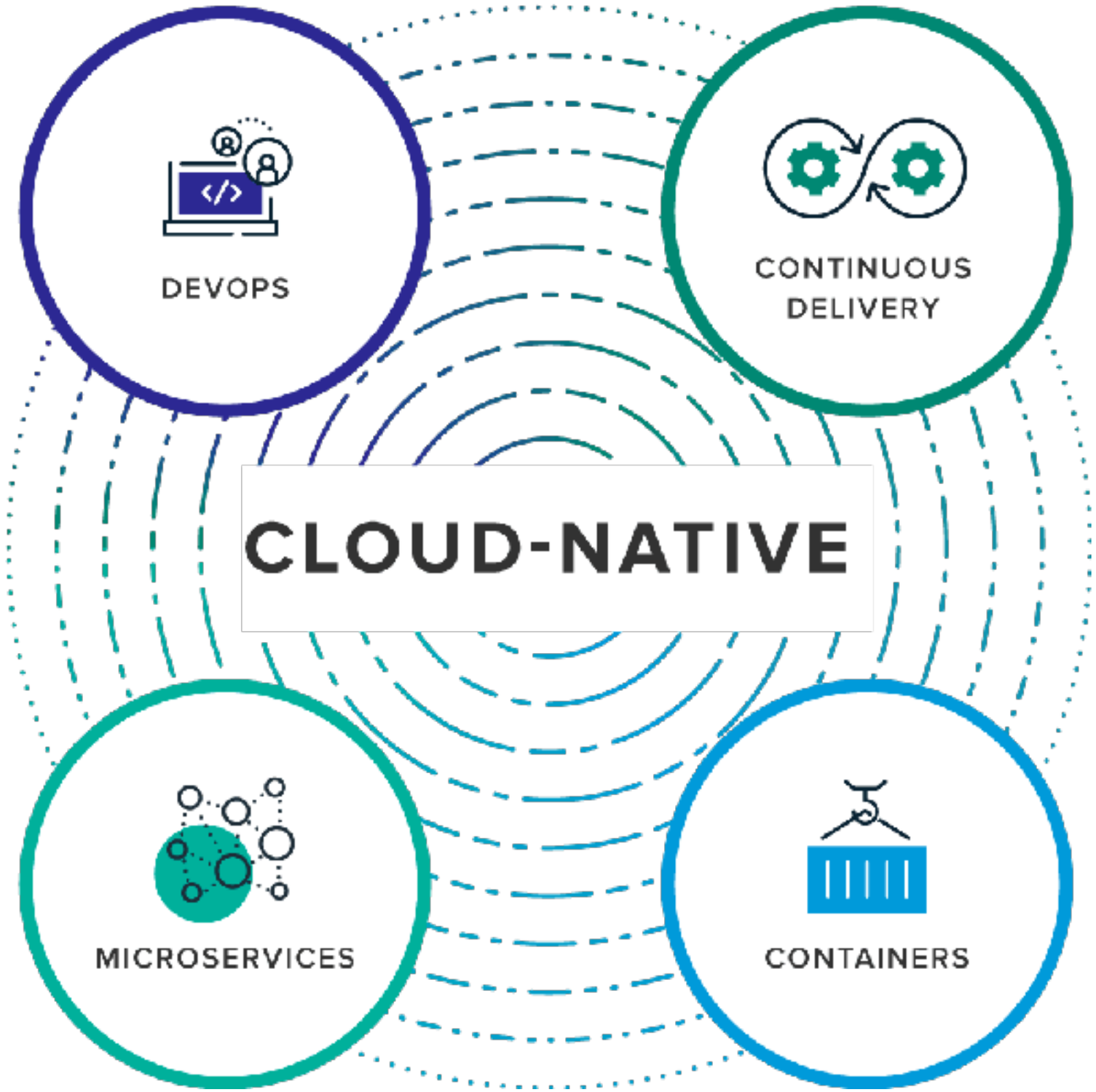


Hierarchical



Mesh

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https://pivotal.io/cloud-native

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Cloud Native ?

Build efficiency
Scalability
Portability
On-premise and On-cloud

Loosely Coupled	Resilience	Manageable	Observable

https://www.somkiat.cc/cloud-native-go-part-1/



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Software design is an exercise in human relationships



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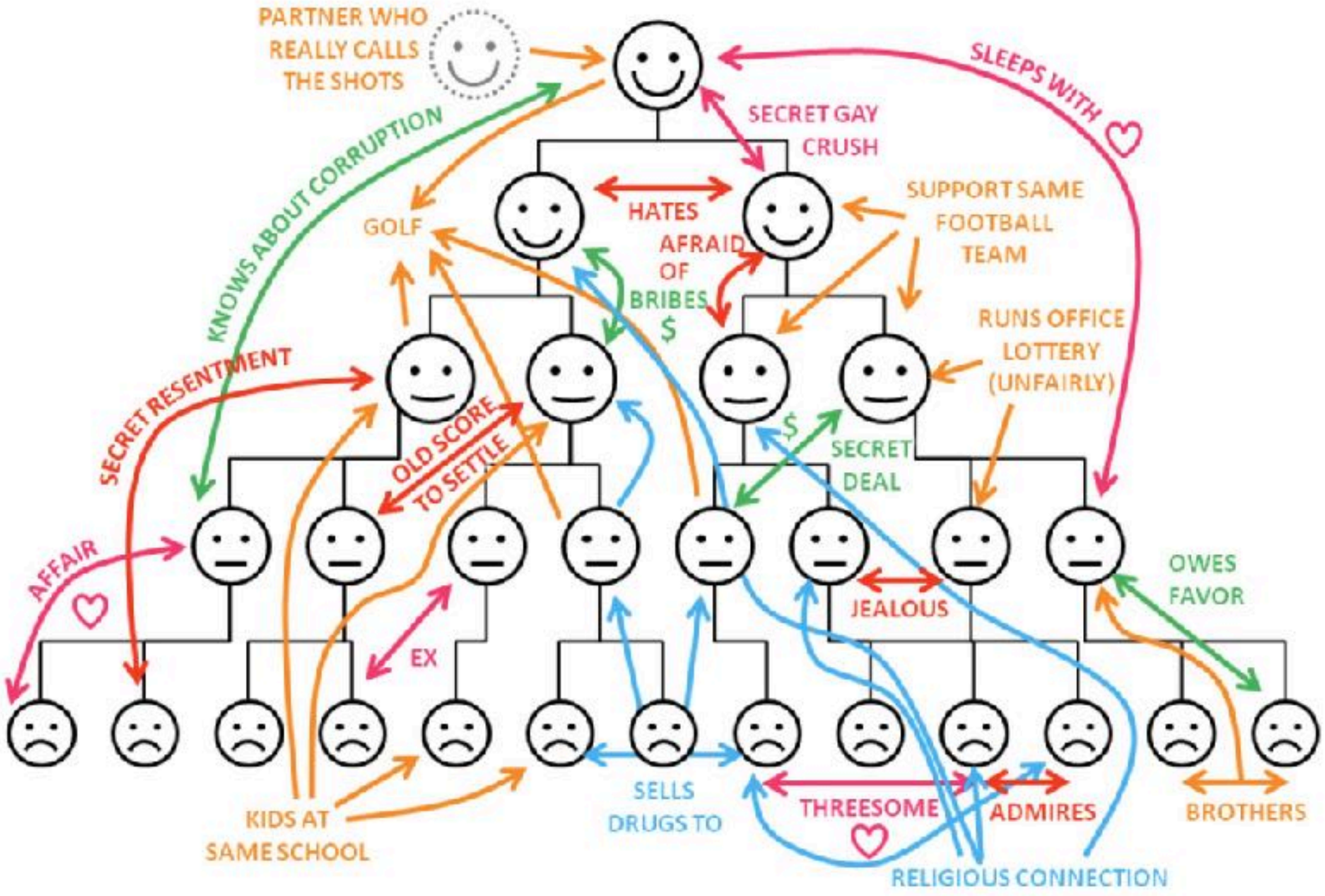
Any organization that designs a system will produce a design whose structure is a copy of the organization's communication structure.

-- Melvin Conway



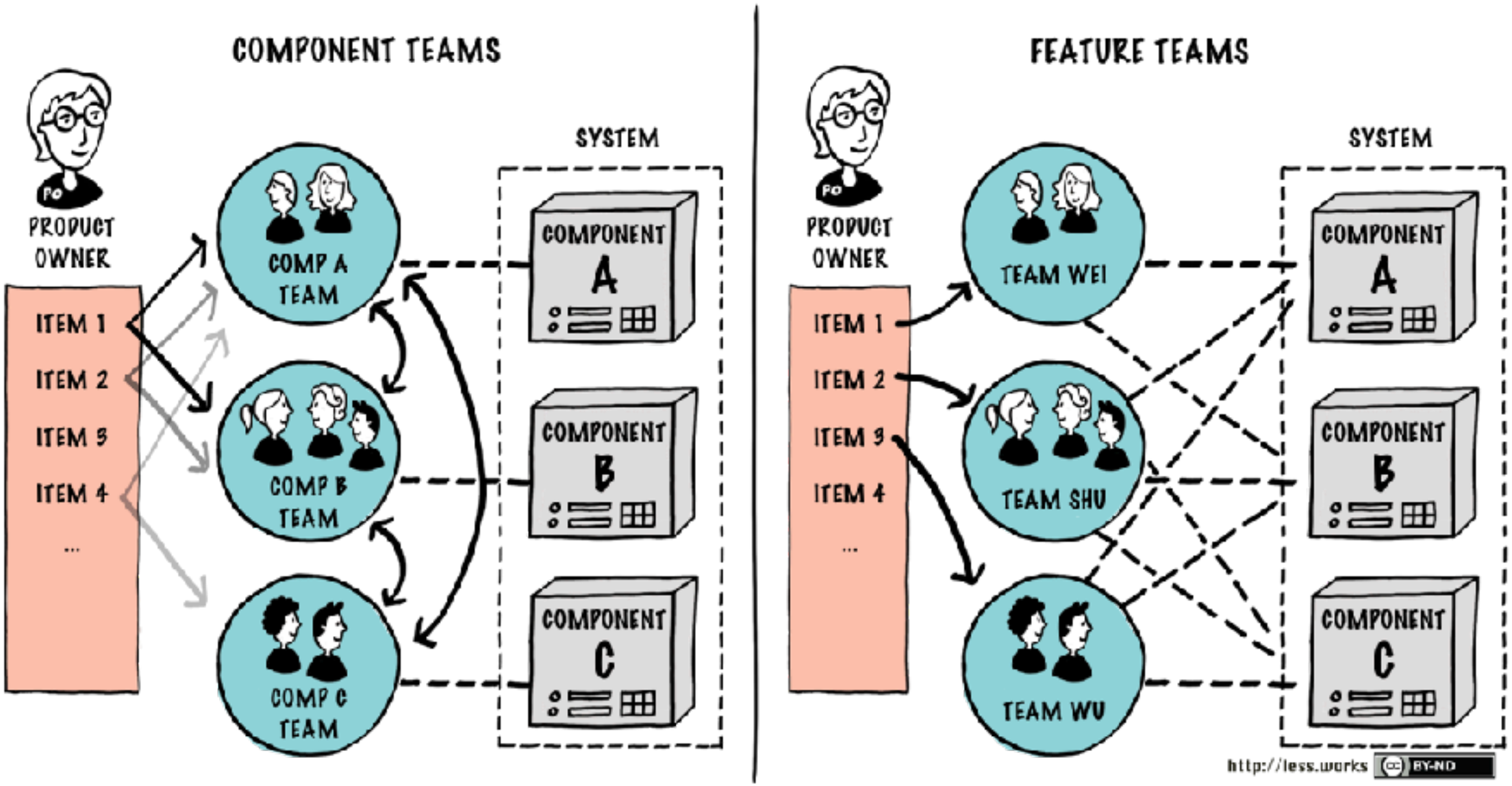
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Organization Structure



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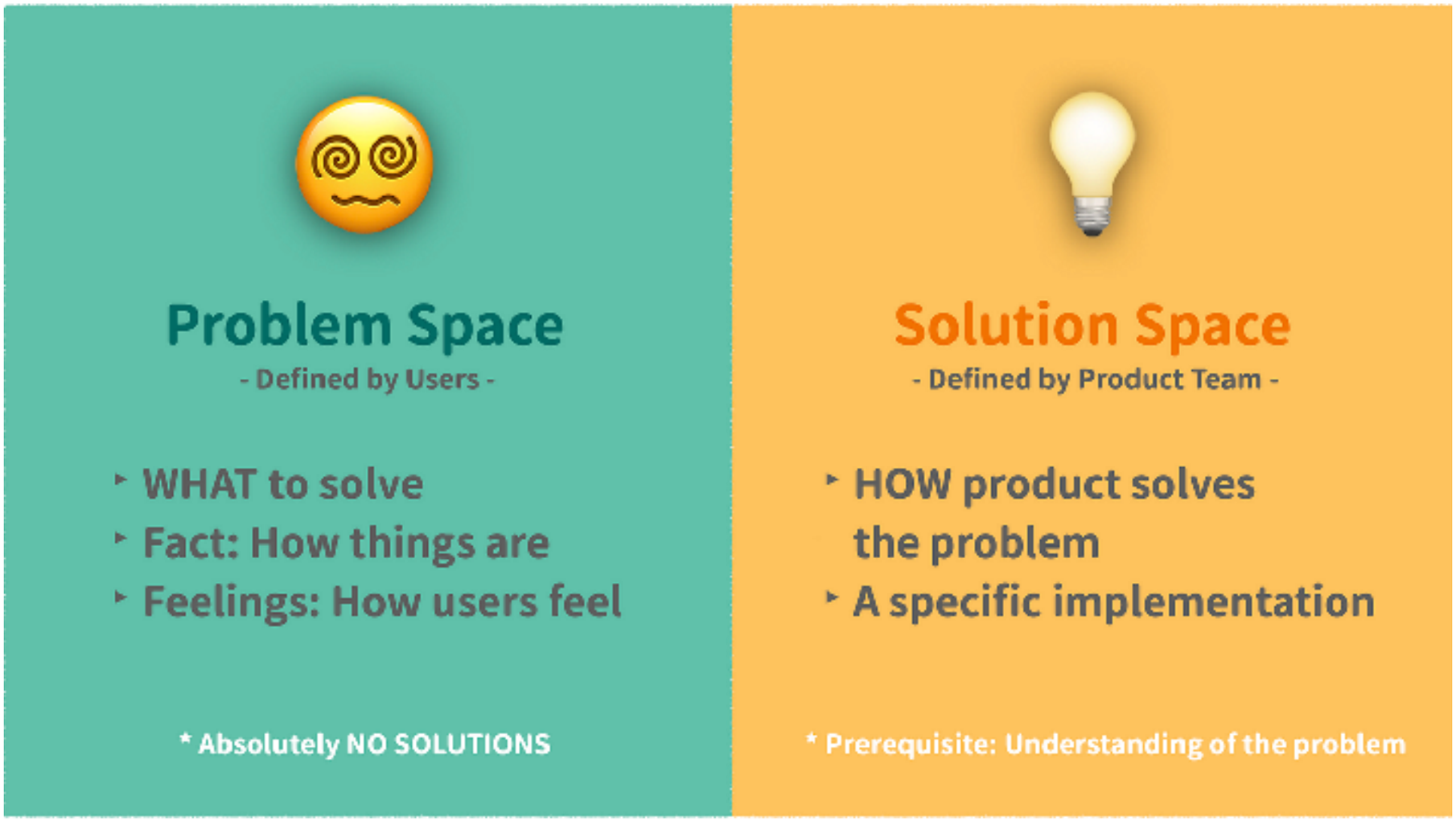
Component vs Feature teams



https://less.works/less/structure/feature-teams

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Problem vs Solution Space

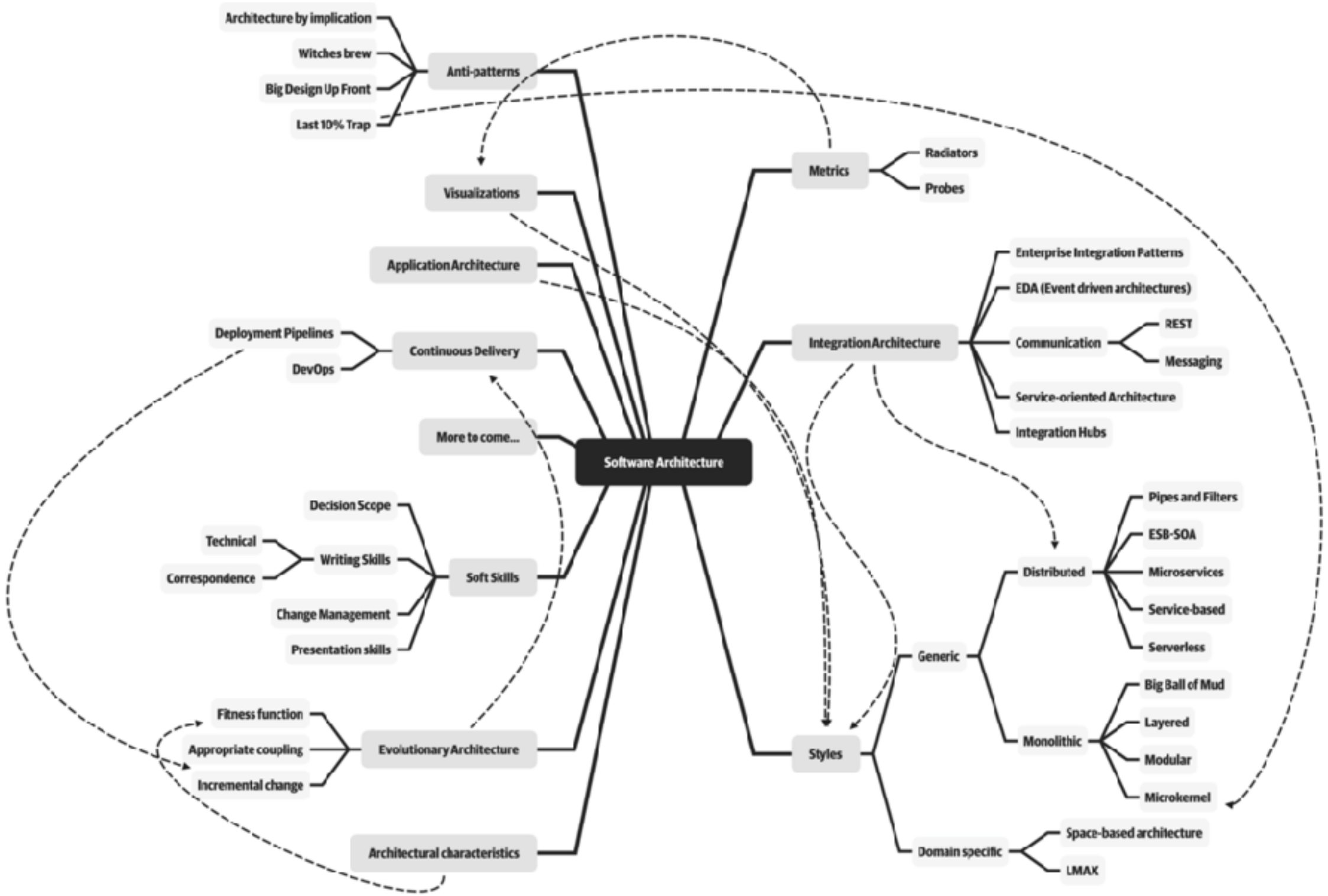


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Let’s Start



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Architecture Styles



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Architecture Styles !!

Layered	Service oriented	Distributed
Domain driven	Event driven	Data centric

more …

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Evolution of Architecture



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WANT TO KNOW THE BEST SOFTWARE ARCHITECTURE?



Pangaea
300M years ago

LOOK AT THE EVOLUTION OF THE EARTH.



nowadays

Daniel Stori {turnoff.us}

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3 Pillars of Evolution Architecture

Solution Structure

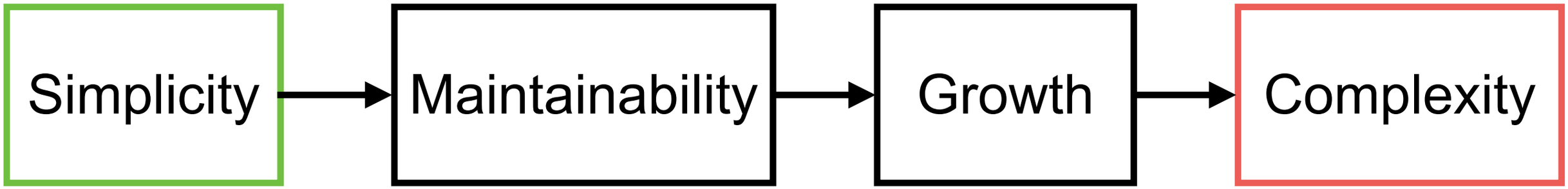
Infrastructure

Deployment strategy



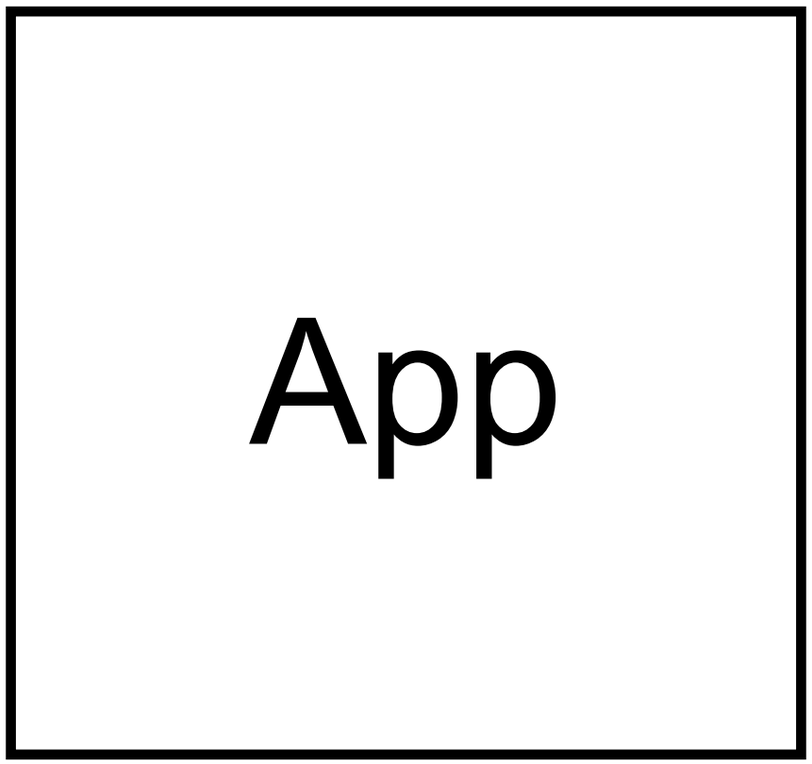
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Steps of Evolutions



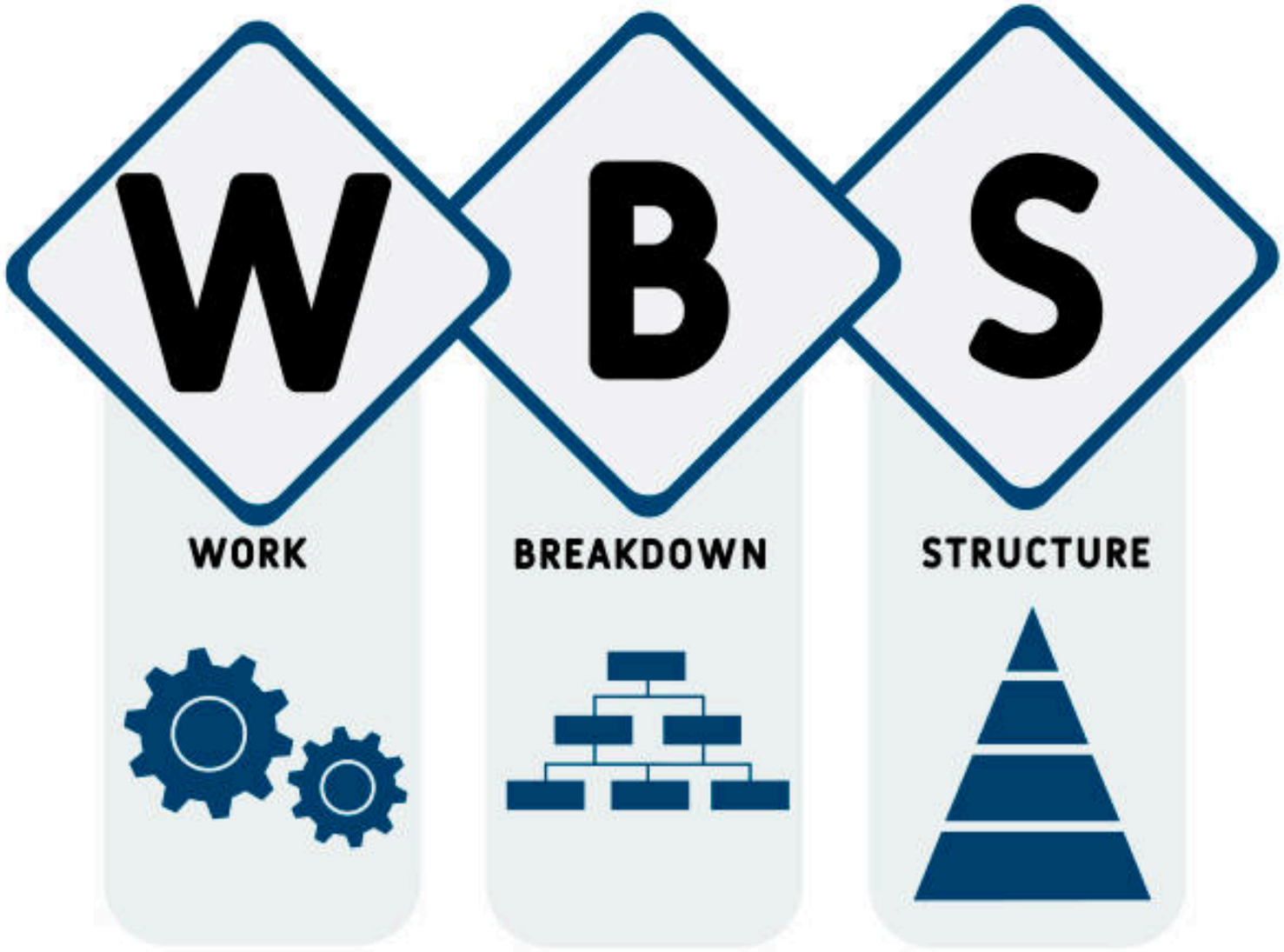
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Monolith



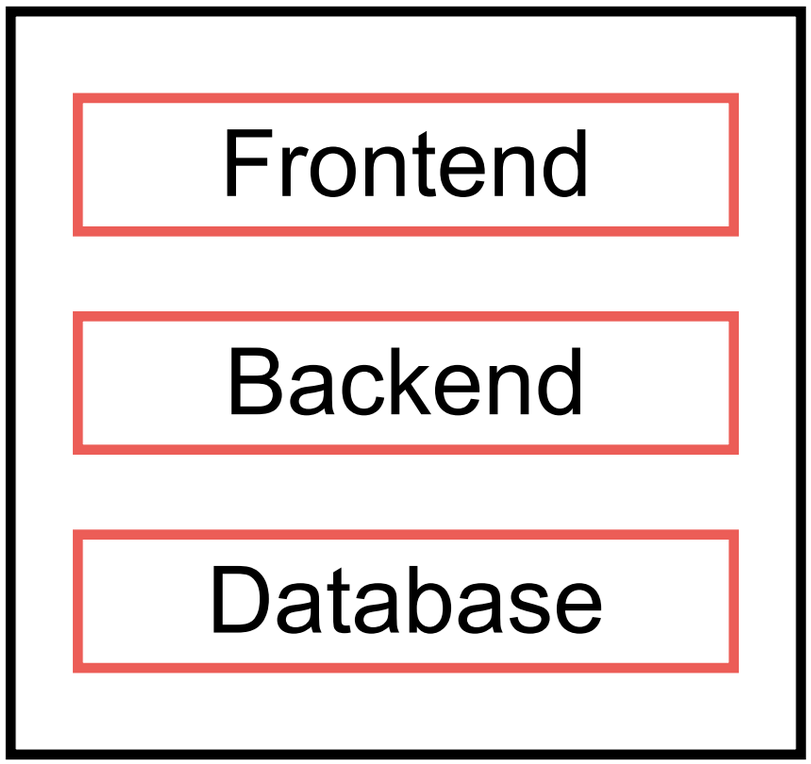
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Work Breakdown Structure



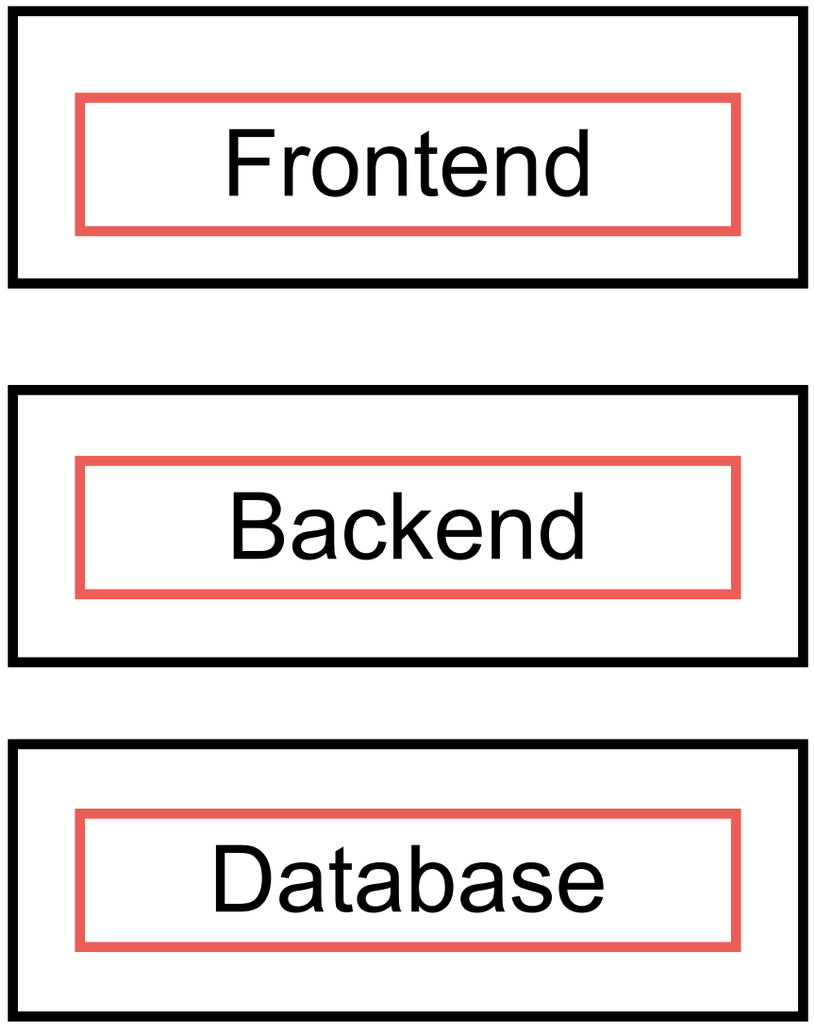
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Layers (inter-process)



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Tiers (out-of-process)



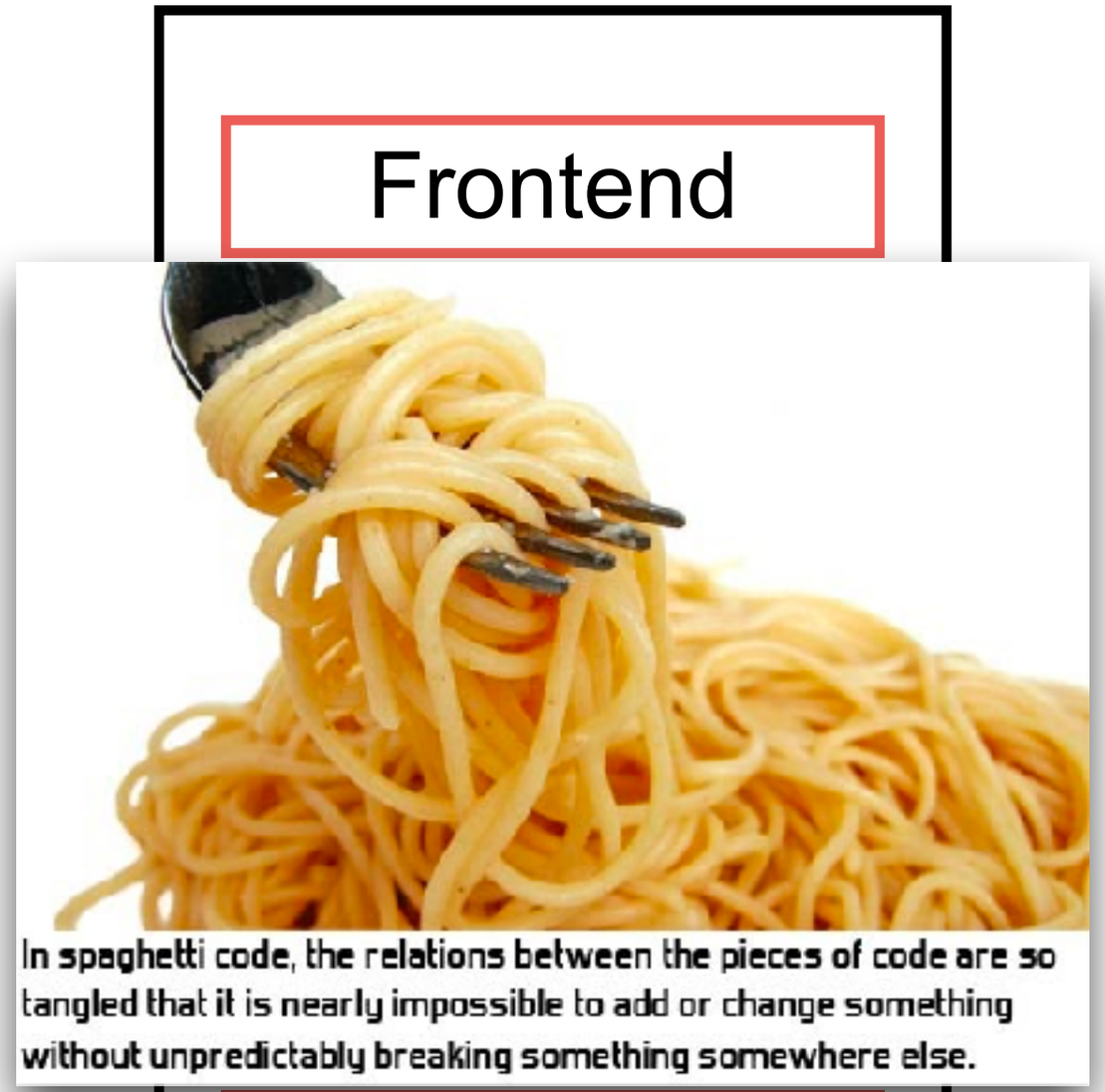
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More features …



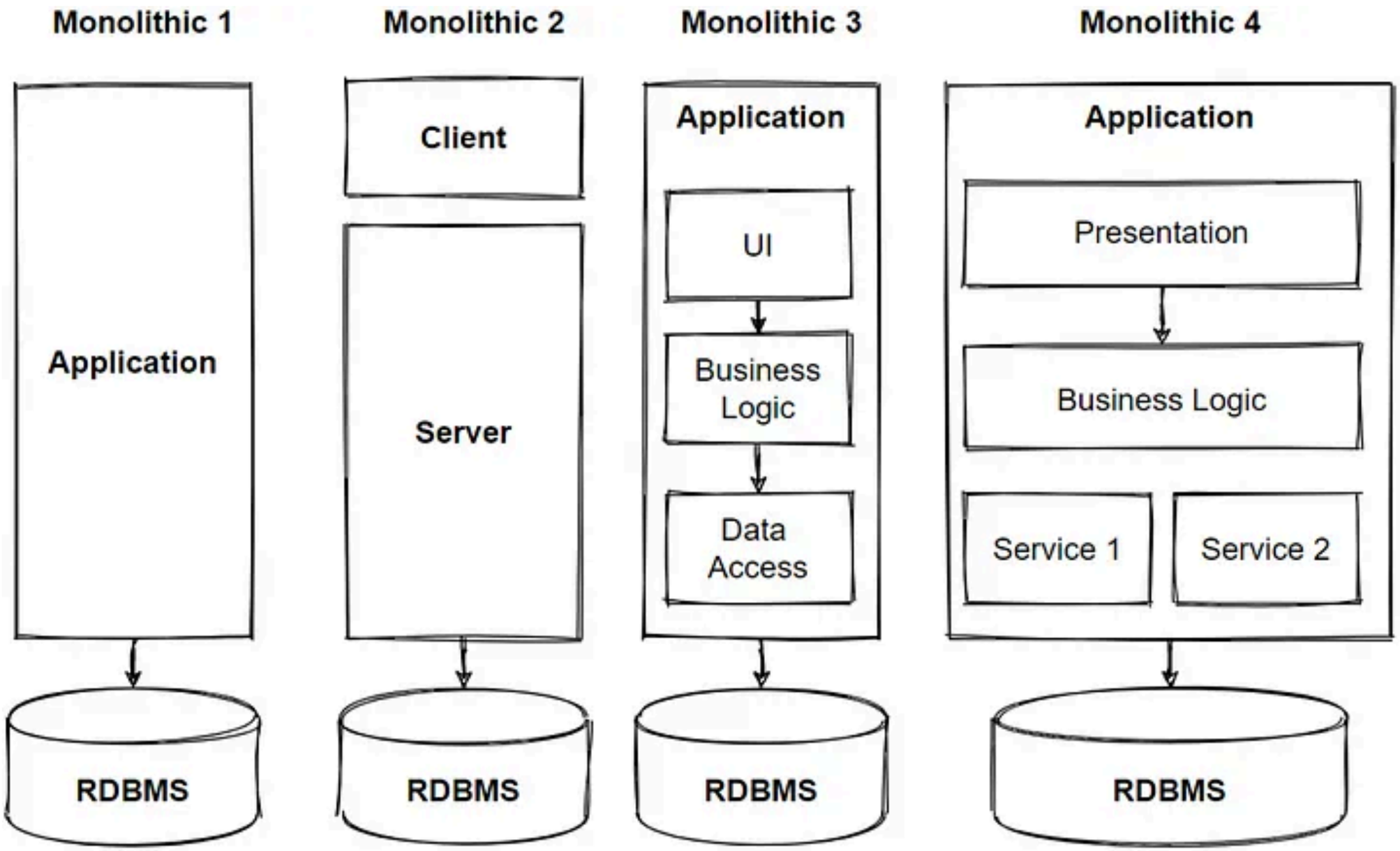
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More features ...



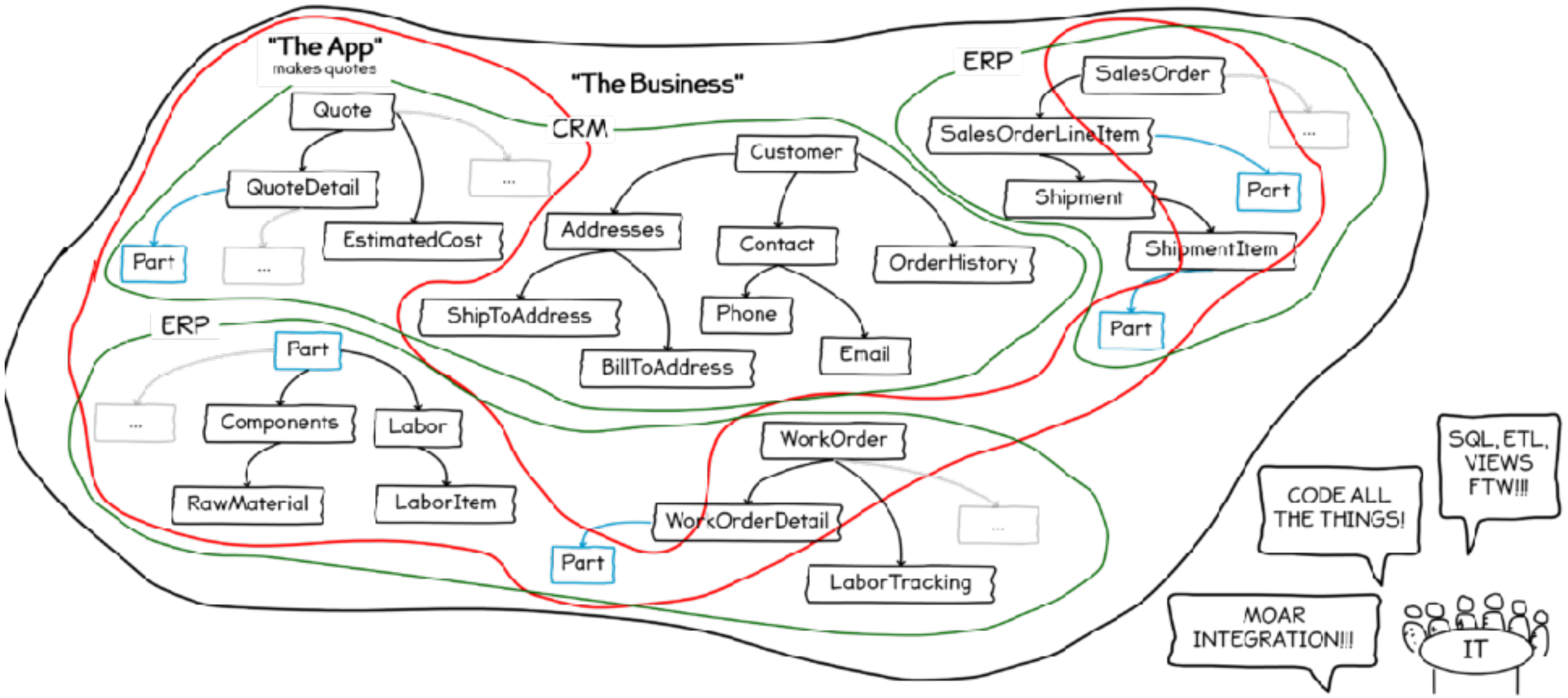
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Evolution of Monolith …

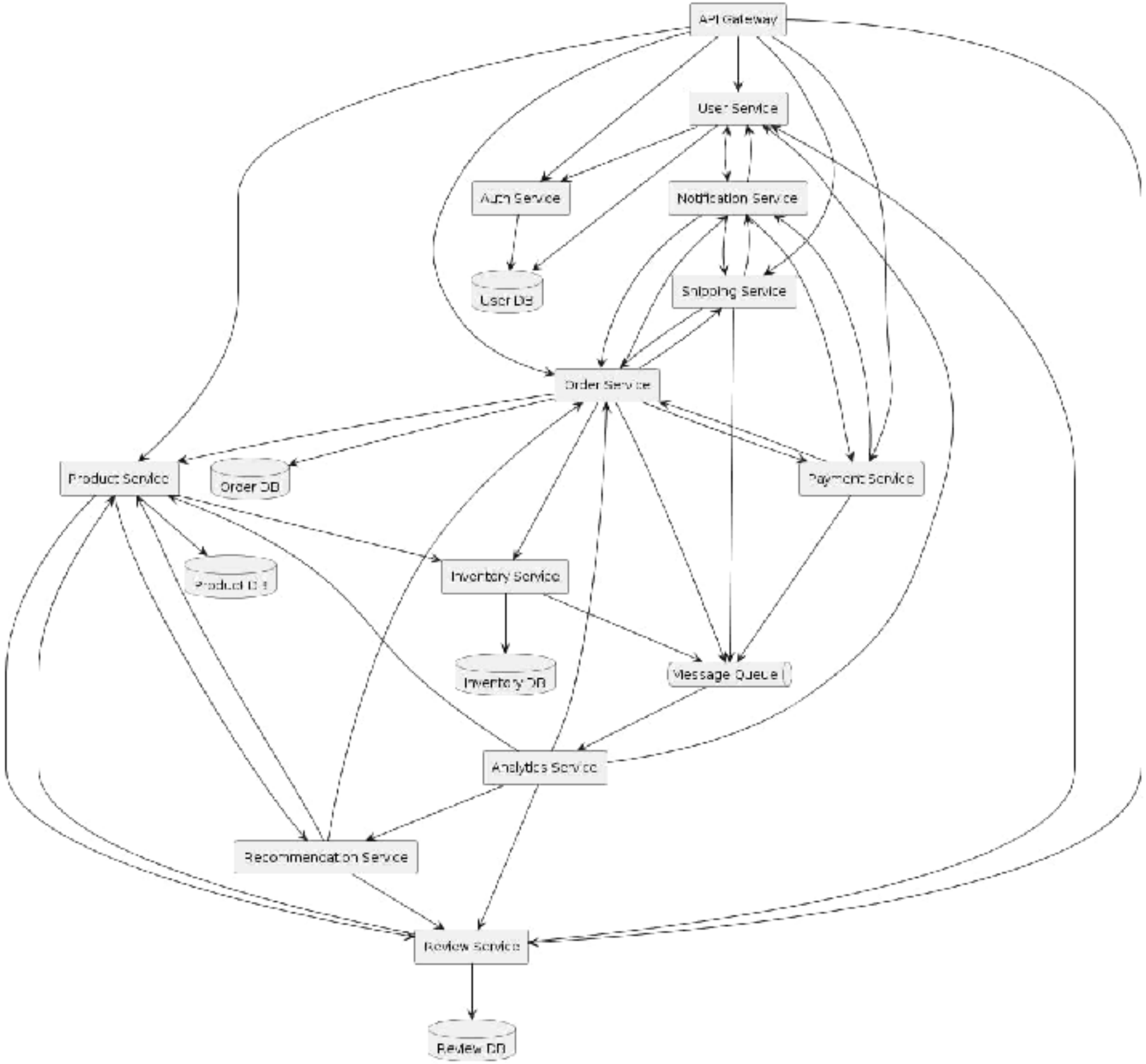


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Monolith Hell

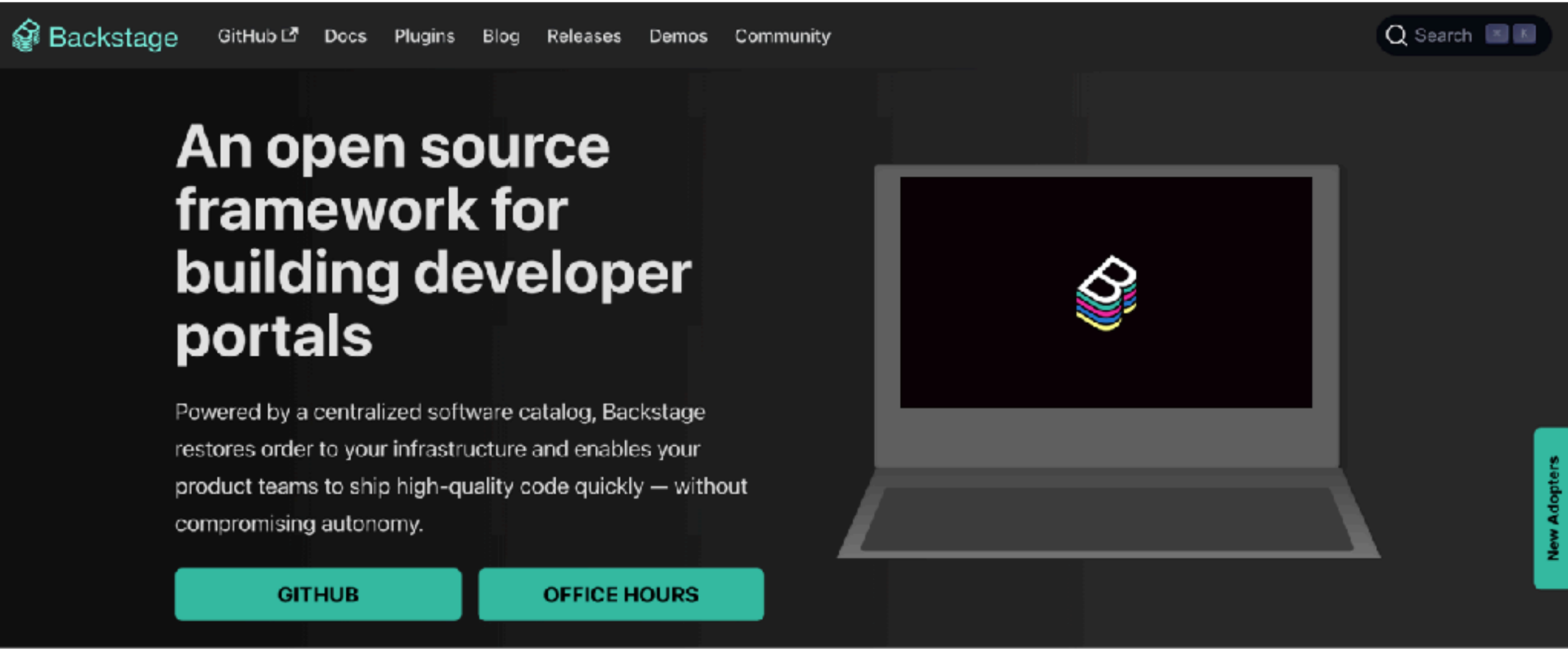


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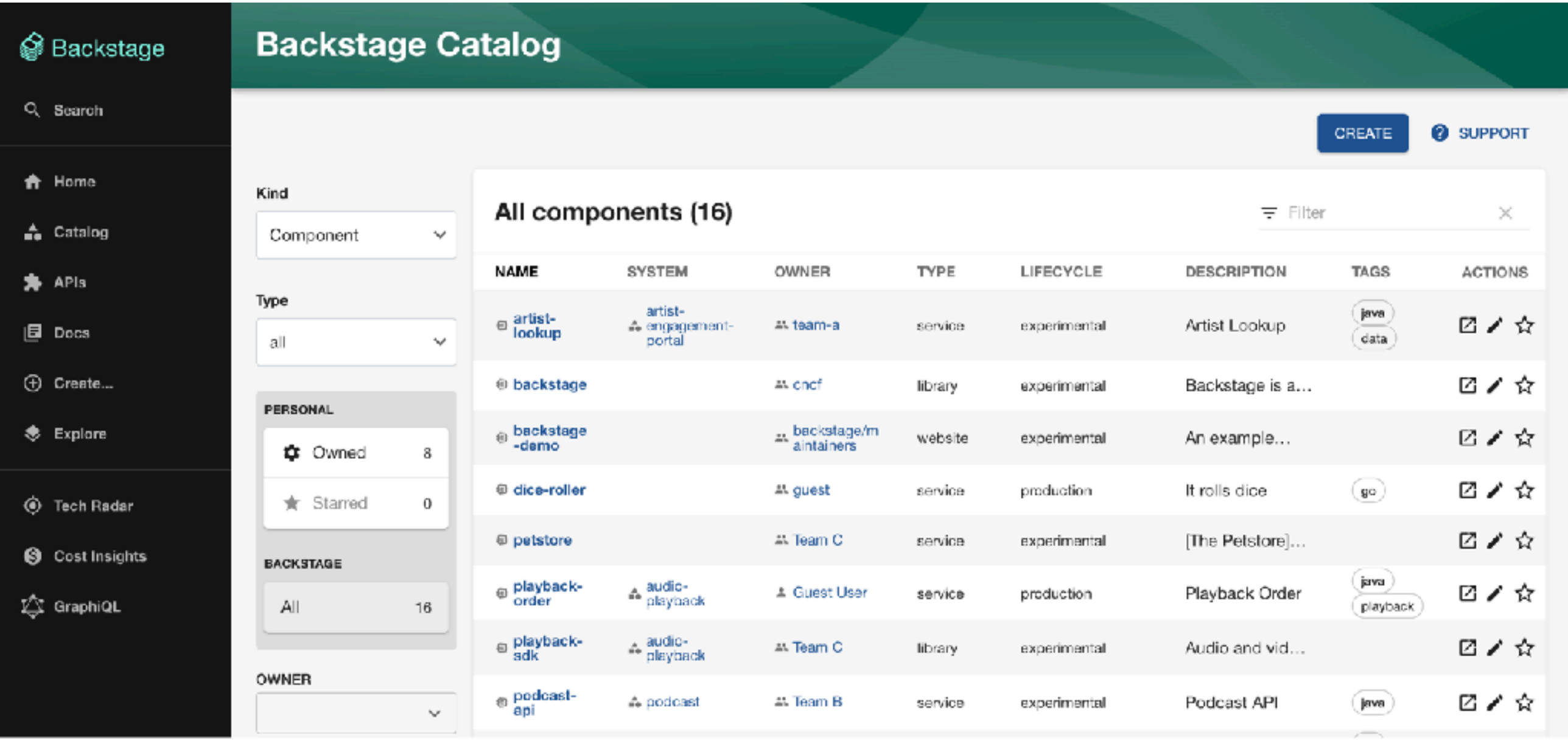
Service Catalog



https://backstage.io/

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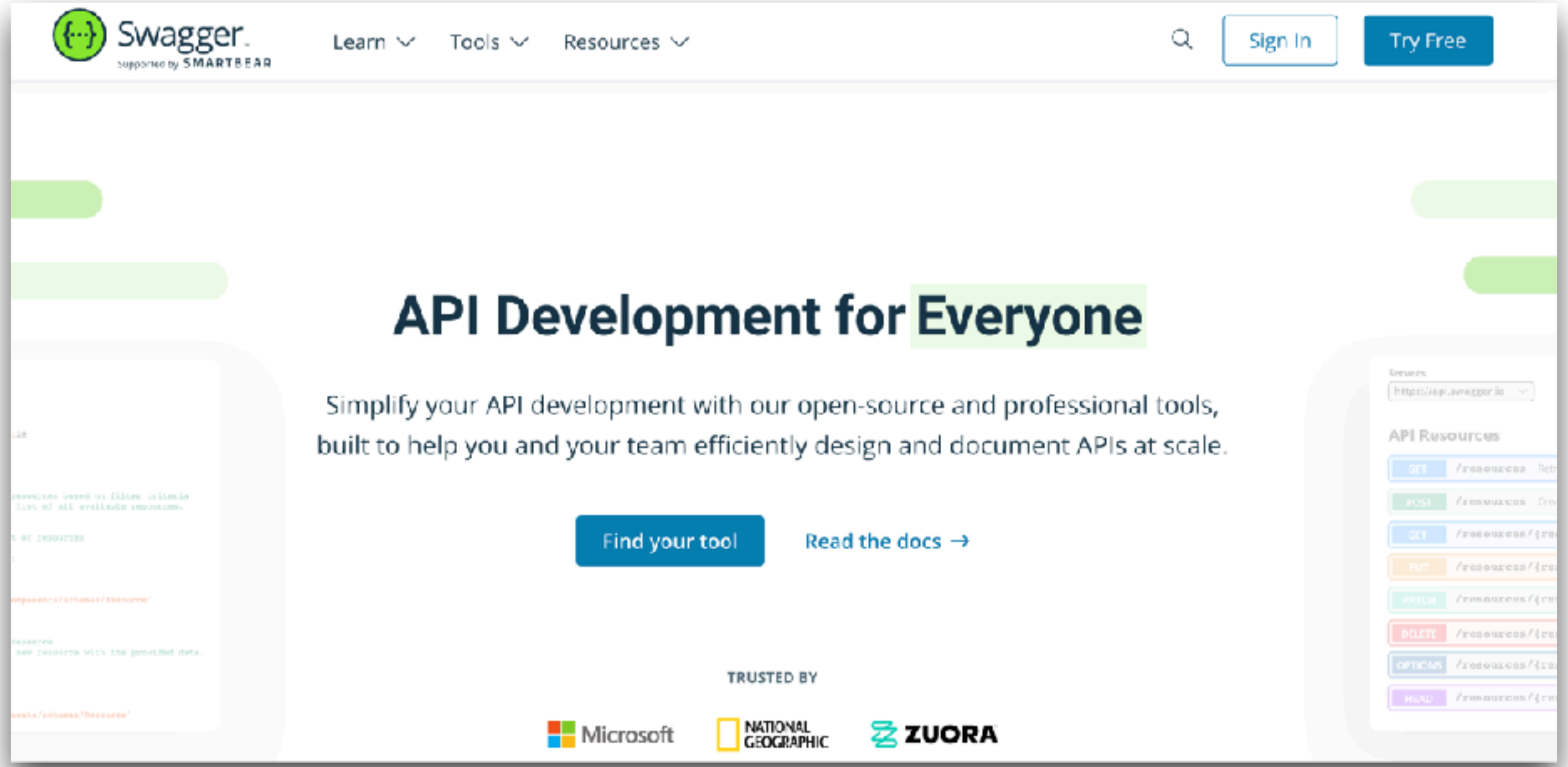
Example of Service Catalog



https://demo.backstage.io/

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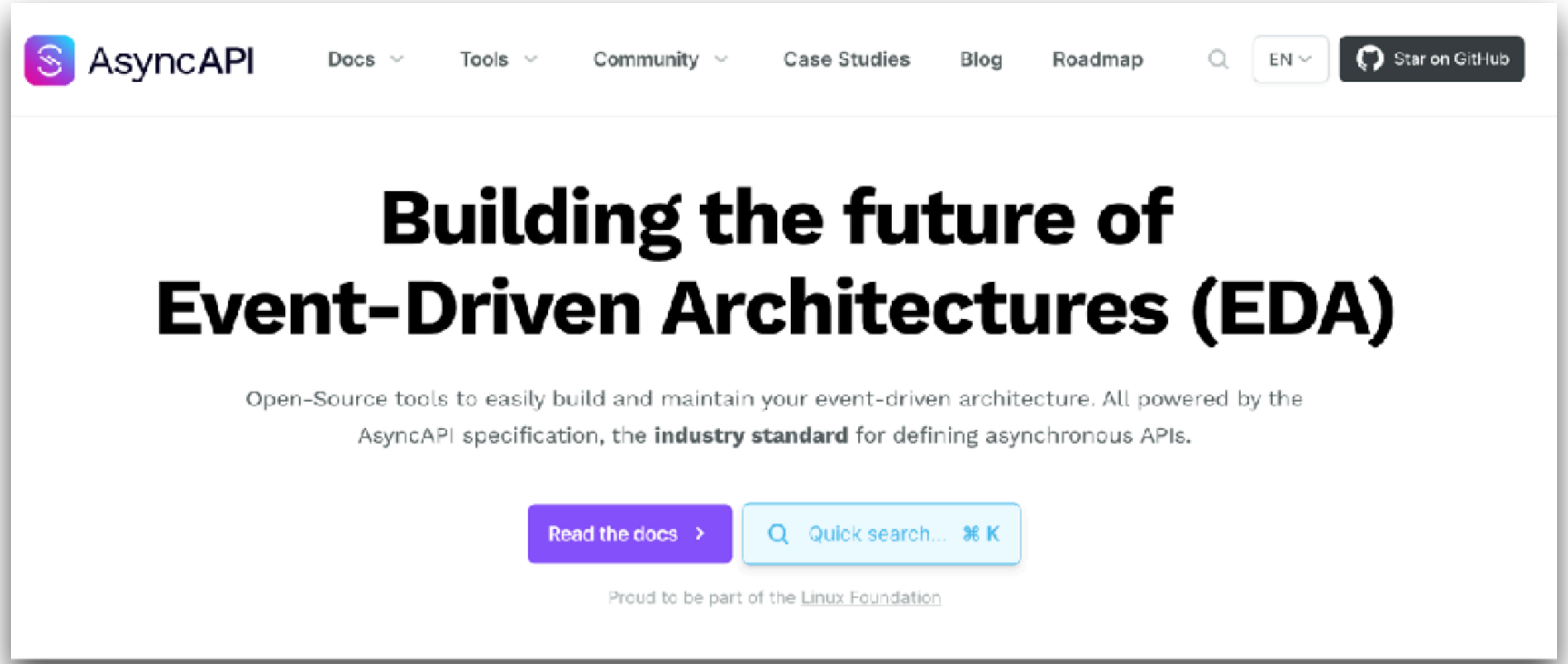
OpenAPI



https://swagger.io/

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AsyncAPI

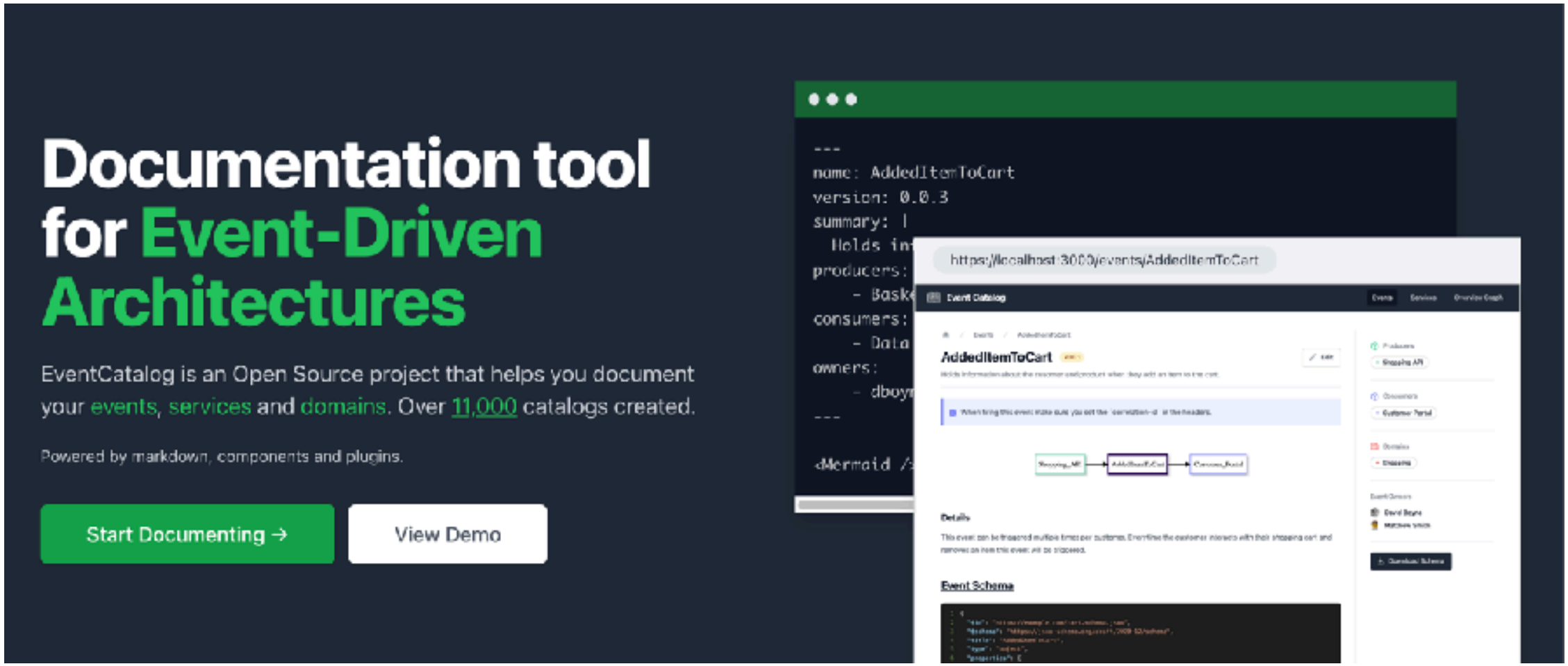


https://www.asyncapi.com/en



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EventCatalog



https://www.eventcatalog.dev/

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Docs as Code

WRITE THE DOCS

Write the Docs is a global community of people who care about documentation. We have a Slack community, conferences on 3 continents, and local meetups!

Home » Software documentation guide »

Docs as Code

author: Eric Holscher & the Write the Docs community

Documentation as Code (*Docs as Code*) refers to a philosophy that you should be writing documentation with the same tools as code:

- Issue Trackers
- Version Control (Git)
- Plain Text Markup (Markdown, reStructuredText, Asciidoc)
- Code Reviews
- Automated Tests

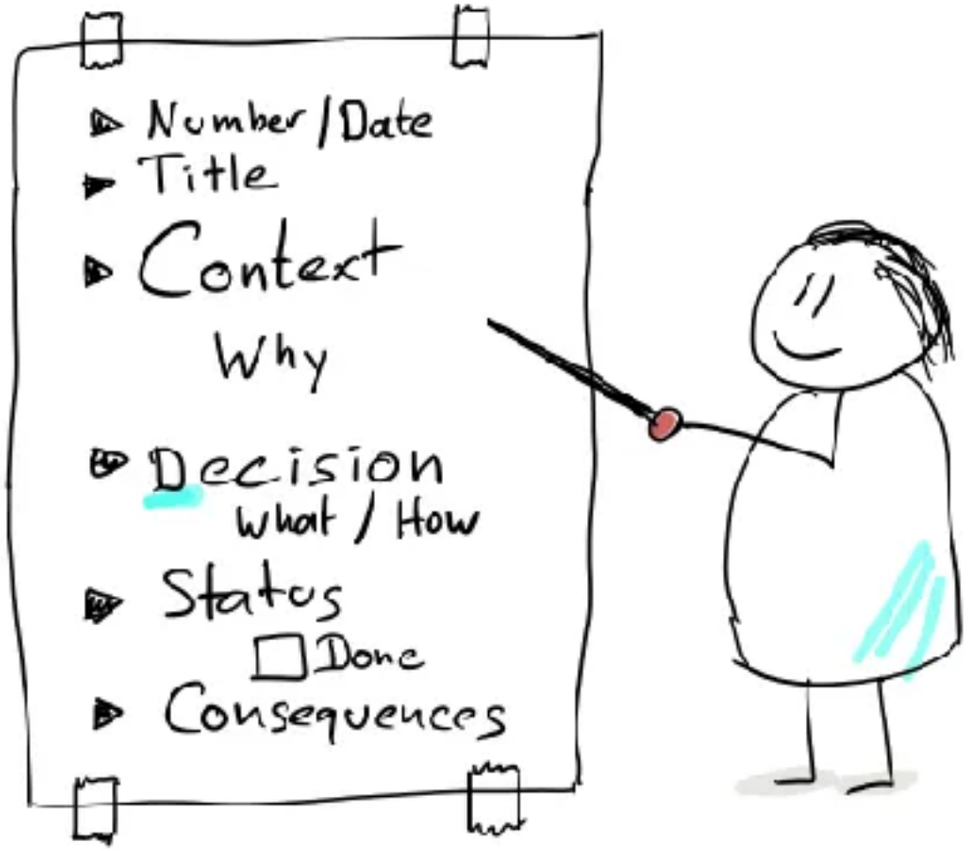
This means following the same workflows as development teams, and being integrated in the product team. It enables a culture where writers and developers both feel ownership of documentation, and work together to make it as good as possible.

https://www.writethedocs.org/guide/docs-as-code/



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Architectural Decision Records (ADR)



https://adr.github.io/

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Overall architecture of system ?

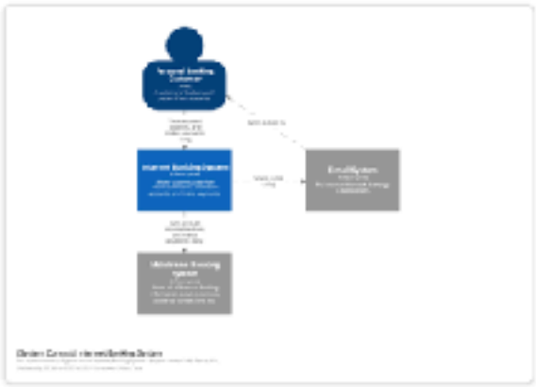


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C4 Model

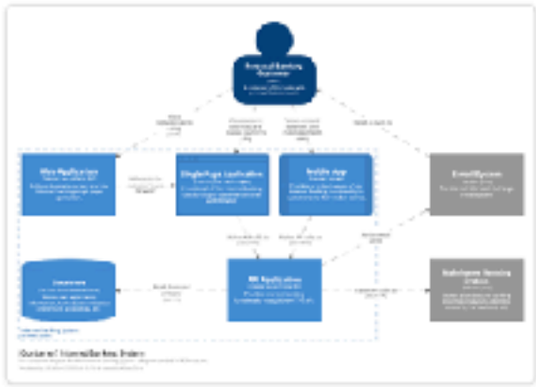
Visualizing software architecture





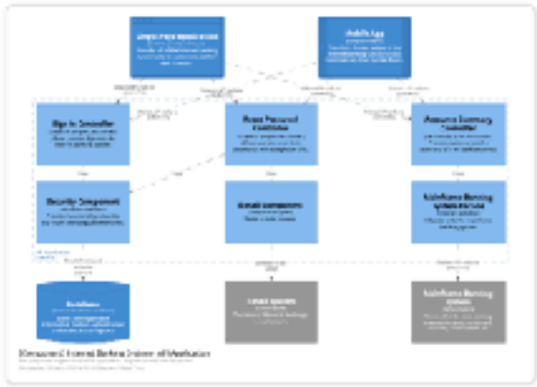
Level 1: A **System Context** diagram provides a starting point, showing how the software system in scope fits into the world around it.





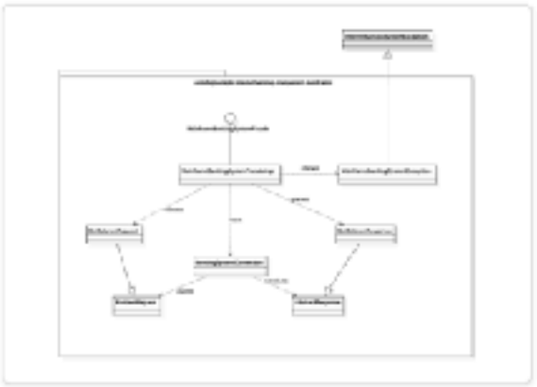
Level 2: A **Container** diagram zooms into the software system in scope, showing the high-level technical building blocks.





Level 3: A **Component** diagram zooms into an individual container, showing the components inside it.





Level 4: A **code** (e.g. UML class) diagram can be used to zoom into an individual component, showing how that component is implemented.

https://c4model.com/

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Monolith problems ?

Build

Test

Deploy

Troubleshooting

Scale out (horizontal)



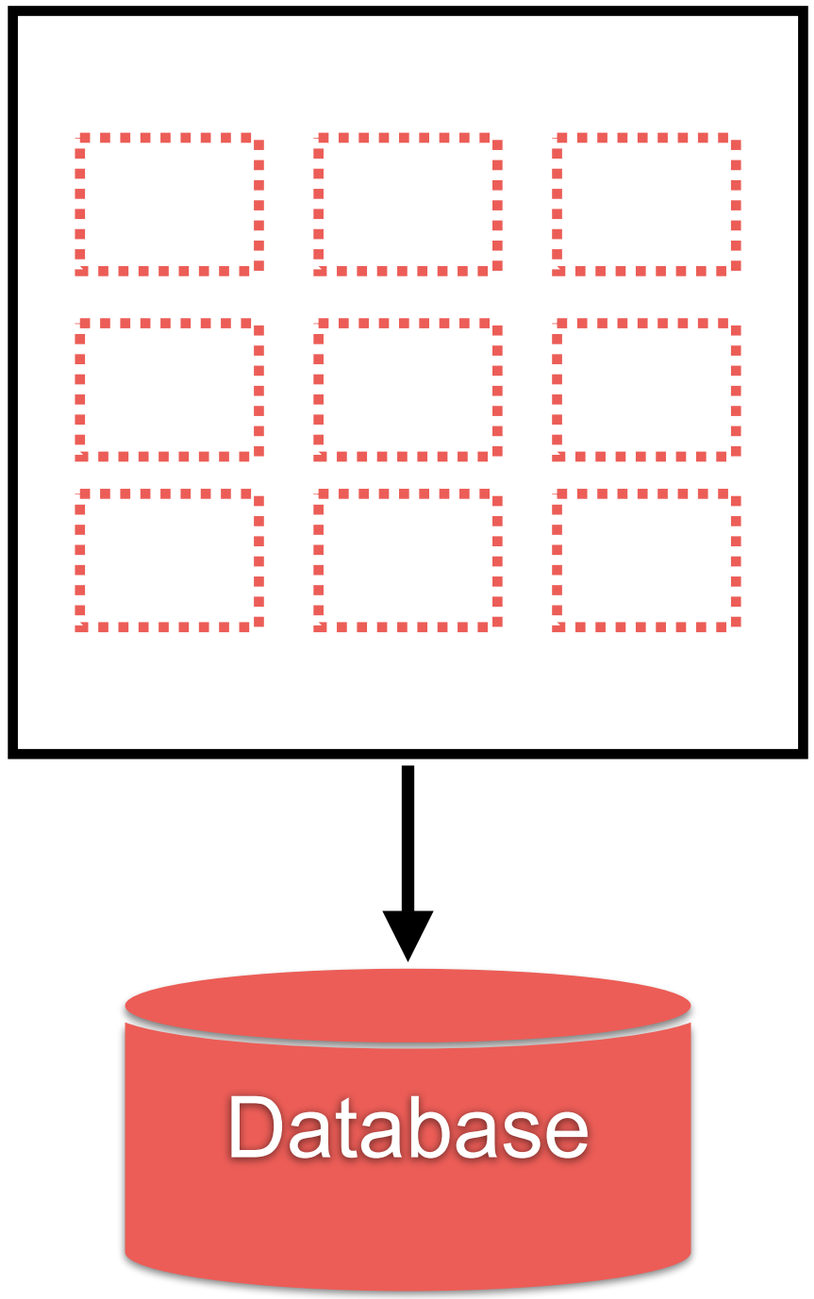
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How to scale system ?

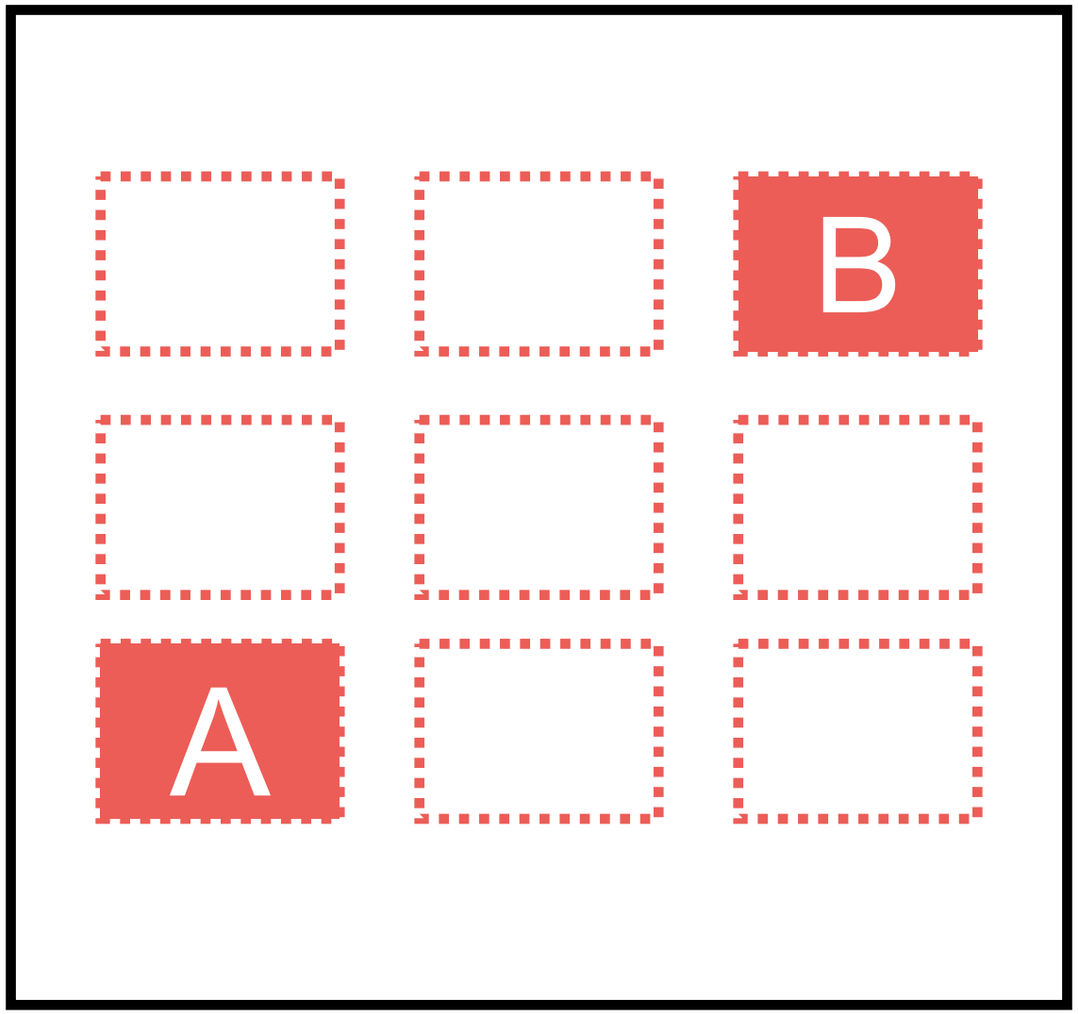


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Service with Modular



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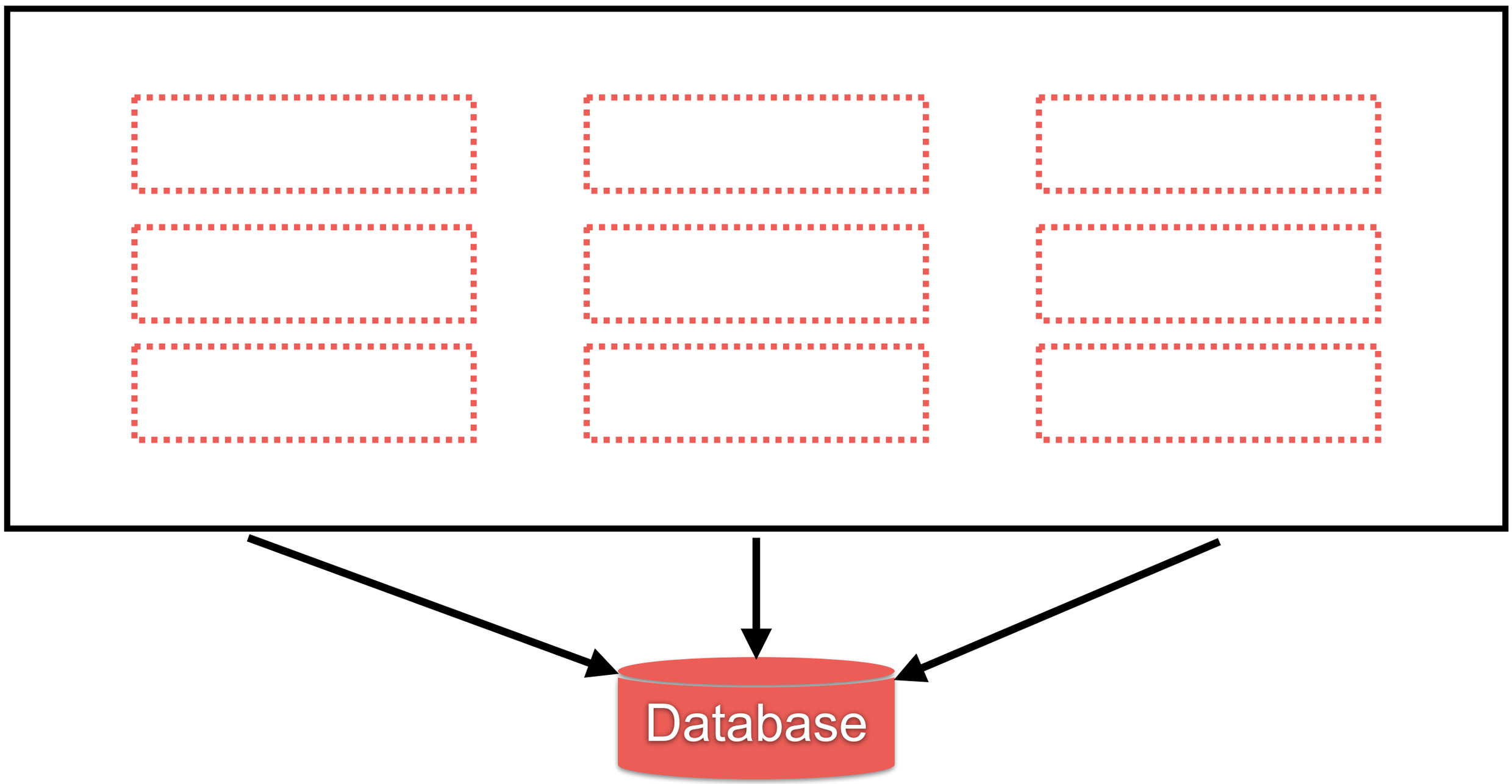
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How to scale ?



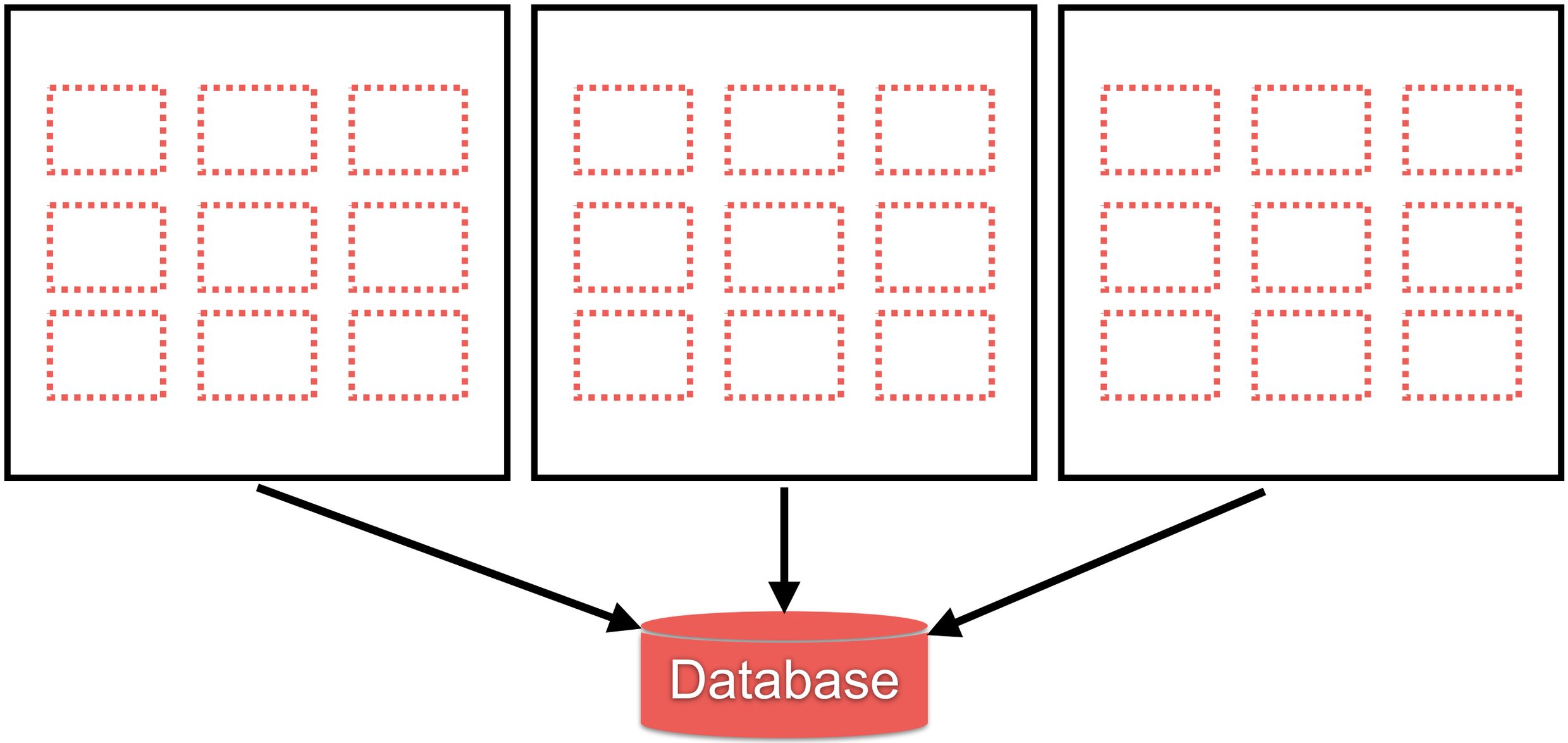
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Scale Up (Vertical)



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Scale Out (Horizontal)



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What happens when we need more servers ?



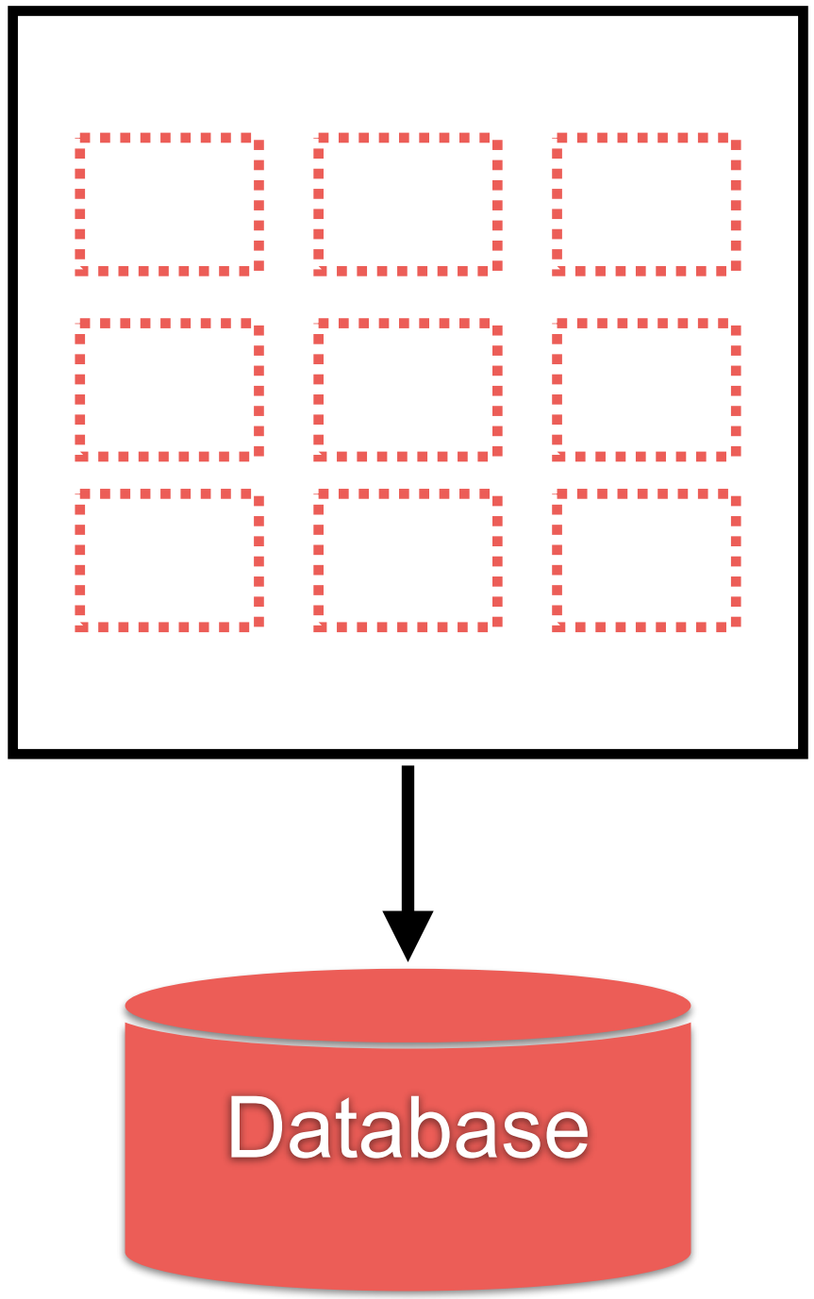
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What if we don’t use all modules equally ?



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Database ?



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How can we update individual models/database ?



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Scale up or Scale out ?



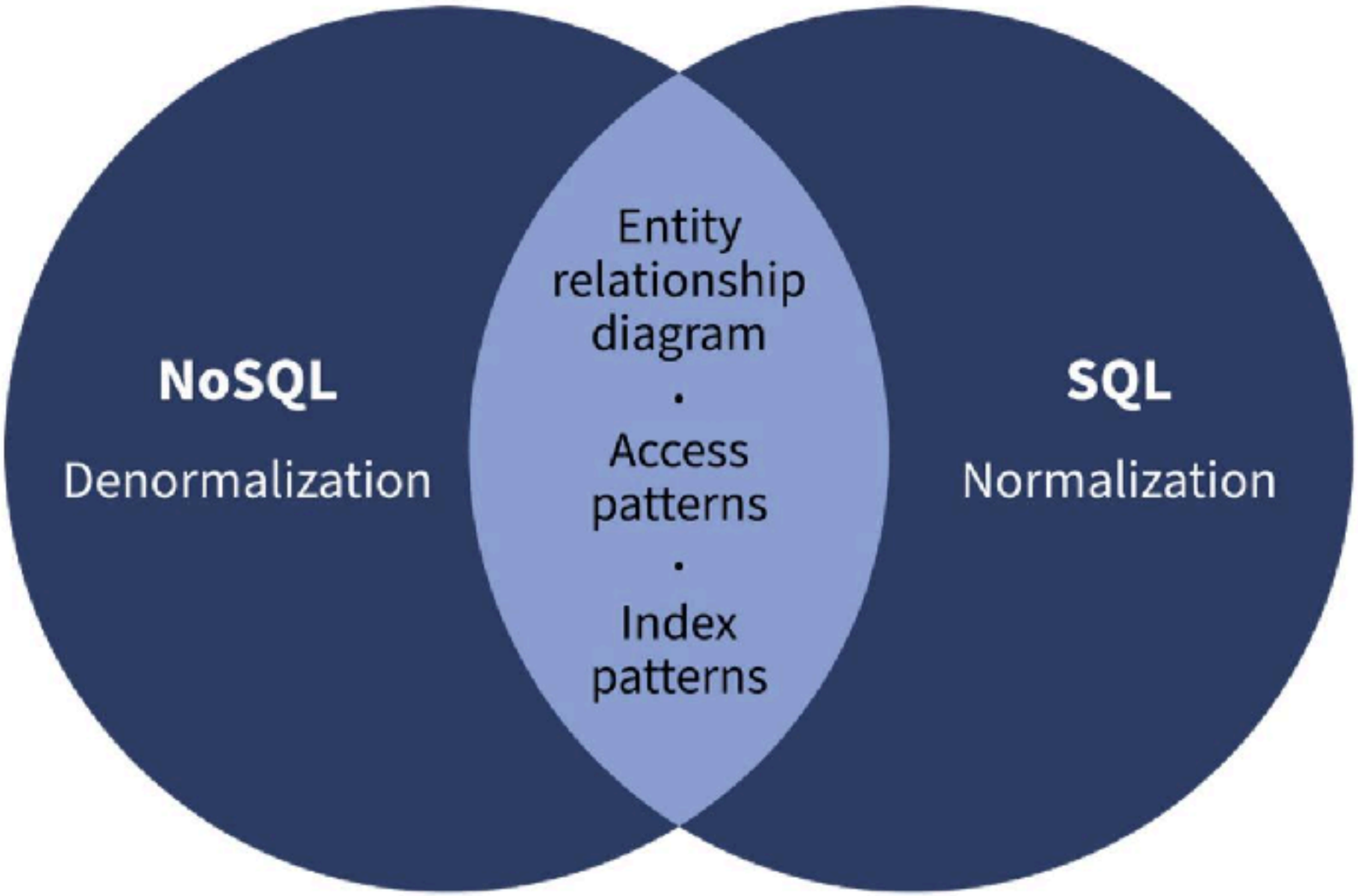
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Do all modules need to use the same Database, Language and Runtime … ?



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Database ?



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Database Models Ranking

424 systems in ranking, June 2025

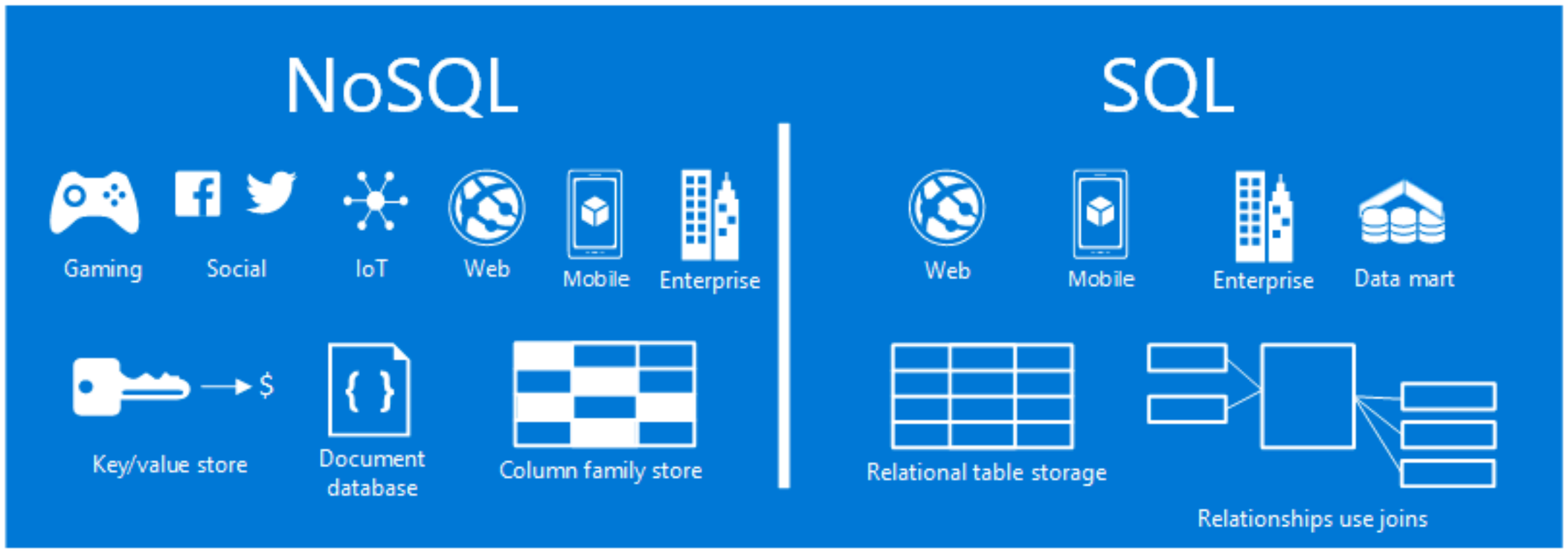
Rank Jun 2025	Rank May 2025	Rank Jun 2024	DBMS	Database Model	Score Jun 2025	Score May 2025	Score Jun 2024
1.	1.	1.	Oracle	Relational, Multi-model	1230.38	+3.82	-13.70
2.	2.	2.	MySQL	Relational, Multi-model	953.57	-11.41	-107.77
3.	3.	3.	Microsoft SQL Server	Relational, Multi-model	776.75	+1.86	-44.81
4.	4.	4.	PostgreSQL	Relational, Multi-model	680.65	+6.34	+44.41
5.	5.	5.	MongoDB	Document, Multi-model	402.85	+0.33	-18.23
6.	6.	↑ 8.	Snowflake	Relational	174.49	+2.48	+44.13
7.	7.	↓ 6.	Redis	Key-value, Multi-model	151.72	-0.47	-4.22
8.	8.	↑ 9.	IBM Db2	Relational, Multi-model	125.13	-1.27	-0.77
9.	9.	↓ 7.	Elasticsearch	Multi-model	121.28	-2.53	-11.55
10.	10.	10.	SQLite	Relational	117.03	-0.74	+5.63
11.	11.	↑ 12.	Apache Cassandra	Wide column, Multi-model	108.27	+0.22	+9.44
12.	12.	↑ 15.	Databricks	Multi-model	104.67	+2.02	+23.59
13.	↑ 14.	13.	MariaDB	Relational, Multi-model	94.54	+0.93	+3.50
14.	↓ 13.	↓ 11.	Microsoft Access	Relational	88.28	-7.63	-12.88
15.	15.	↑ 17.	Amazon DynamoDB	Multi-model	83.34	+4.04	+8.90

https://db-engines.com/en/ranking



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Database Models



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Database design

Design for read
Design for write
Pre-join tables or pre-aggregate data
Manage data size
Partitioning and Sharding
Caching data



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We need to improve

We need to get better



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SOA

1990s and earlier

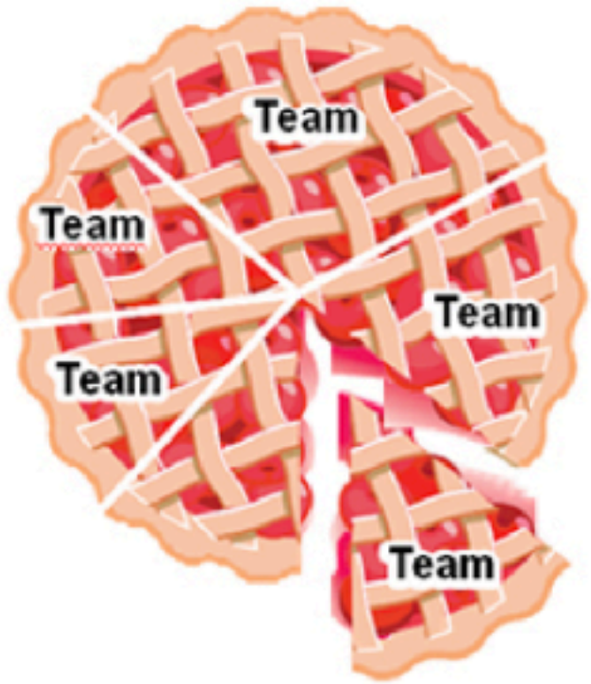
Pre-SOA (monolithic)
Tight coupling



For a monolith to change, all must agree on each change. Each change has unanticipated effects requiring careful testing beforehand.

2000s

Traditional SOA
Looser coupling



Elements in SOA are developed more autonomously but must be coordinated with others to fit into the overall design.

2010s

Microservices
Decoupled



Developers can create and activate new microservices without prior coordination with others. Their adherence to MSA principles makes continuous delivery of new or modified services possible.

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Service-Oriented Architecture



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What is service ?

Standalone and loosely couple
Independently deployable software component
Implement some useful functionality



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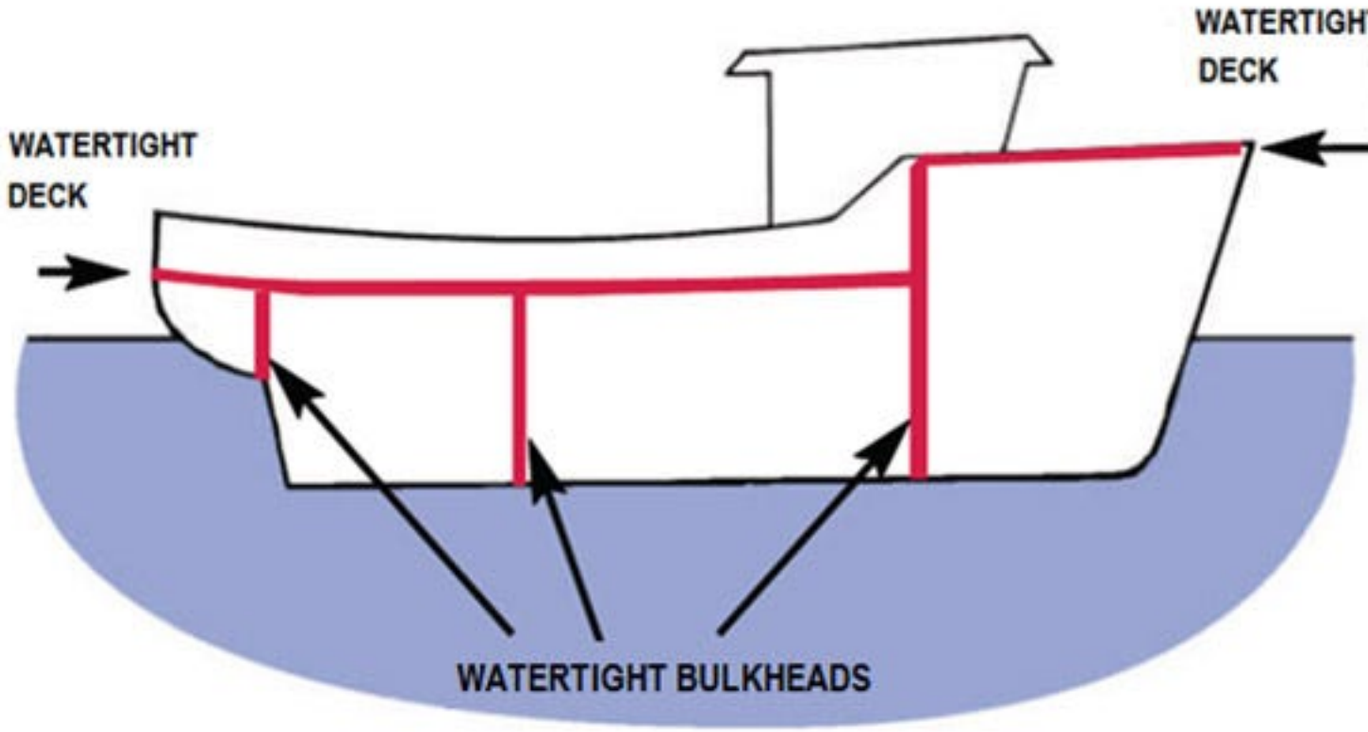
Good service is invisible

Bad service is imposible to ignore



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Bulk Head



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Service-Oriented Architecture

SOA Manifesto

Service orientation is a paradigm that frames what you do.
Service-oriented architecture (SOA) is a type of architecture
that results from applying service orientation.

We have been applying service orientation to help organizations consistently deliver sustainable business value, with increased agility and cost effectiveness, in line with changing business needs.

http://www.soa-manifesto.org/



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Service-Oriented Architecture

Through our work we have come to prioritize:

Business value over technical strategy

Strategic goals over project-specific benefits

Intrinsic interoperability over custom integration

Shared services over specific-purpose implementations

Flexibility over optimization

Evolutionary refinement over pursuit of initial perfection



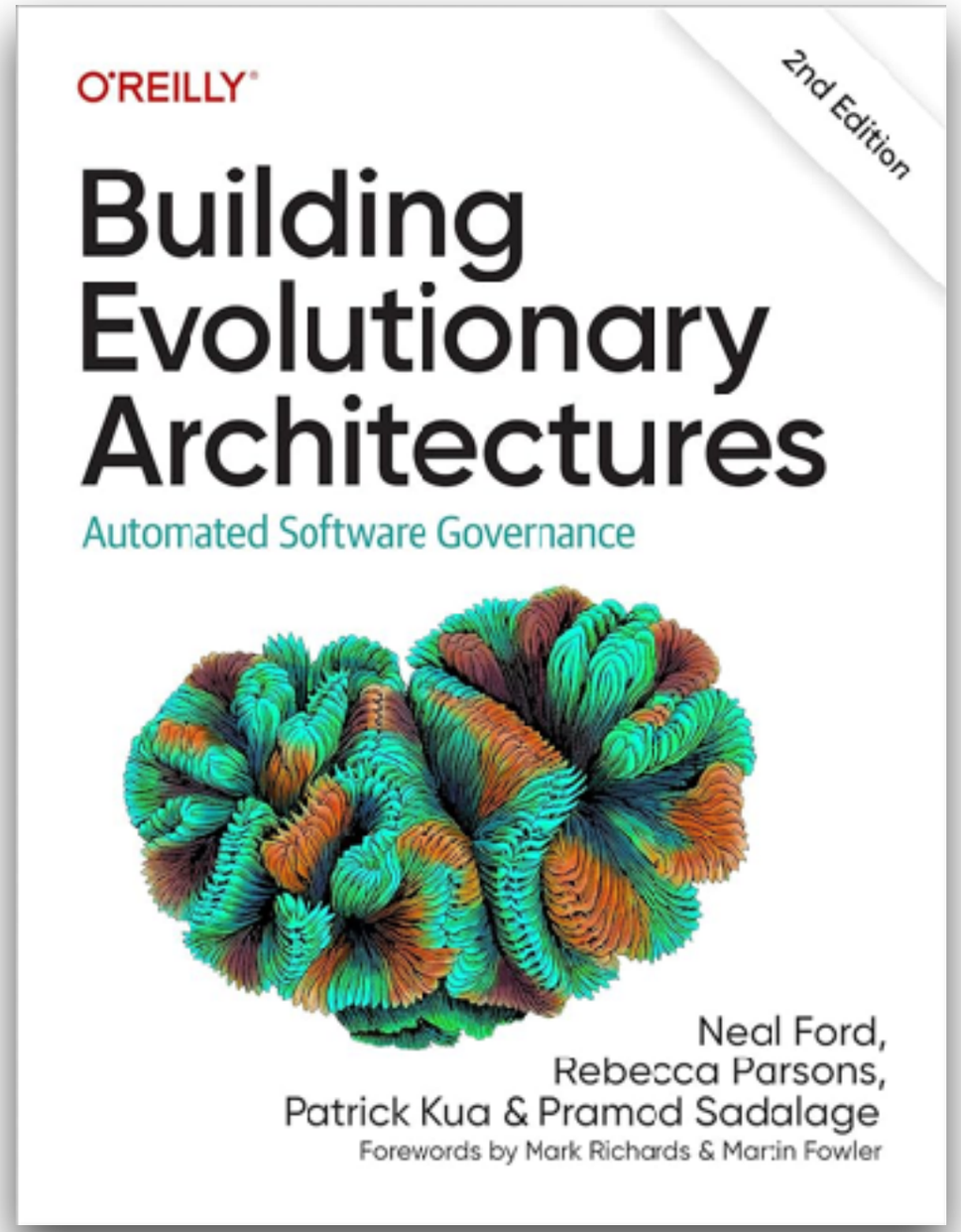
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Evolution design and continuous improvement



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Building Evolutionary Architecture



https://www.amazon.com/Building-Evolutionary-Architectures-Automated-Governance/dp/1492097543

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Service-Oriented Architecture

Through our work we have come to prioritize:

Business value over technical strategy

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Shared services over specific-purpose implementations

Flexibility over optimization

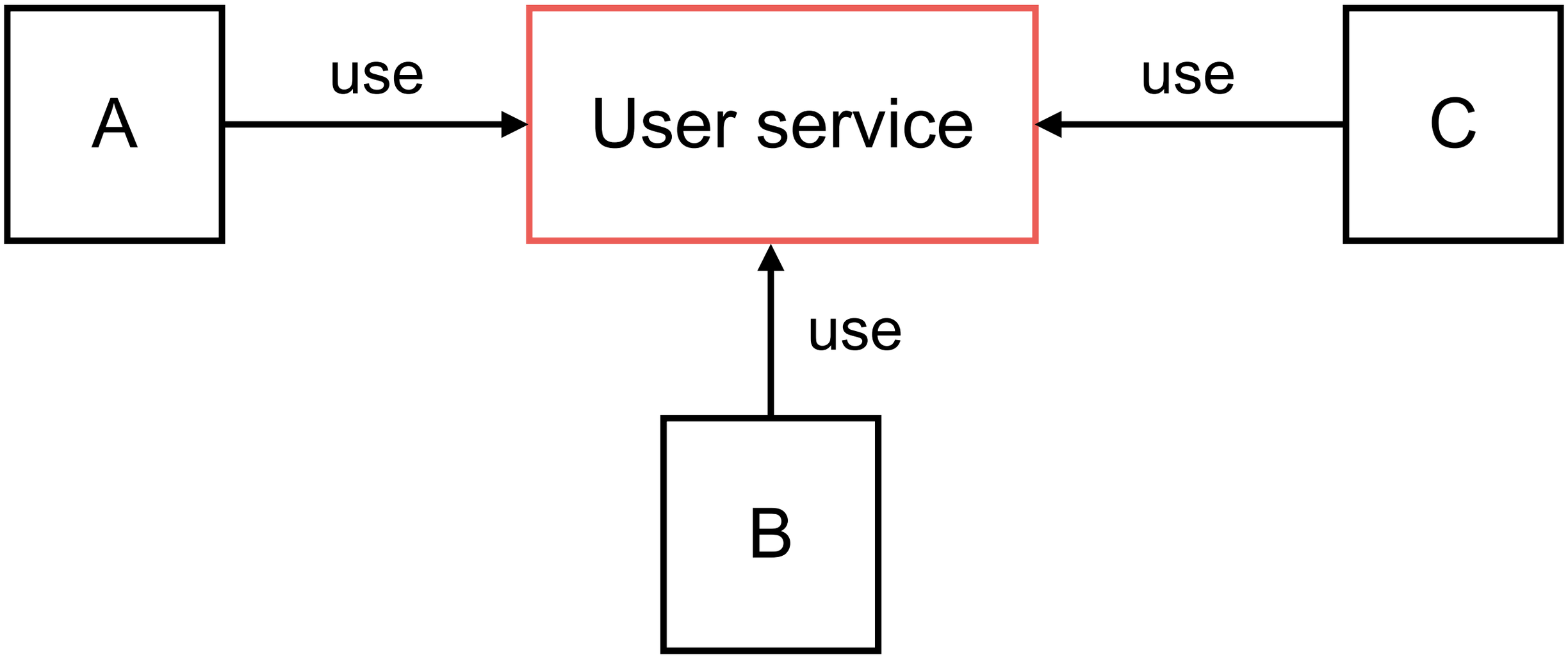
Evolutionary refinement over pursuit of initial perfection

http://www.soa-manifesto.org/



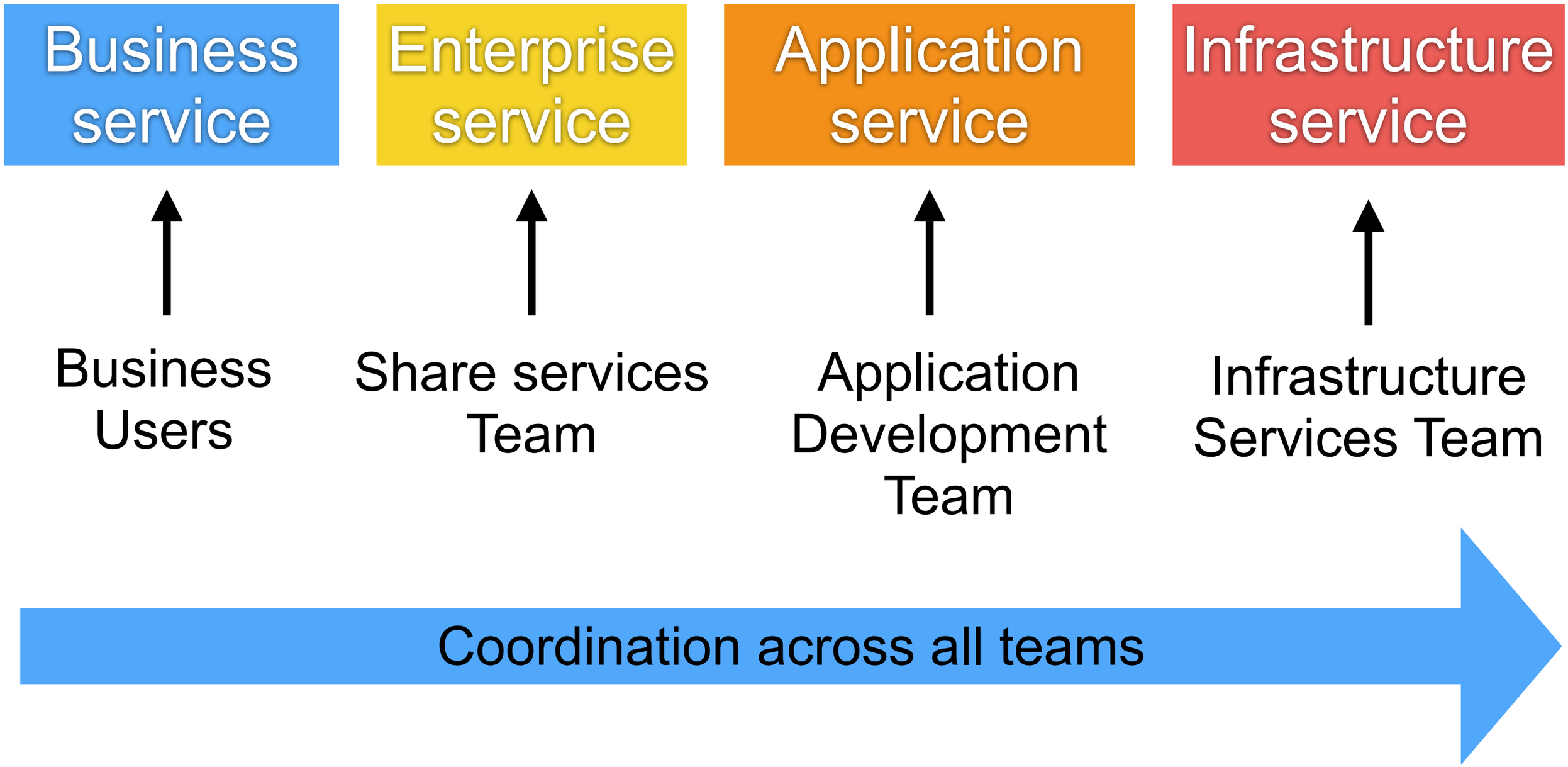
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Shared service ?



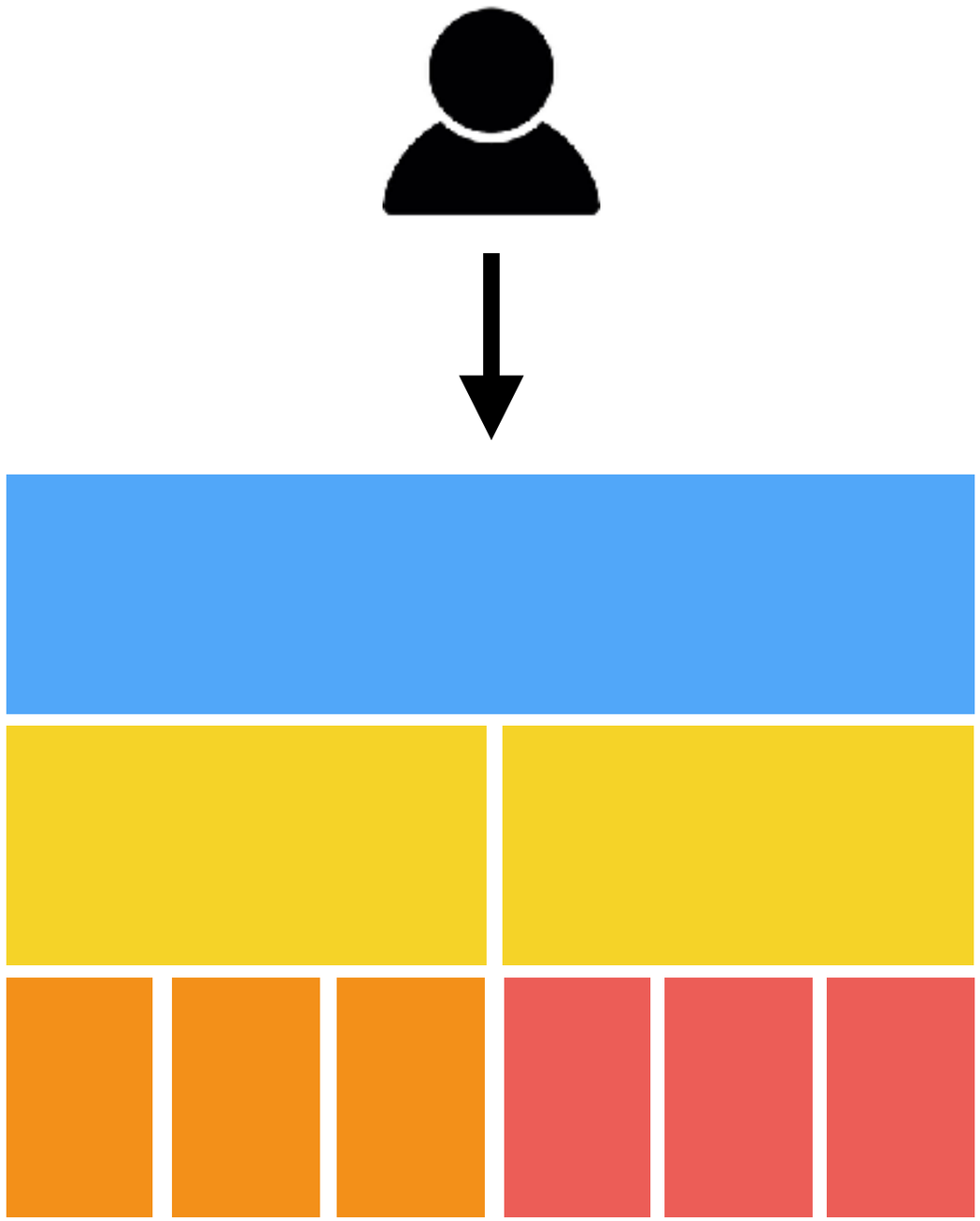
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Service-Oriented Architecture



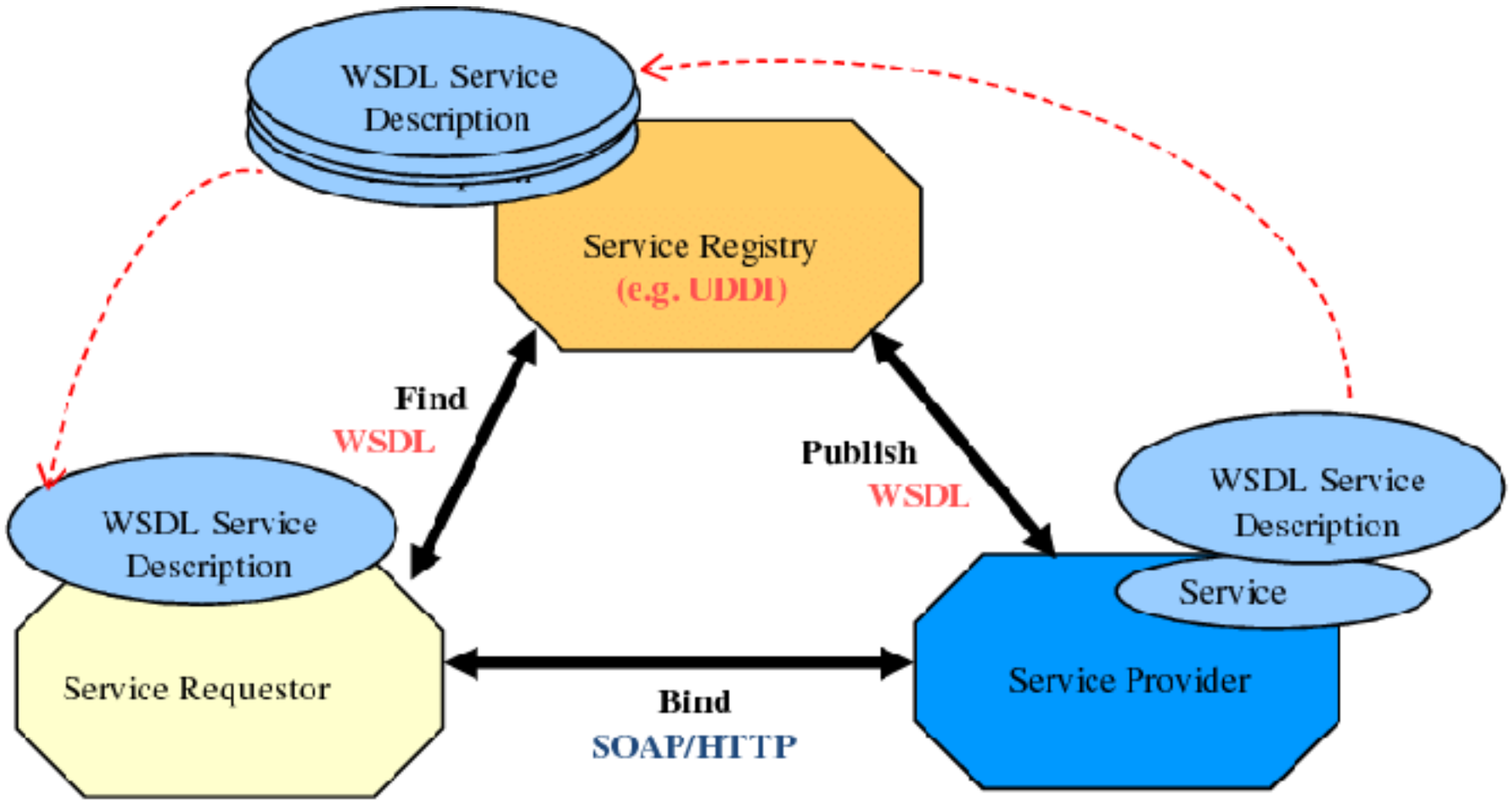
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Service-Oriented Architecture



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WebService



https://www.w3.org/TR/ws-arch/



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We need to improve

We need to get better



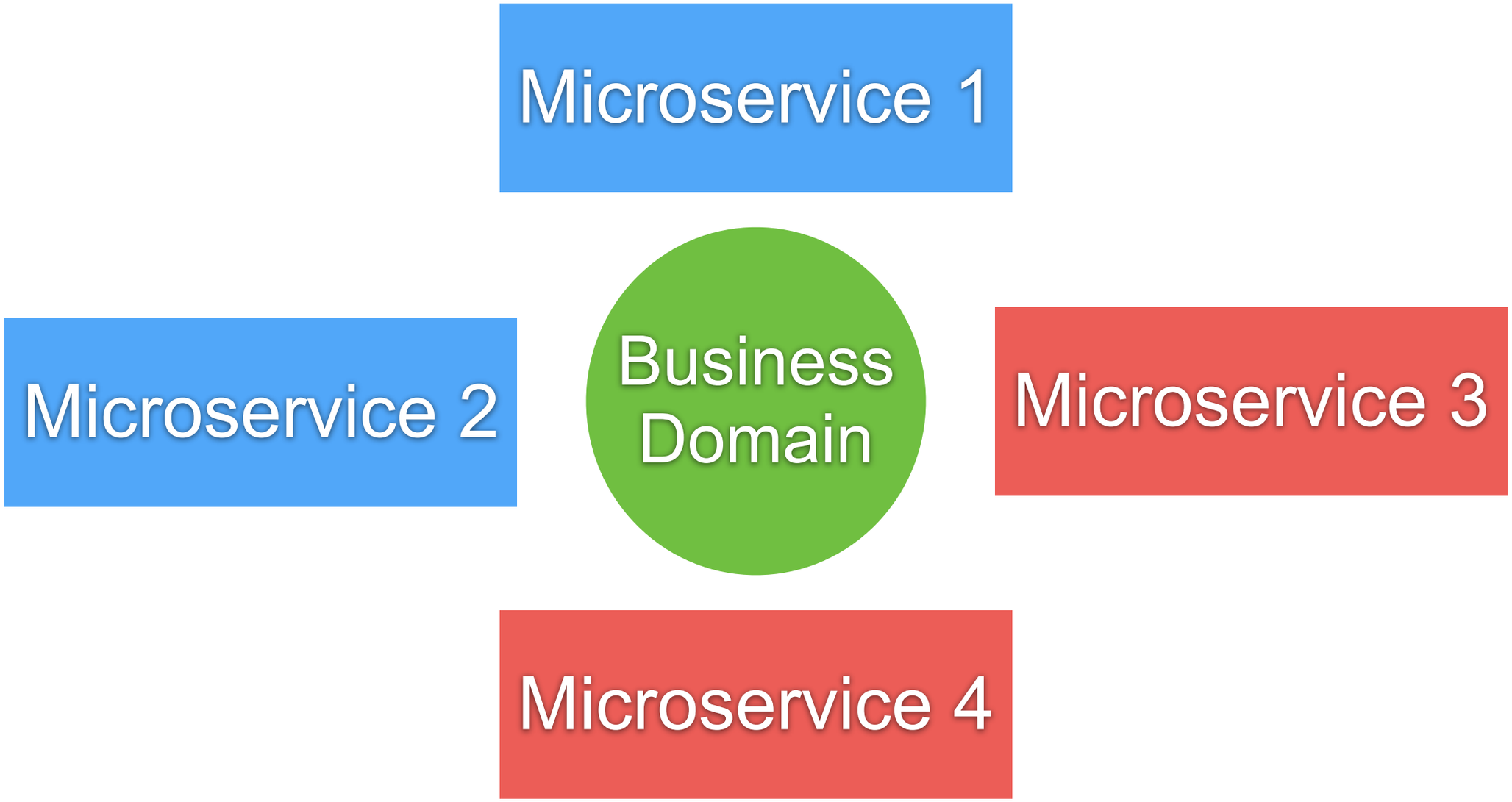
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Microservice



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Microservice



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Microservice

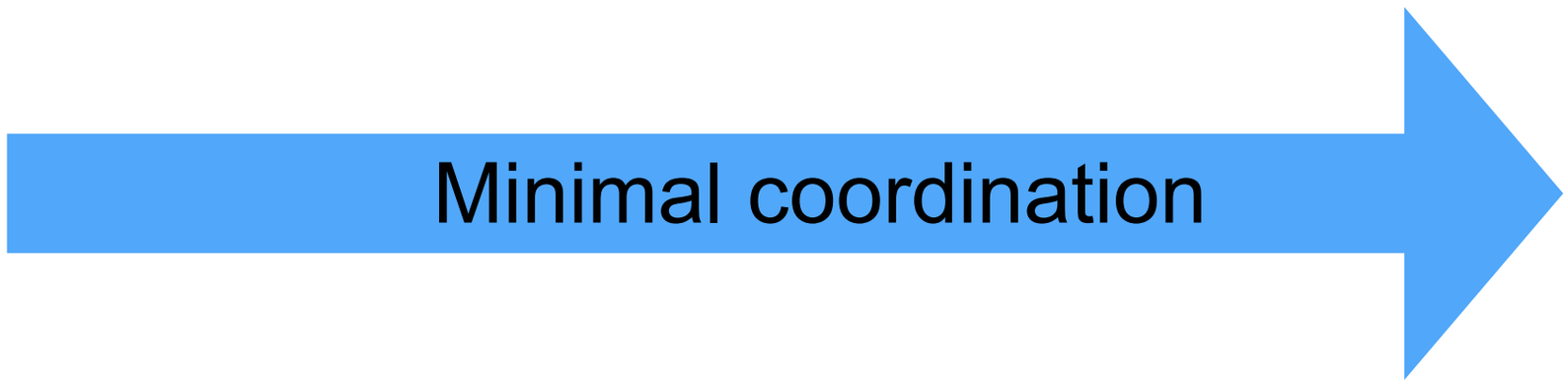
Business service

Self-service



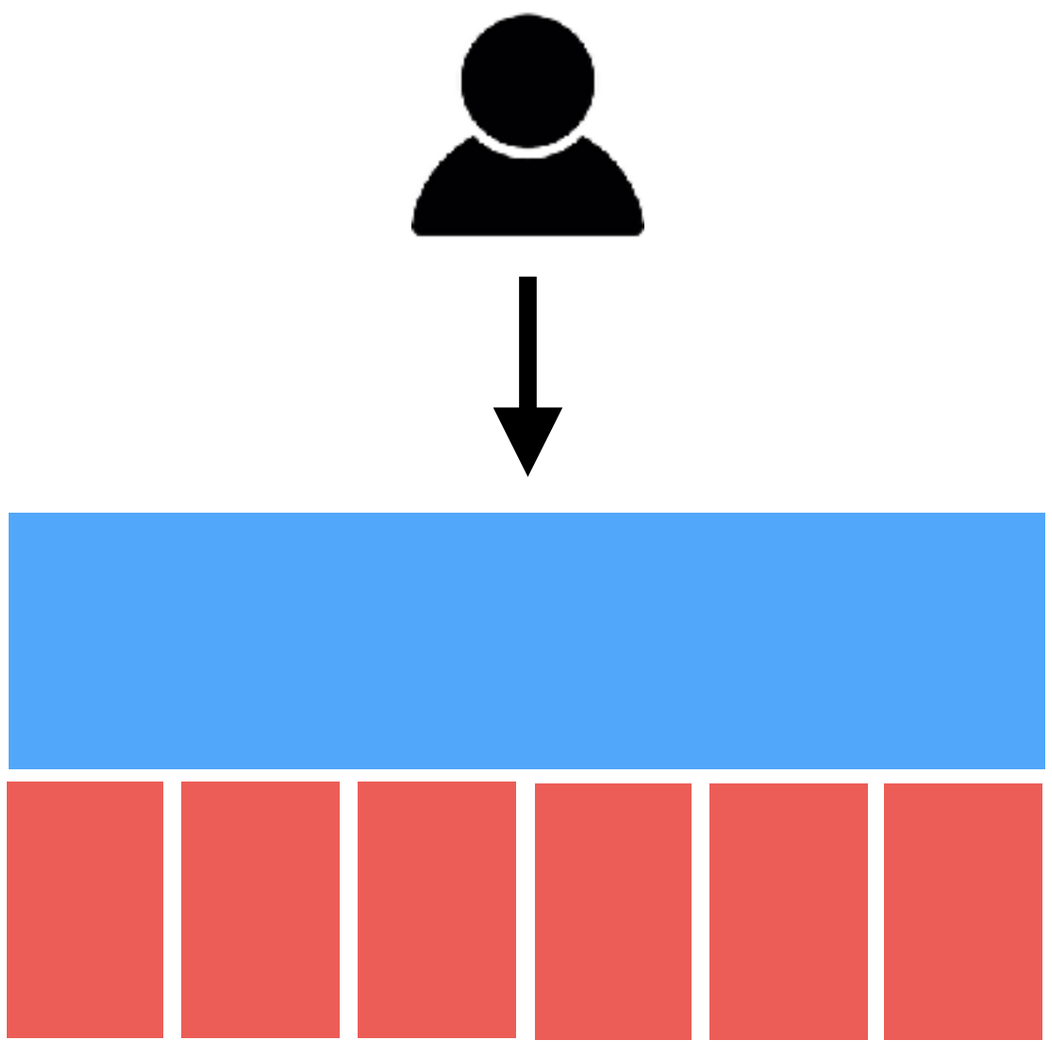


Application Development Team

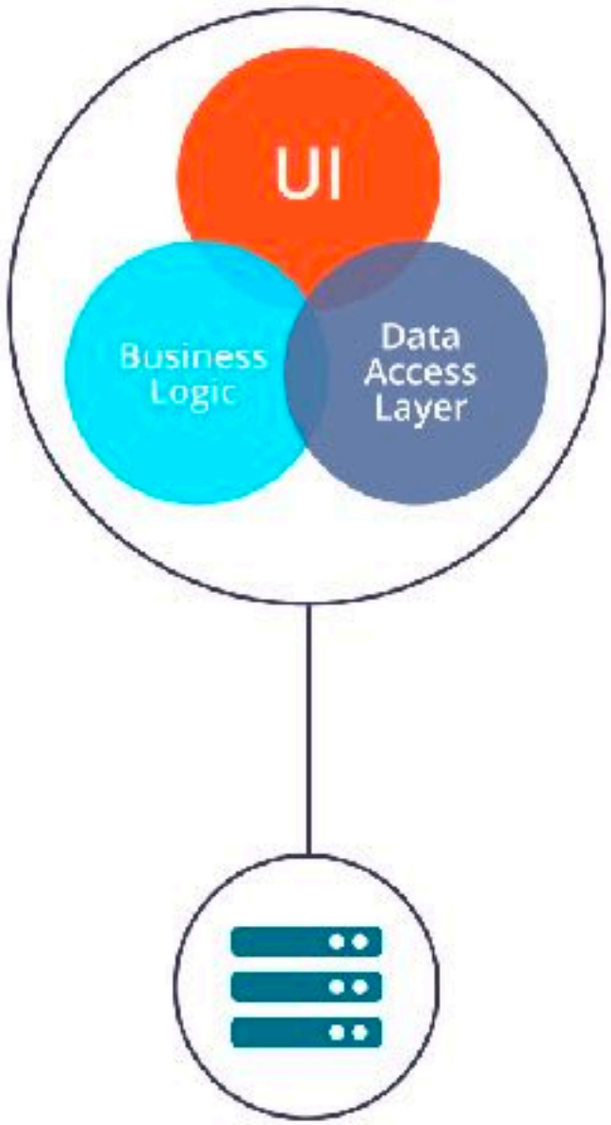


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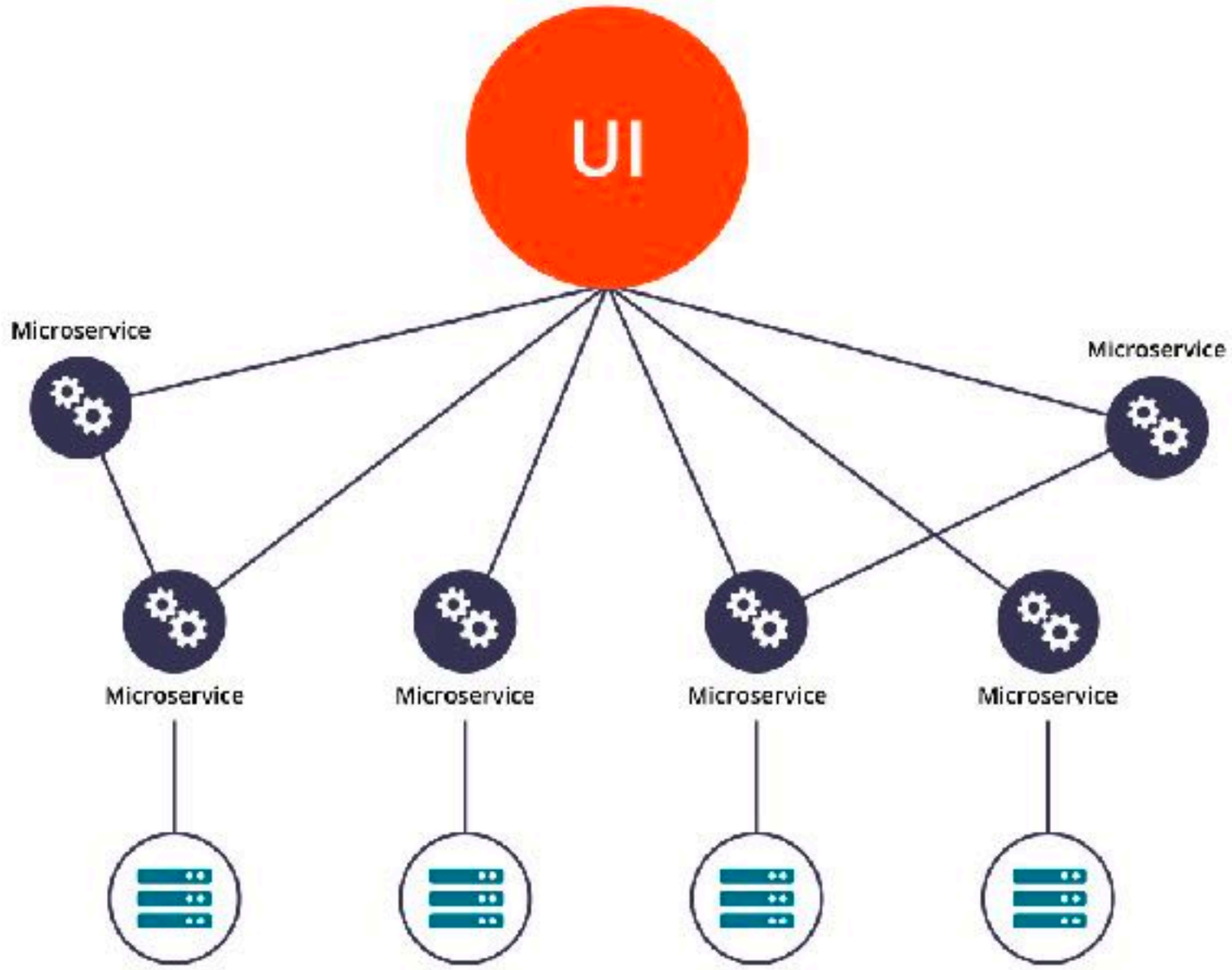
Microservice



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Monolithic Architecture



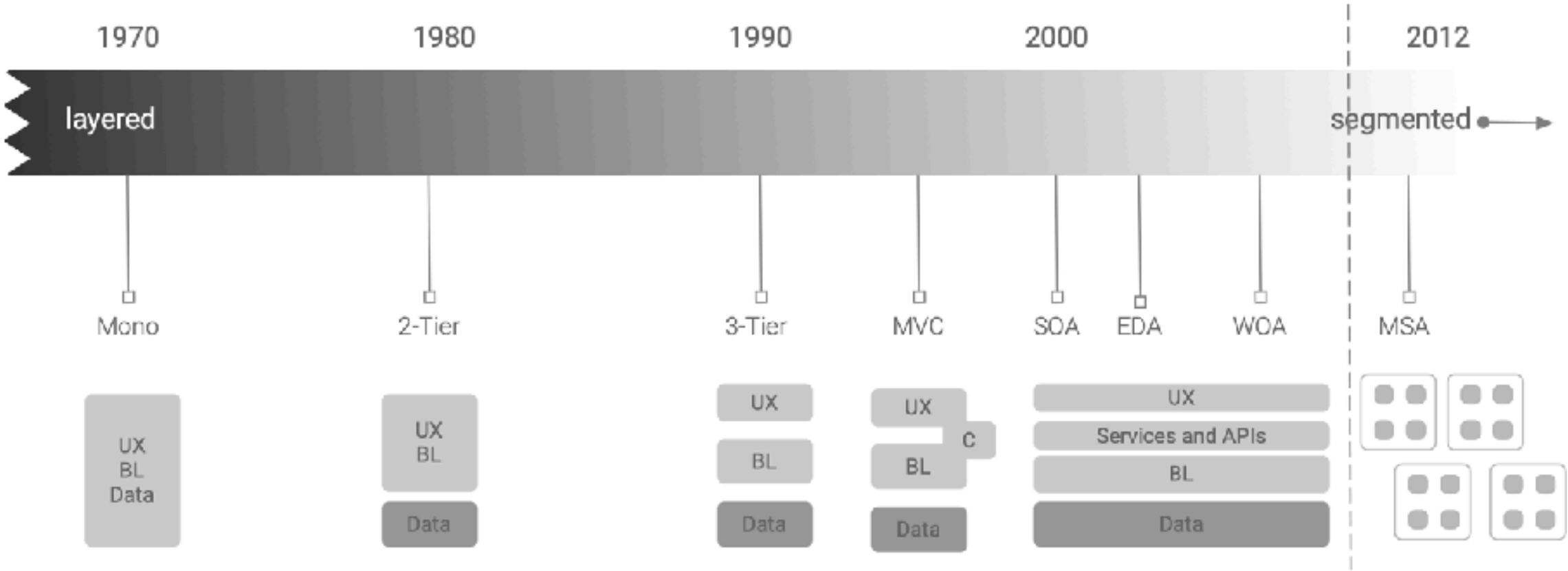
Microservice Architecture

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Summary

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Evolution of Architecture

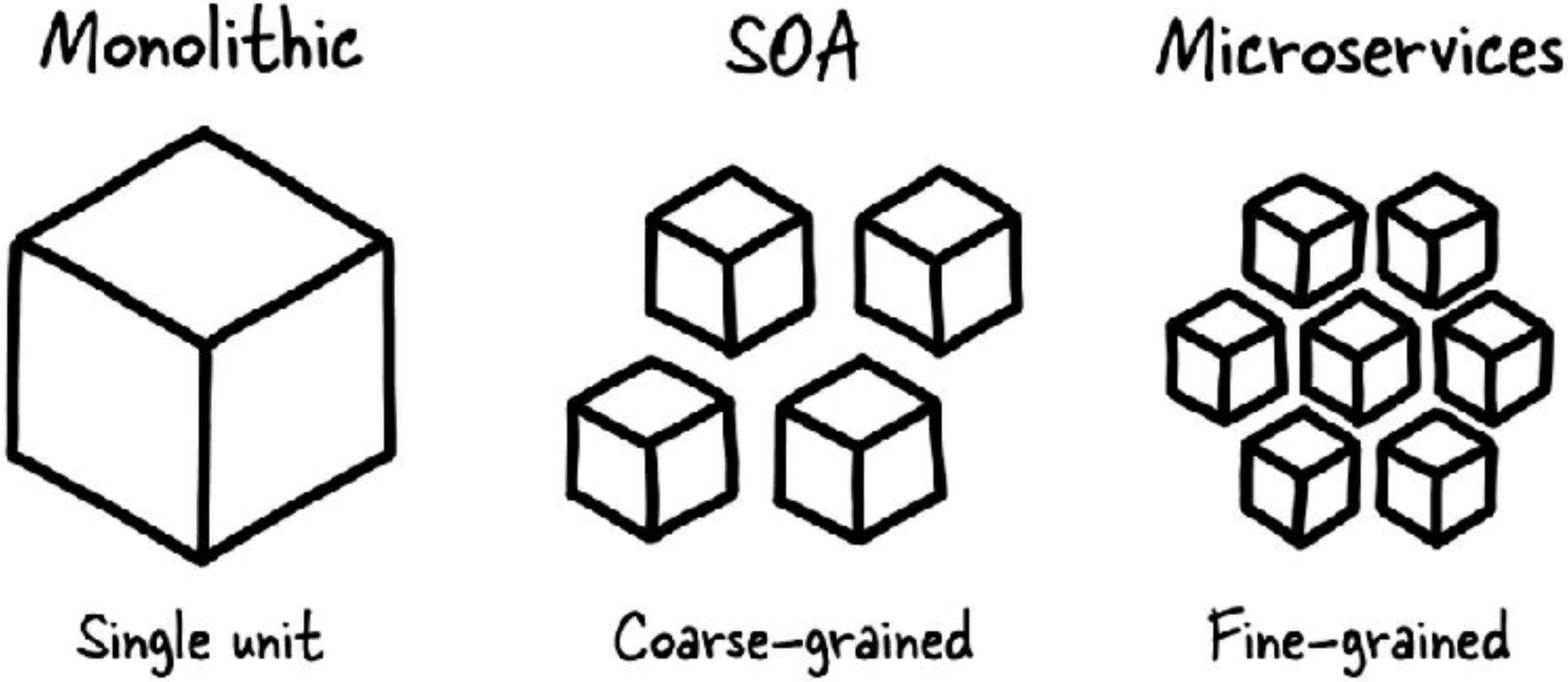


https://github.com/wso2/reference-architecture/blob/master/reference-architecture-layered-segmented.md



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GRANULARITY OF SYSTEMS

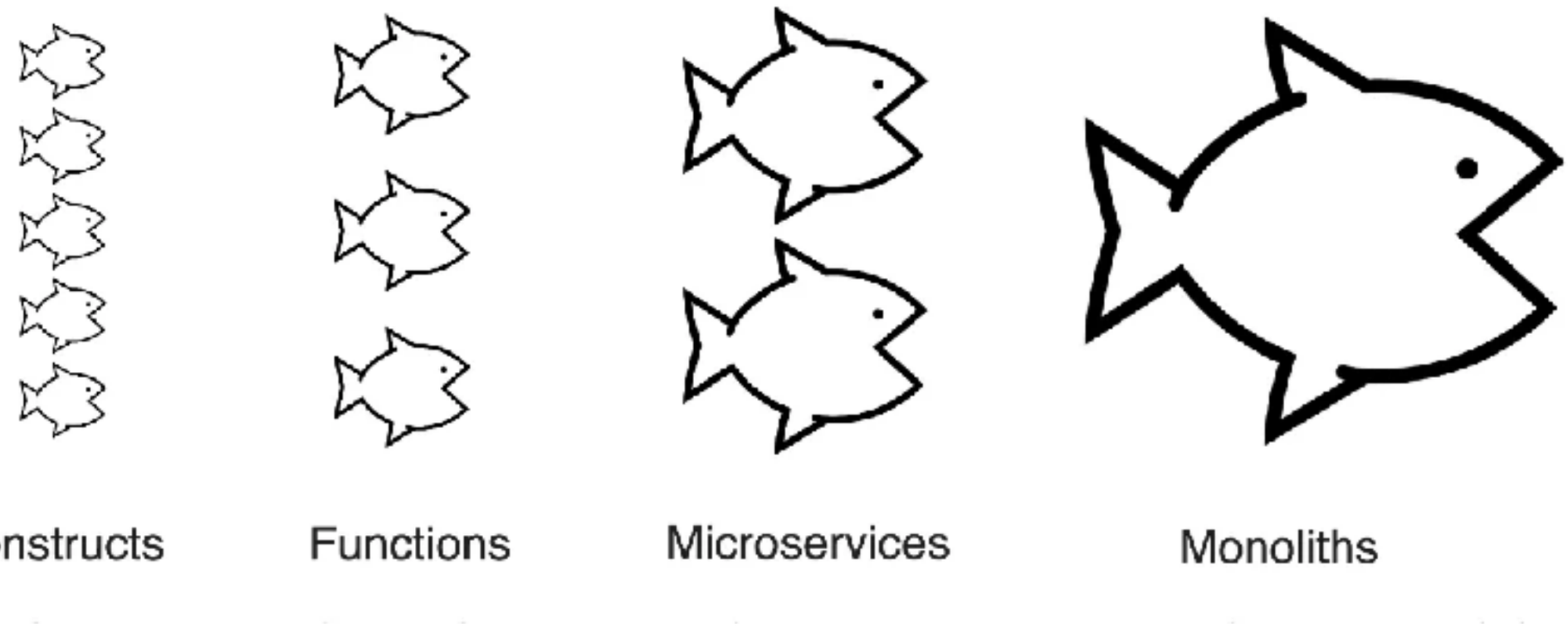


https://architecturenotes.co/p/granularity-of-systems

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Next !!!

FaaS → Cloud Constructs



https://www.infoq.com/articles/cloud-computing-post-serverless-trends/

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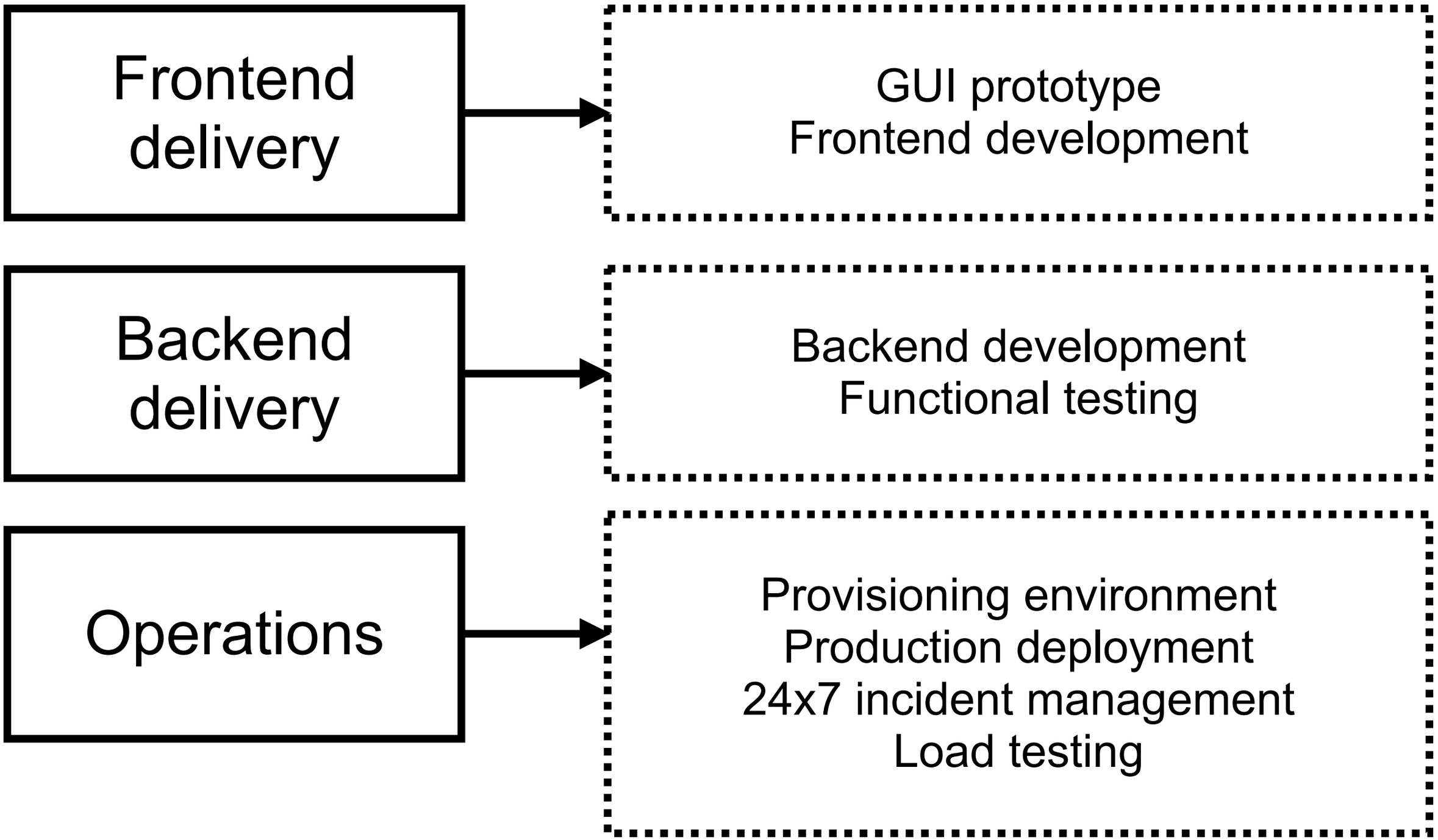
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Delivery Problems



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Delivery Responsibilities



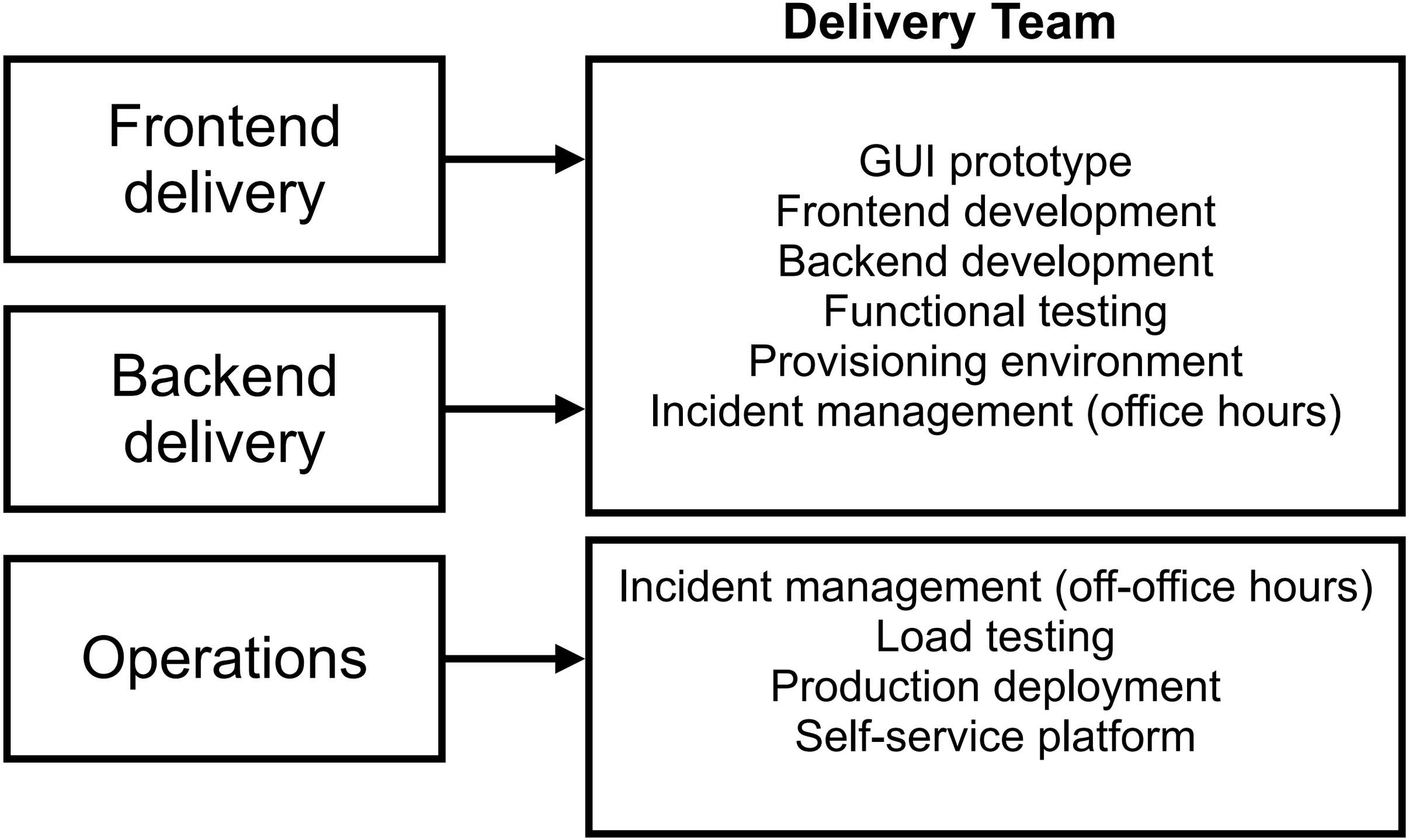
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Delivery Improvement ?



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Better ?



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Microservice's History



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History

ZUUL
Ribbon
Hystrix
Eureka

Agile Manifesto **2001**

Java EE 6
DevOps **2009**

Microservices **2012**

Spring Boot **2013**

Kubernetes **2014**



1990
Continuous **I**ntegration

2006
Amazon EC2

2010
Netflix + AWS

2013
Container **Docker**

2014
Adopt Microservices

2017
Istio
Linkerd



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Microservice

Small, Do one thing (Single Responsibility)

Modular

Easy to **understand**

Easy to **develop**

Easy to **deploy**

Easy to **maintain**

Scale independently



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Design Microservice ?

Interface segregation

Deployability

Event-driven

Loose coupling

Single responsibility

Availability over consistency

https://www.infoq.com/articles/microservices-design-ideals/



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Goals

Increase the **velocity** of application release
By **decomposing** app to **small** services
Autonomous services
Deploy **independently**



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Independent

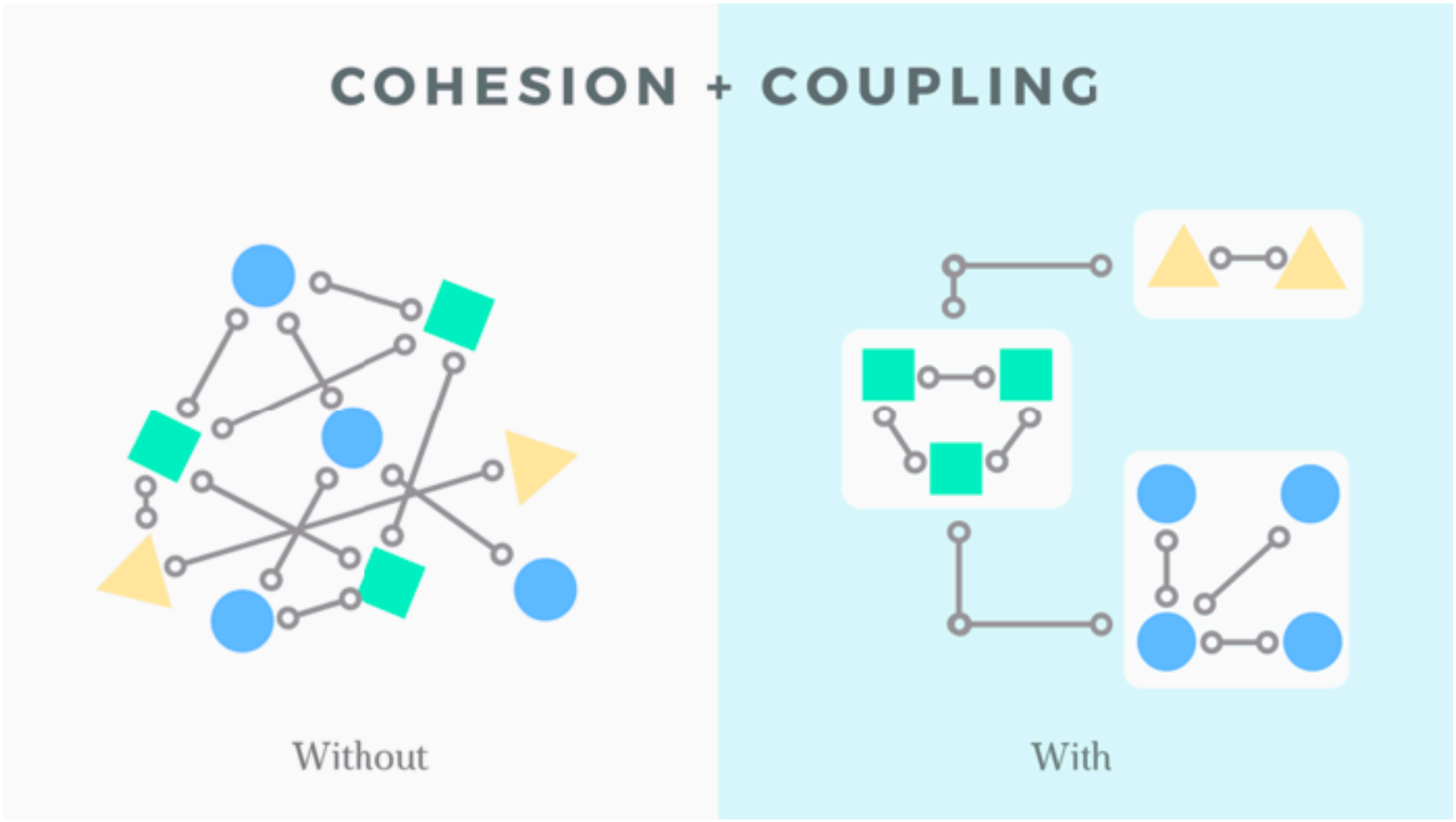
Replicable
Updatable
Scalable
Deployable



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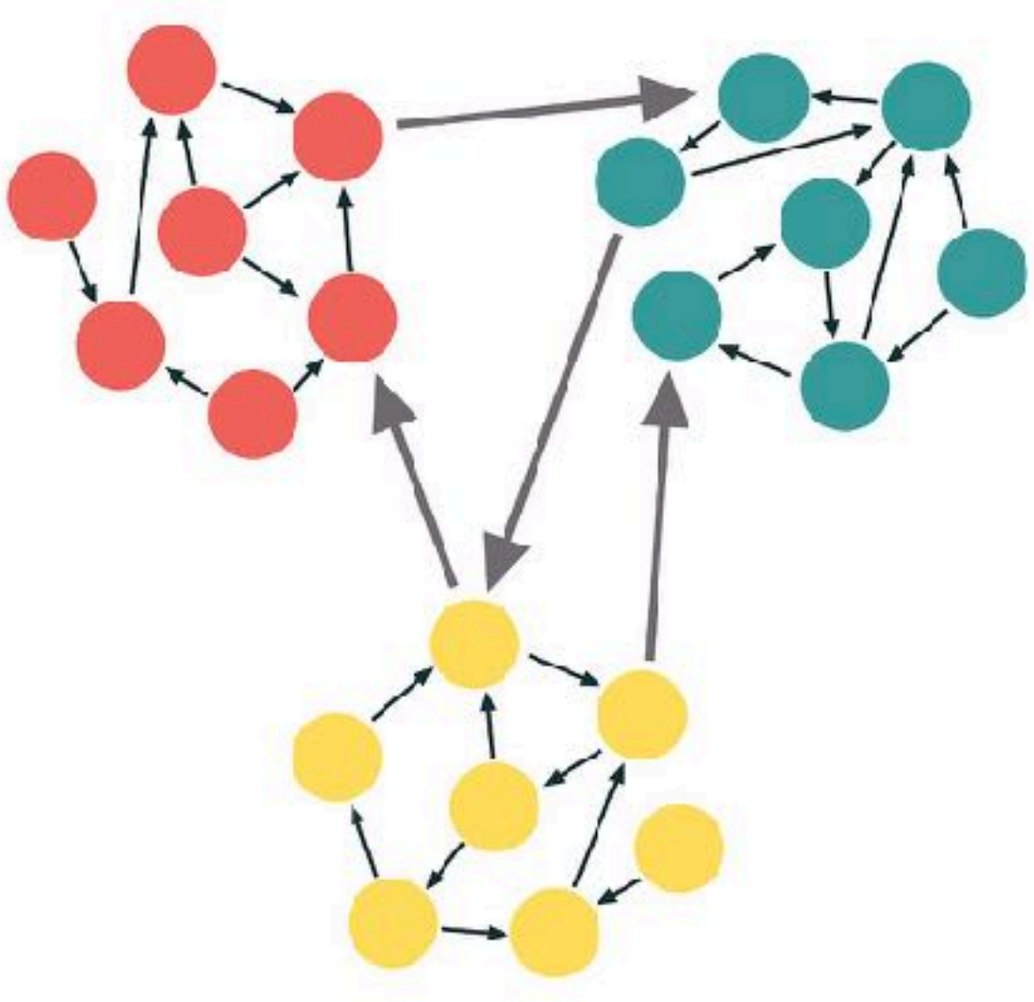
Better Architecture

Loose coupling
High cohesion



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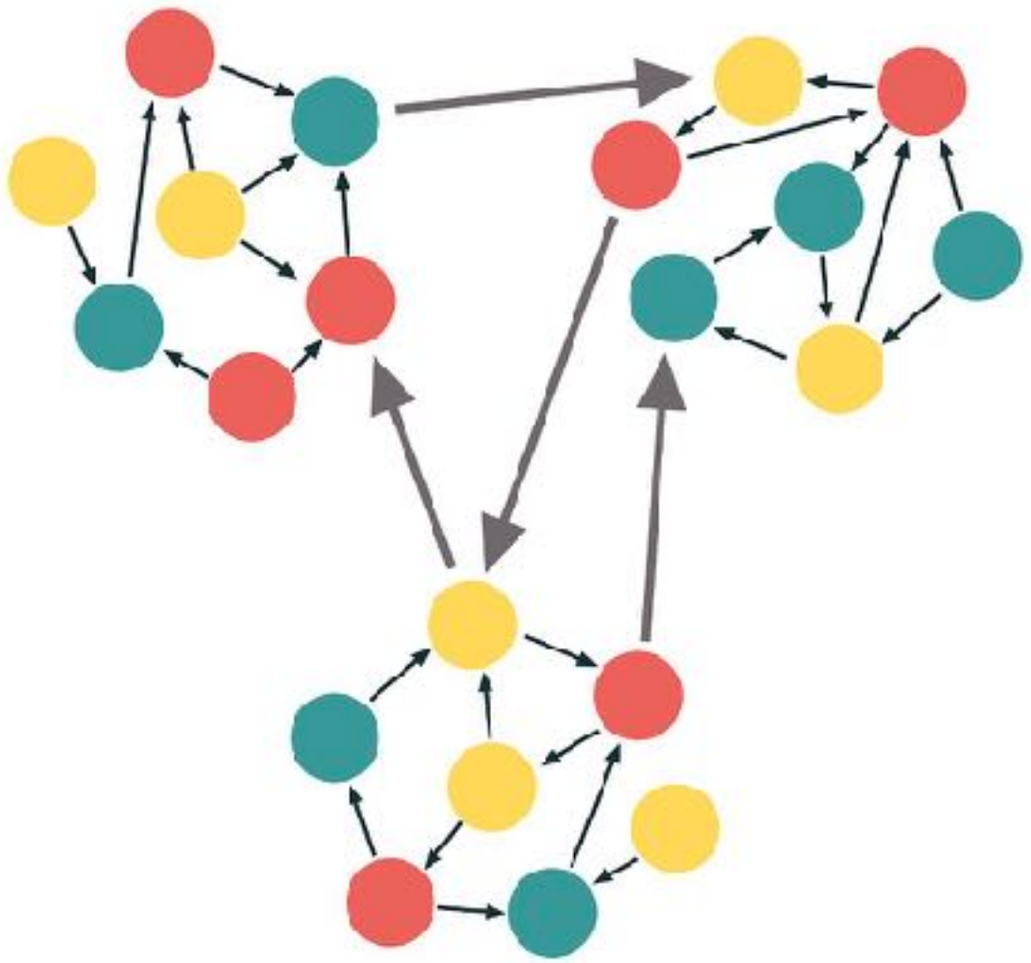
Coupling (Idea)



https://enterprisecraftsmanship.com/posts/cohesion-coupling-difference/

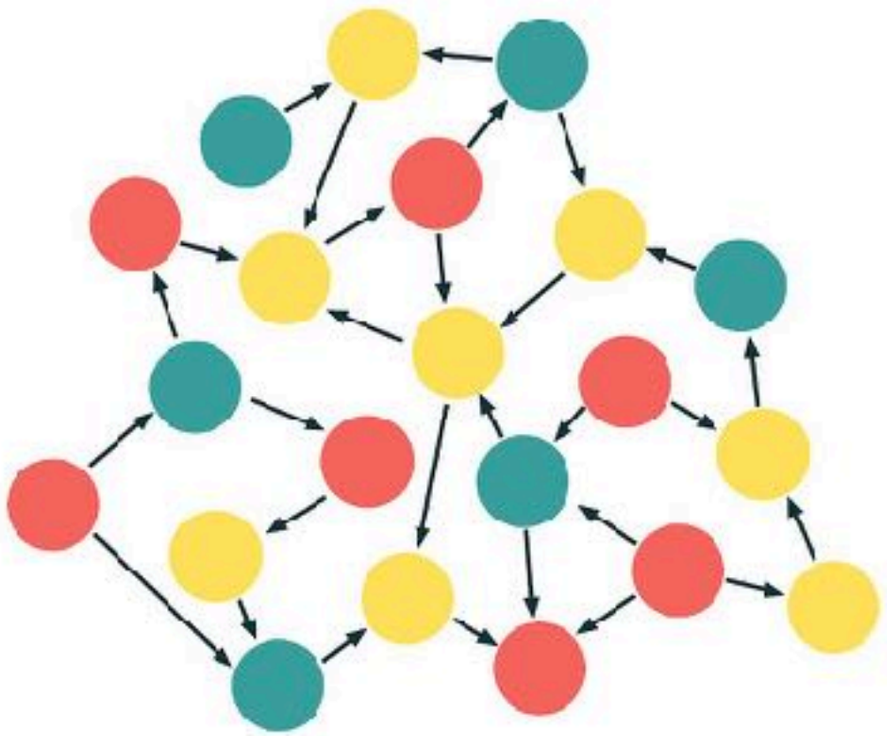
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Coupling (Poorly boundary)



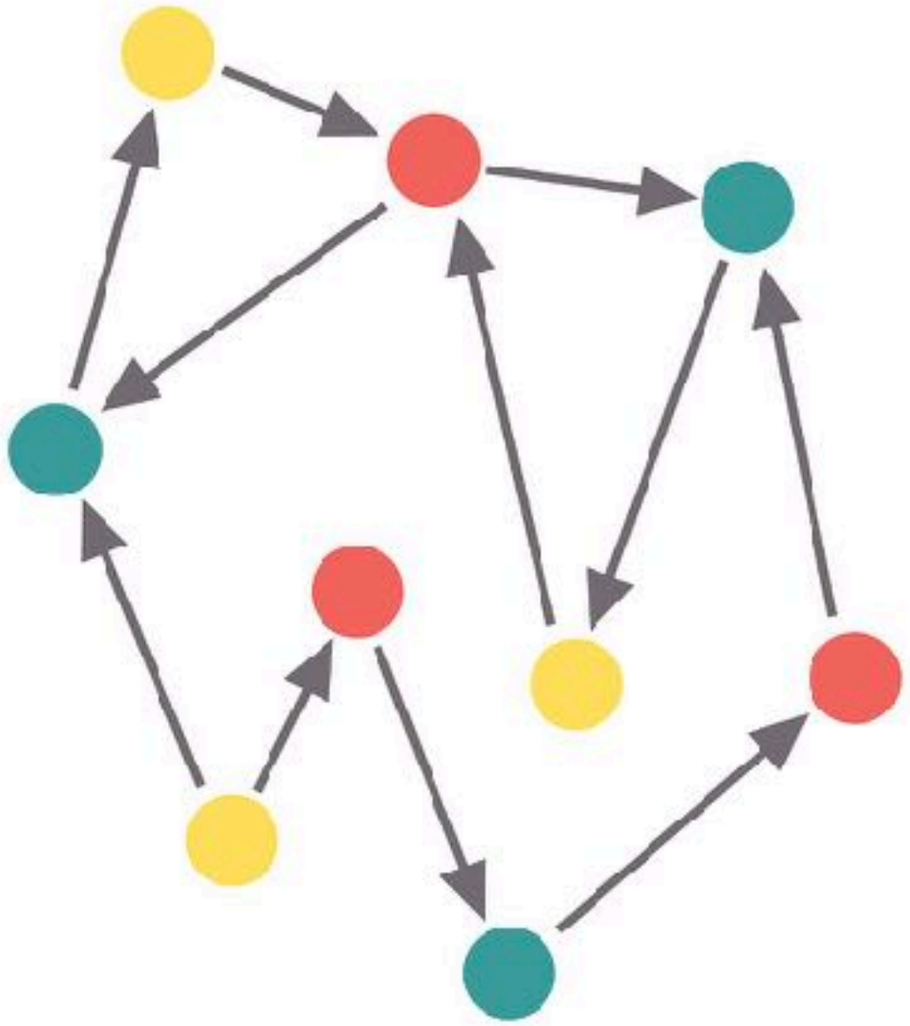
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Coupling (God)



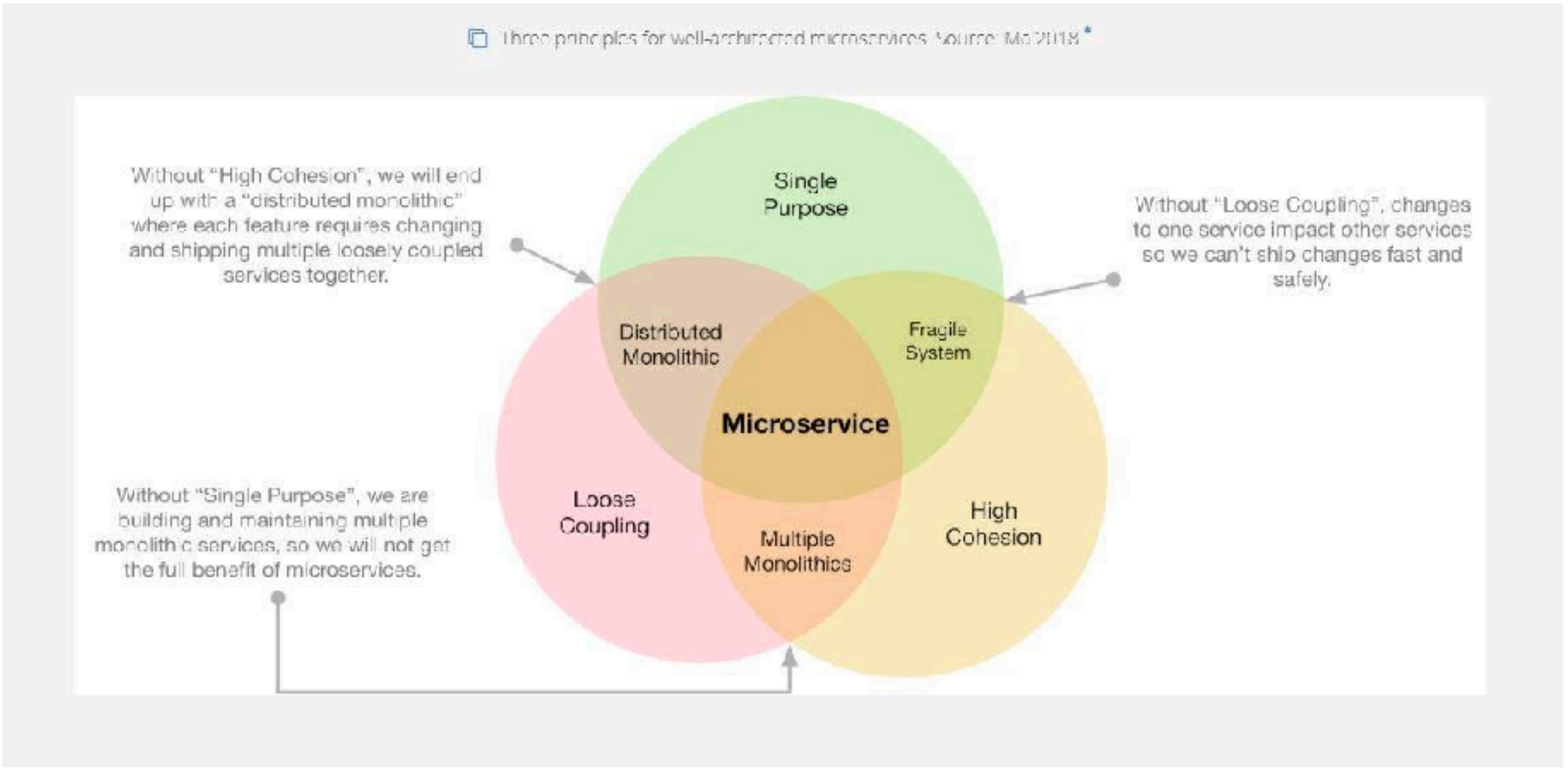
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Coupling (Destructive decouple)



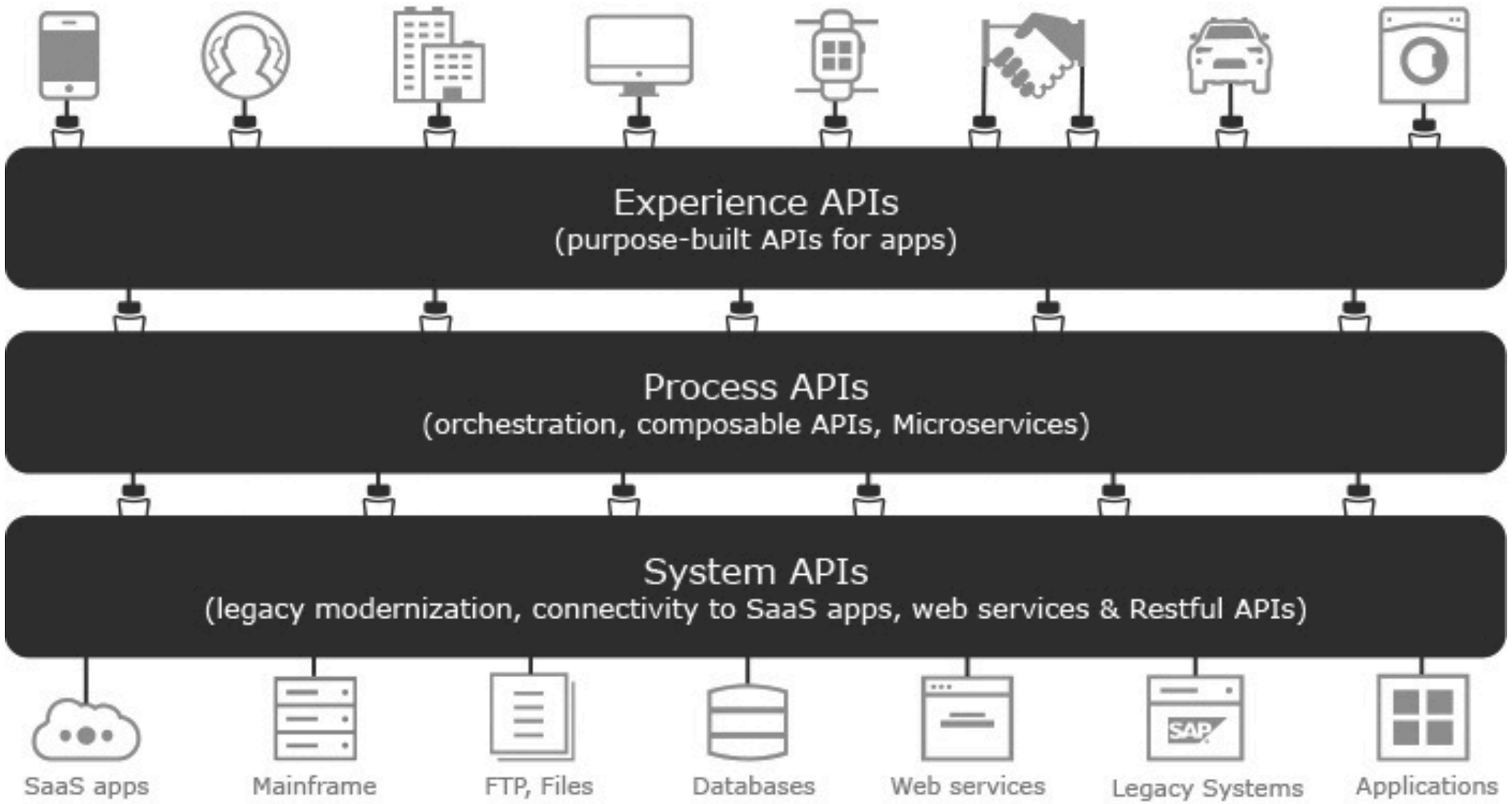
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Better Architecture



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Real World Architecture



https://blogs.mulesoft.com/dev-guides/how-to-tutorials/api-templates-reusable-system-process-apis/

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Key Benefits

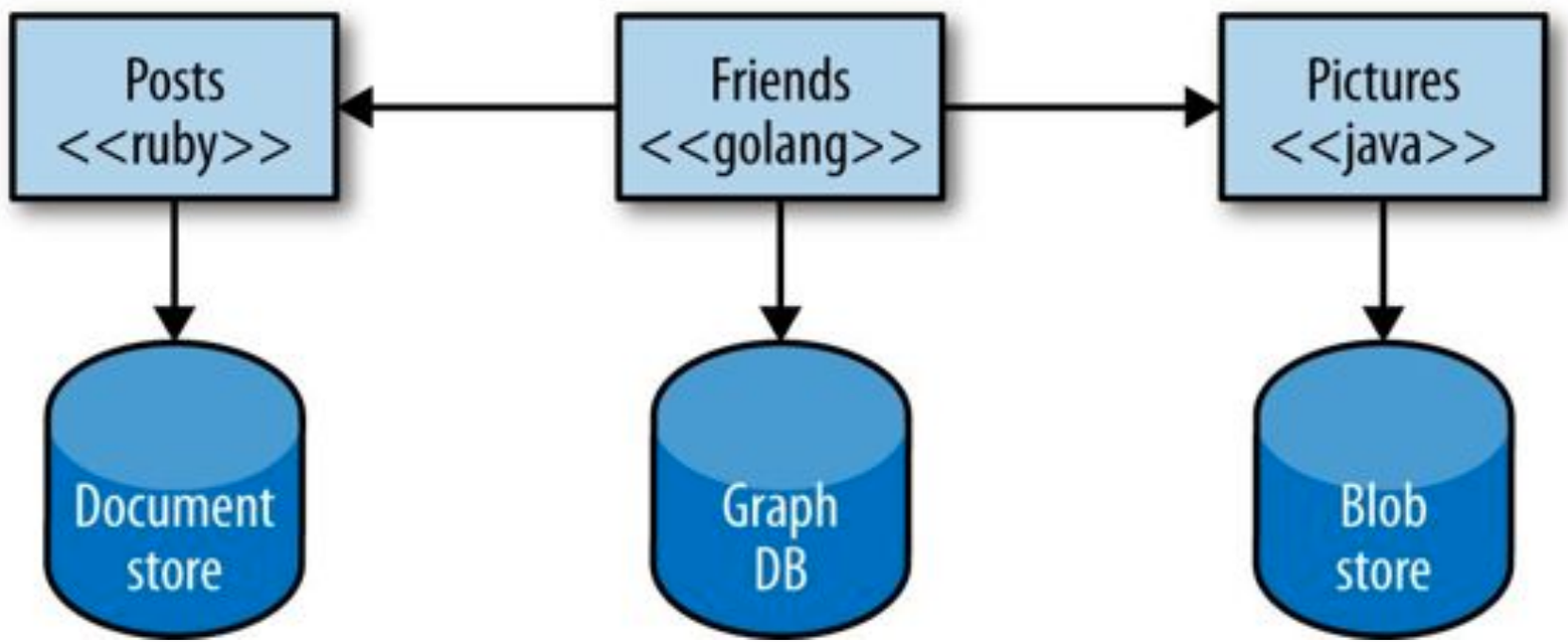


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1. Technology heterogeneous

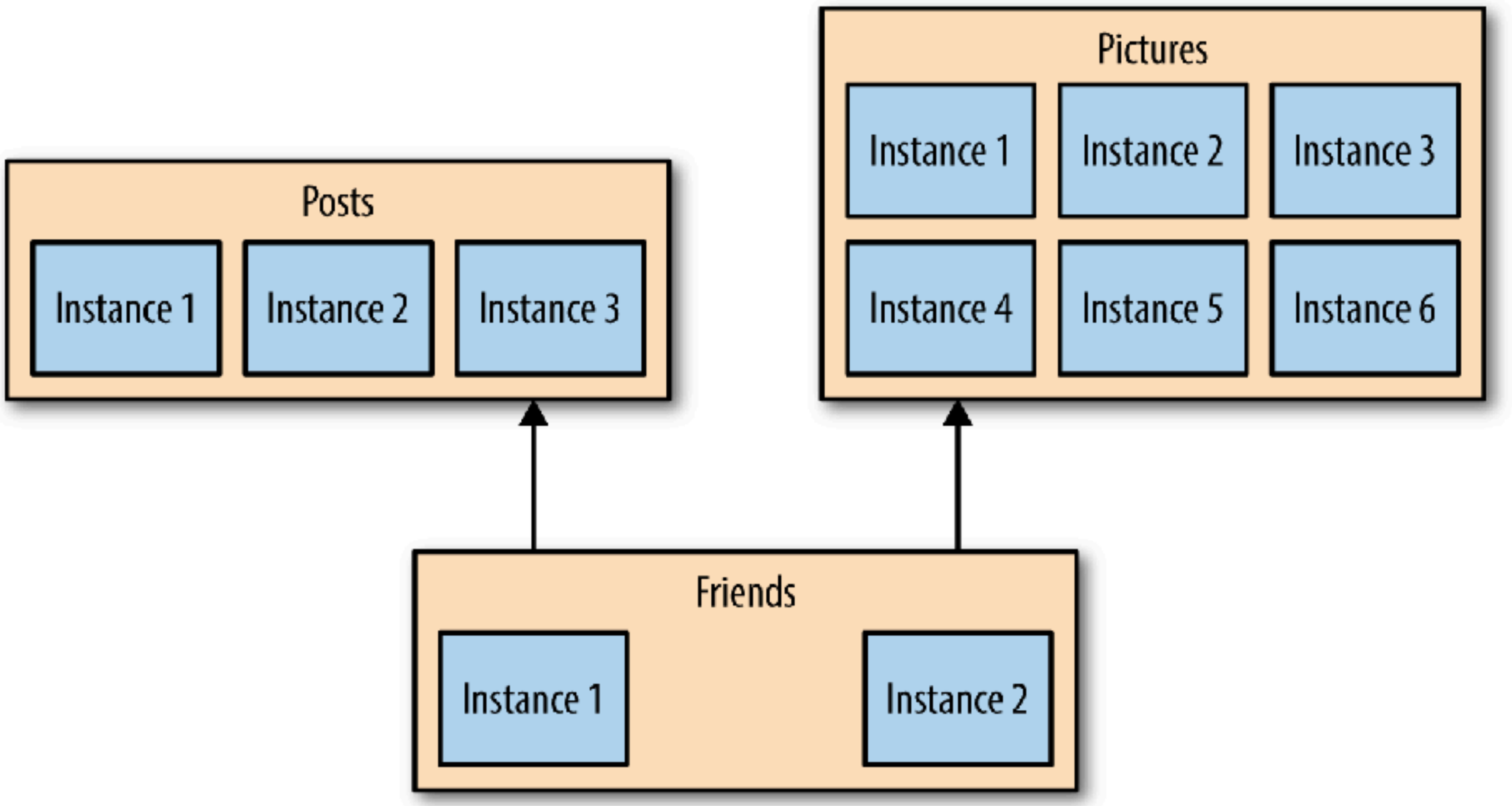
The right tool for each job

Allow easy experimenting and adoption of new technology



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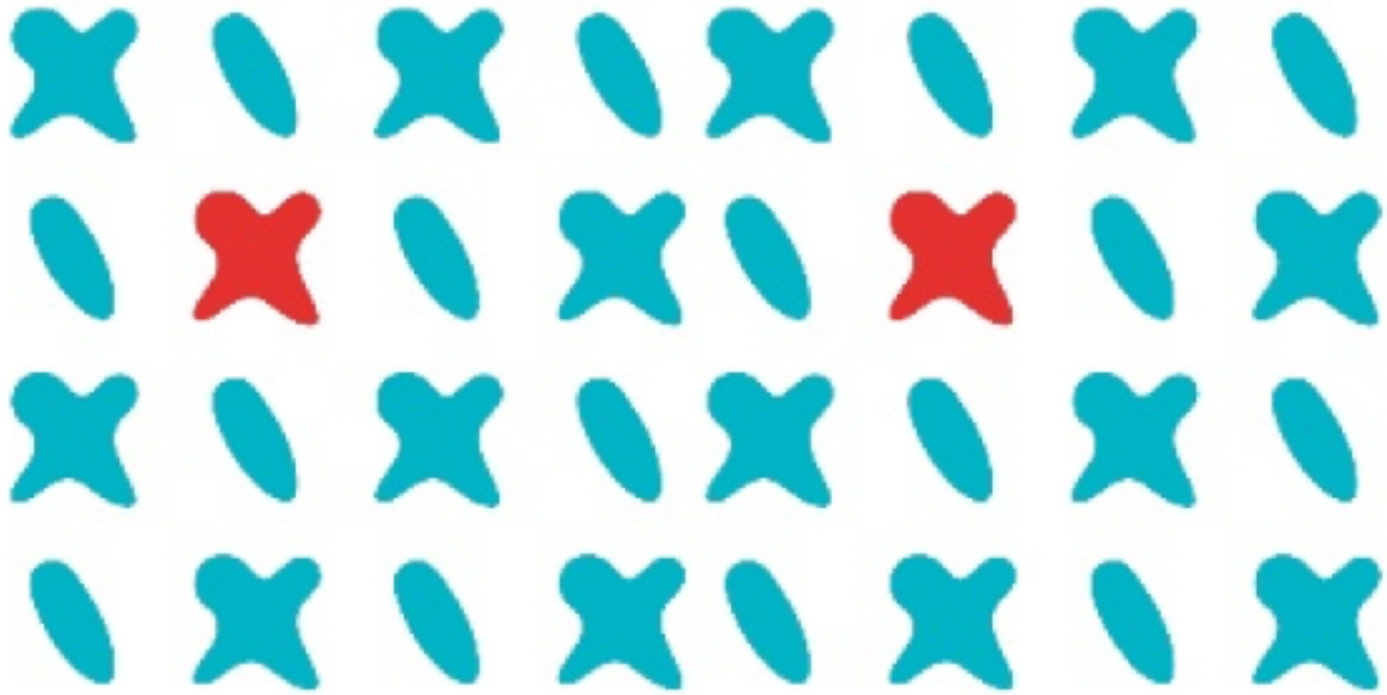
2. Scaling



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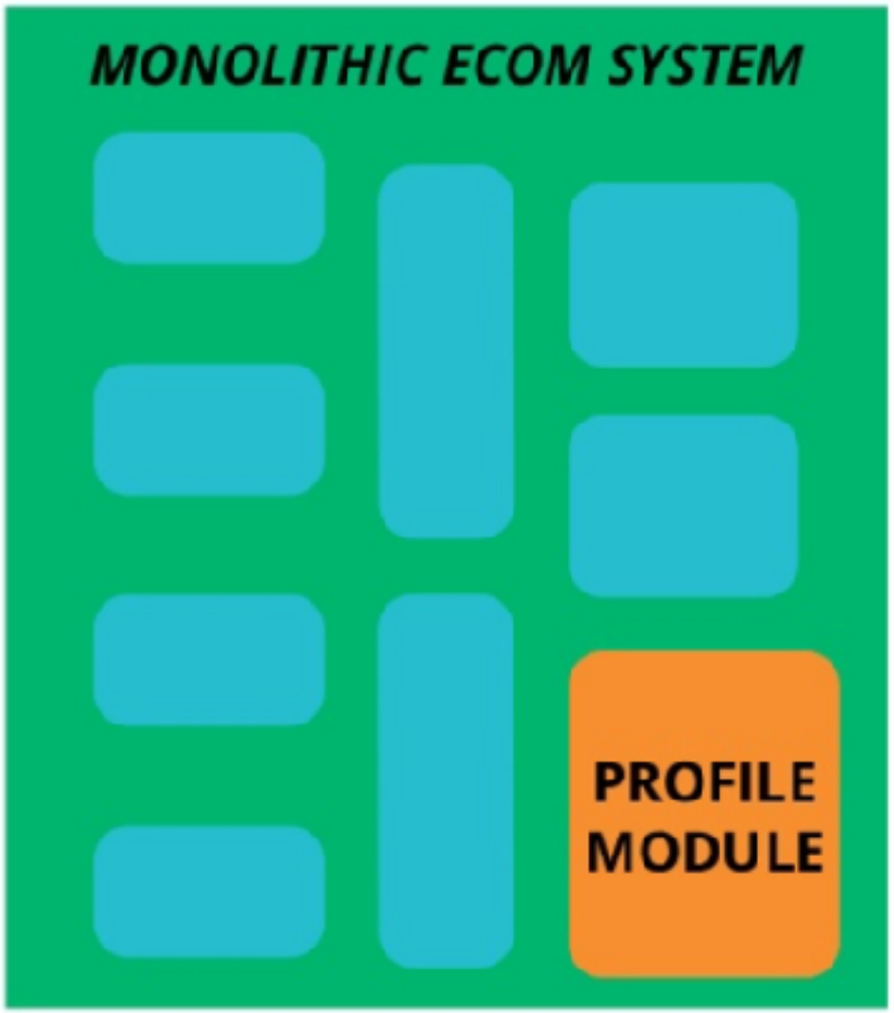
3. Ease of deployment

Deploys are faster, independent and problems can be isolated more easily



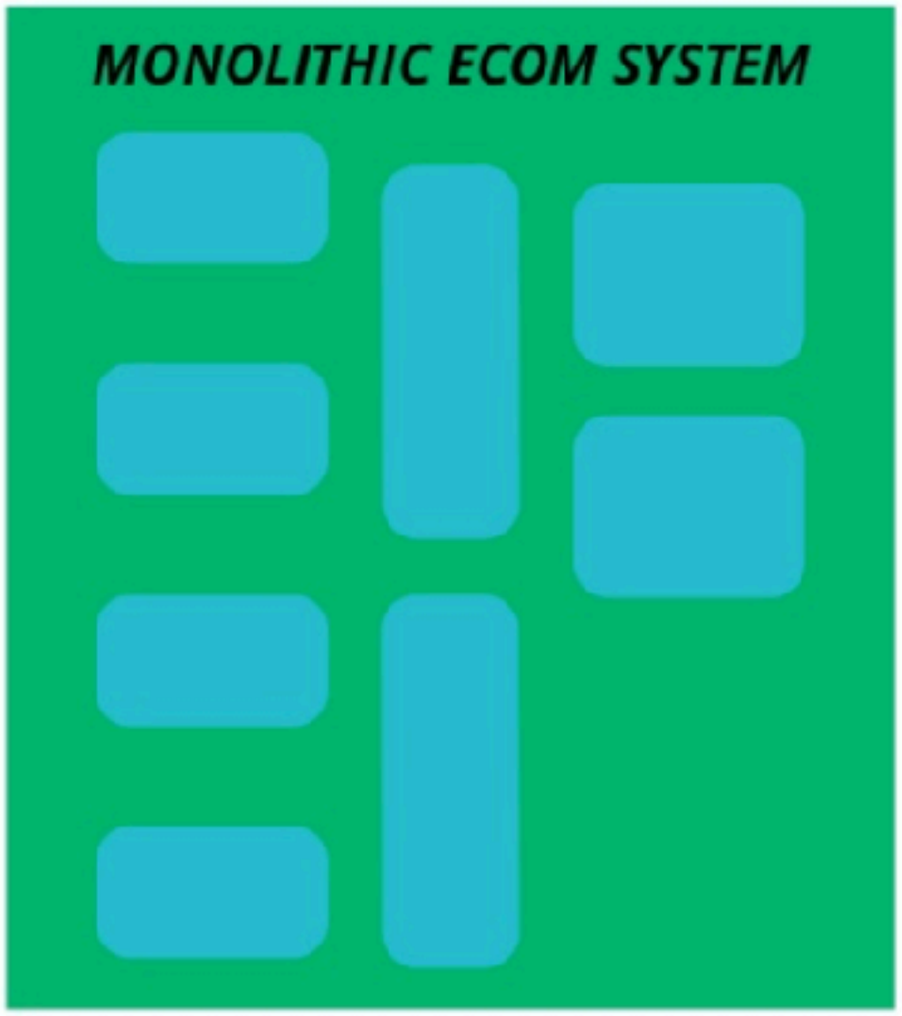
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4. Composability and replaceability



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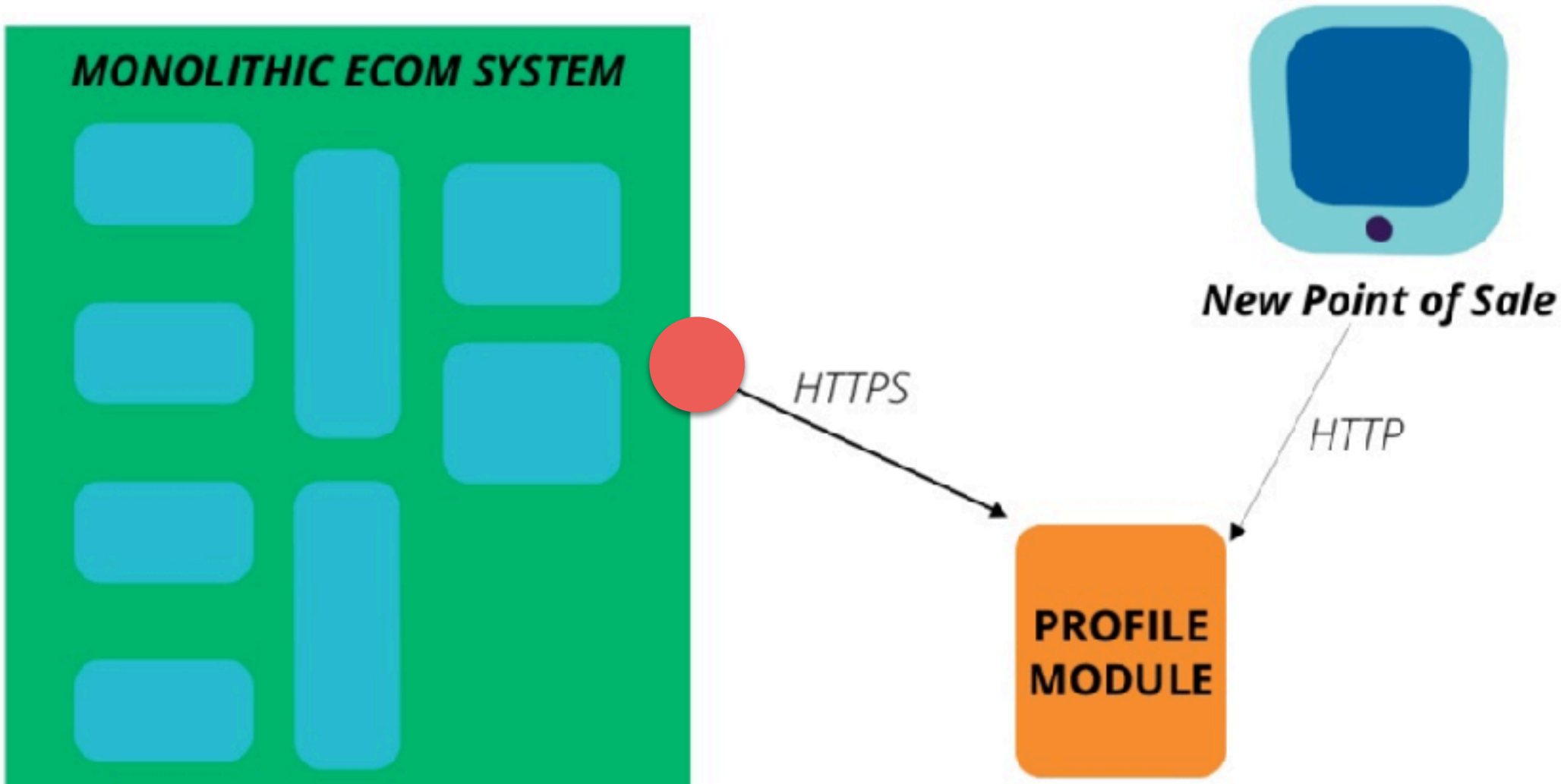
4. Composability and replaceability





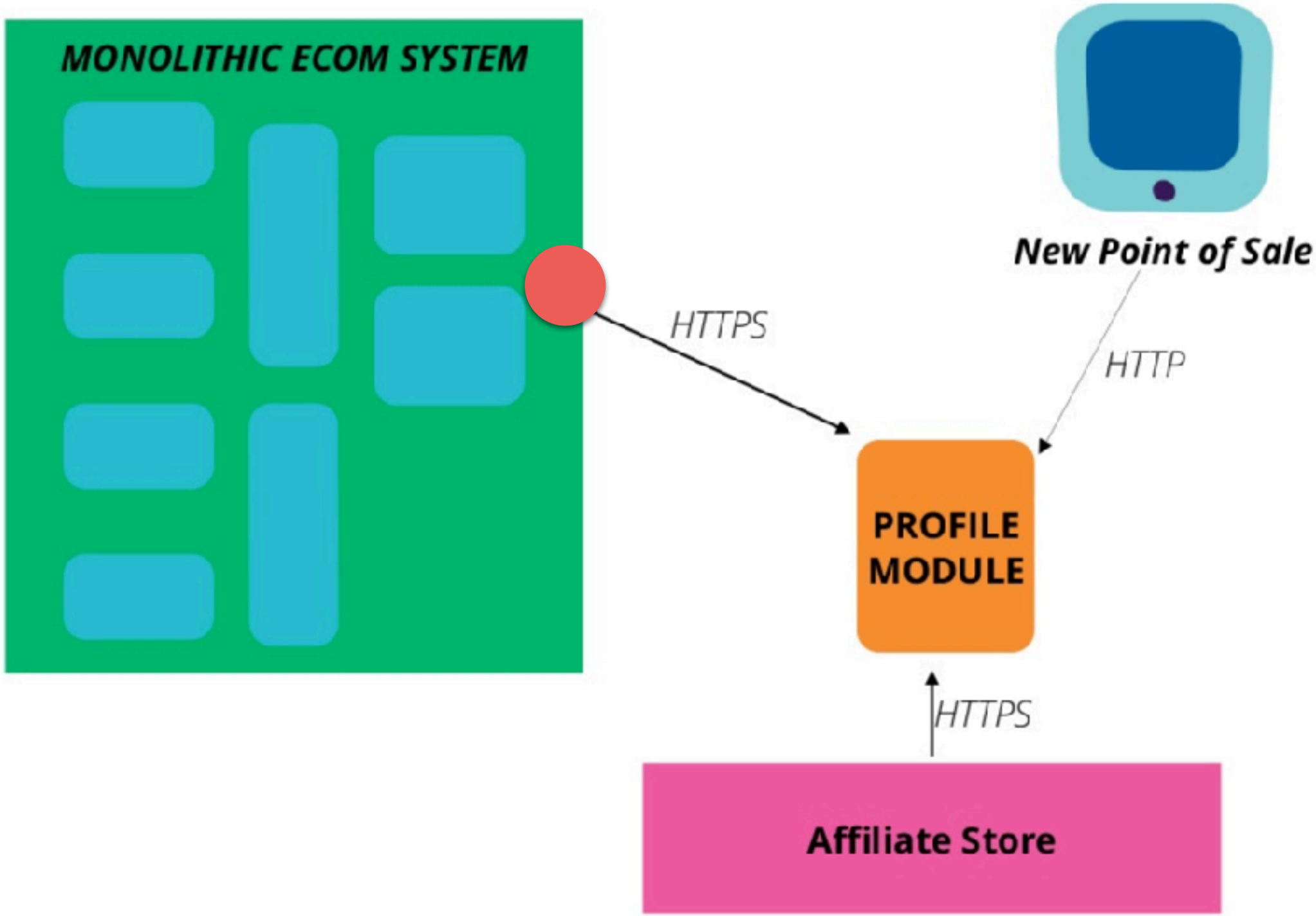
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4. Composability and replaceability



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4. Composability and replaceability



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5. Organization alignment

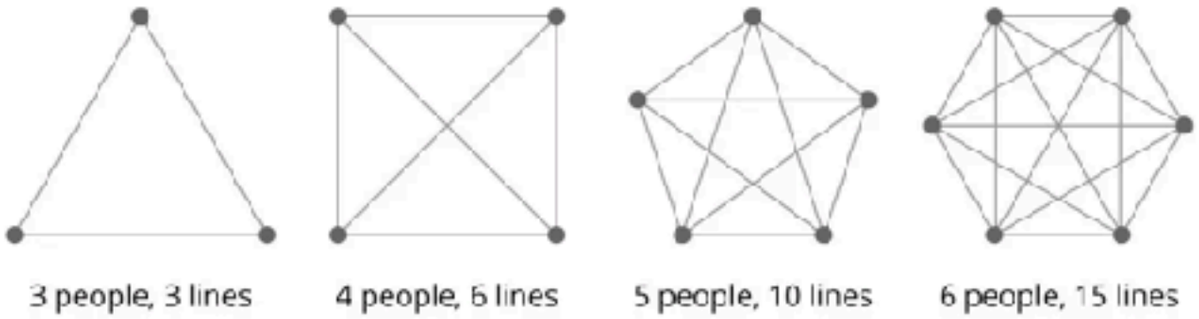
Small teams and smaller codebases



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Small team !!

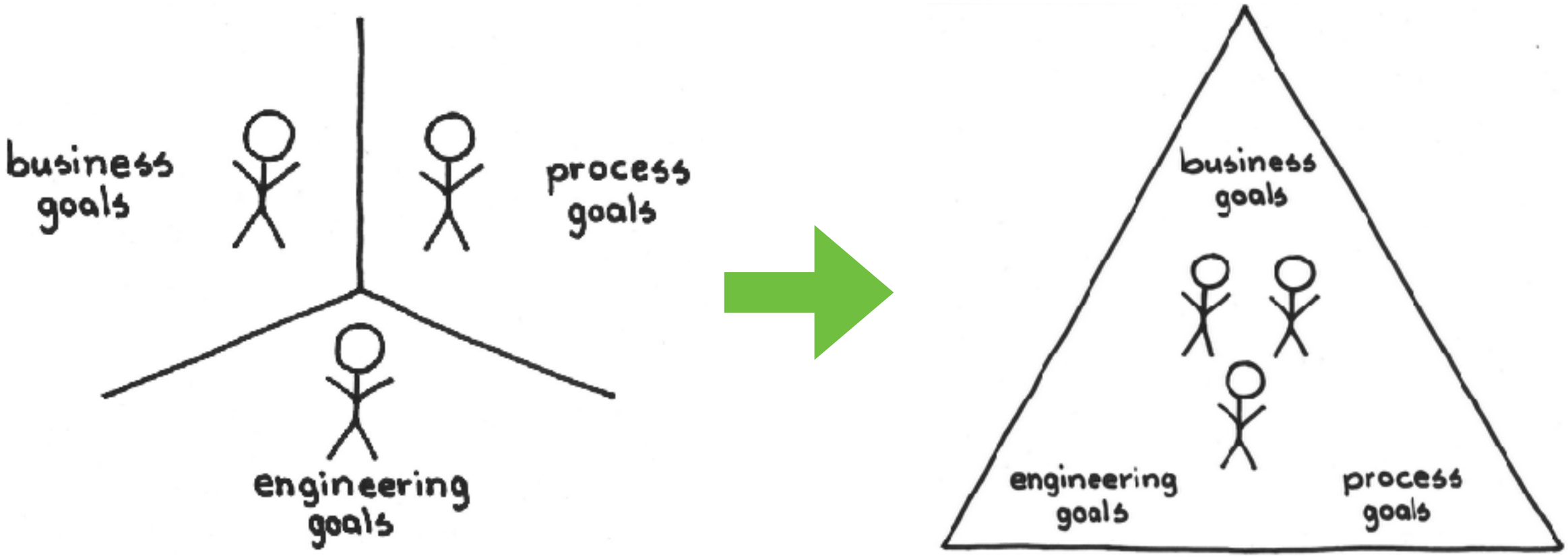
Line of communication and team size



https://www.leadingagile.com/2018/02/lines-of-communication-team-size-applying-brooks-law

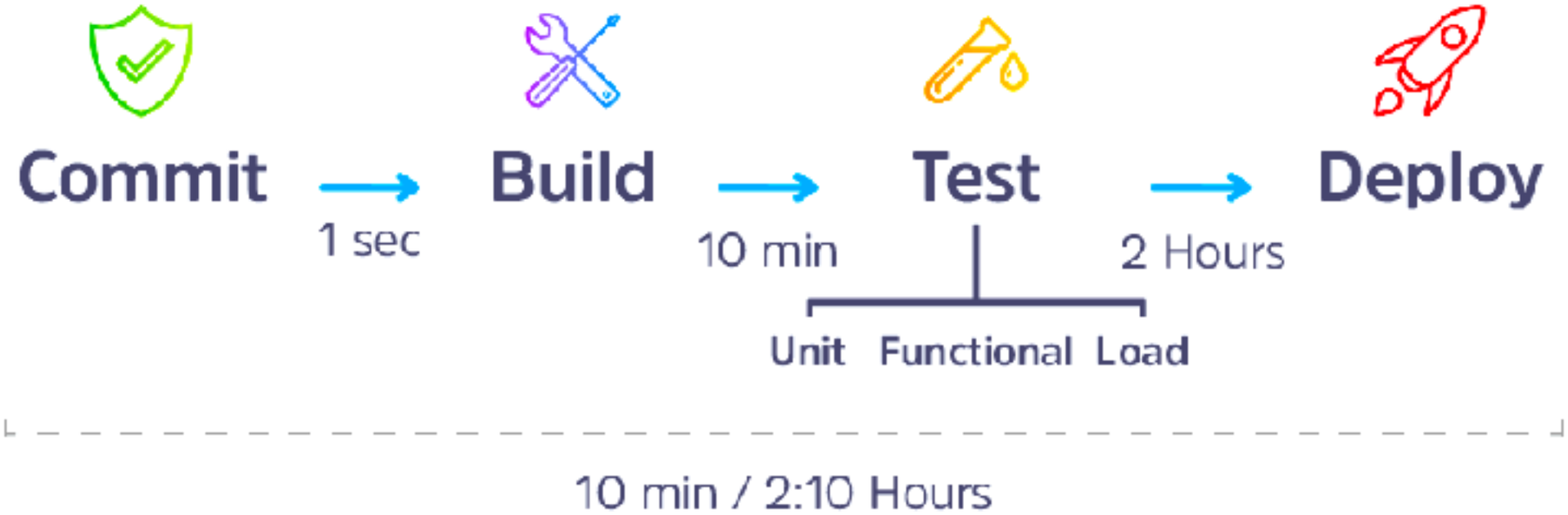
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Autonomous team



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Autonomous team



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6. Enable Continuous Delivery

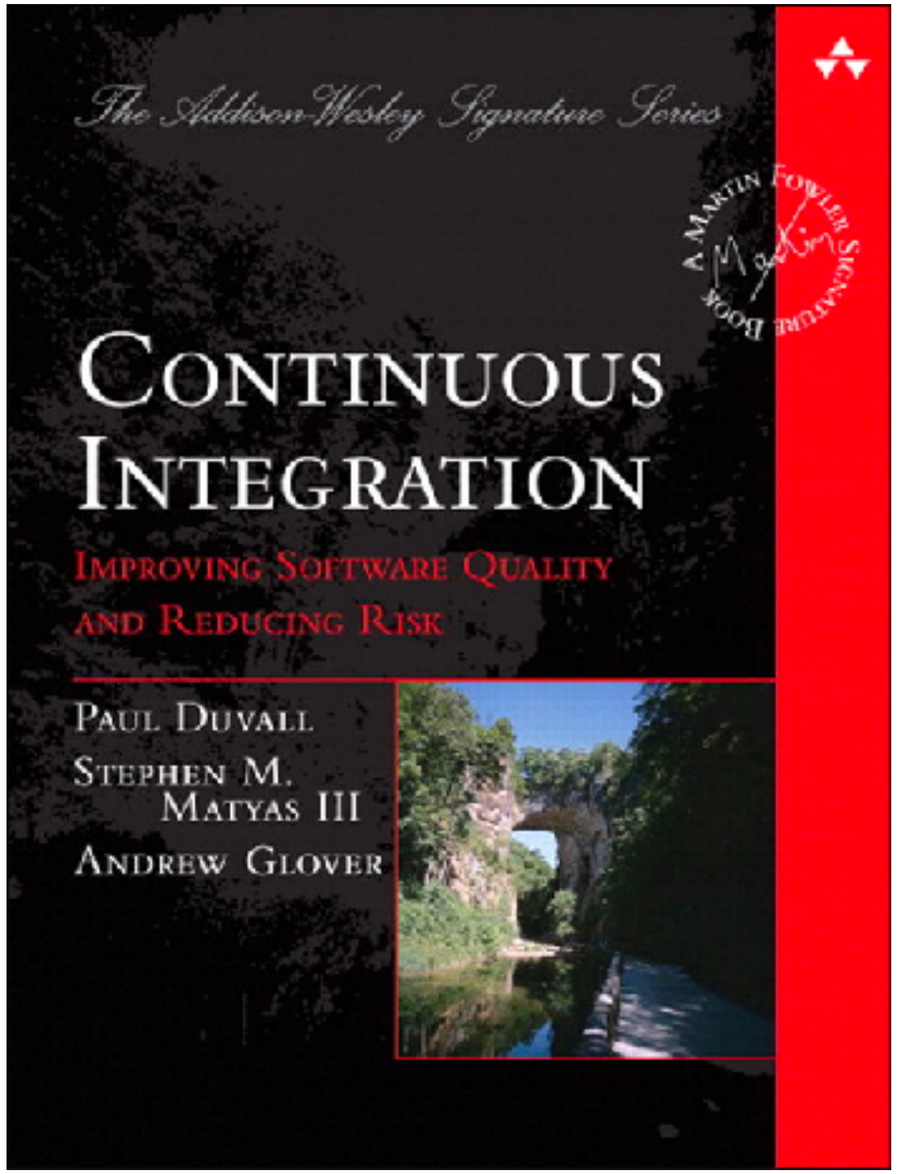
A part of **DevOps**

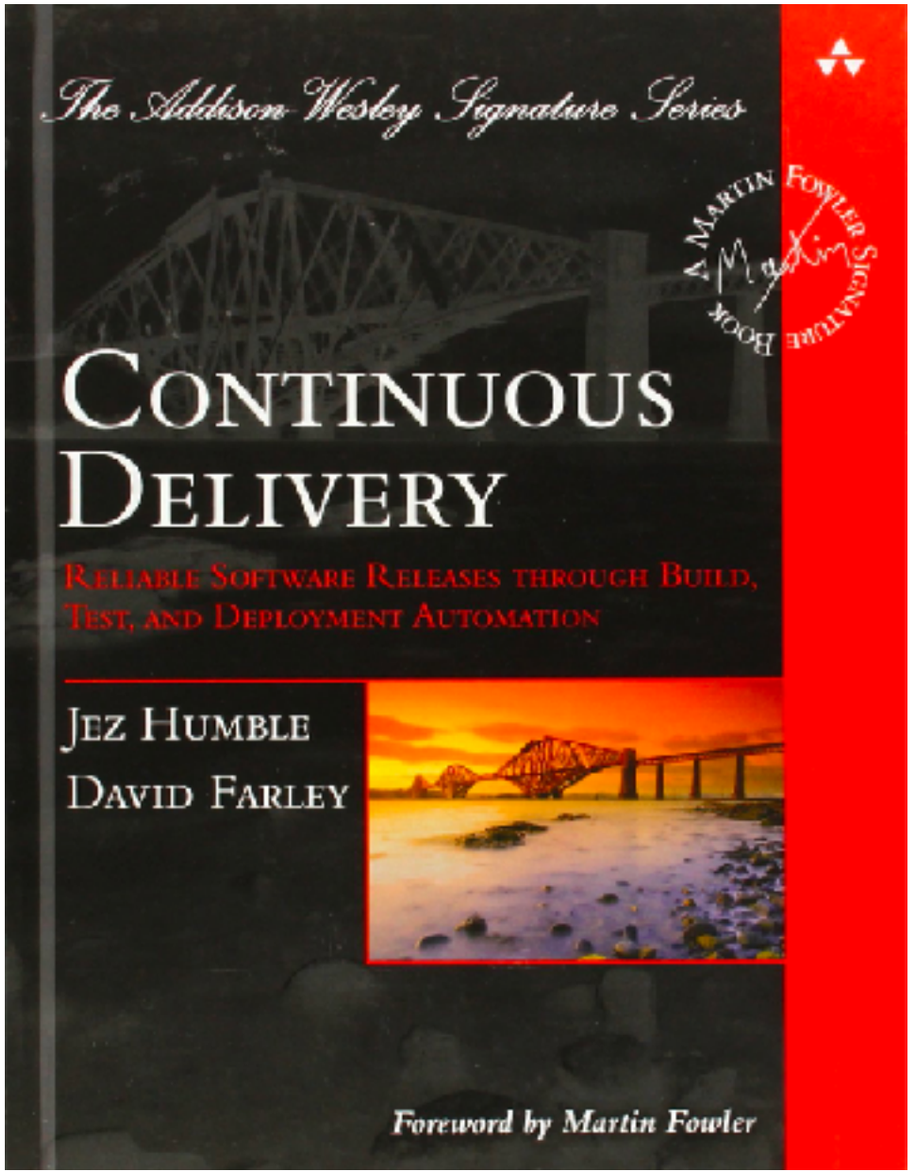
Set of practices for the **rapid**, **frequent** and **reliable delivery** software



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Continuous Delivery/Deployment





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Continuous Delivery/Deployment

Testability
Deployability
Autonomous team and loose coupling



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Benefits of Continuous Delivery

Reduce time to market

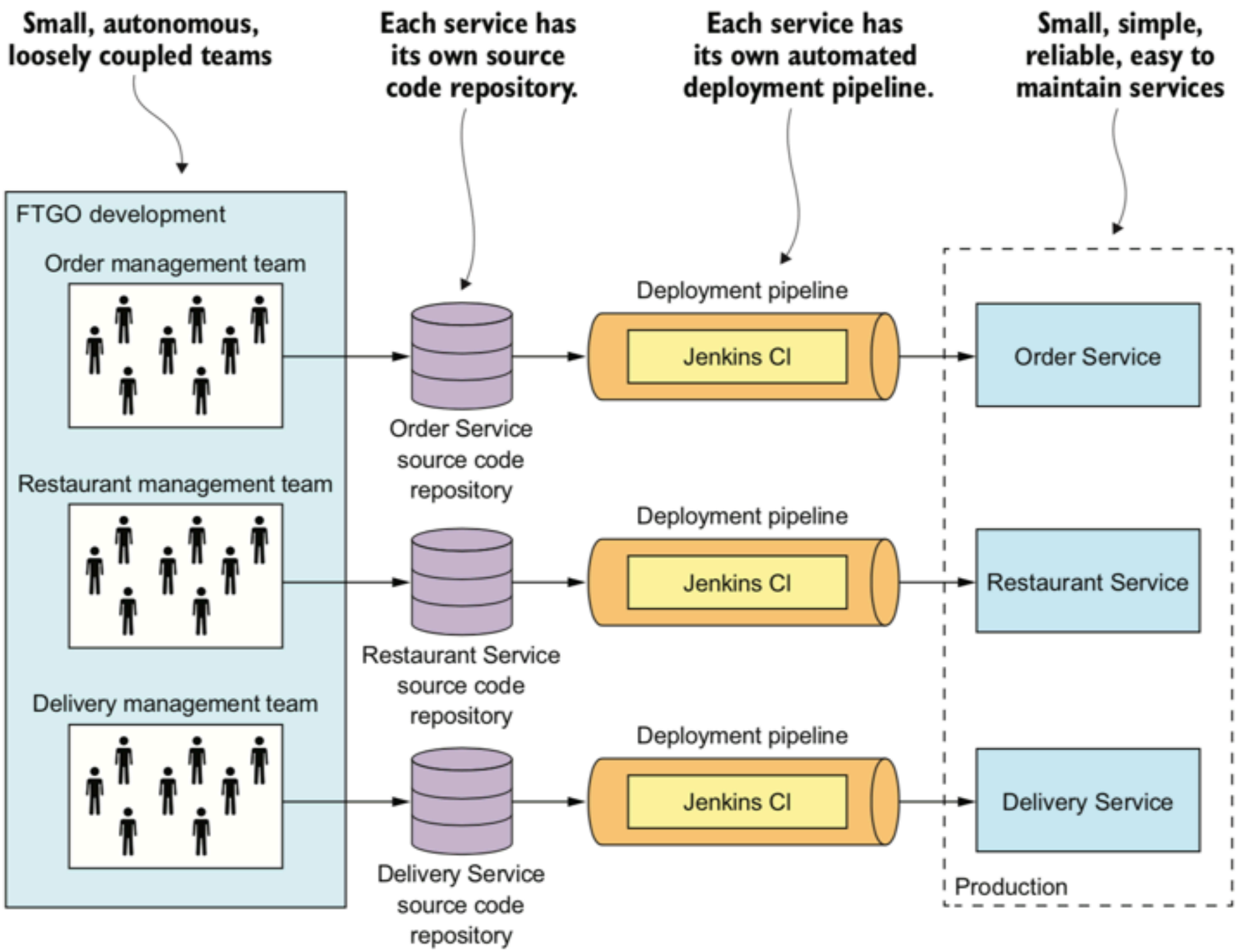
Enable **business** to provide reliable service

Employee satisfaction is higher



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Continuous Delivery for service



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Drawbacks of Microservice



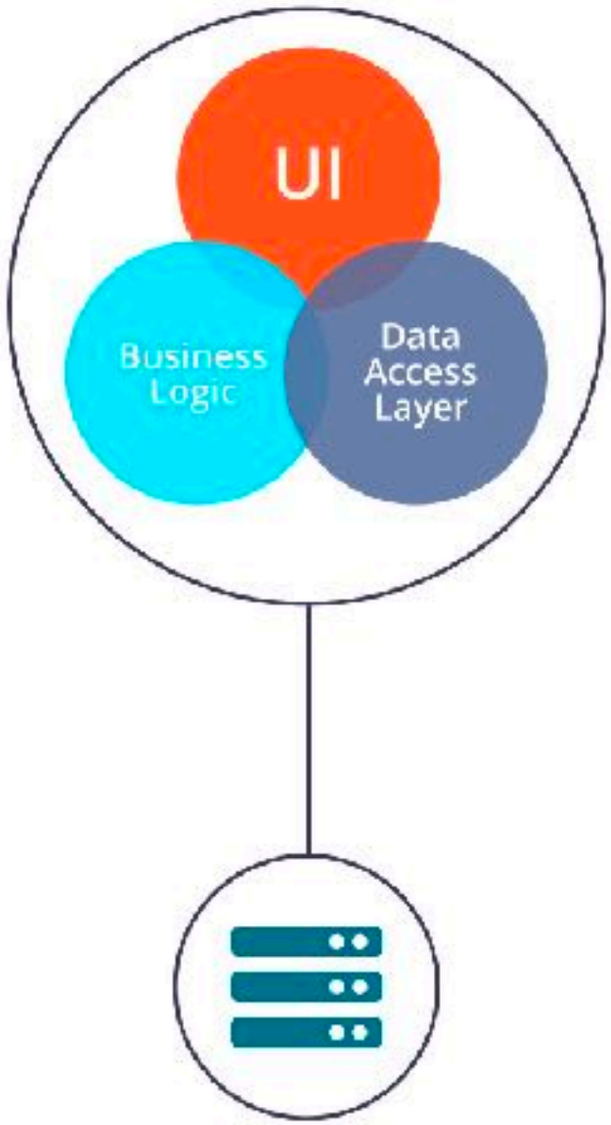
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Drawbacks of Microservice

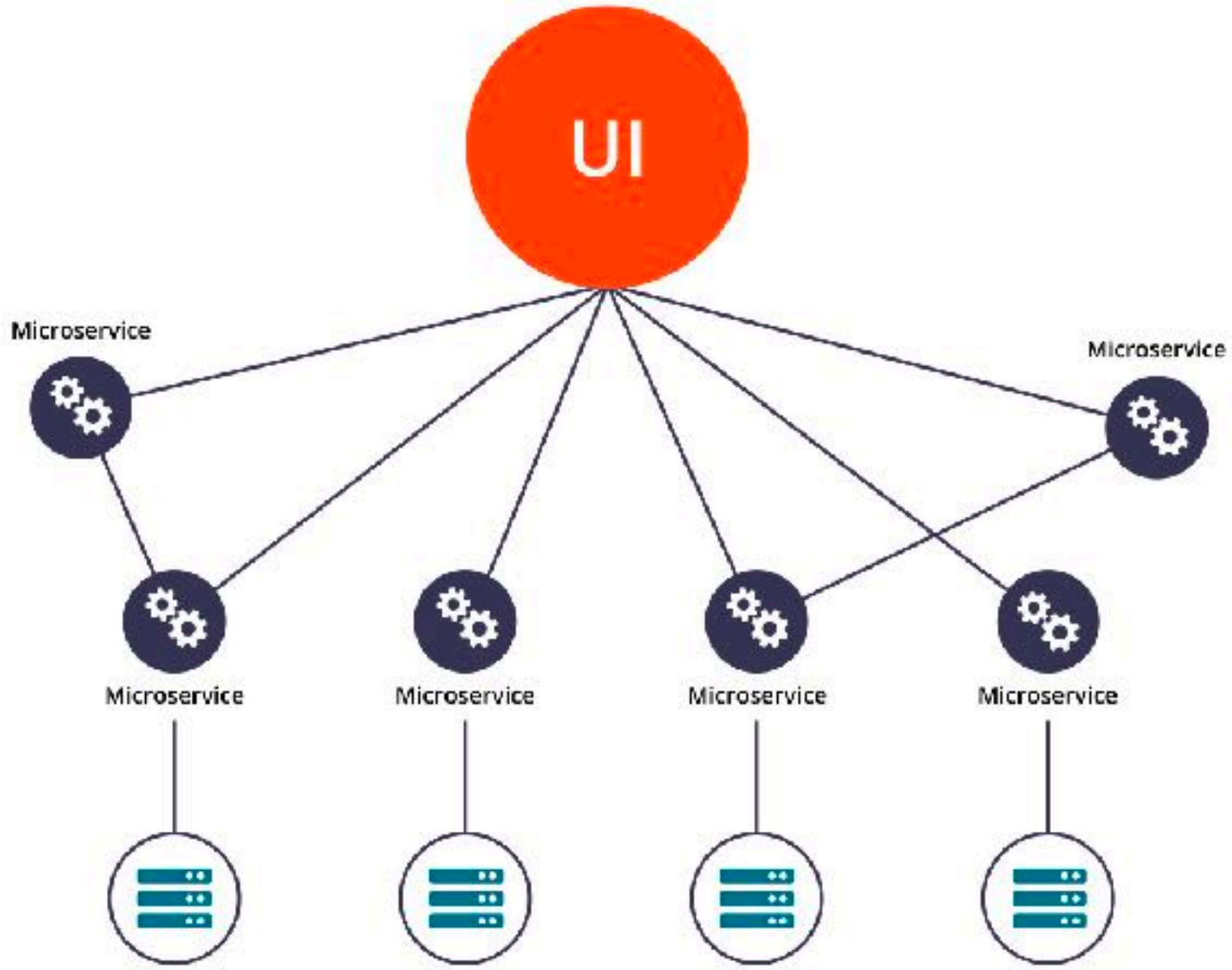
Find the right set of services
Distributed systems are complex
How to develop, testing and deploy ?



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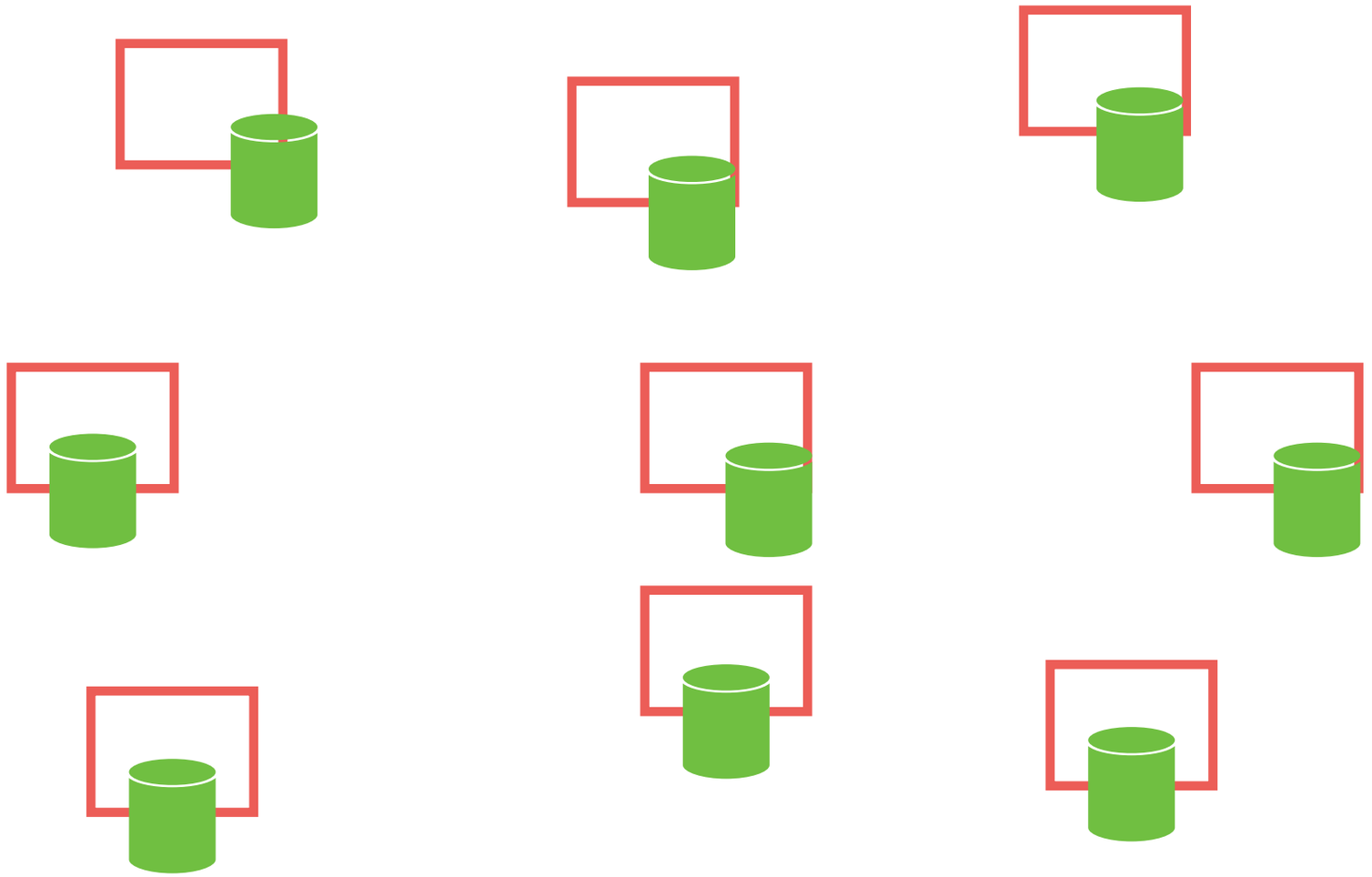
Monolithic Architecture



Microservice Architecture

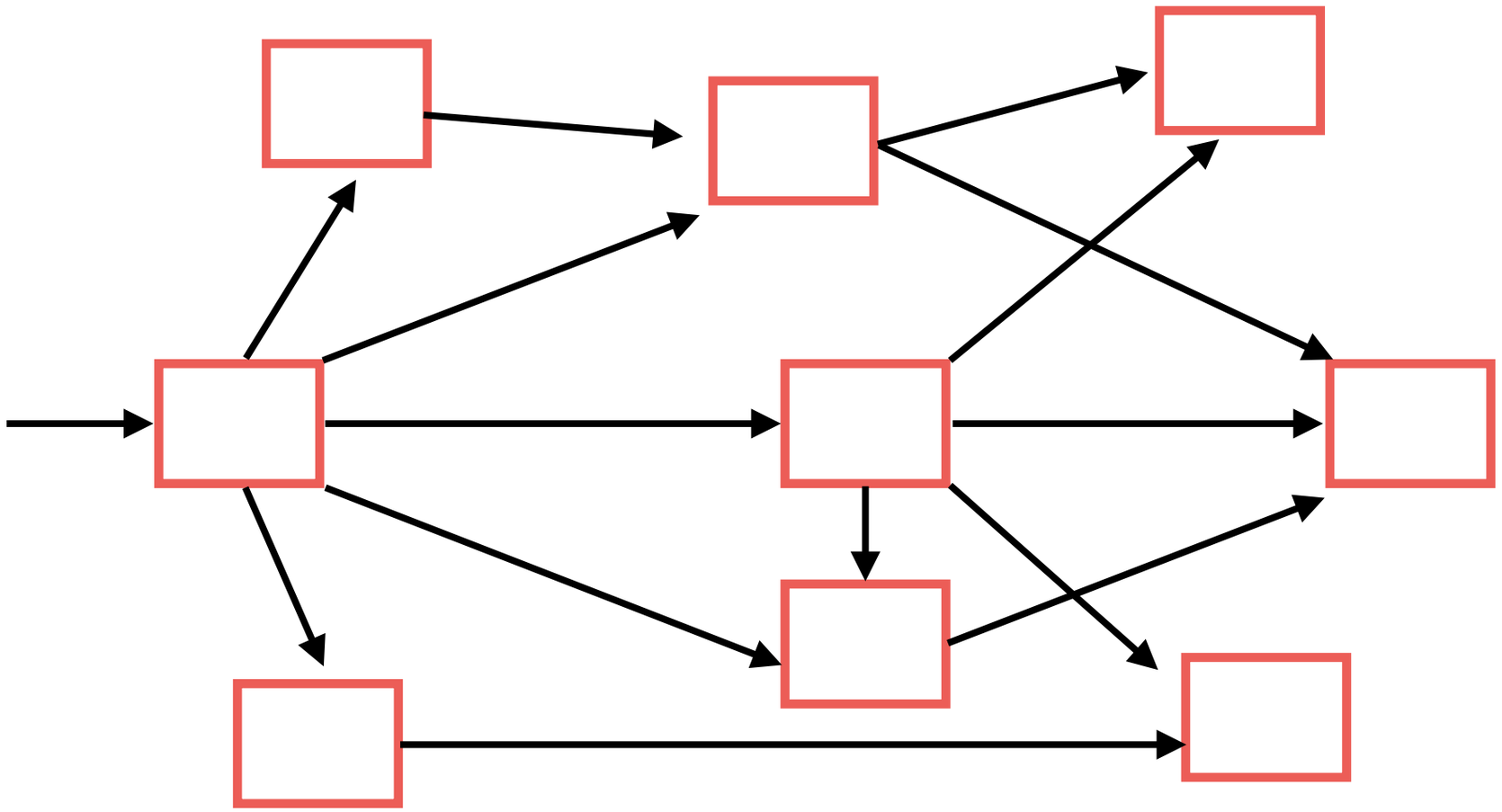
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Microservice (Loosely coupled)



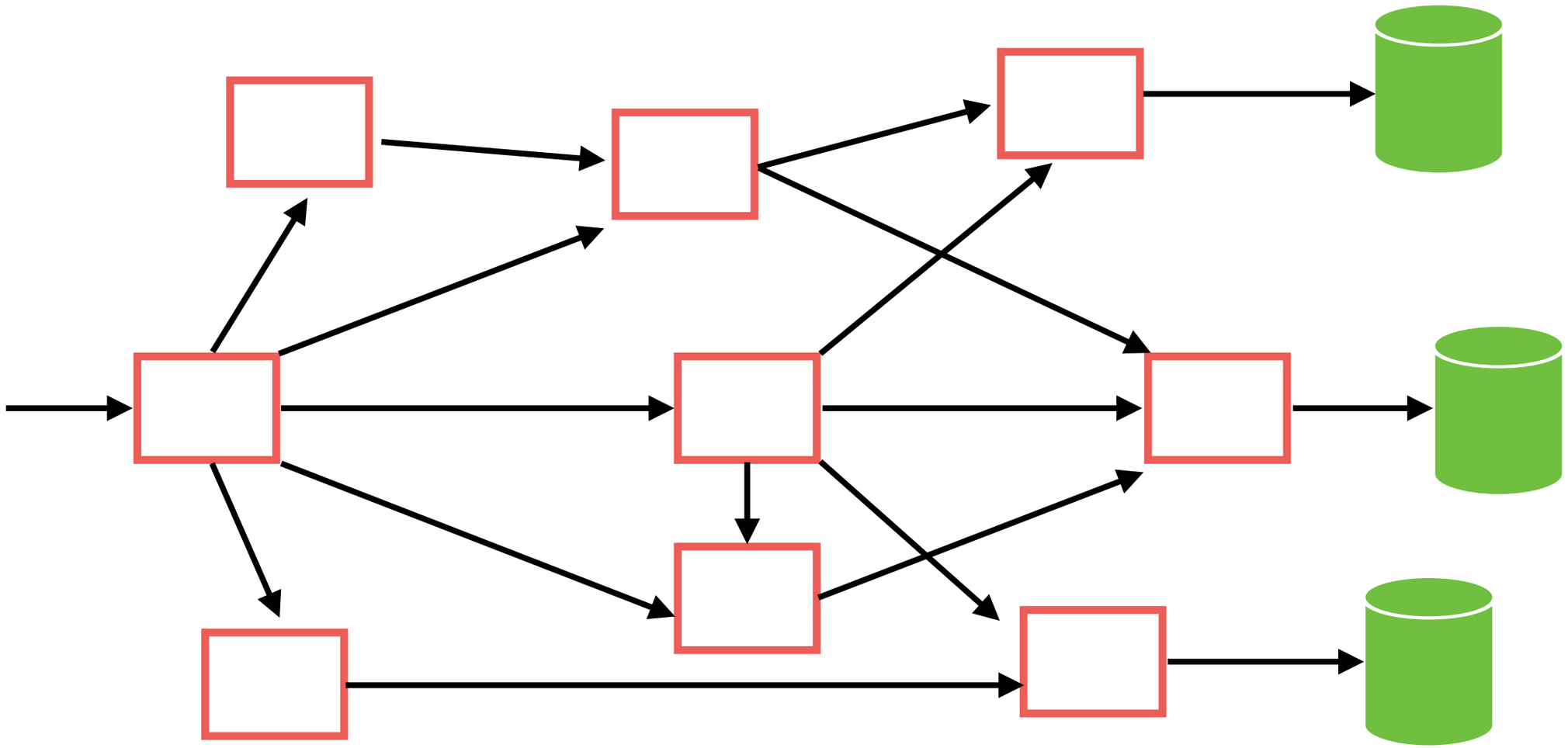
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Microservice (Tightly coupled)



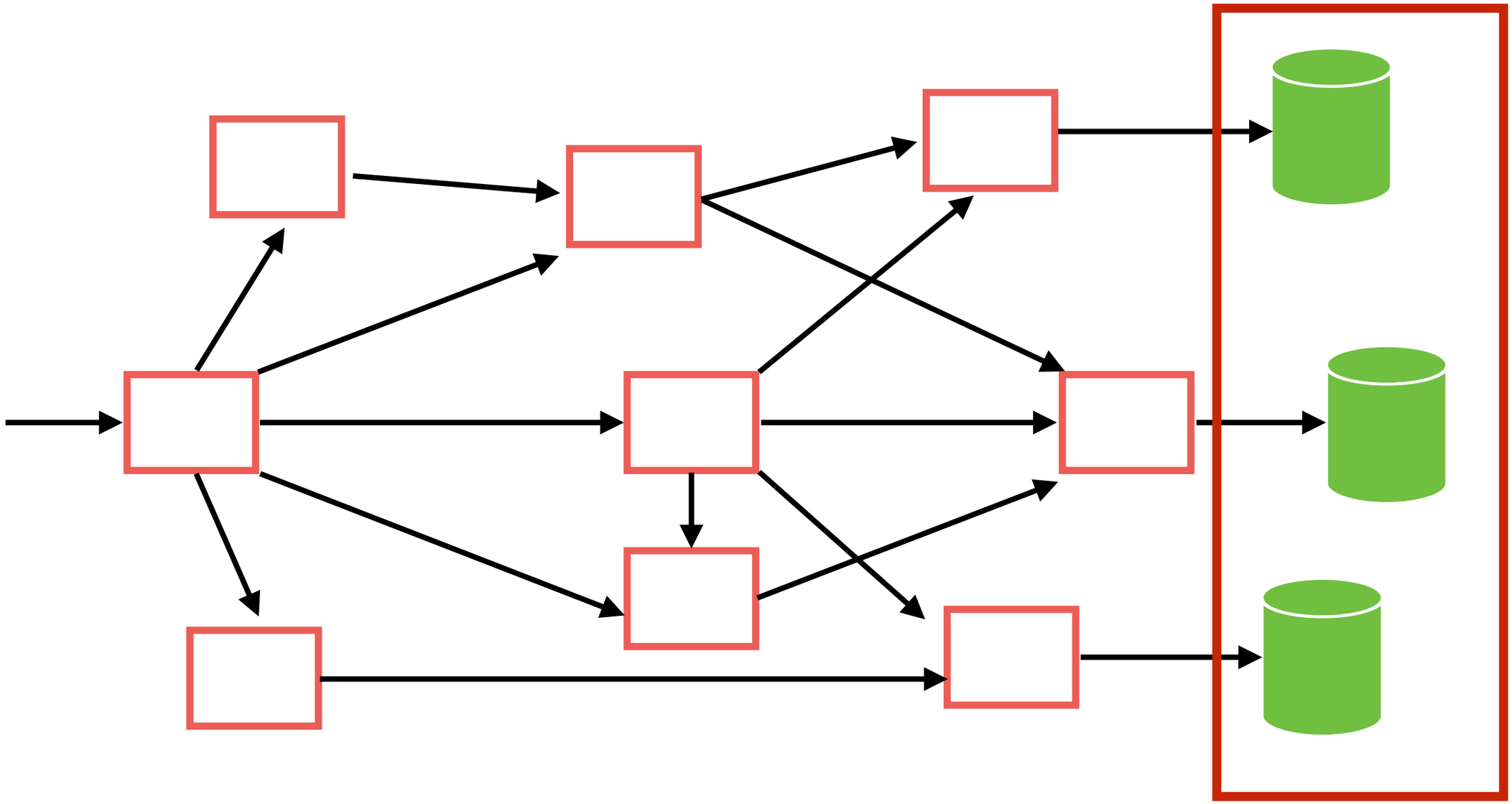
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Own data



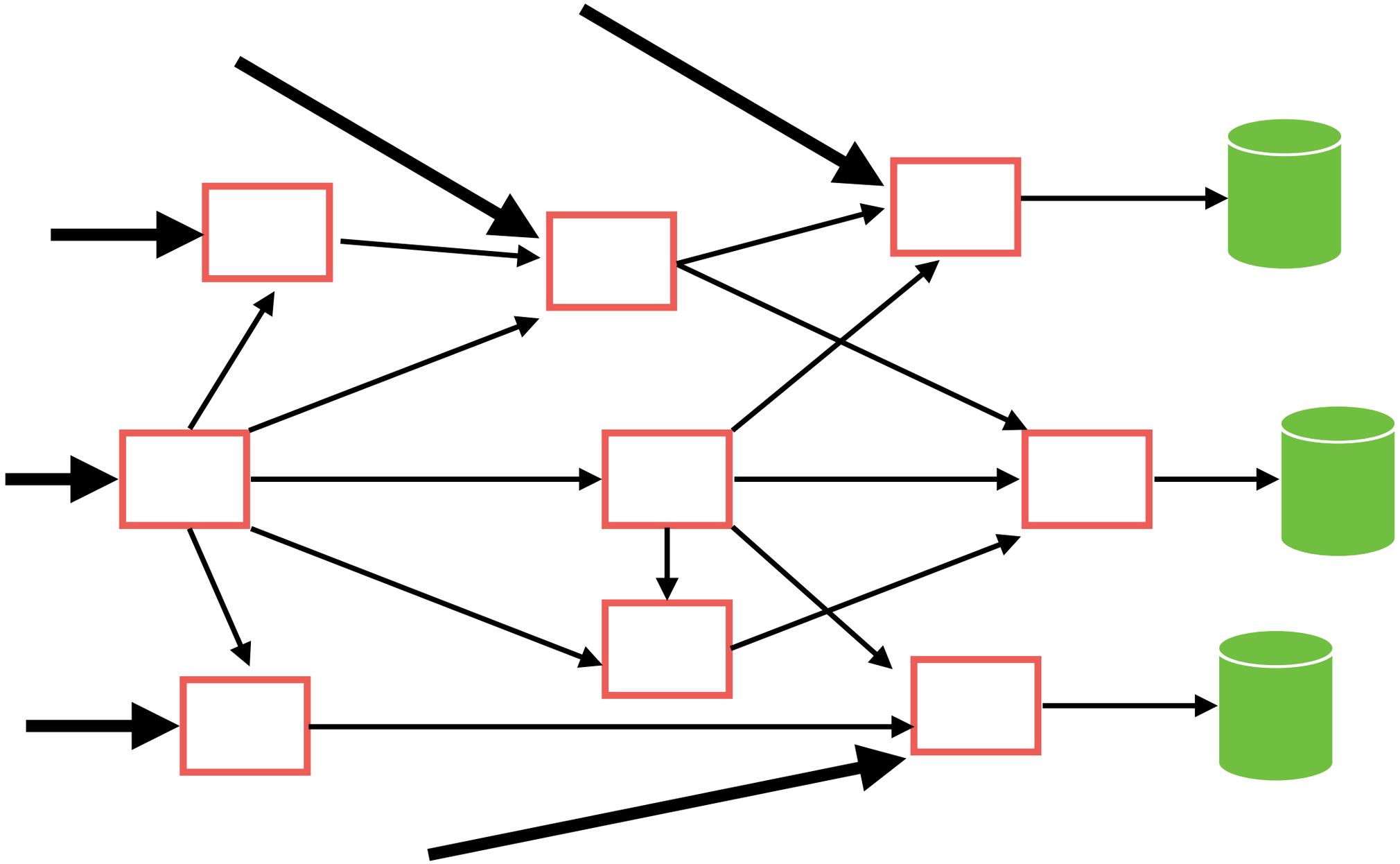
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Data consistency !!



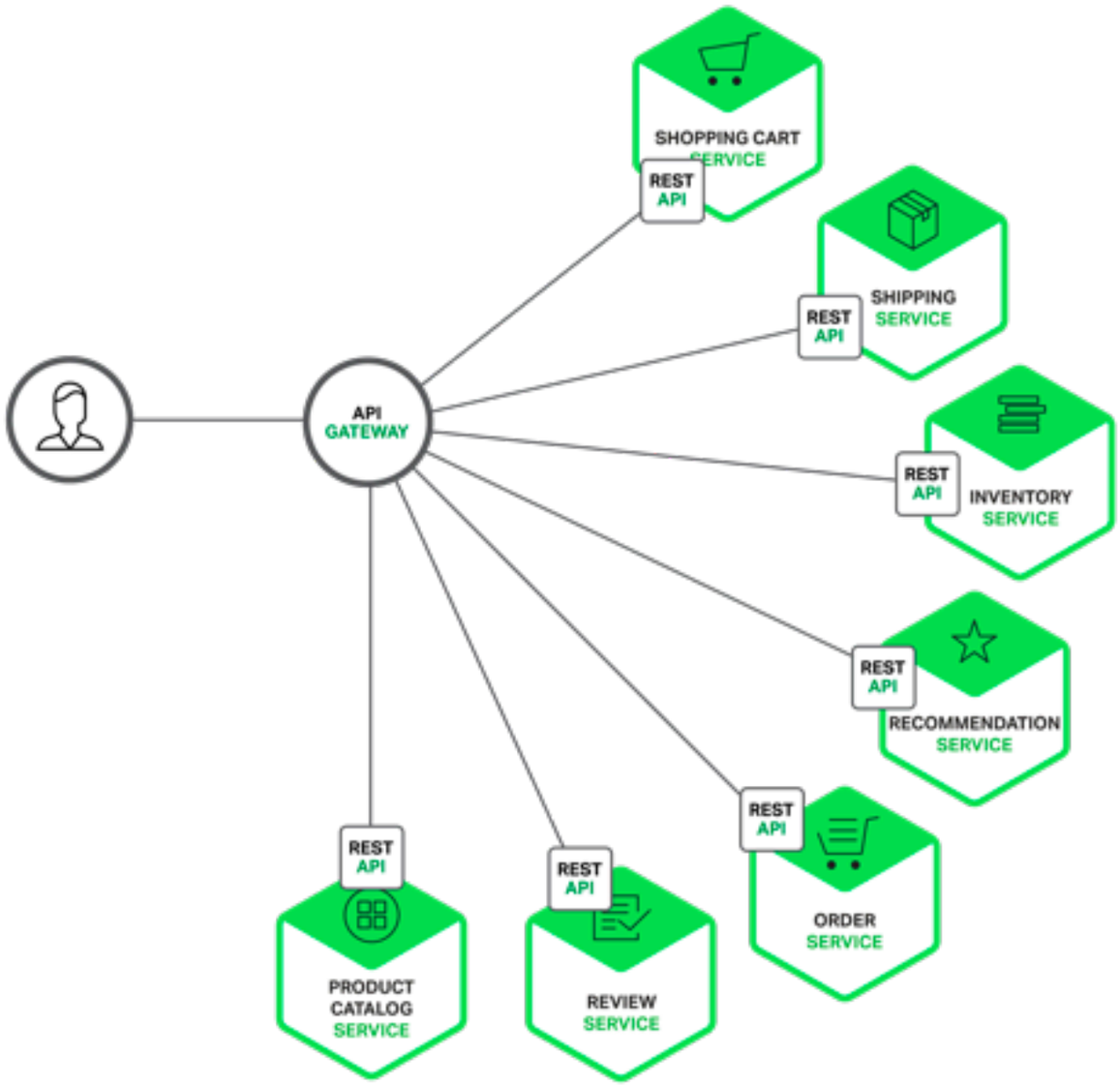
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Multiple entry points



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Working with API gateway



https://www.nginx.com

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Features of API gateway

Authentication
Authorization
Rate limiting
Caching
Metrics collection
Request logging
Load balancing
Circuit breaking



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Benefits

Encapsulation interface structure
Client talk to service via gateway
Reduce round trip between service



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Drawbacks

Development/Performance bottleneck

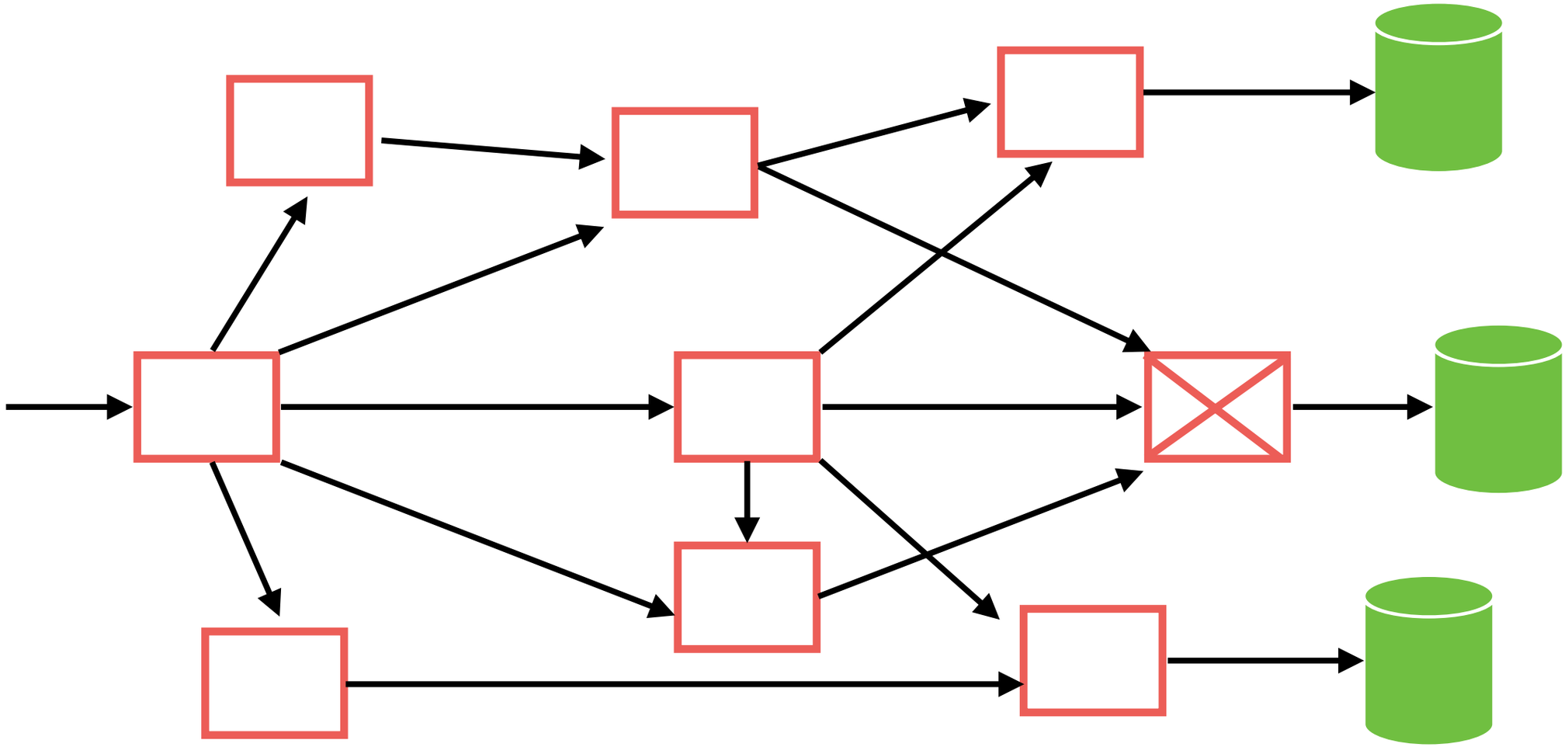
Single point of failure

Waiting for update data in gateway



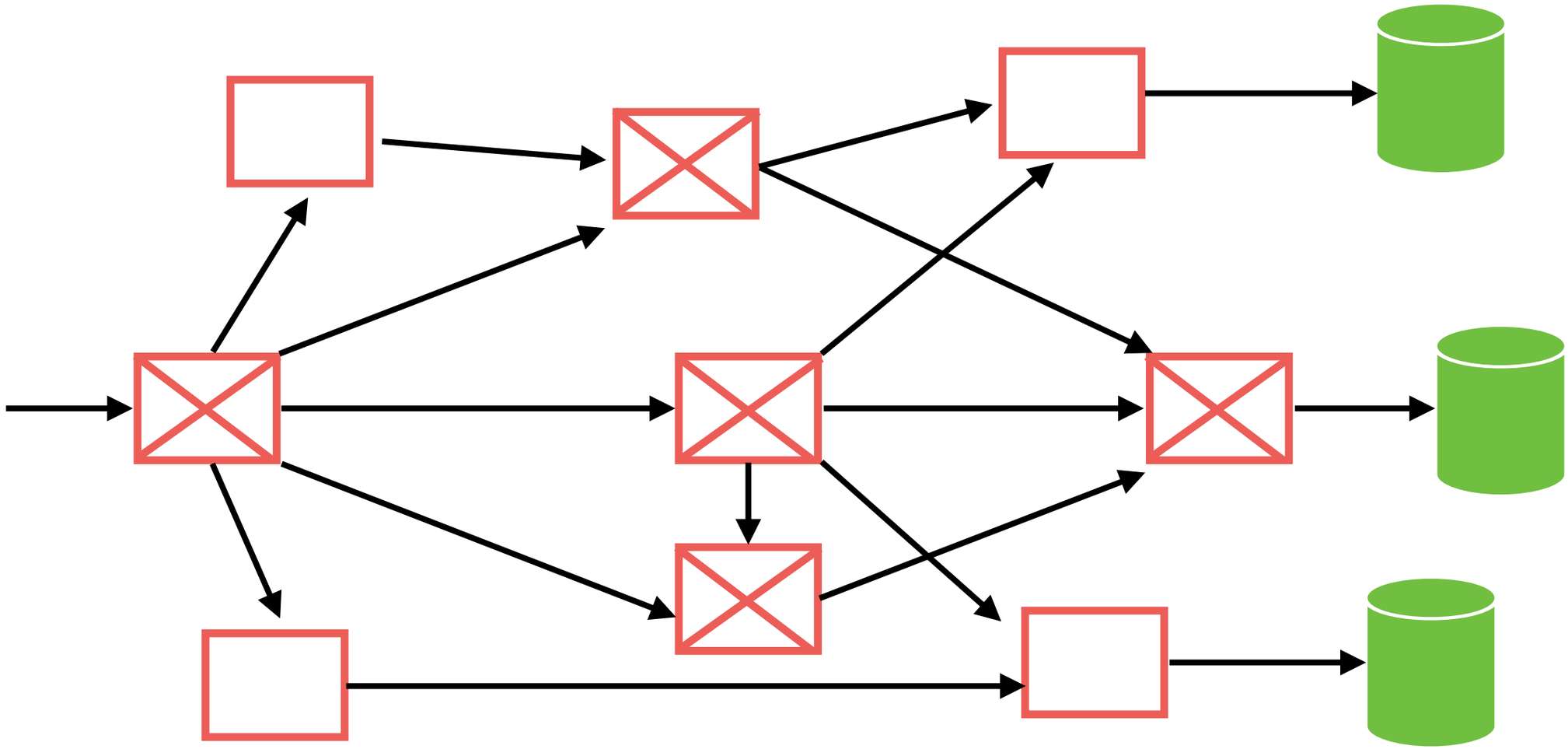
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Failure !!



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Cascading Failure !!



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Solutions for Resilience ?

Timeout
Retry
Rate Limit
Queue messaging
Circuit breaker
Bulkhead
Auto-scale

https://learn.microsoft.com/en-us/azure/well-architected/resiliency/reliability-patterns

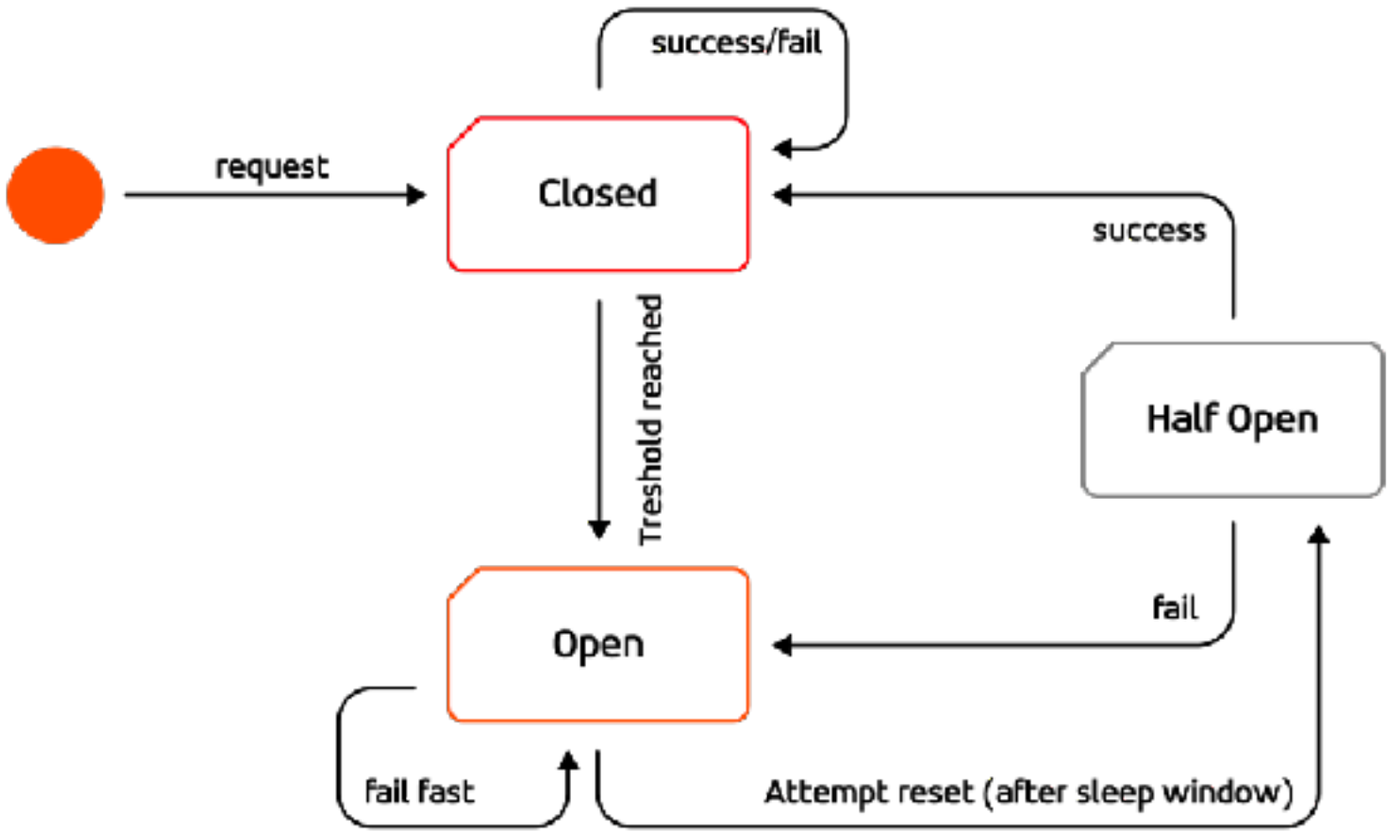


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Circuit Breaker pattern

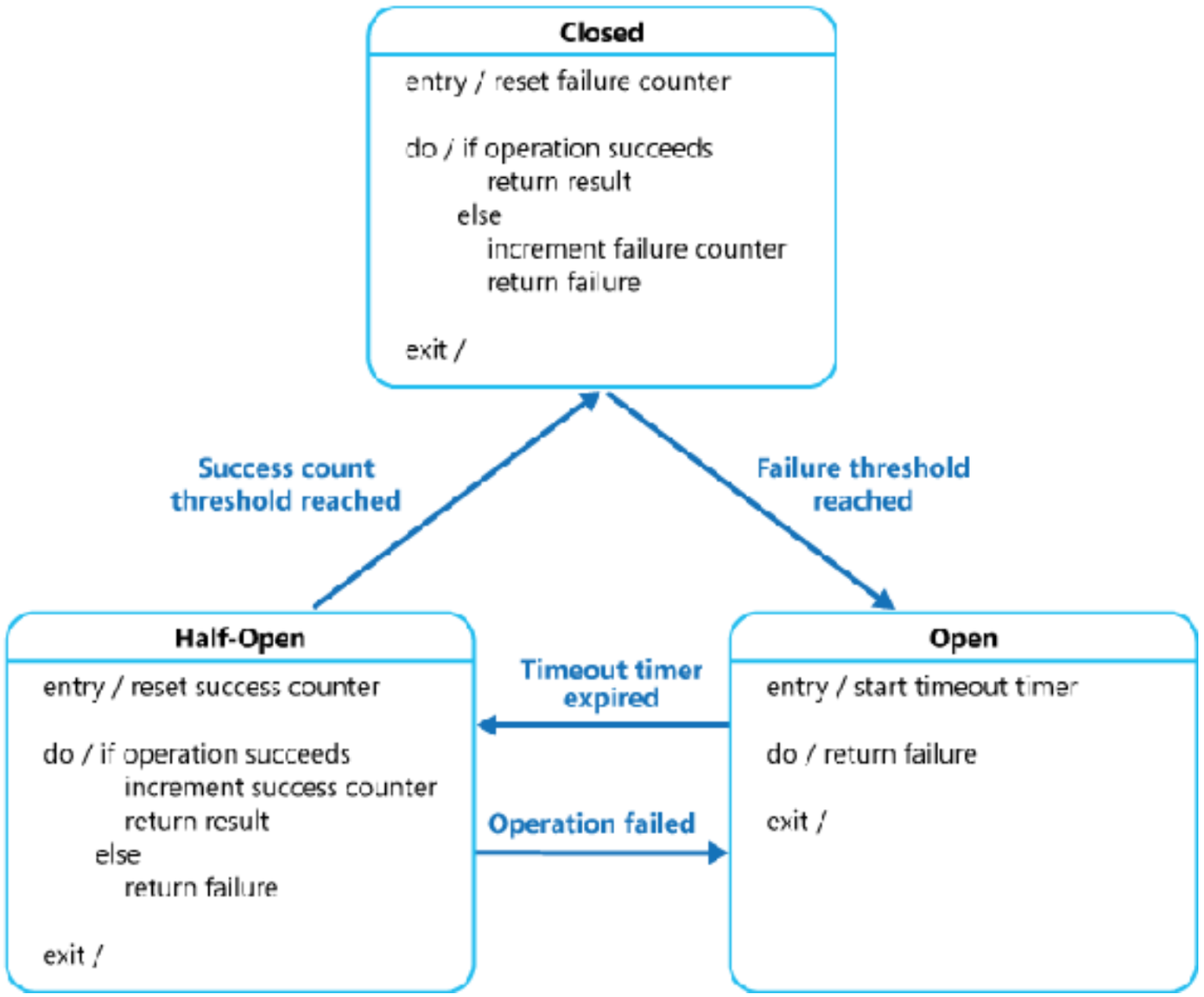
Track the number of success and failure

If error rate exceed some threshold then enable circuits breaker



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Circuit Breaker pattern



https://learn.microsoft.com/en-us/azure/architecture/patterns/circuit-breaker

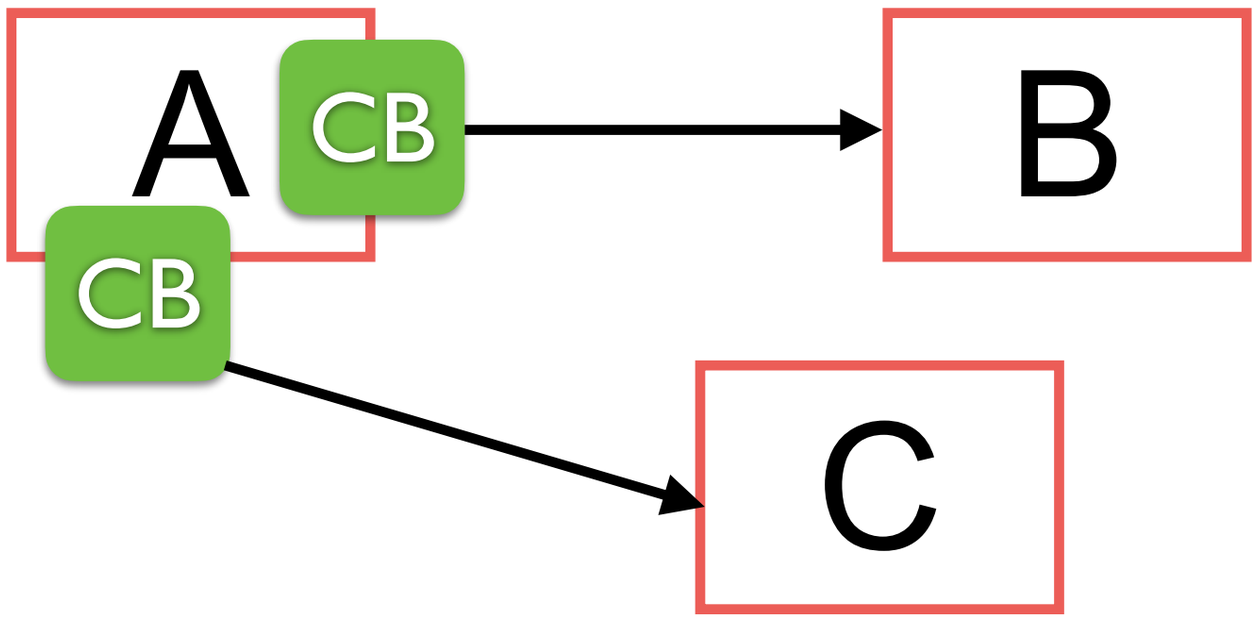


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Circuit Breaker pattern

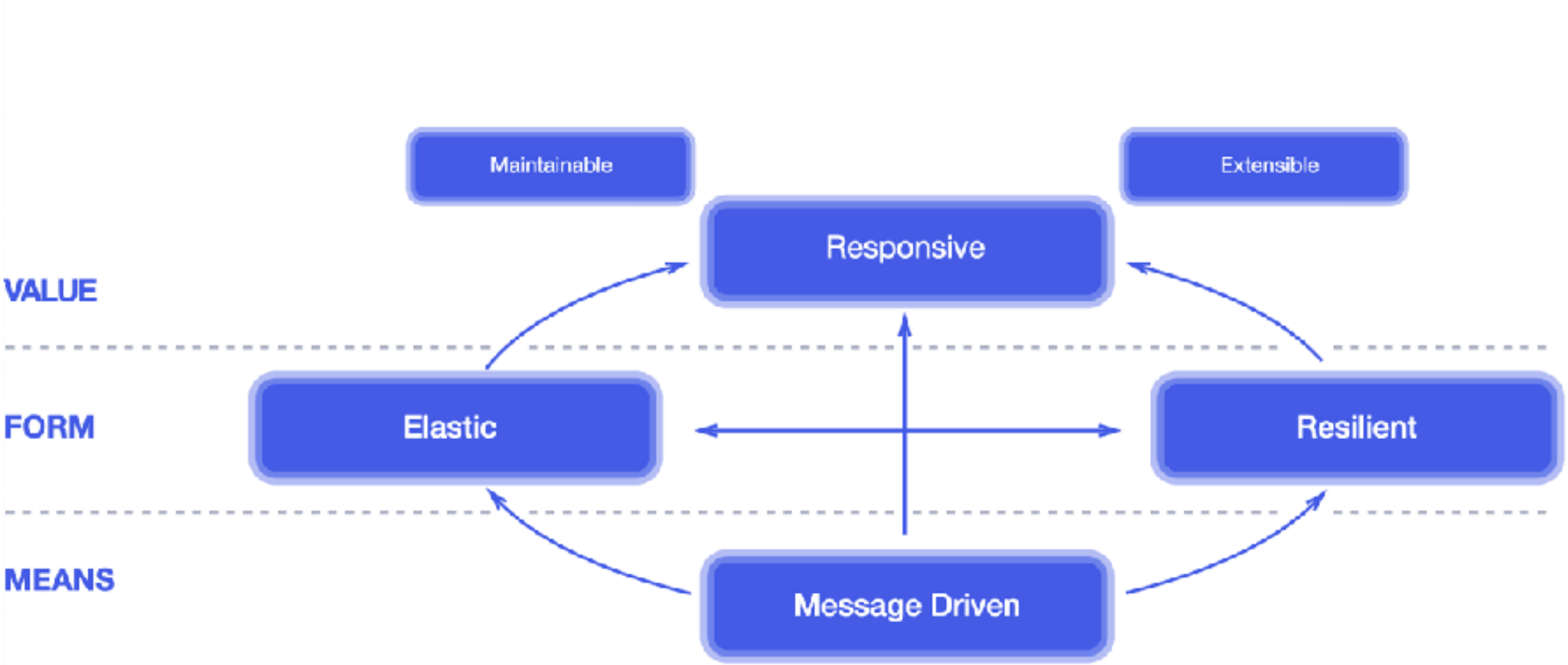
How to handling partial failure ?





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Reactive Architecture



https://www.reactivemanifesto.org/

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Service Principles



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Service Principles

Creation

Interface

Testing

Operation

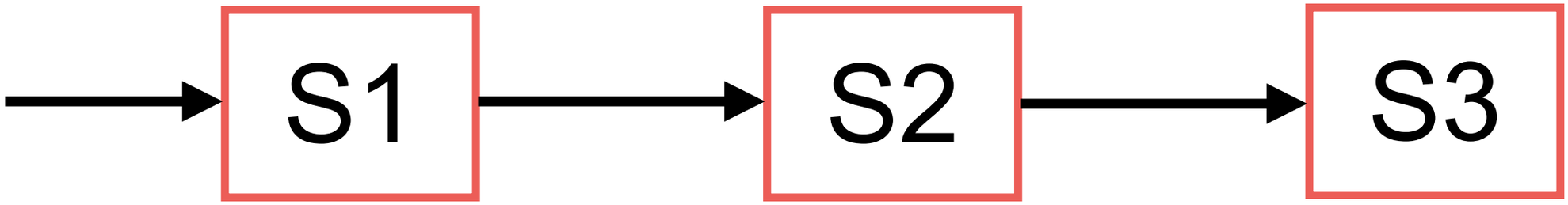
https://github.com/Yelp/service-principles



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Drawbacks of Microservice

Deploy feature that required multiple service ?

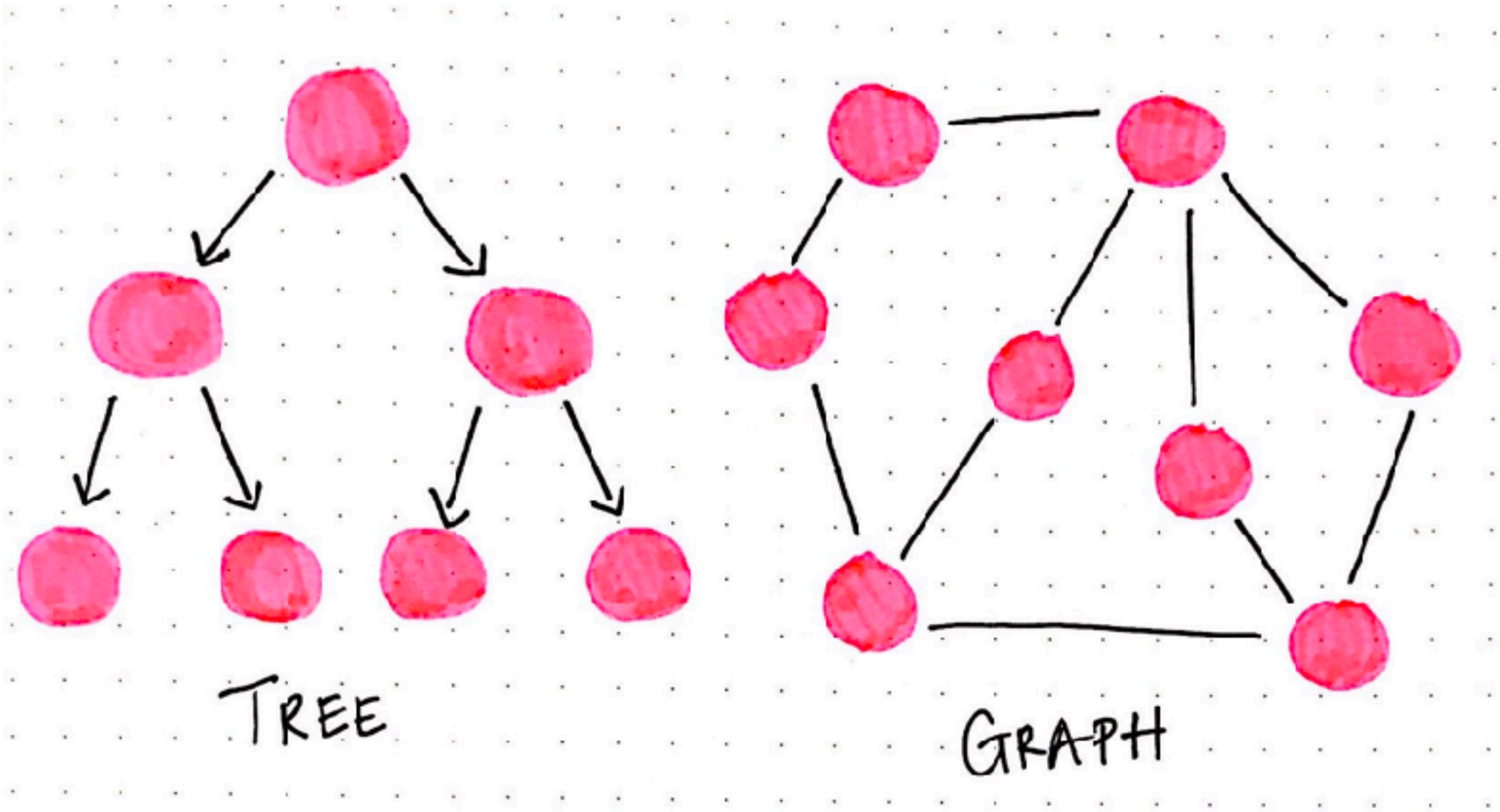




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Service Principles

Minimize the depth of the service call-graph



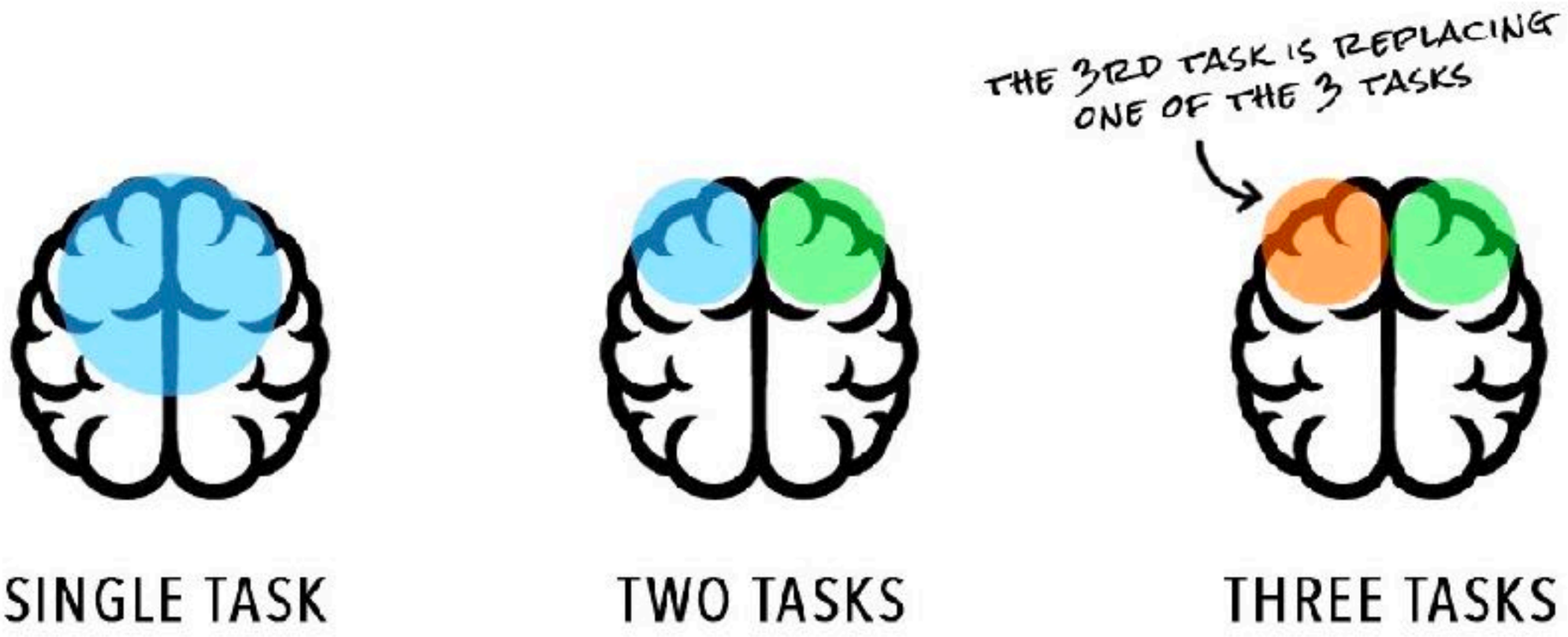
https://github.com/Yelp/service-principles



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Service Principles

Minimize the number of services owned by your team



https://github.com/Yelp/service-principles

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Service Interfaces

APIs => HTTP, Messaging

Backward compatibility

Versioning

Design standard

JSON Schema

Kafka Schema Registry



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Testing strategies for Services

Unit testing
Integration testing
Component testing
Contract testing
End-to-End testing
Chaos testing

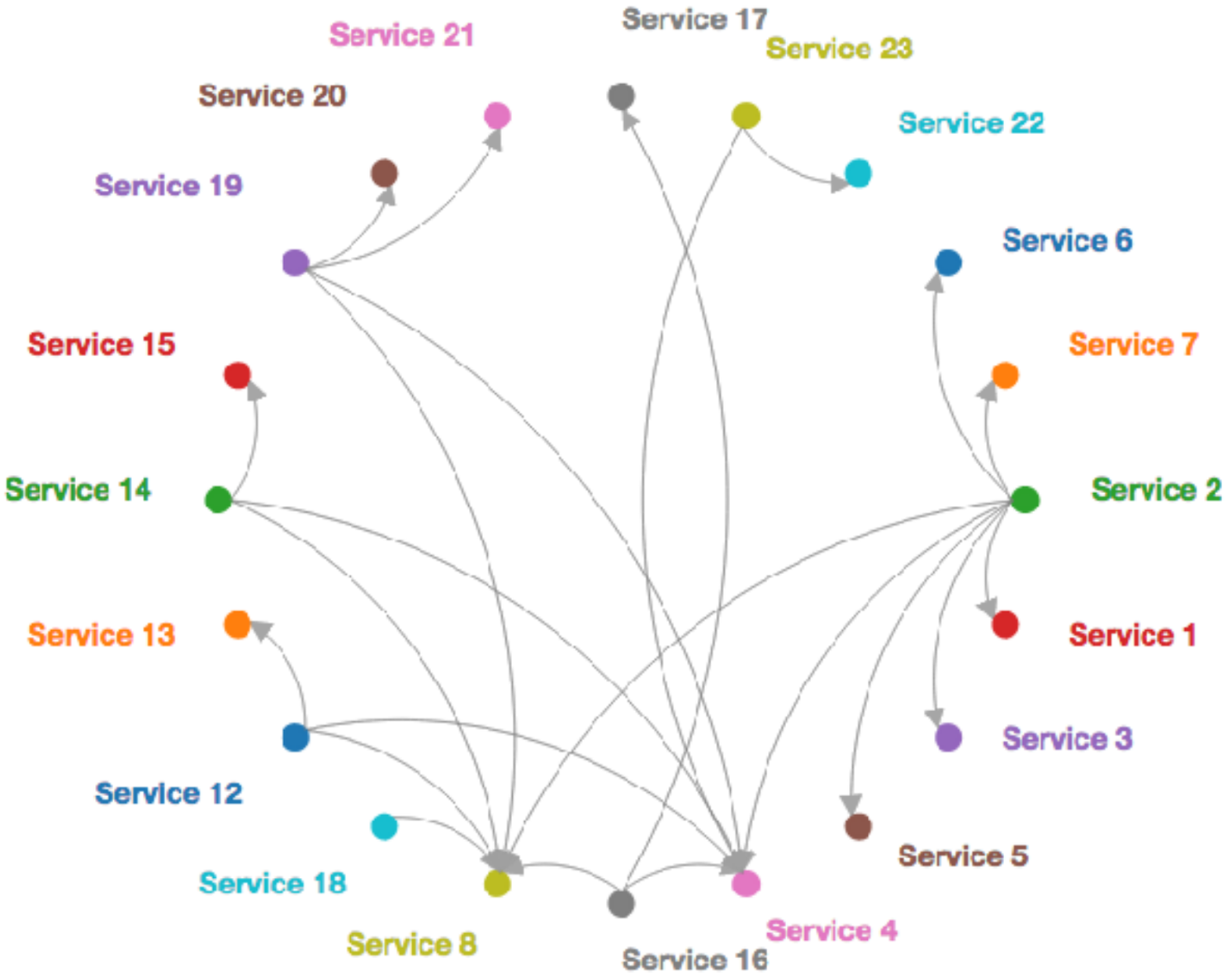
Performance testing

Security testing



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Testing strategies for Services



https://docs.pact.io/pact_broker

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Operation for Services

How to deploy ?
Observable services
Plan for failures !!



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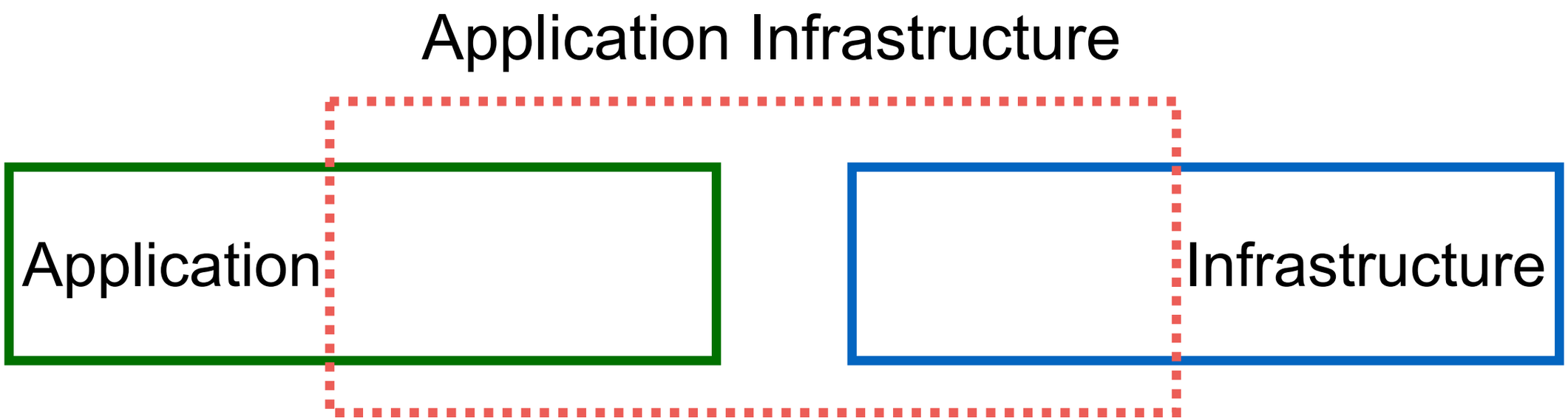
Microservice architecture patterns



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3 Layers of service patterns

Application patterns
Application infrastructure pattern
Infrastructure pattern



https://microservices.io/

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Decompose application into small services !!



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Premature splitting is the root of all **evil**.



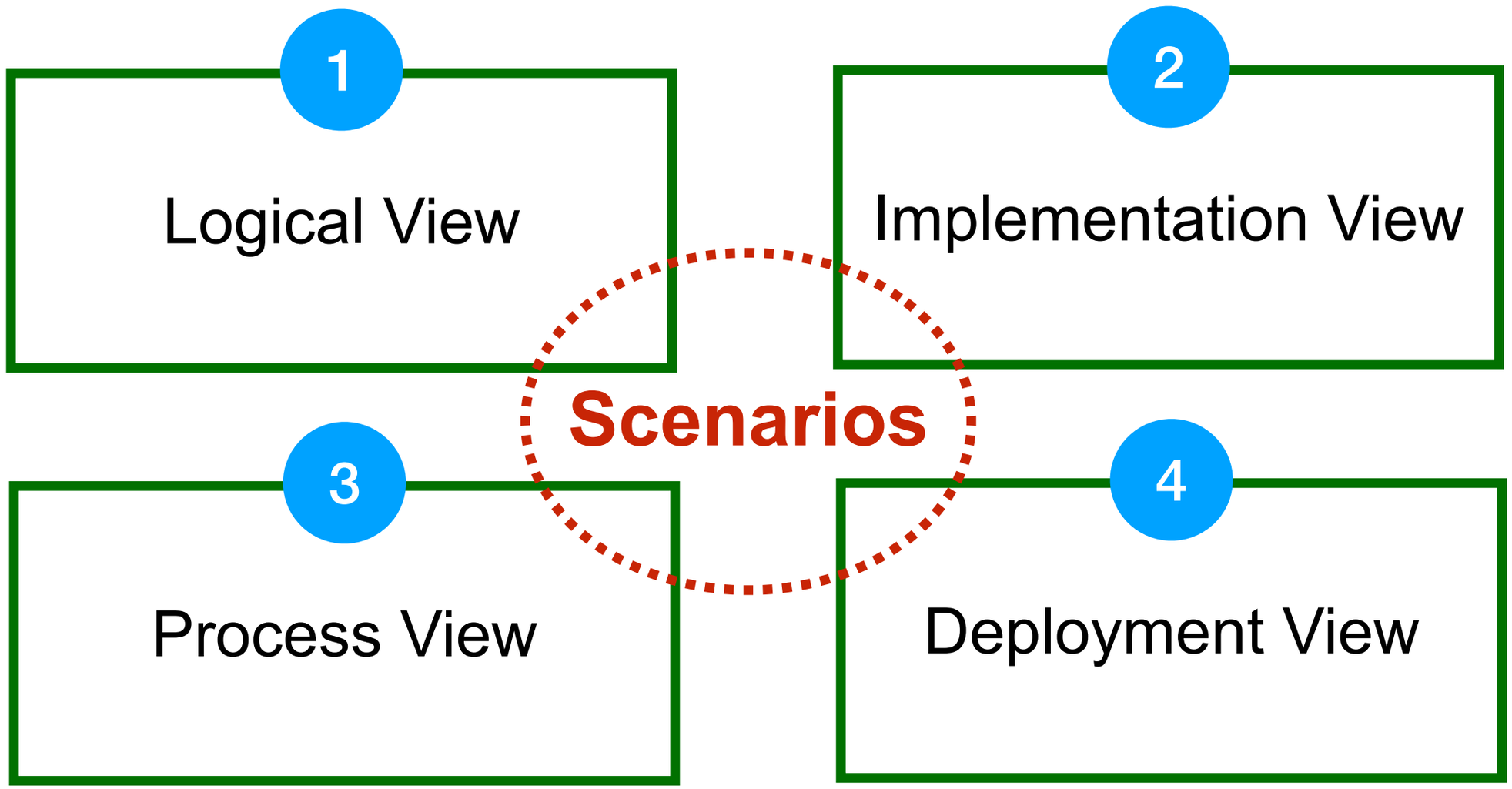
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Every time you make the decision to split out a new microservice, there's a **risk** of ending up with a bloated app.



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4 View model of Software Architecture



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Decompose application into services

By business capability ?
By subdomain/technical ?
By team ?



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Step to define services

1. Identify system operations
2. Identify services
3. Define service APIs and collaboration



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1. Identify system operations

Start with functional requirements

User story

1

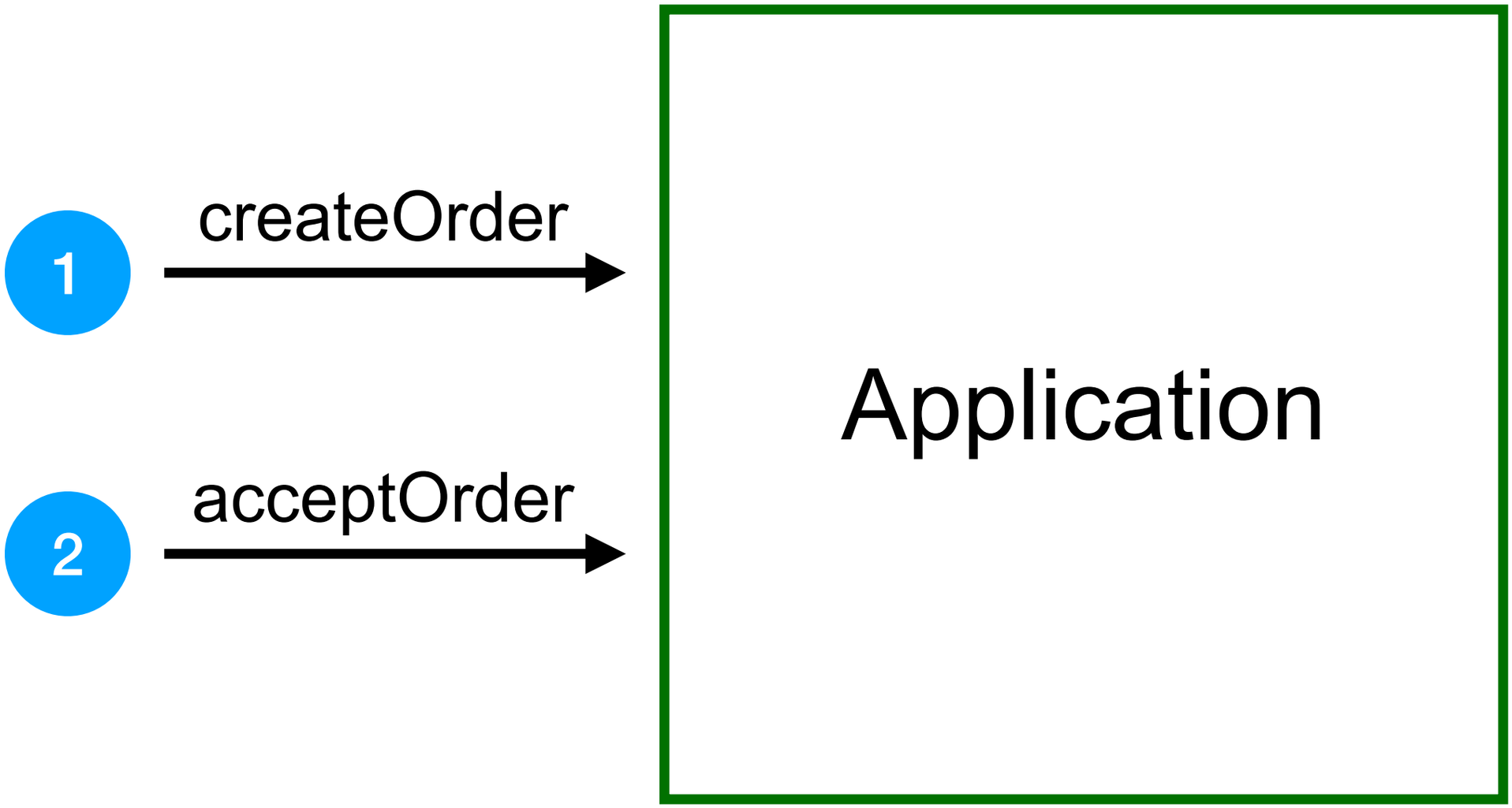
ลูกค้าต้องการสั่งซื้ออาหาร

2

ร้านอาหารรับคำสั่งซื้อ

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1. Identify system operations



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2. Identify services

Try to decomposition into services

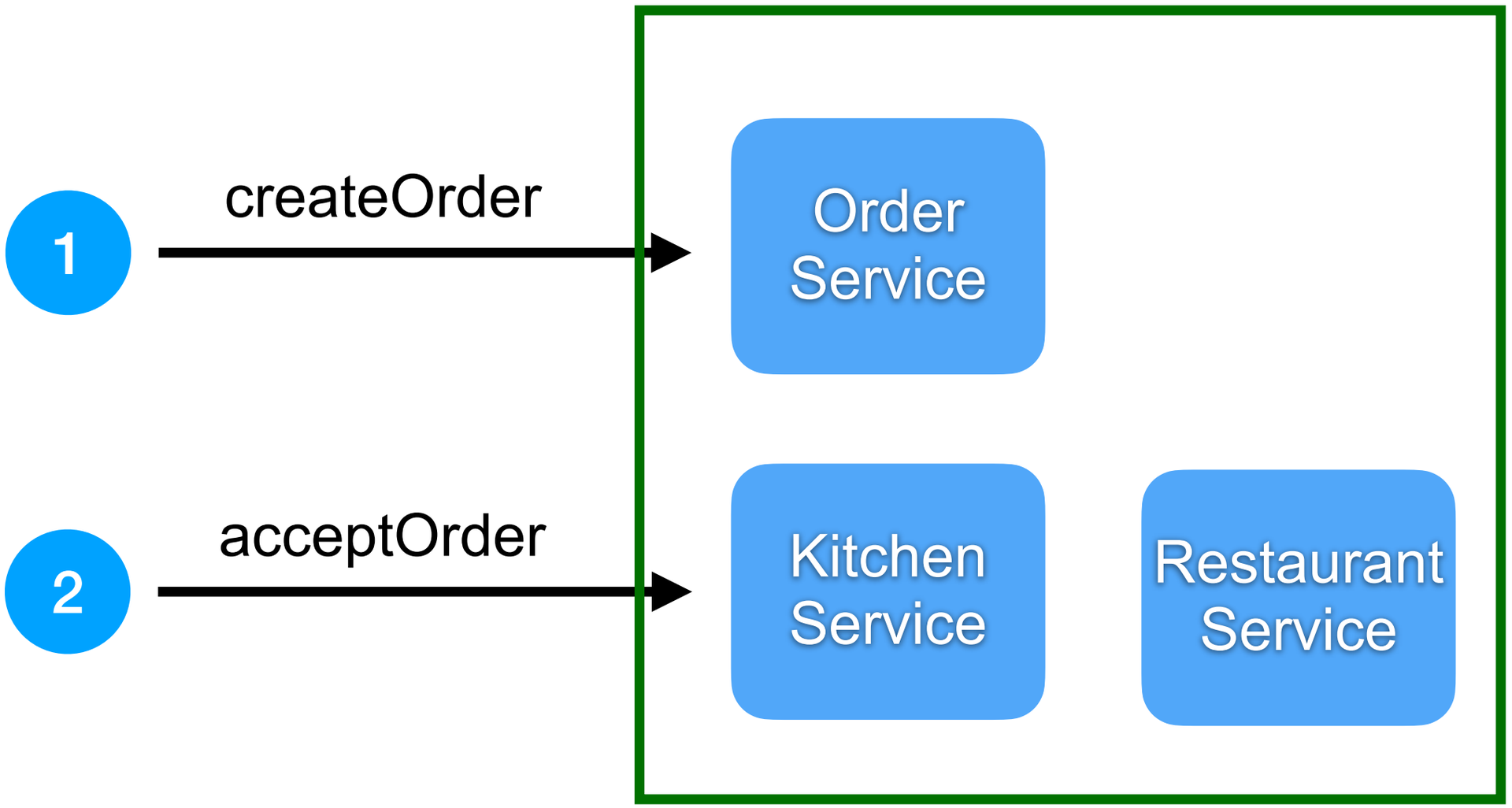
Business > Technical concept

What > How



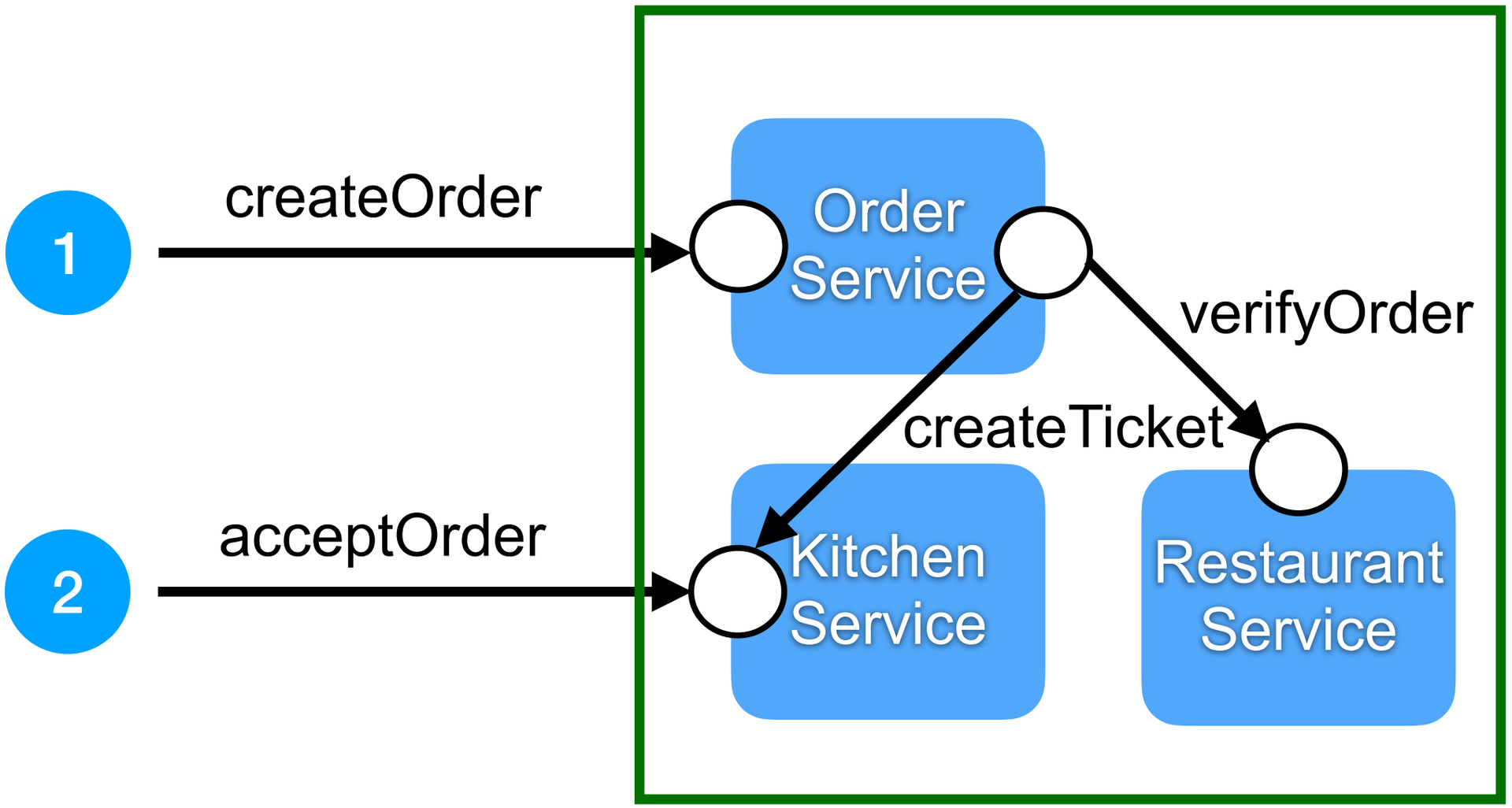
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2. Identify services



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3. Define service APIs and collaborations



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Problems when decompose services ?

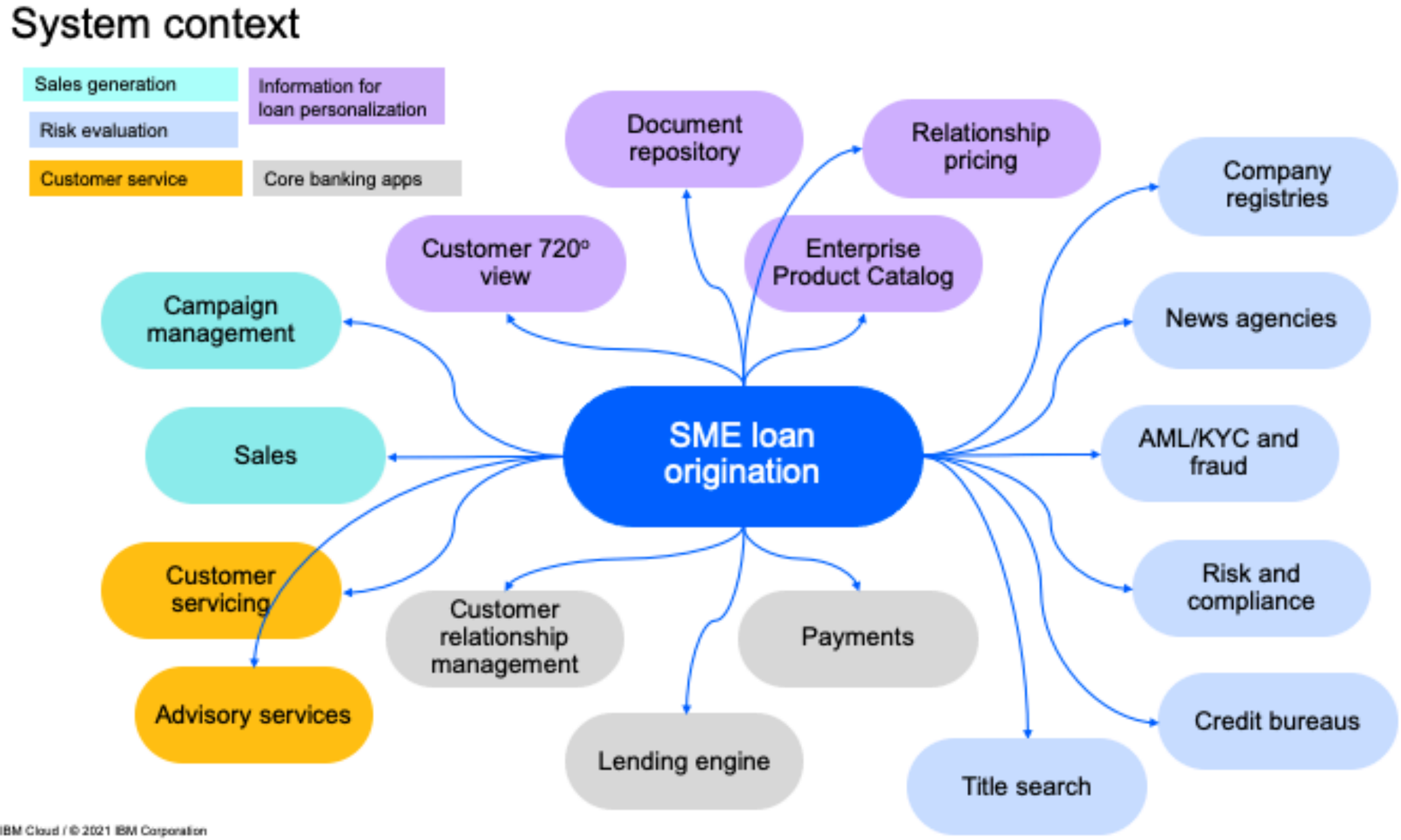
Network latency
Reliability of communication
Maintain data consistency

God services !!
Microlith !!



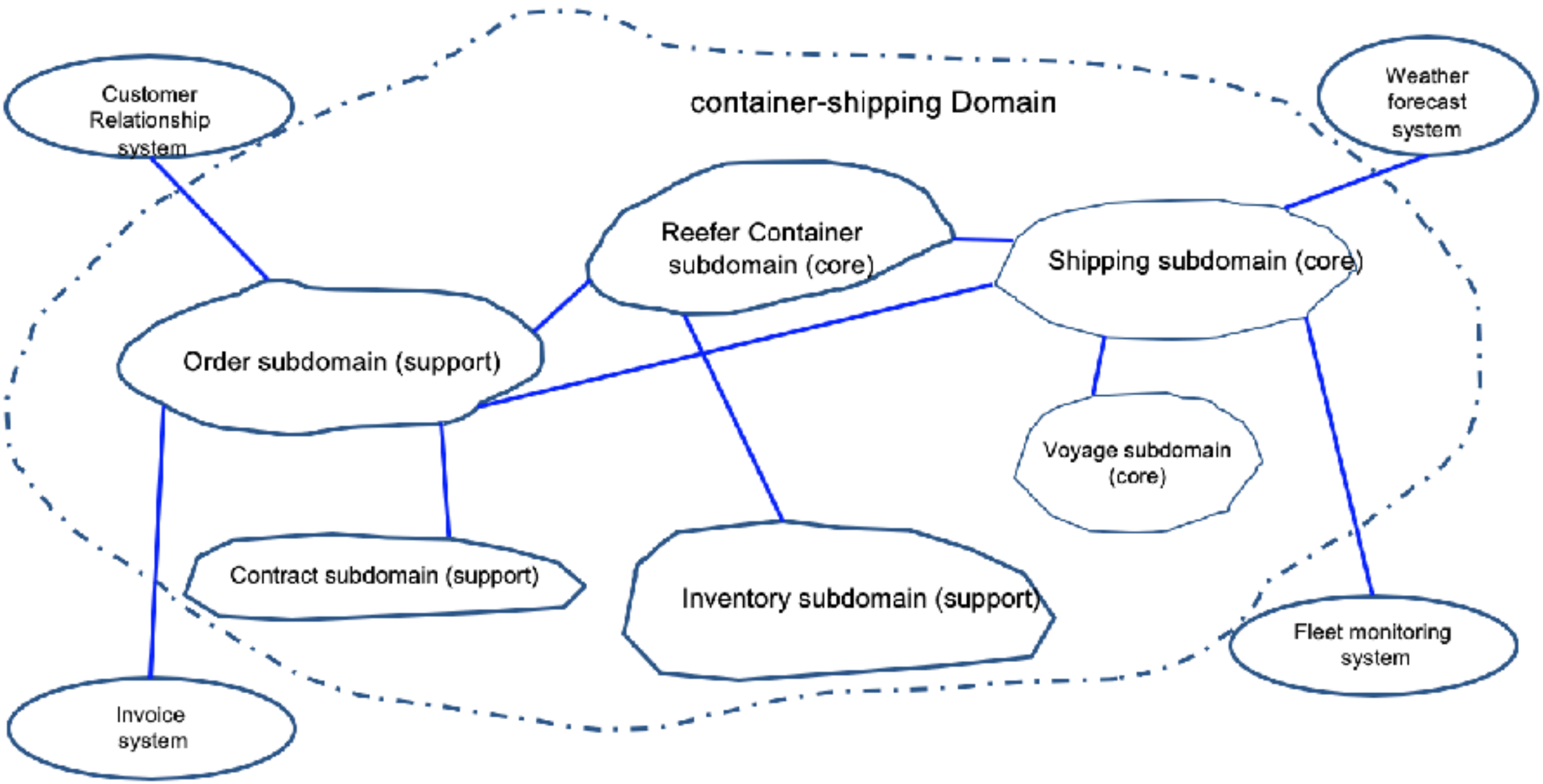
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Example of System Context



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Assess domains and sub-domains



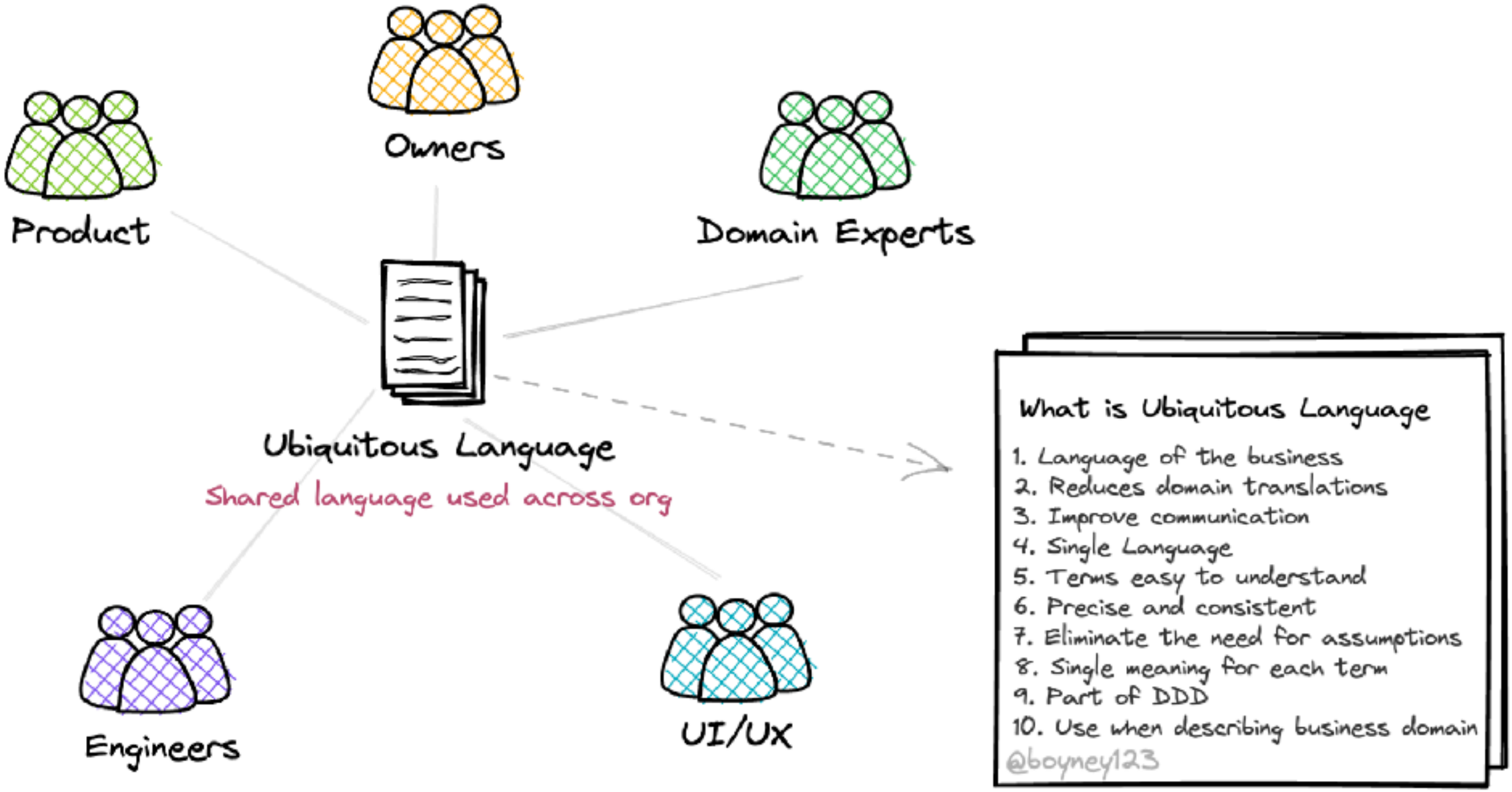
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Define the ubiquitous language



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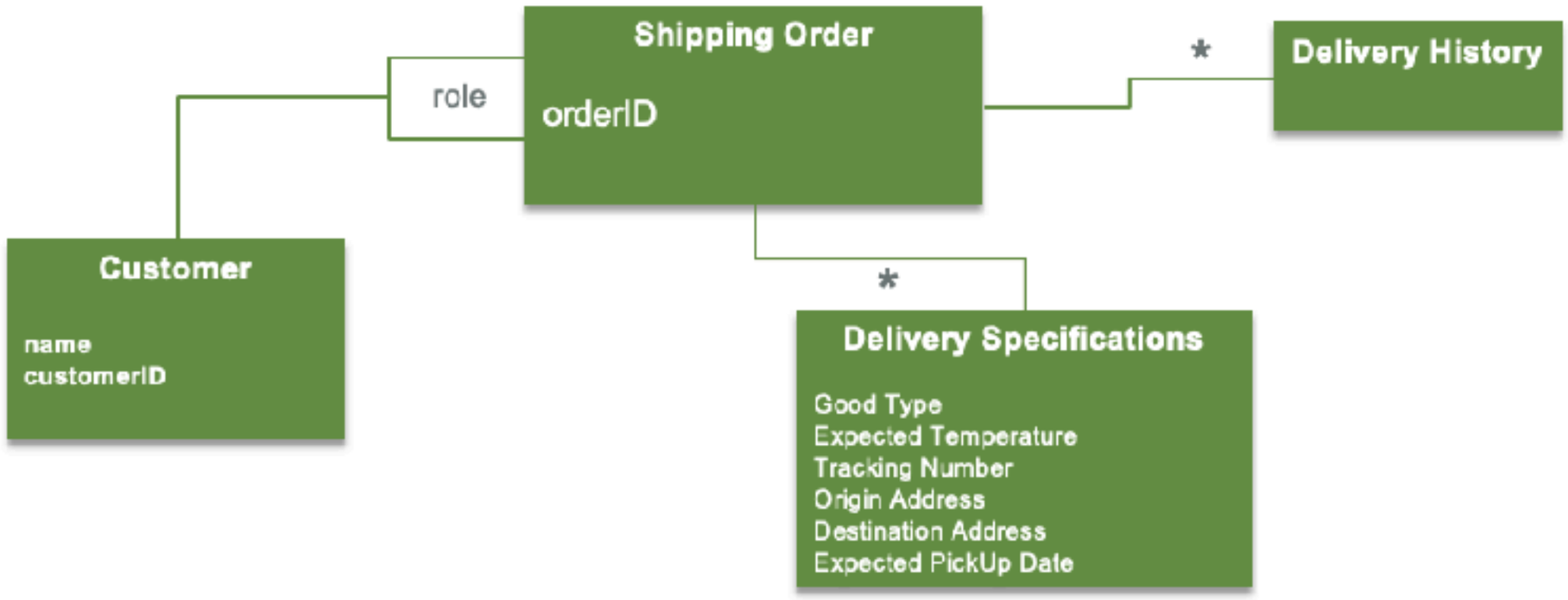
Define the ubiquitous language



https://serverlessland.com/event-driven-architecture/visuals/ubiquitous-language

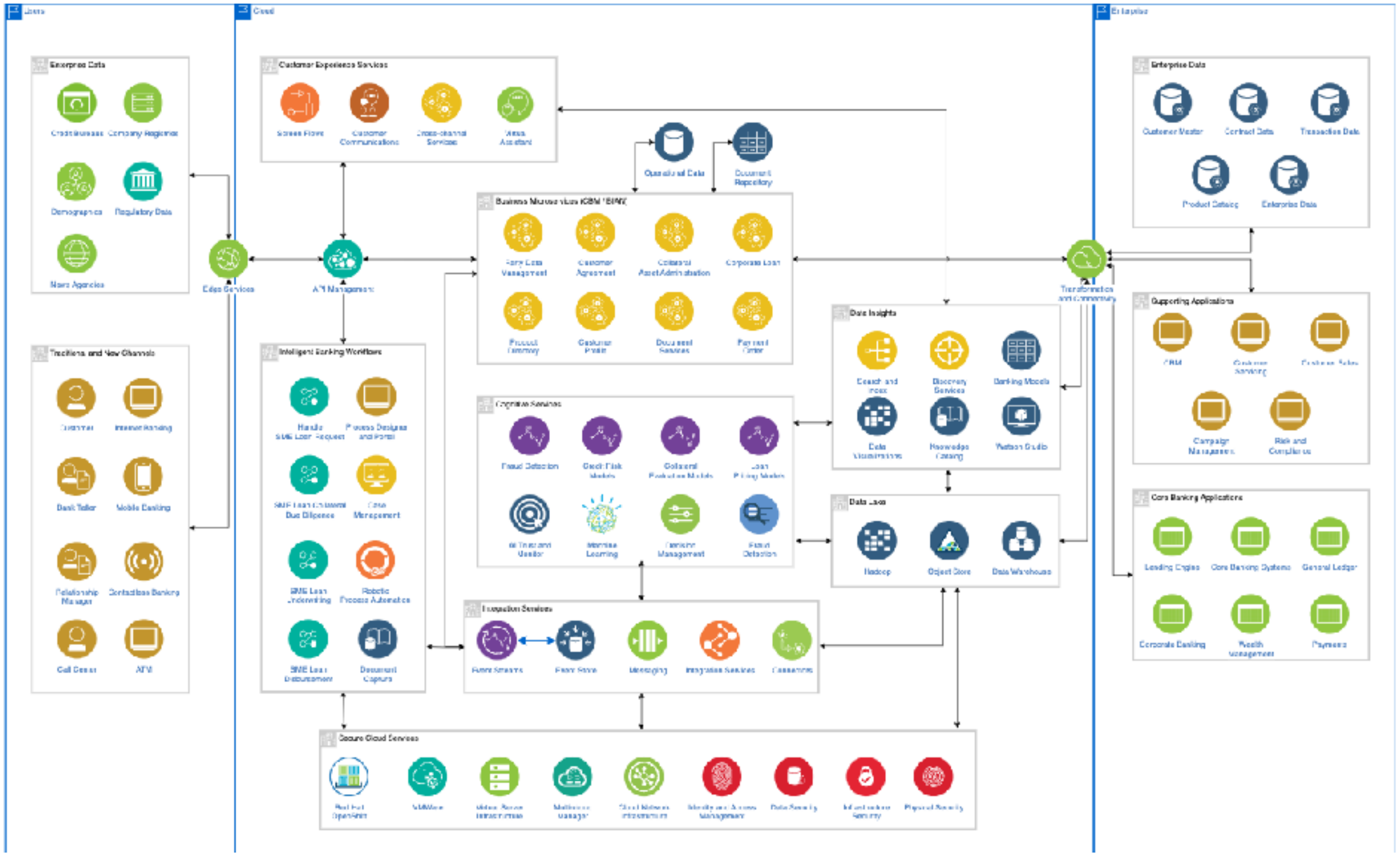
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Structure of data



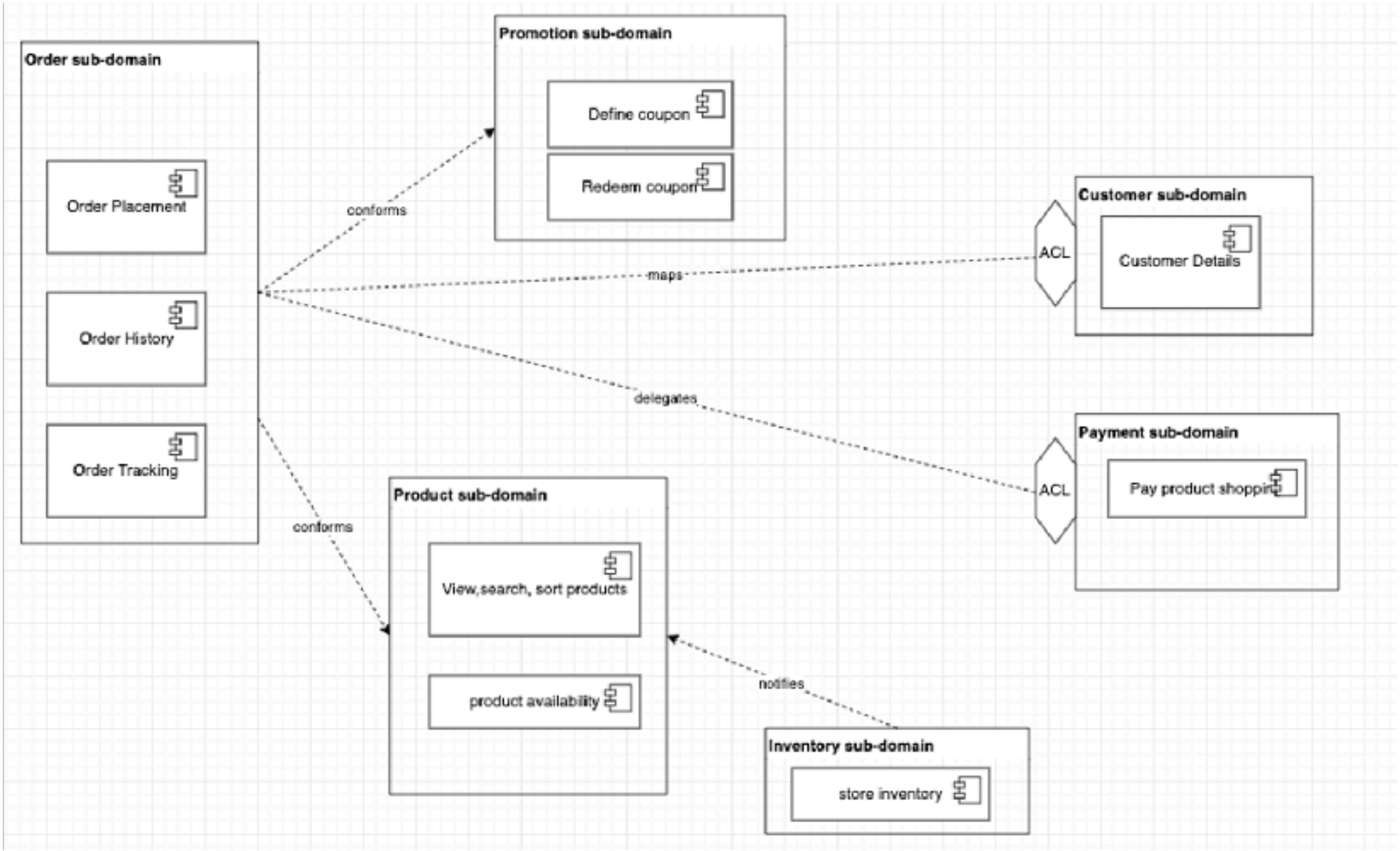
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Grouping services



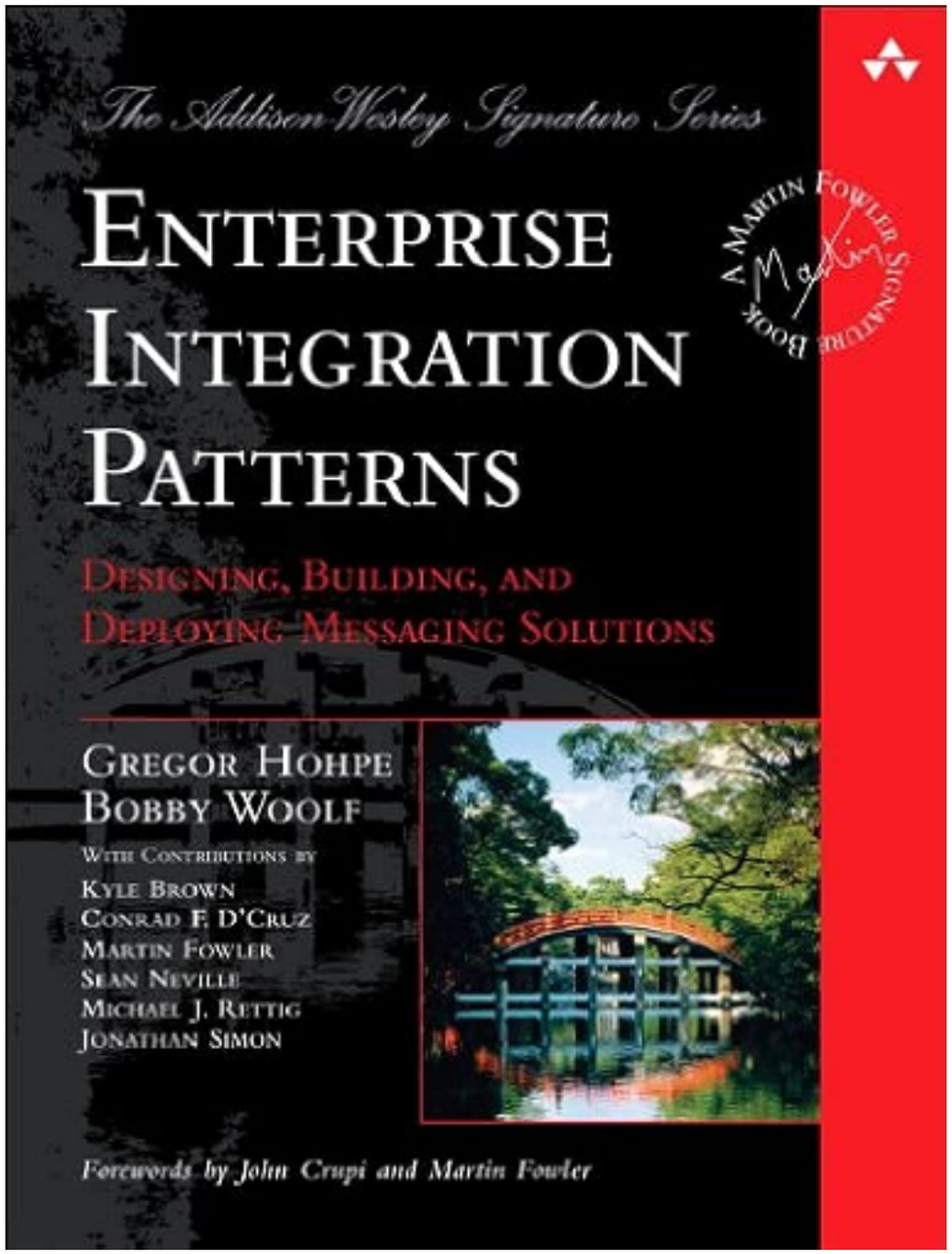
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Communication types ?



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Enterprise Integration Patterns

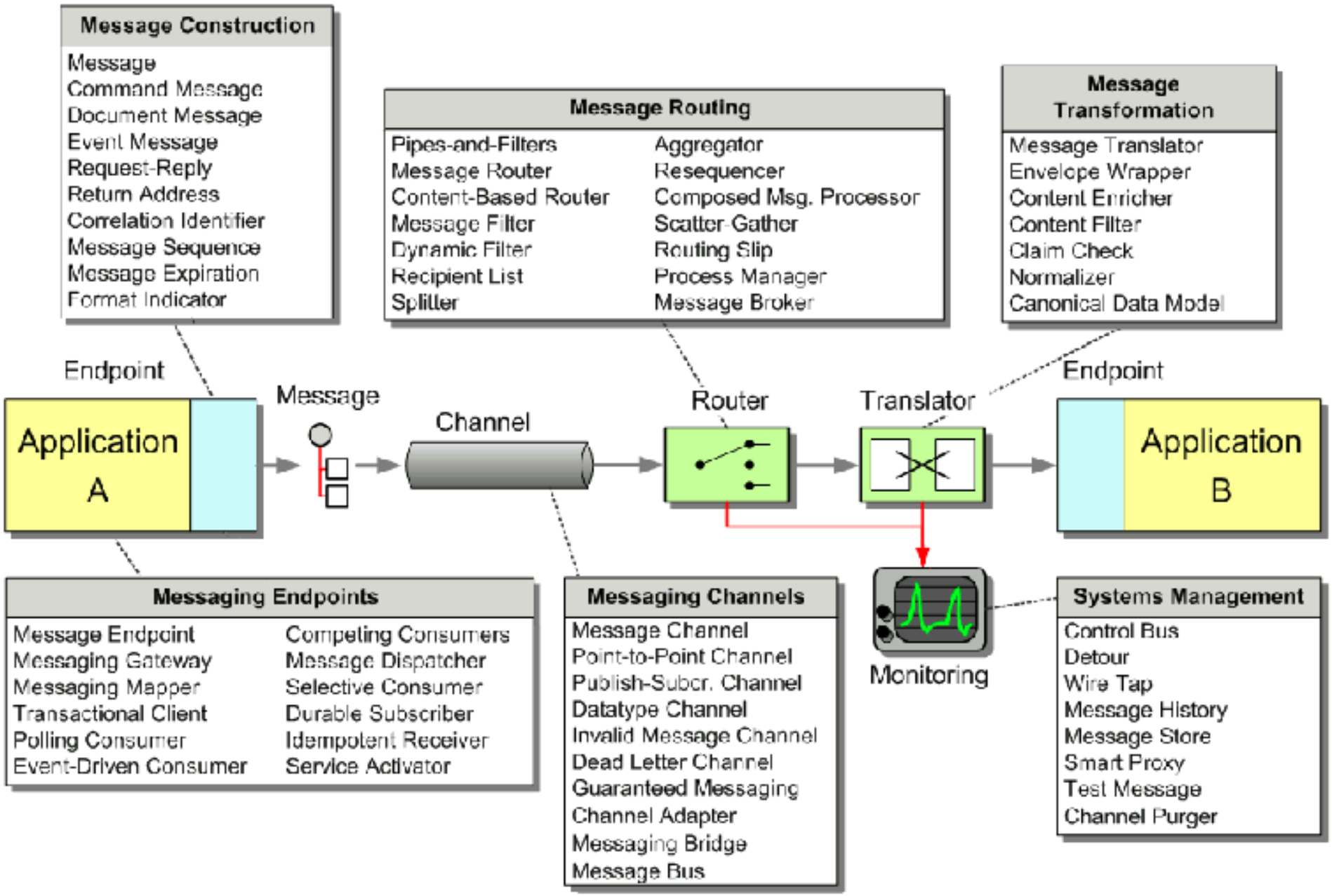


https://www.amazon.com/Enterprise-Integration-Patterns-Designing-Deploying/dp/0321200683



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Integration Pattern Language



https://www.enterpriseintegrationpatterns.com/patterns/messaging/



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Communication between services



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Interaction Styles

	One-to-one	One-to-many
Synchronous	Request/response	-
Asynchronous	Async request/response	Publish/subscribe
	Notification	Publish/async response



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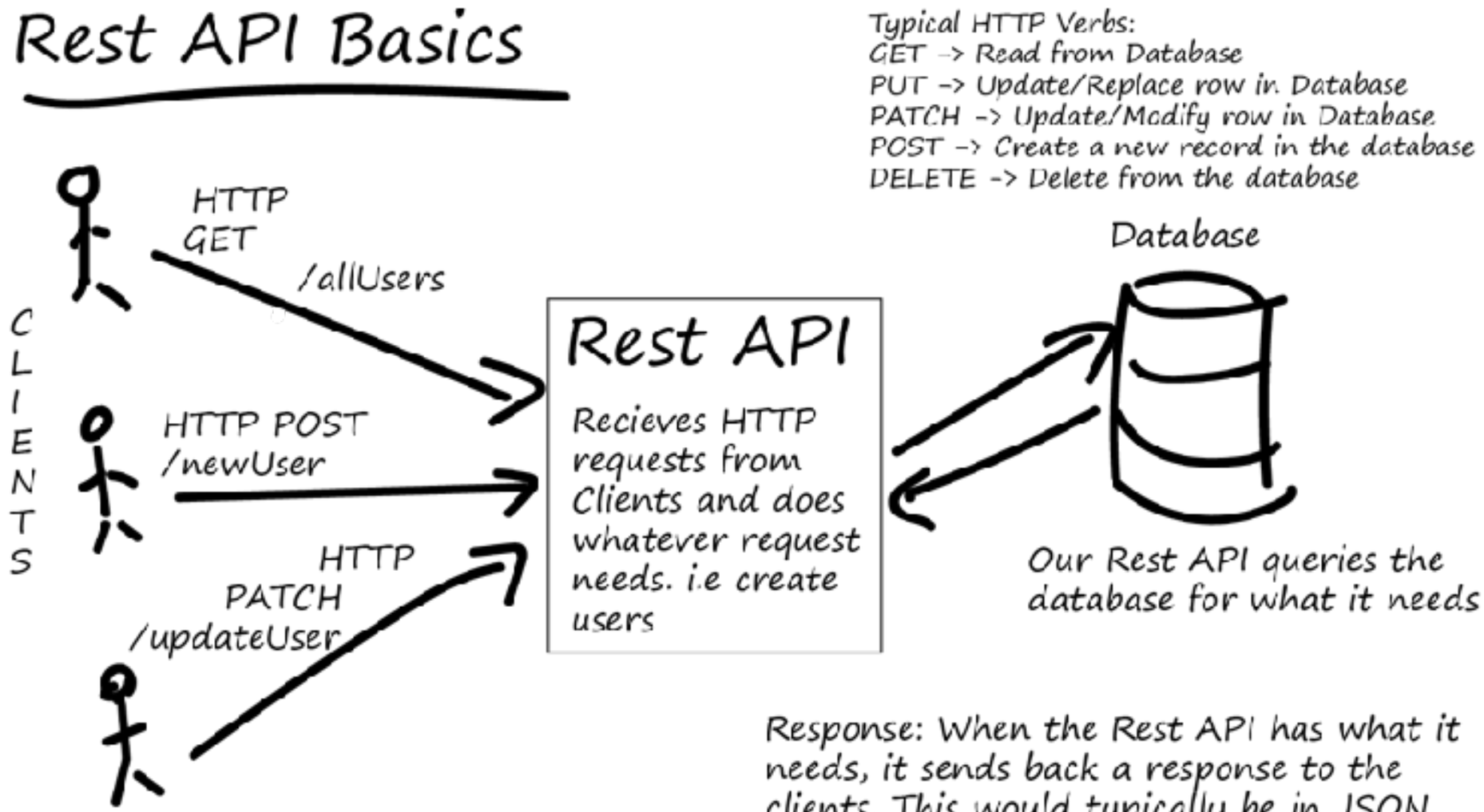
Synchronous Communication

REST
gRPC
Handling failure with Circuit breaker
Service discovery



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REpresentational **S**tate **T**ransfer



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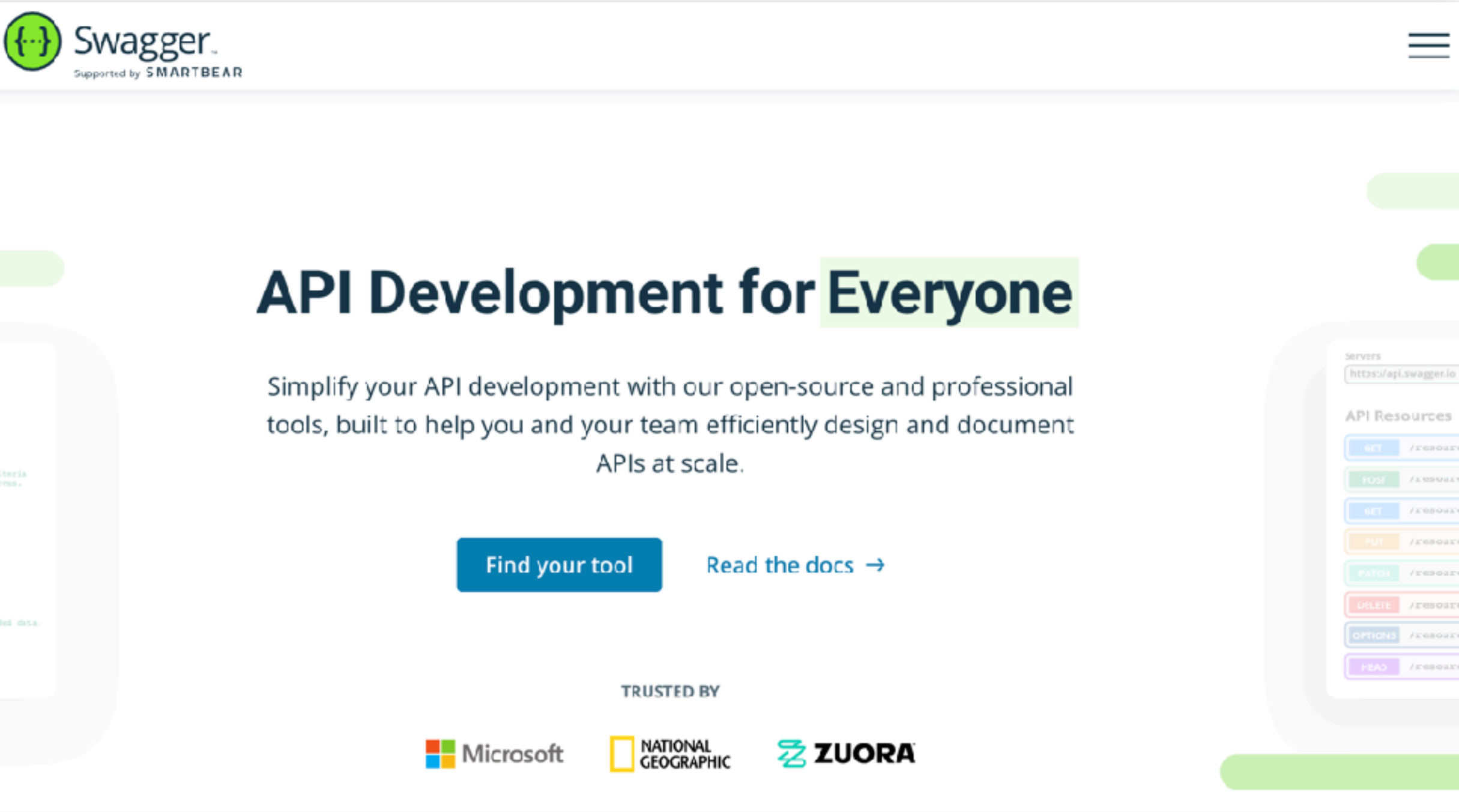
REST :: mapping with HTTP verbs

Activity	HTTP Method
Retrieve data	GET
Create new data	POST
Update data	PUT
Delete data	DELETE



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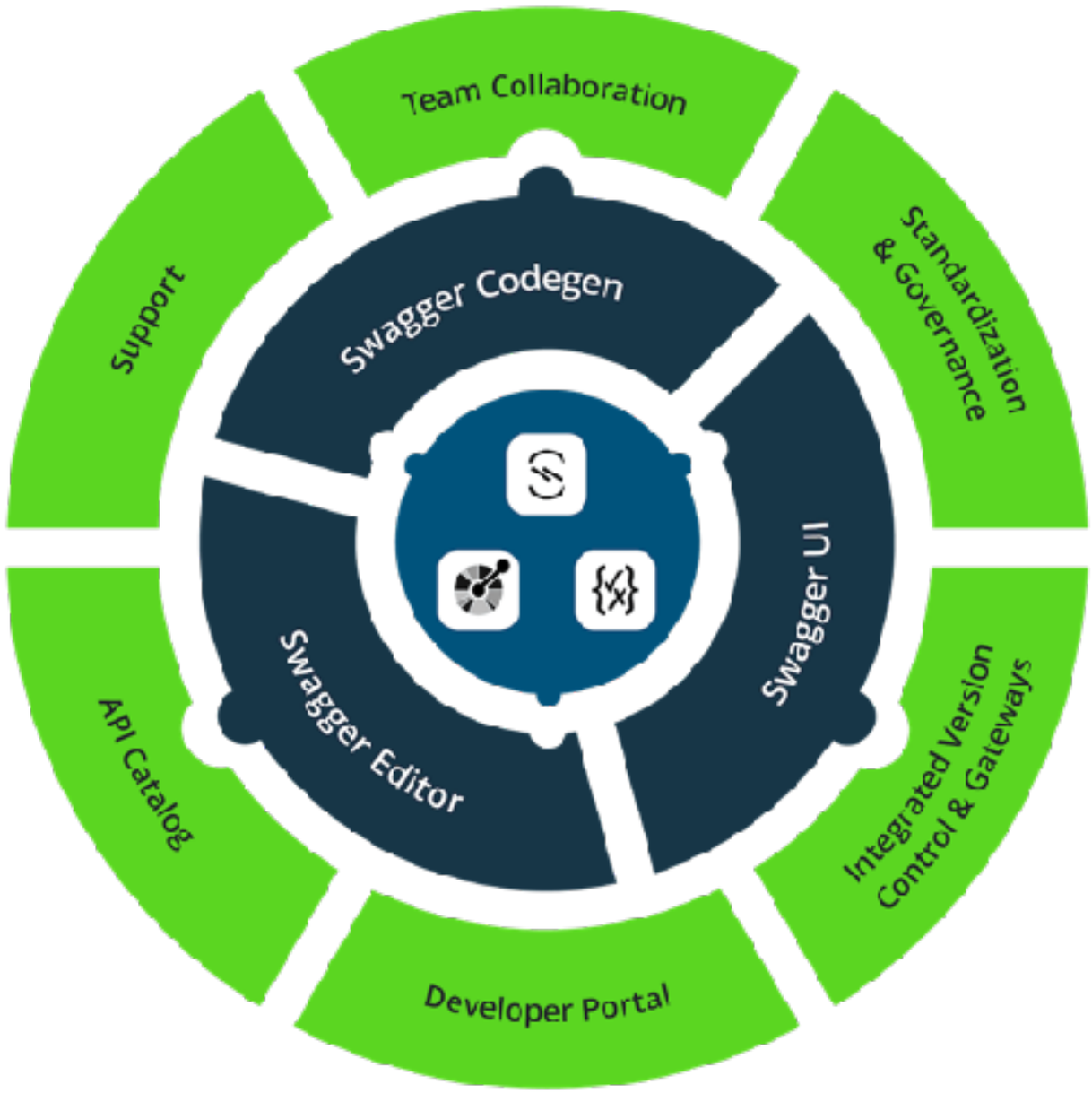
API Documentation



https://swagger.io/

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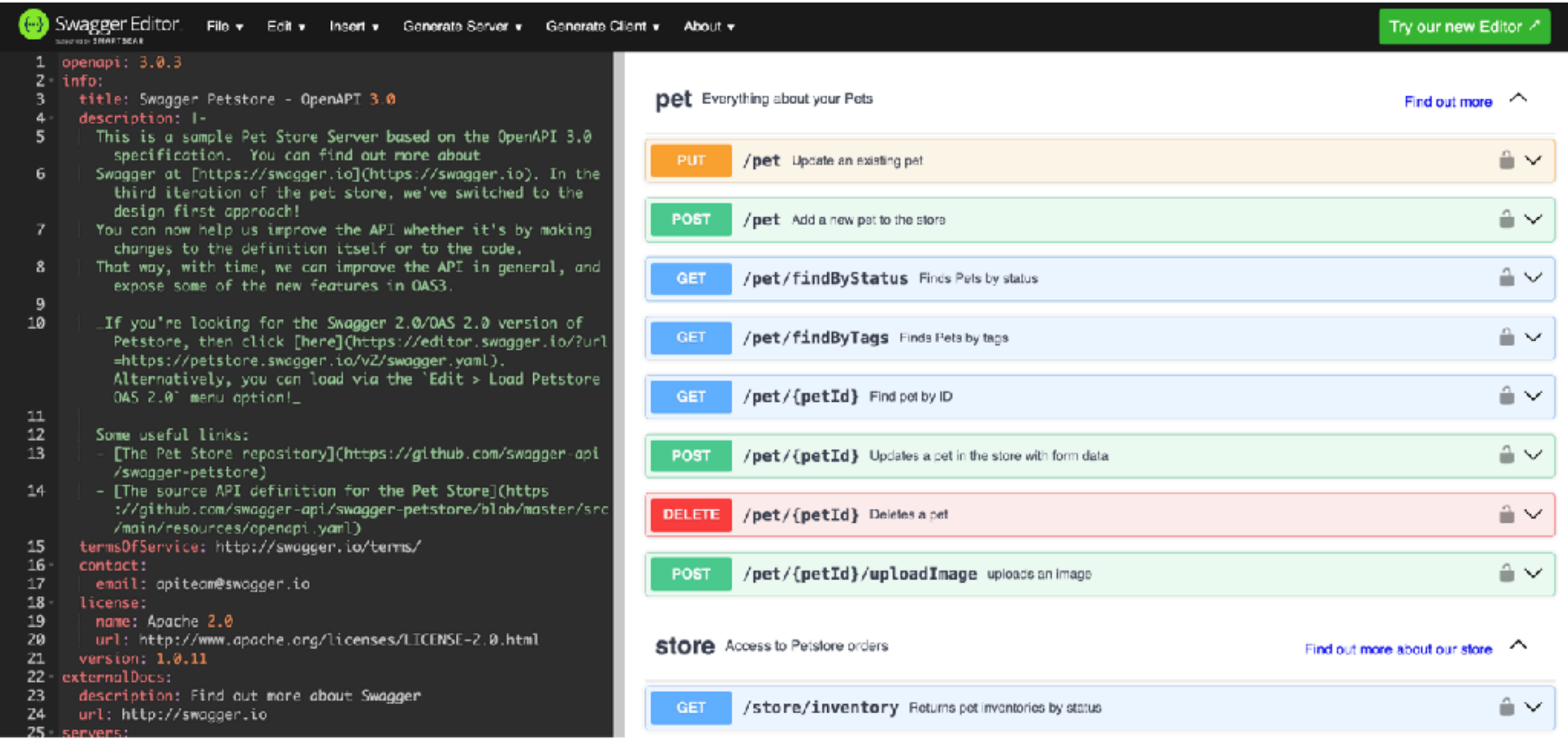
Swagger/OpenAPI



https://swagger.io/

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Swagger/OpenAPI



https://swagger.io/



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How to design your API ?

Design first

Code first



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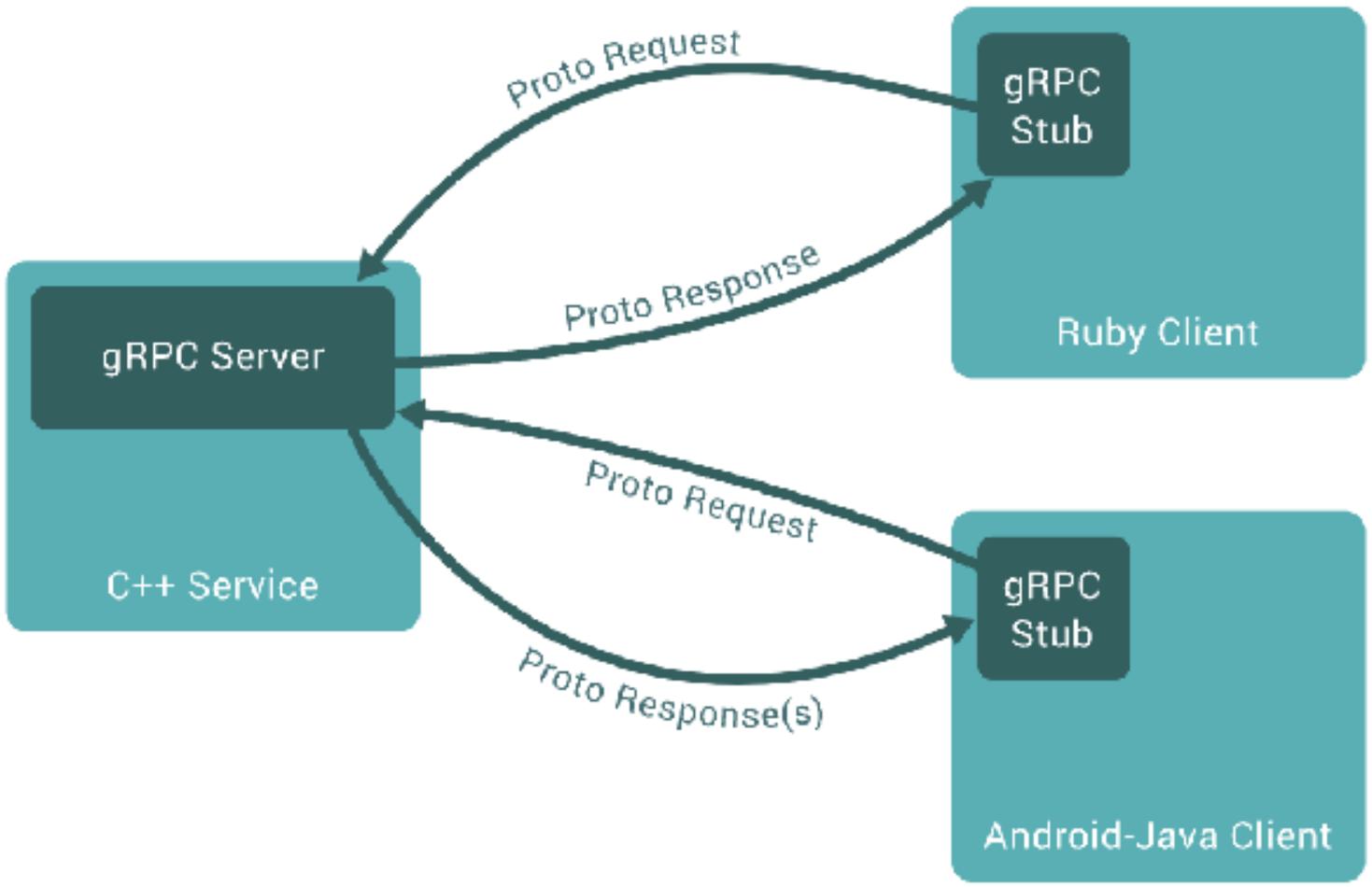
gRPC



https://grpc.io/

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gRPC

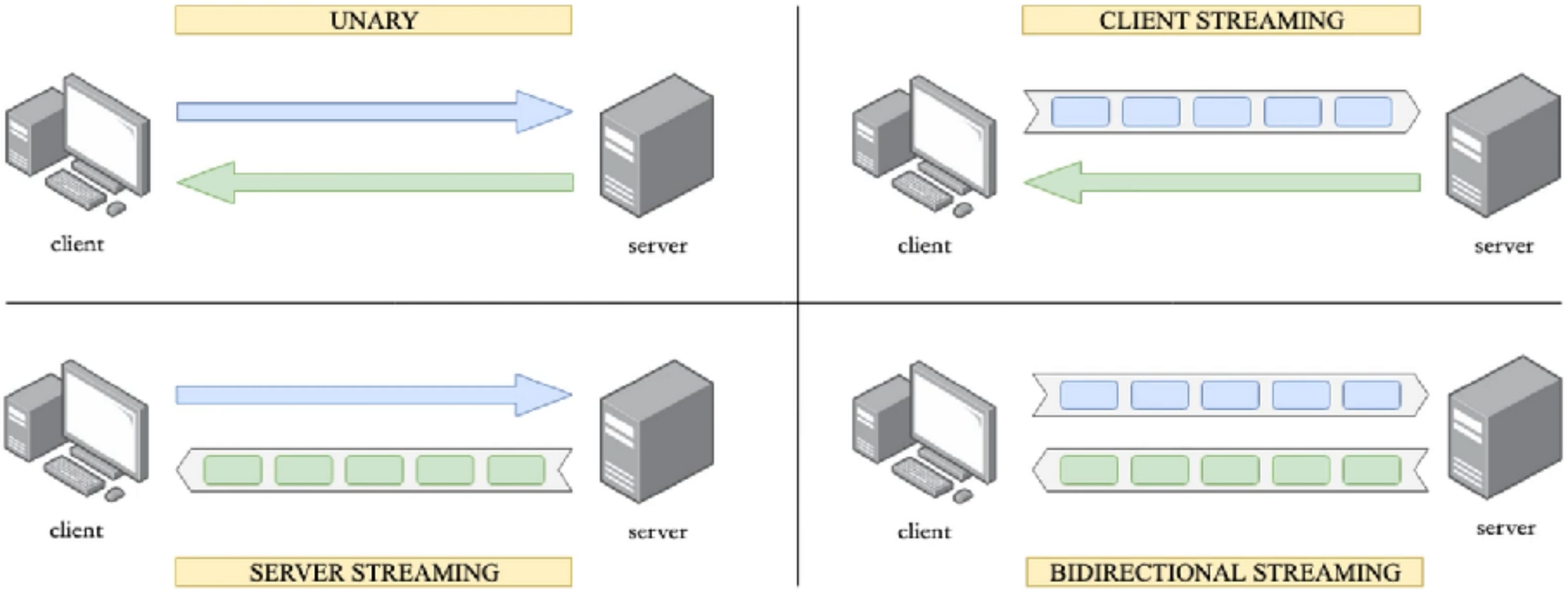


https://grpc.io/



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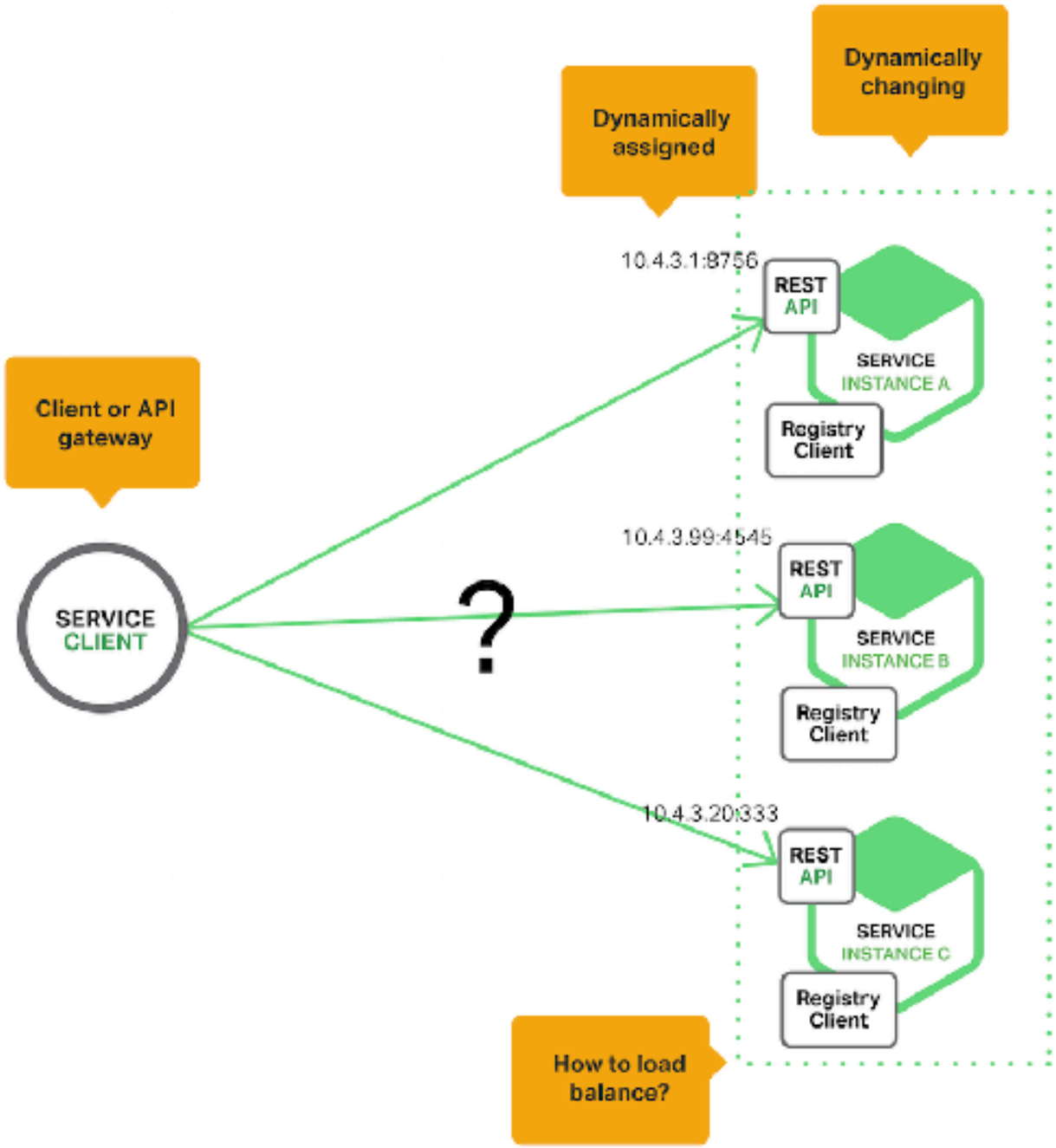
gRPC



https://grpc.io/

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Service Discovery ?



https://www.nginx.com/blog/service-discovery-in-a-microservices-architecture/



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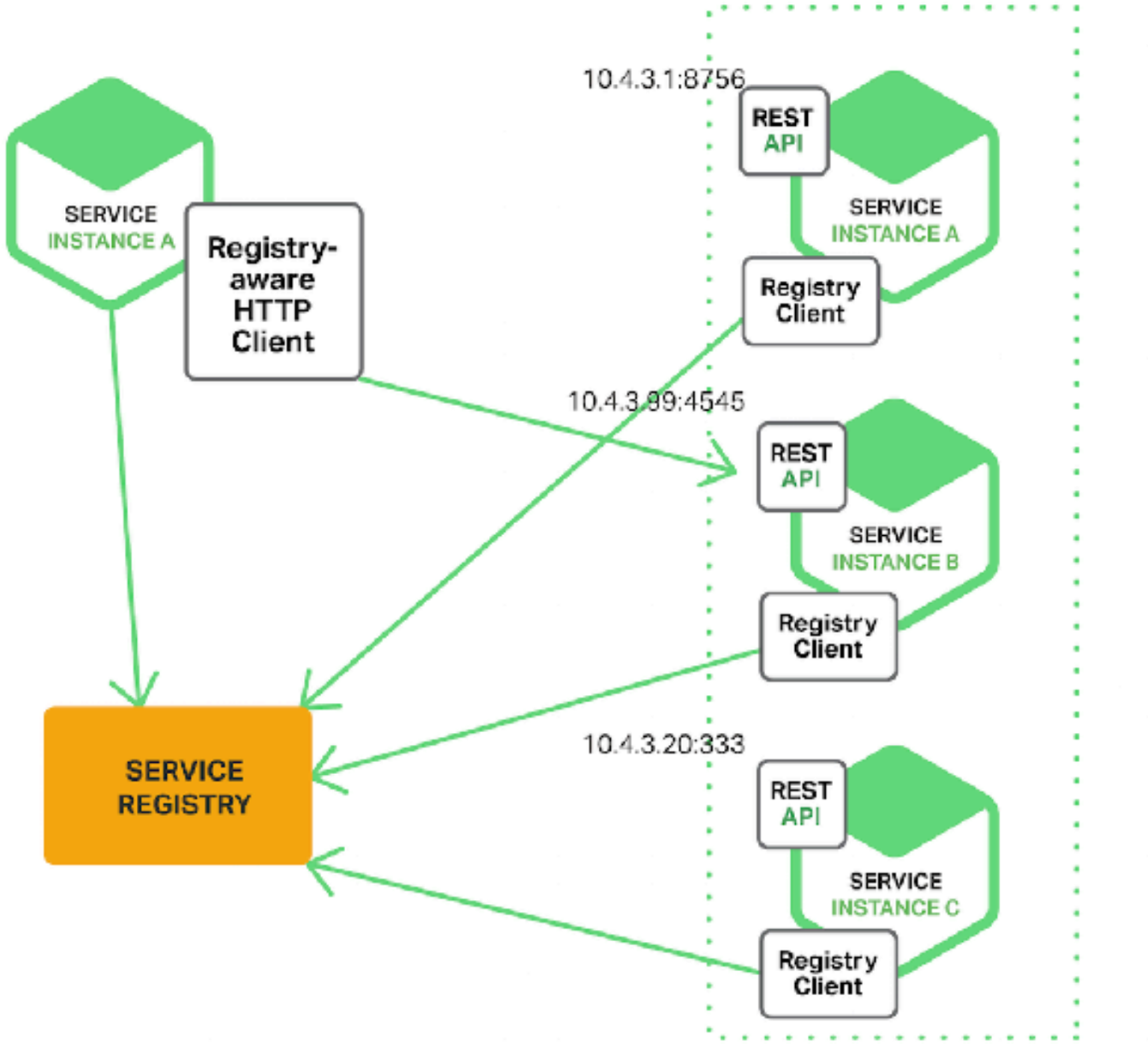
Service Discovery ?

Client-side
Server-side



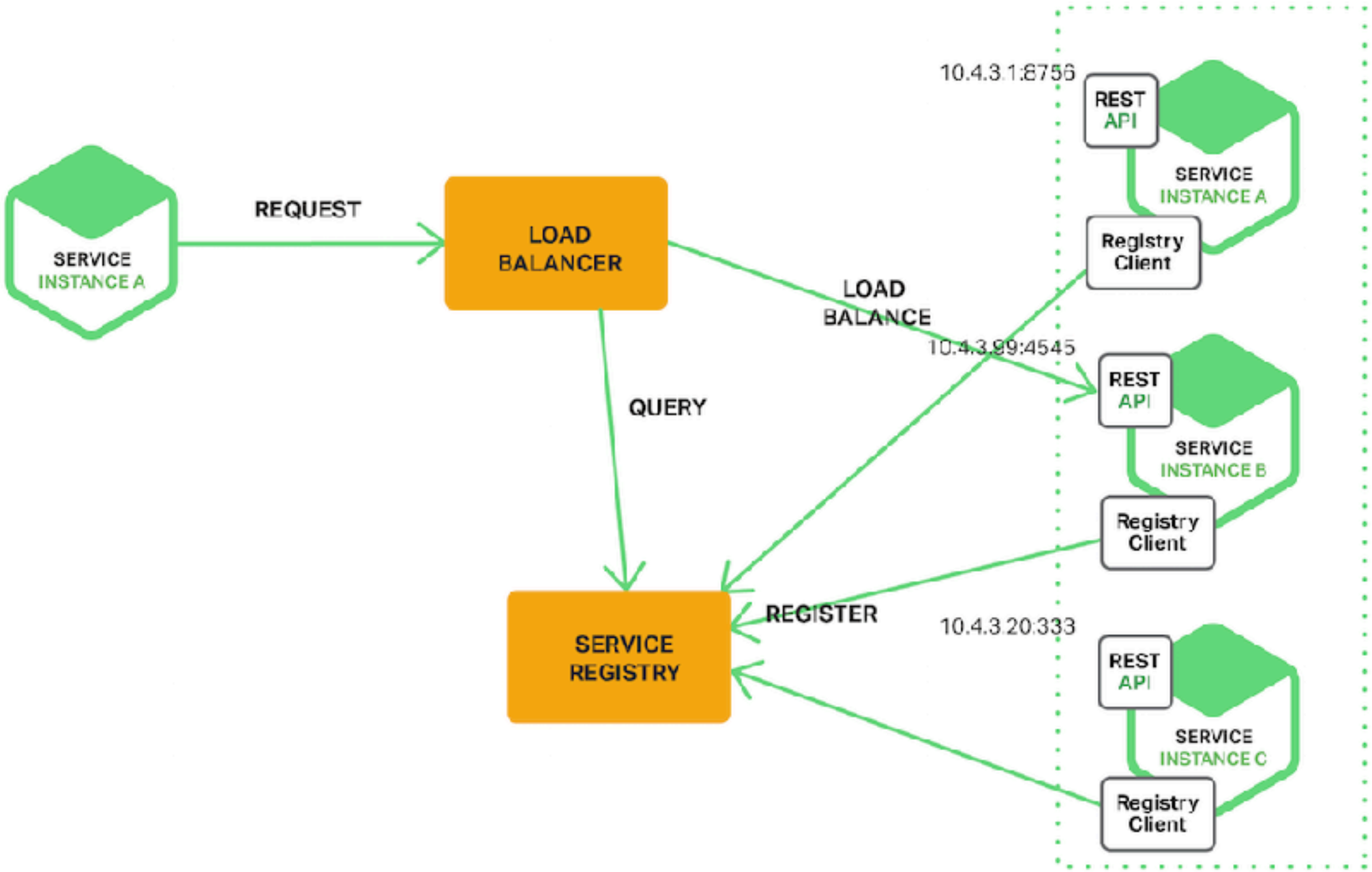
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Client-side Discovery



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Server-side Discovery



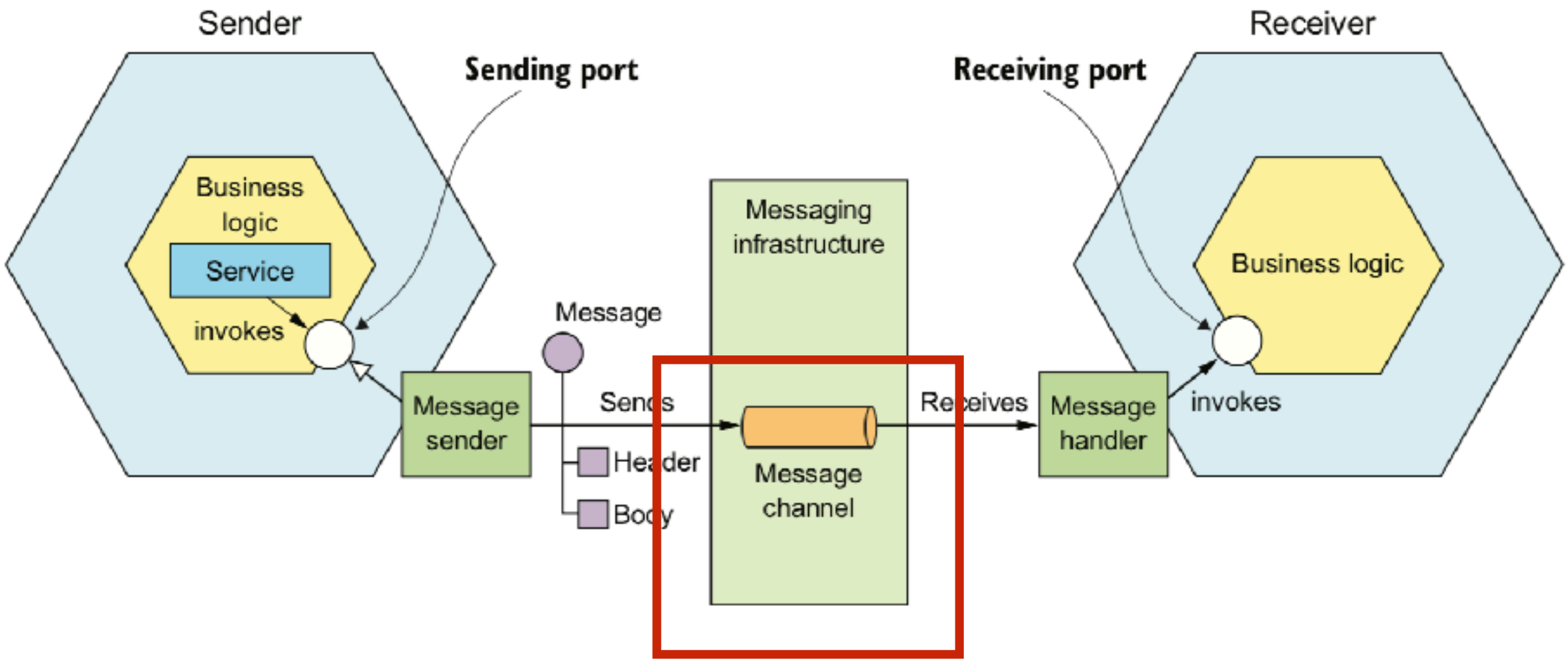
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Asynchronous communication



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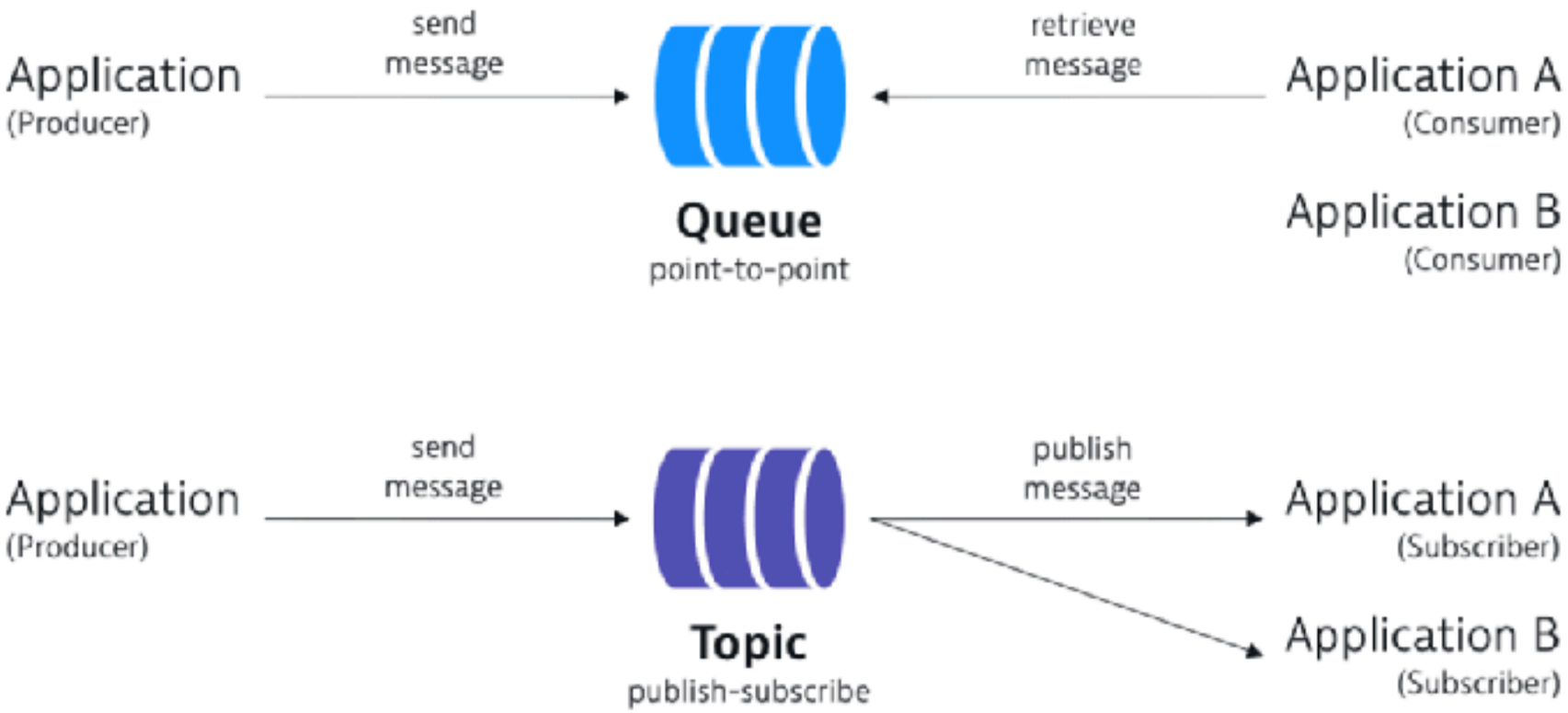
Asynchronous messaging pattern



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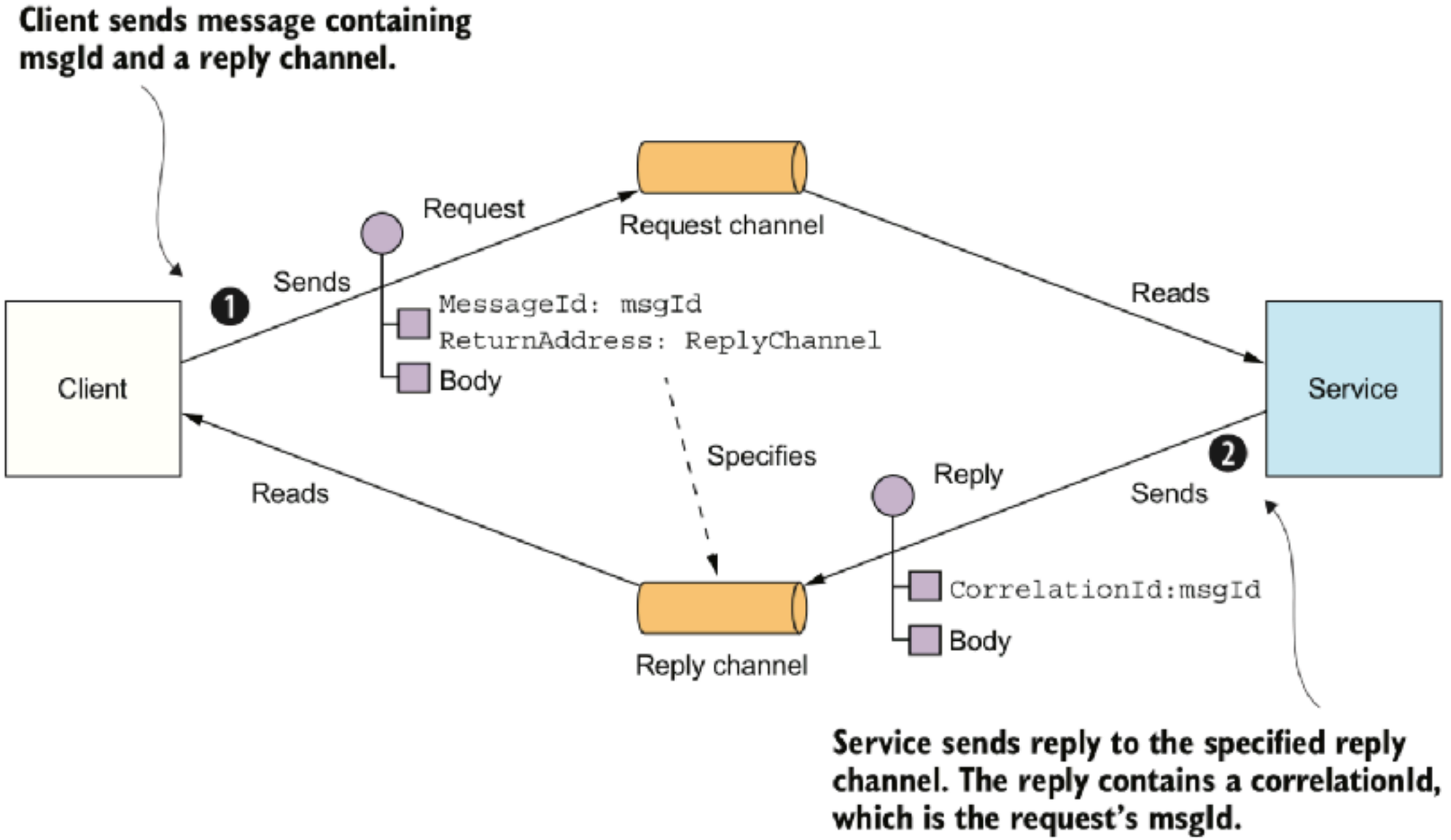
Messaging channels

Publish-subscribe
Point-to-point



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Async request/response (1)



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Message brokers











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Benefits

Loose-coupling
Message buffer
Flexible communication



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Drawbacks

Performance bottleneck
Single point of failure
Operational complexity
Testing
Monitoring



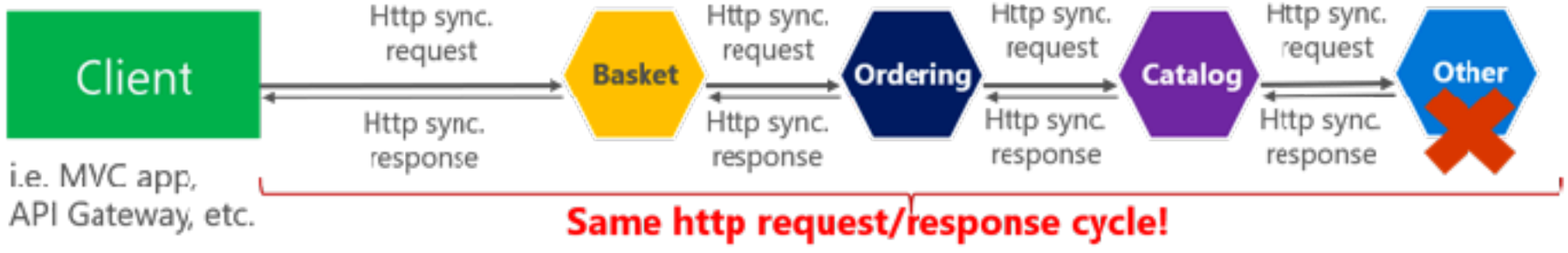
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Communication !!

Synchronous vs. async communication across microservices

Anti-pattern

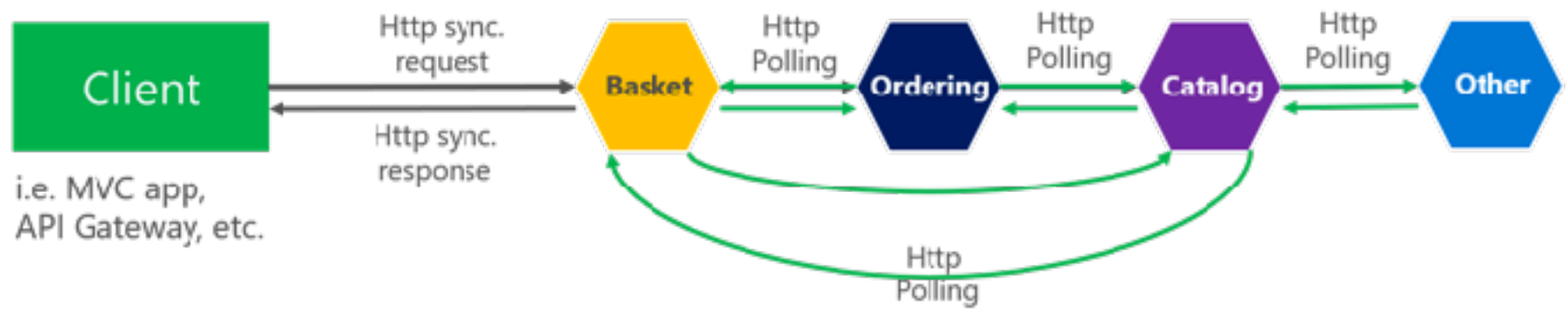
Synchronous
all req./resp. cycle



Asynchronous
Comm. across
internal microservices
(EventBus: i.e. **AMPQ**)

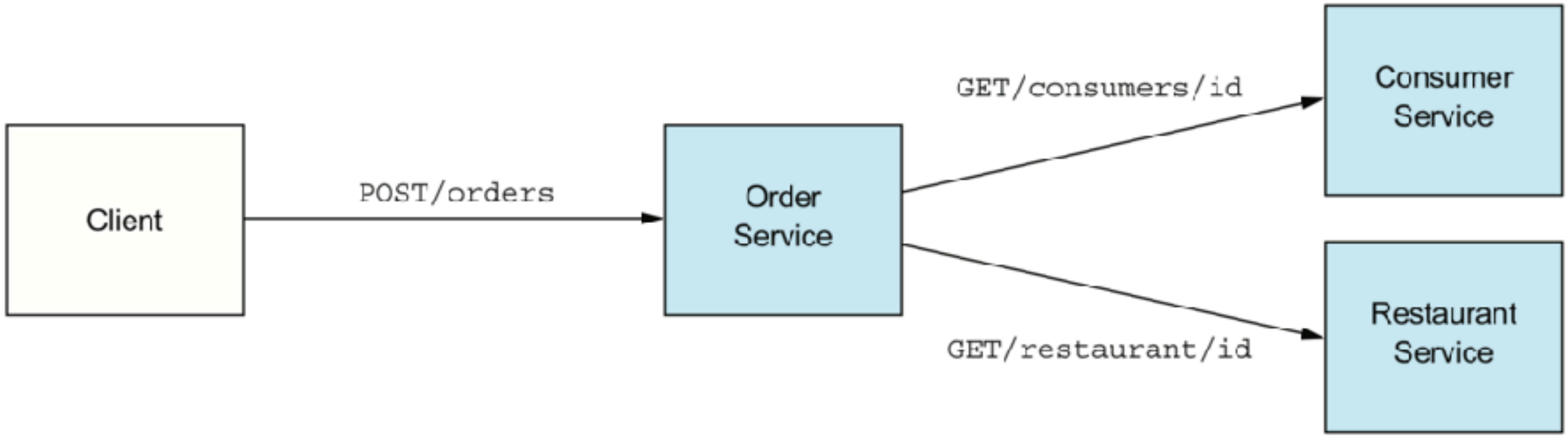


"Asynchronous"
Comm. across
internal microservices
(Polling: **Http**)



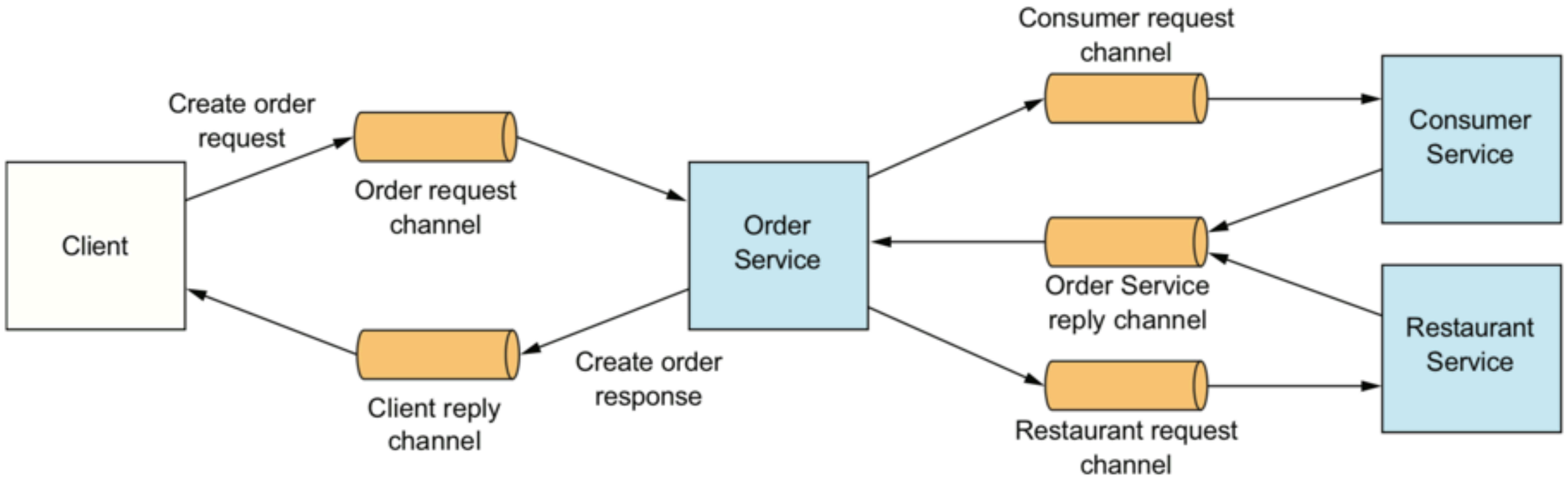
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Synchronous reduce availability



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Replace with asynchronous



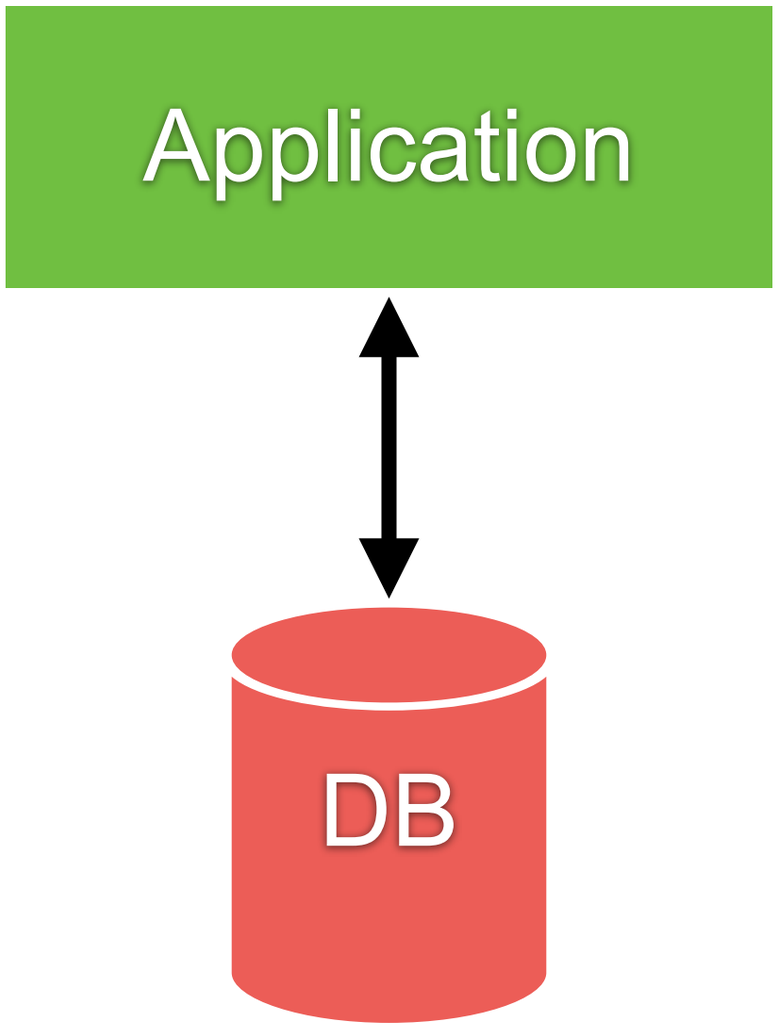
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Manage Data Consistency



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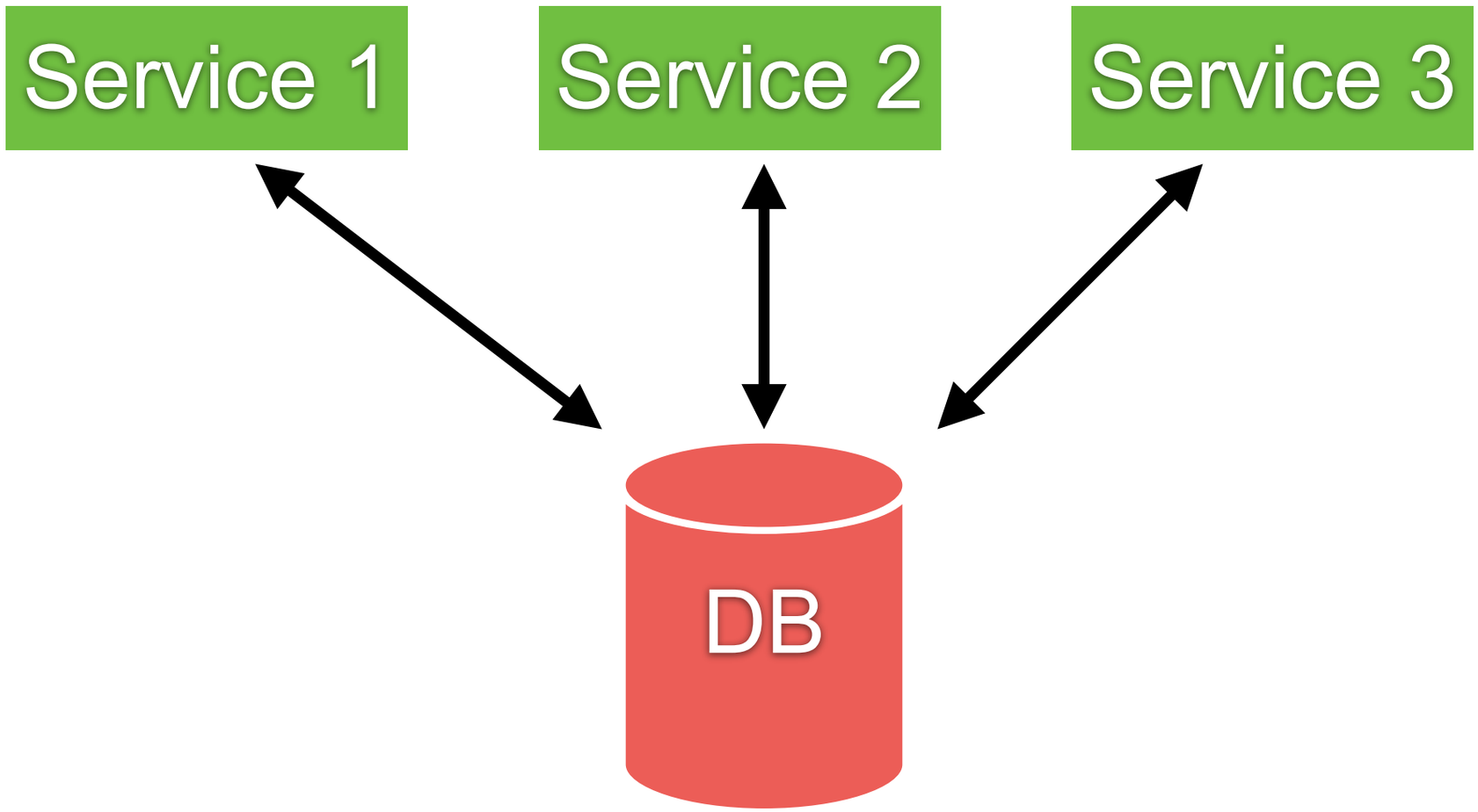
Monolithic



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Microservices

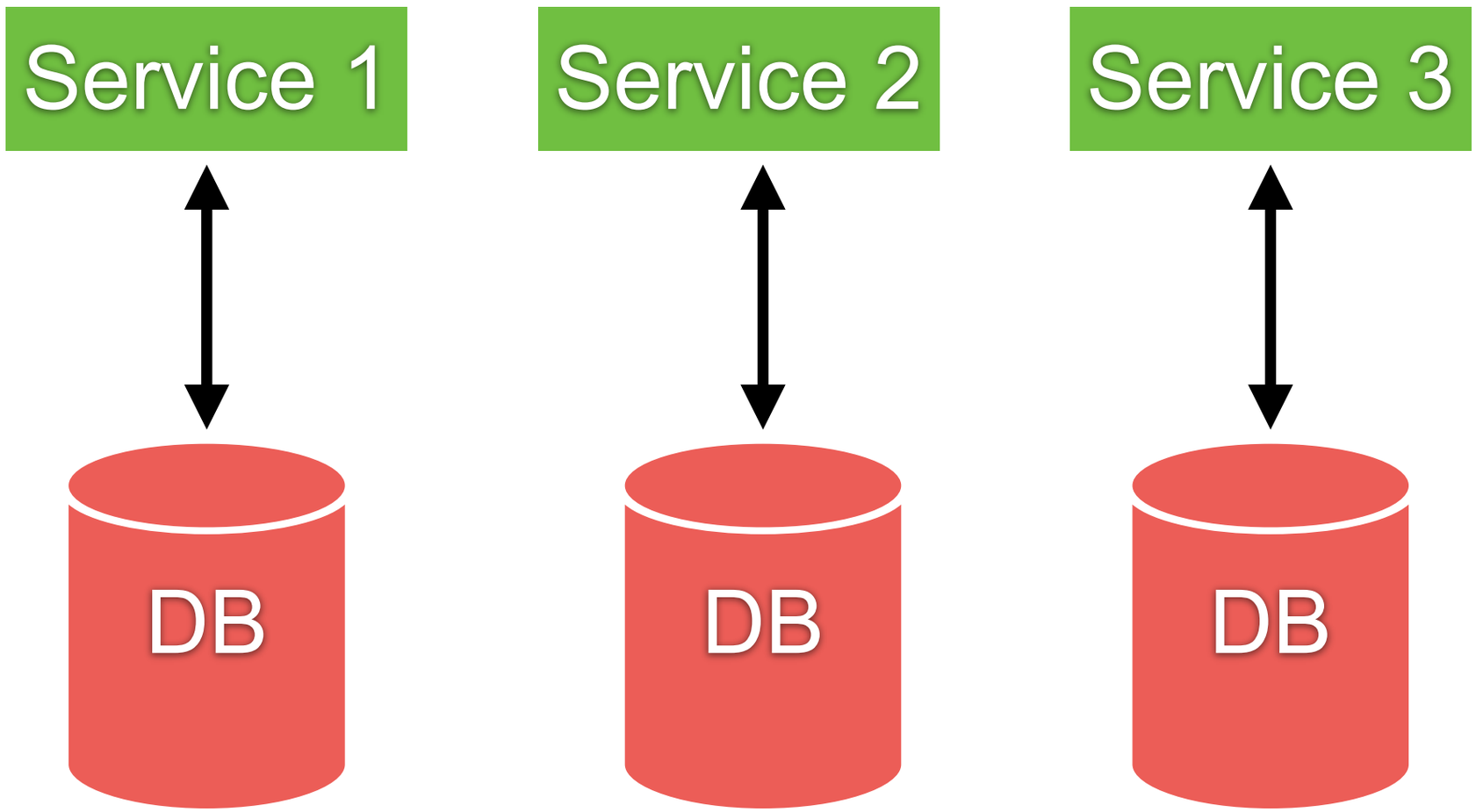
Shared database



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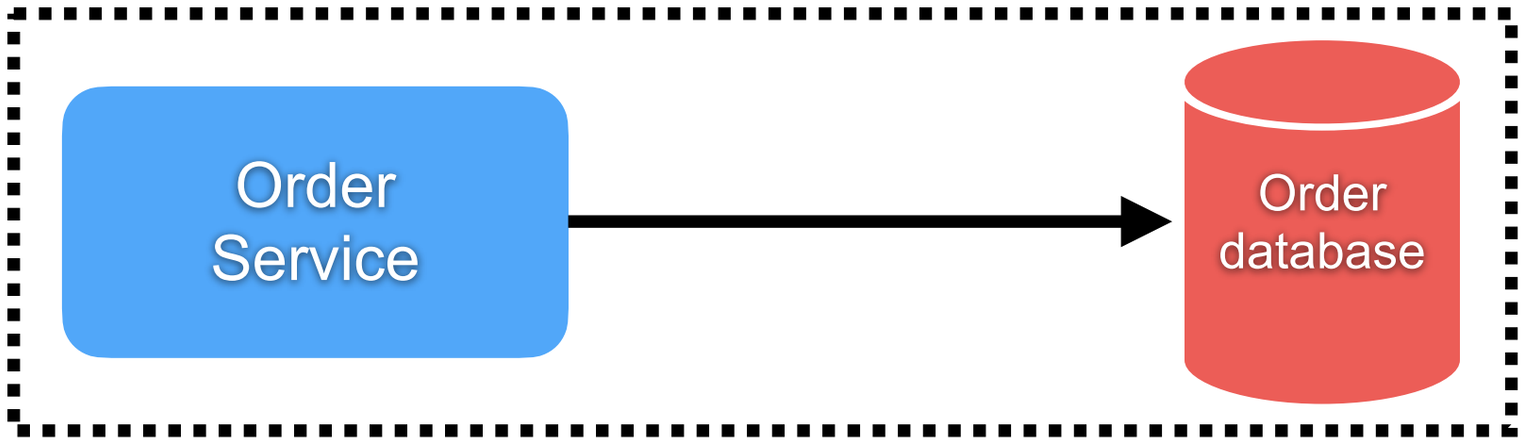
Microservices

Database per service

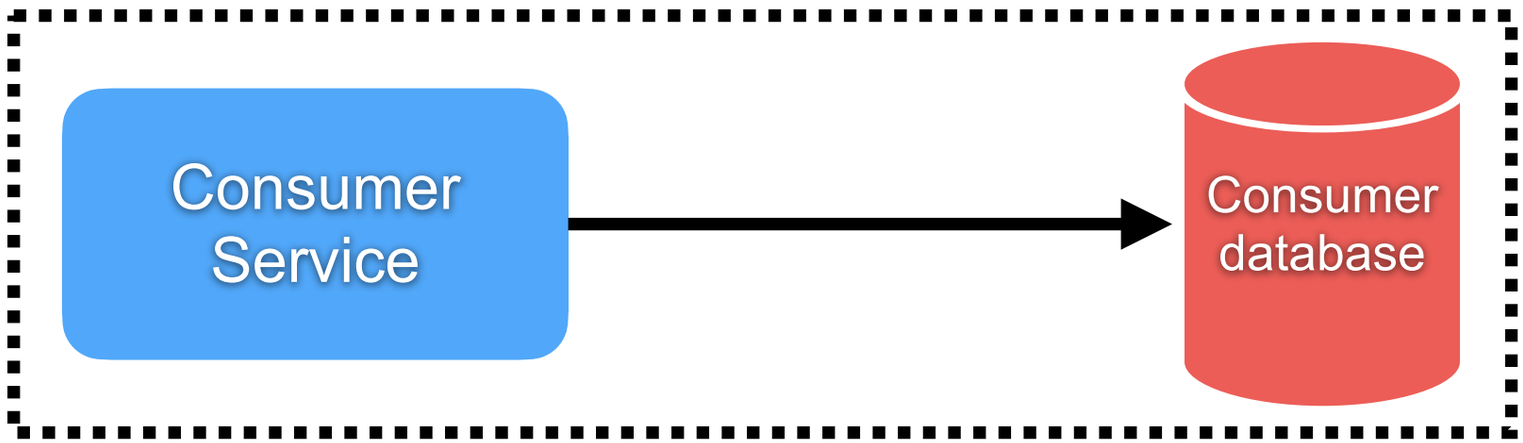


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Database per service

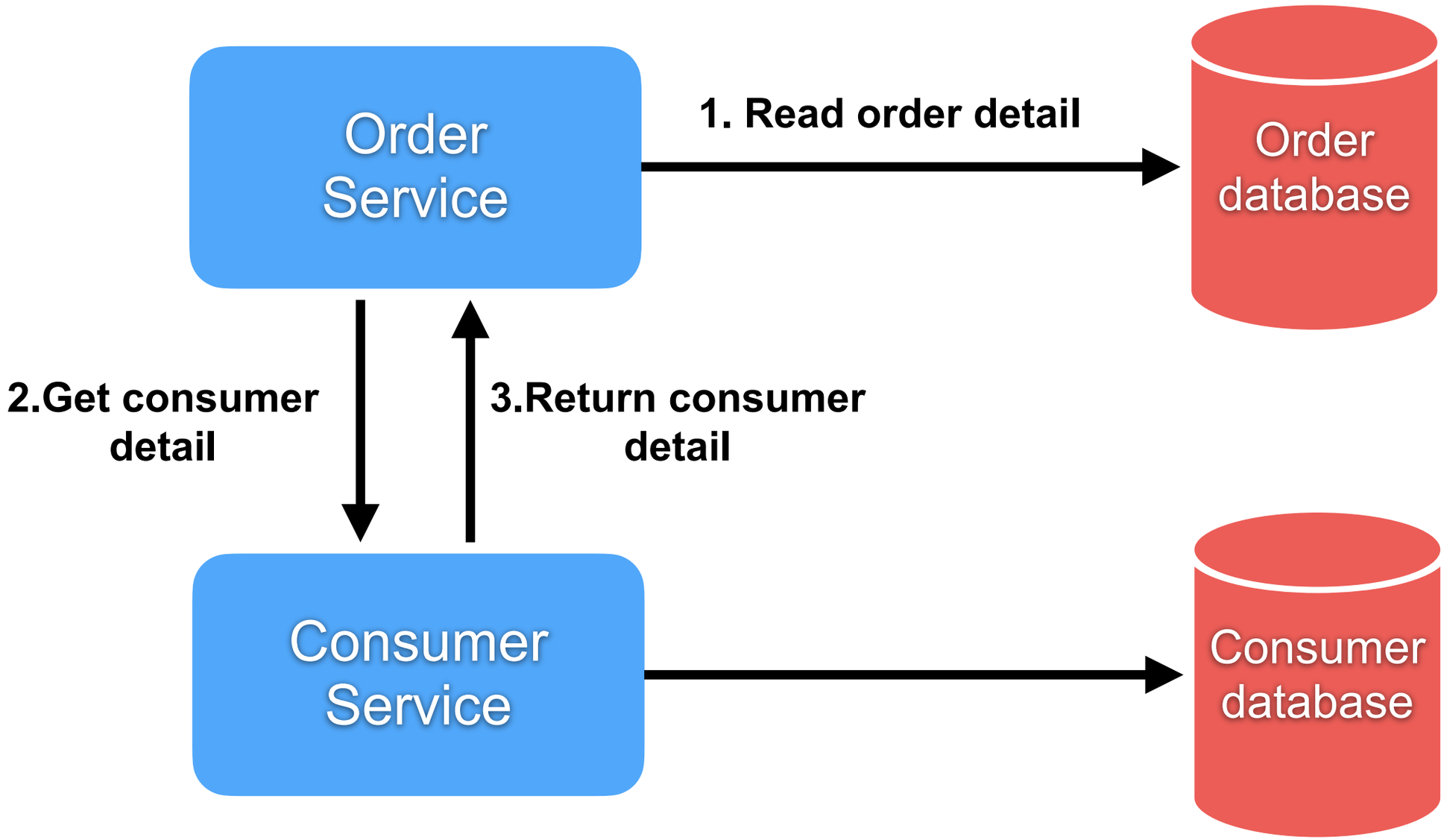


Strong Boundary



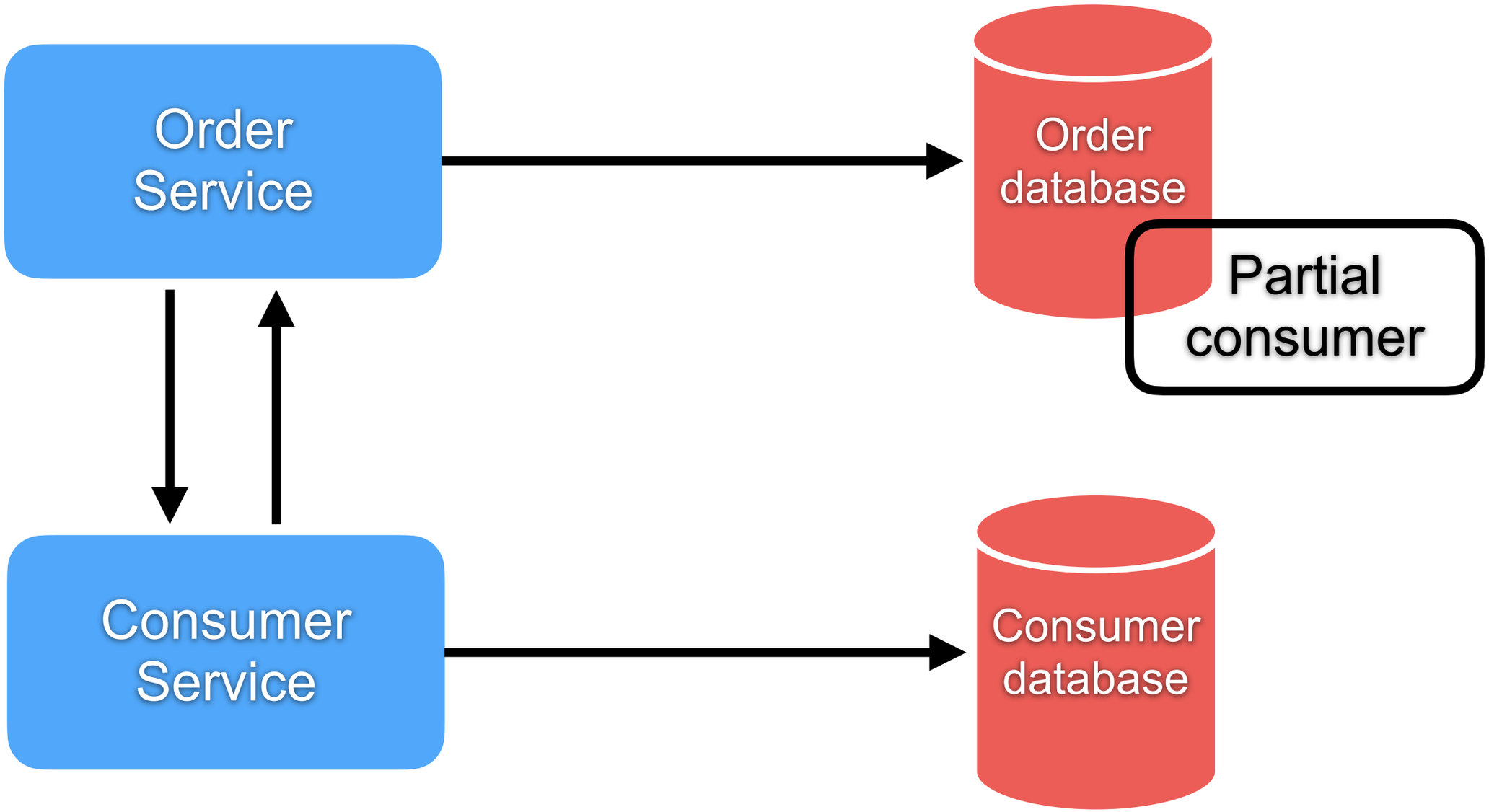
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Loose coupling = encapsulation data



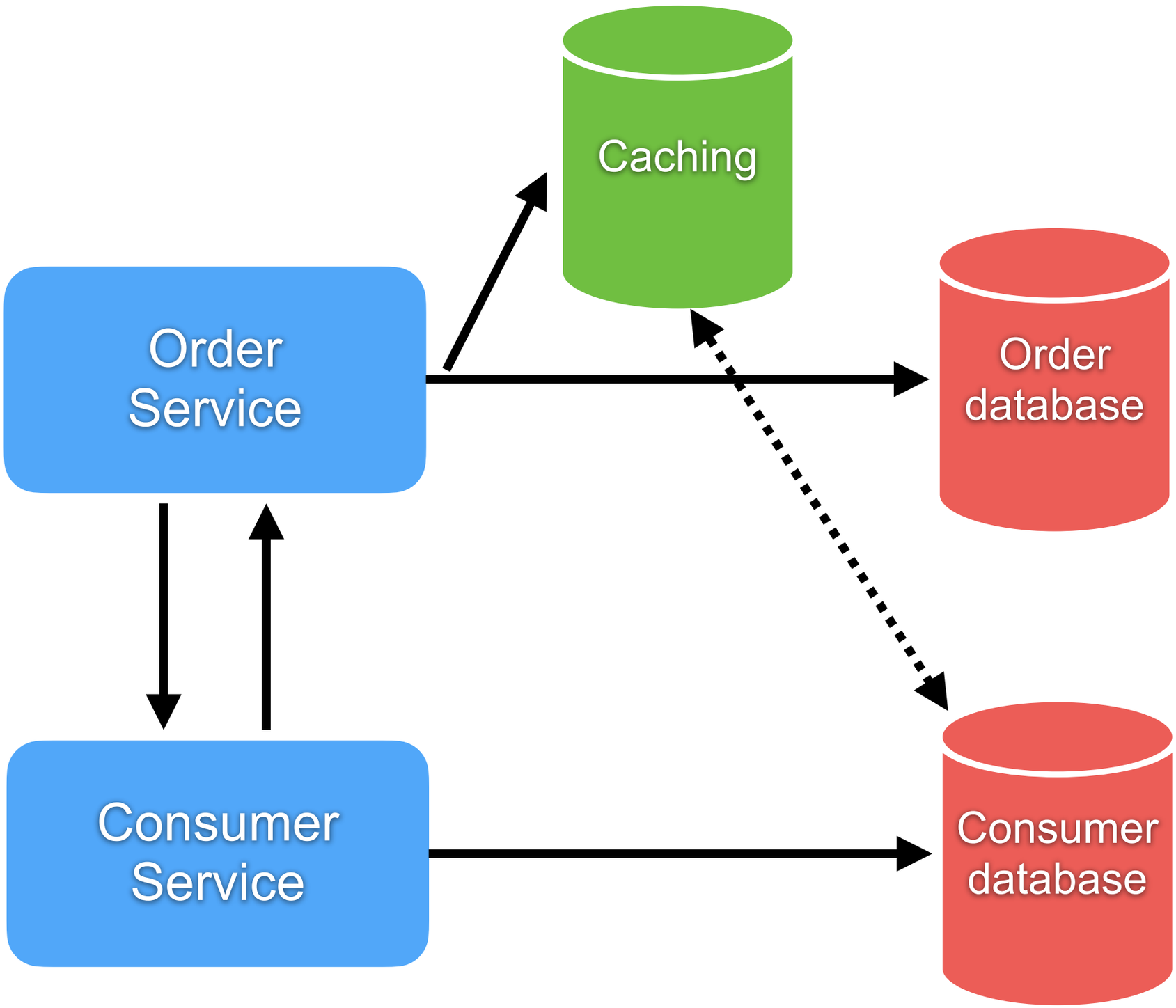
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Store partial data



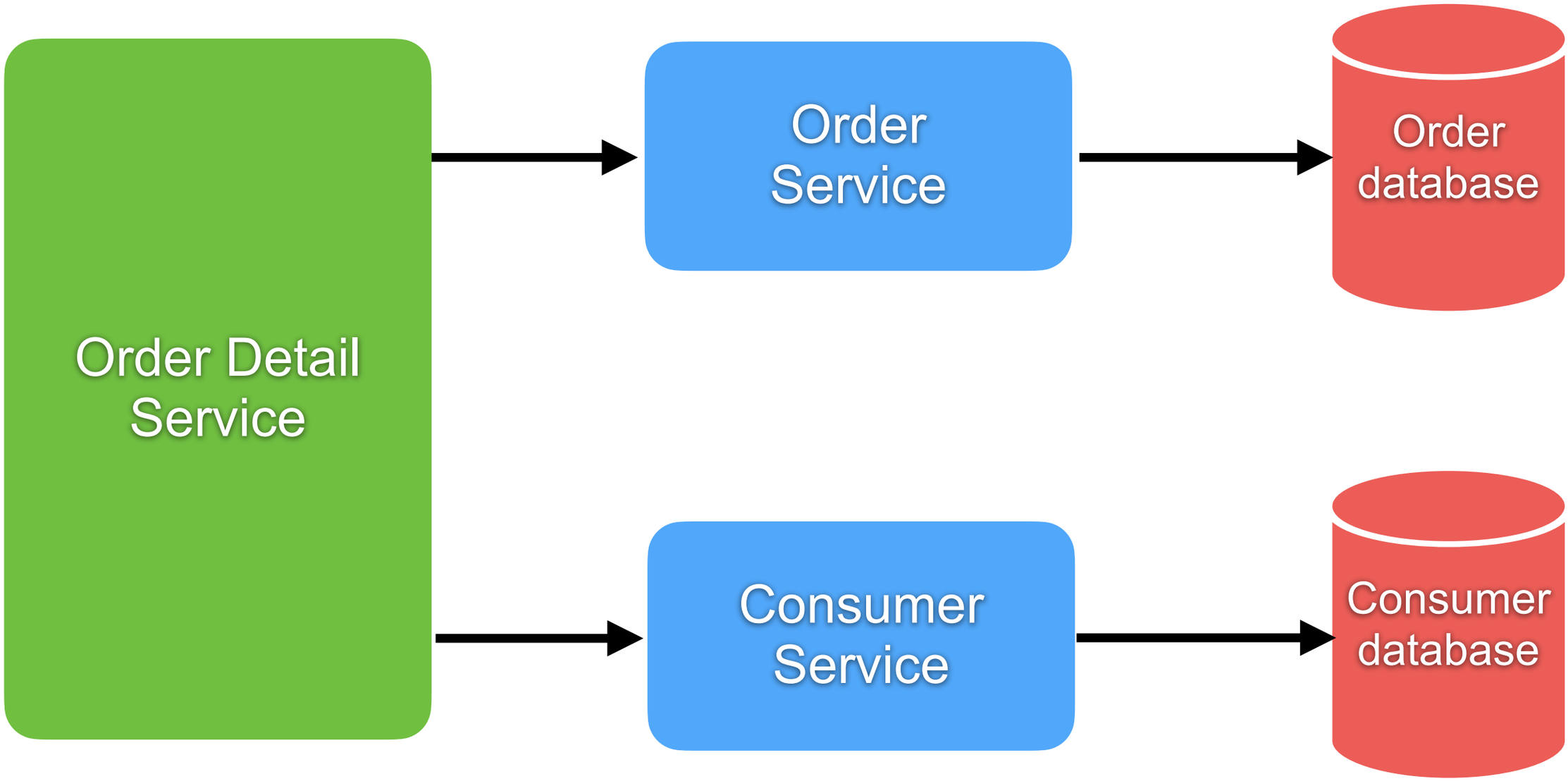
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Caching data



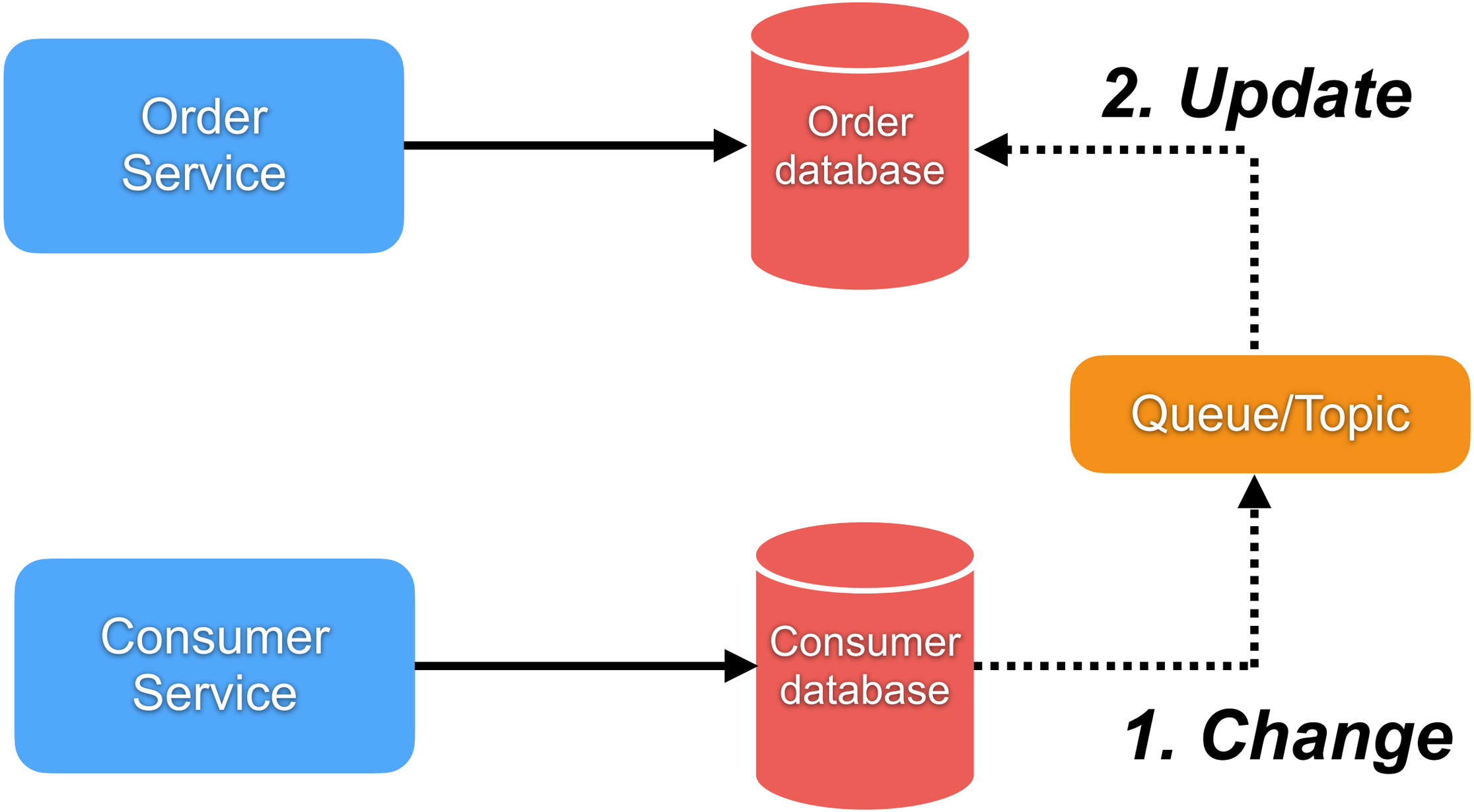
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Order detail orchestration service



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Use messaging to shared data



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Database per service

High complexity

Query data across multiple databases

Challenge “How to join data ?”

Maintain database consistency



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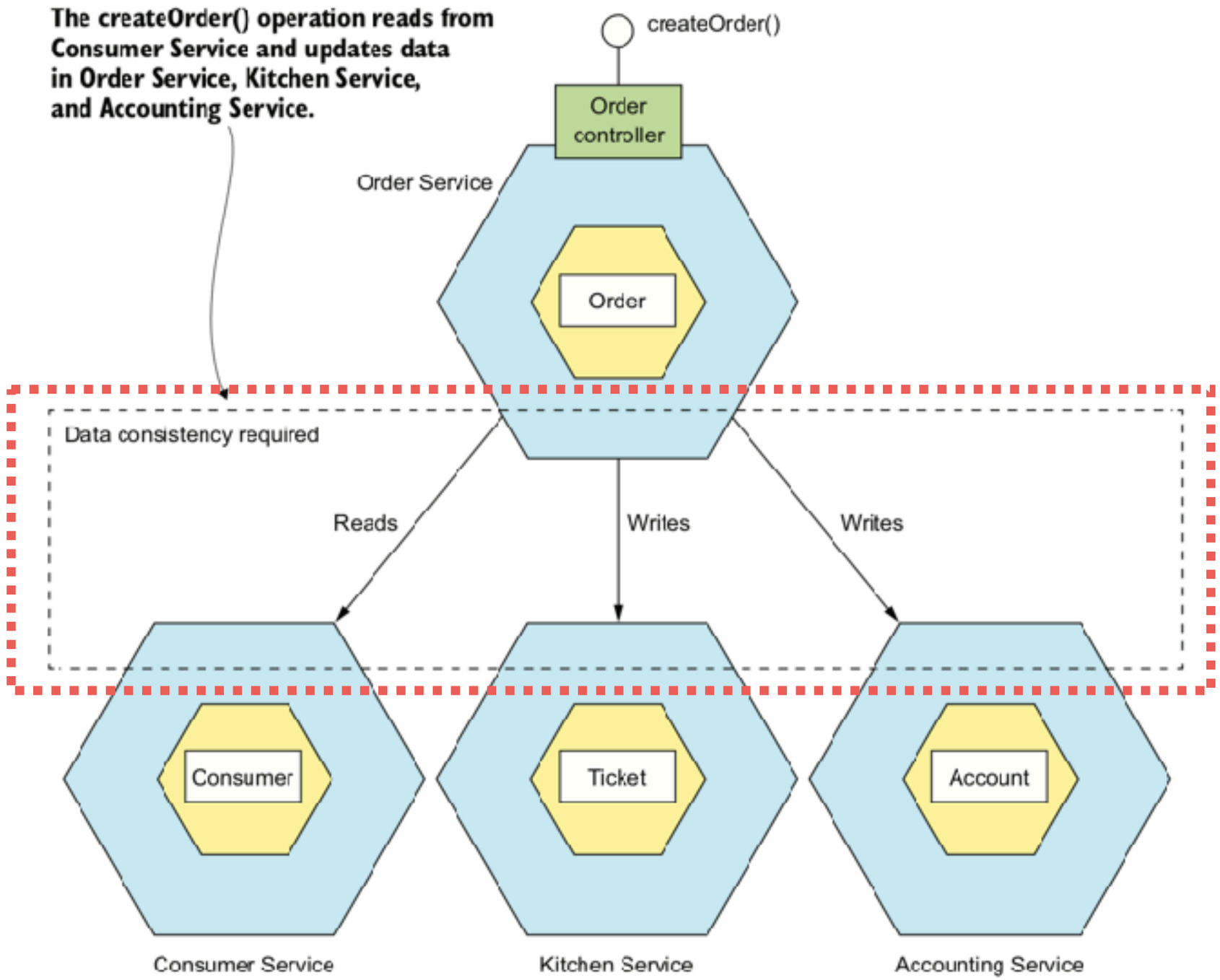
Manage transaction ?

Manage data consistency ?



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Distributed process failure !!



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Can’t use ACID transaction !!



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A - Atomicity

All or Nothing Transactions

C - Consistency

Guarantees Committed Transaction State

I - Isolation

Transactions are Independent

D – Durability

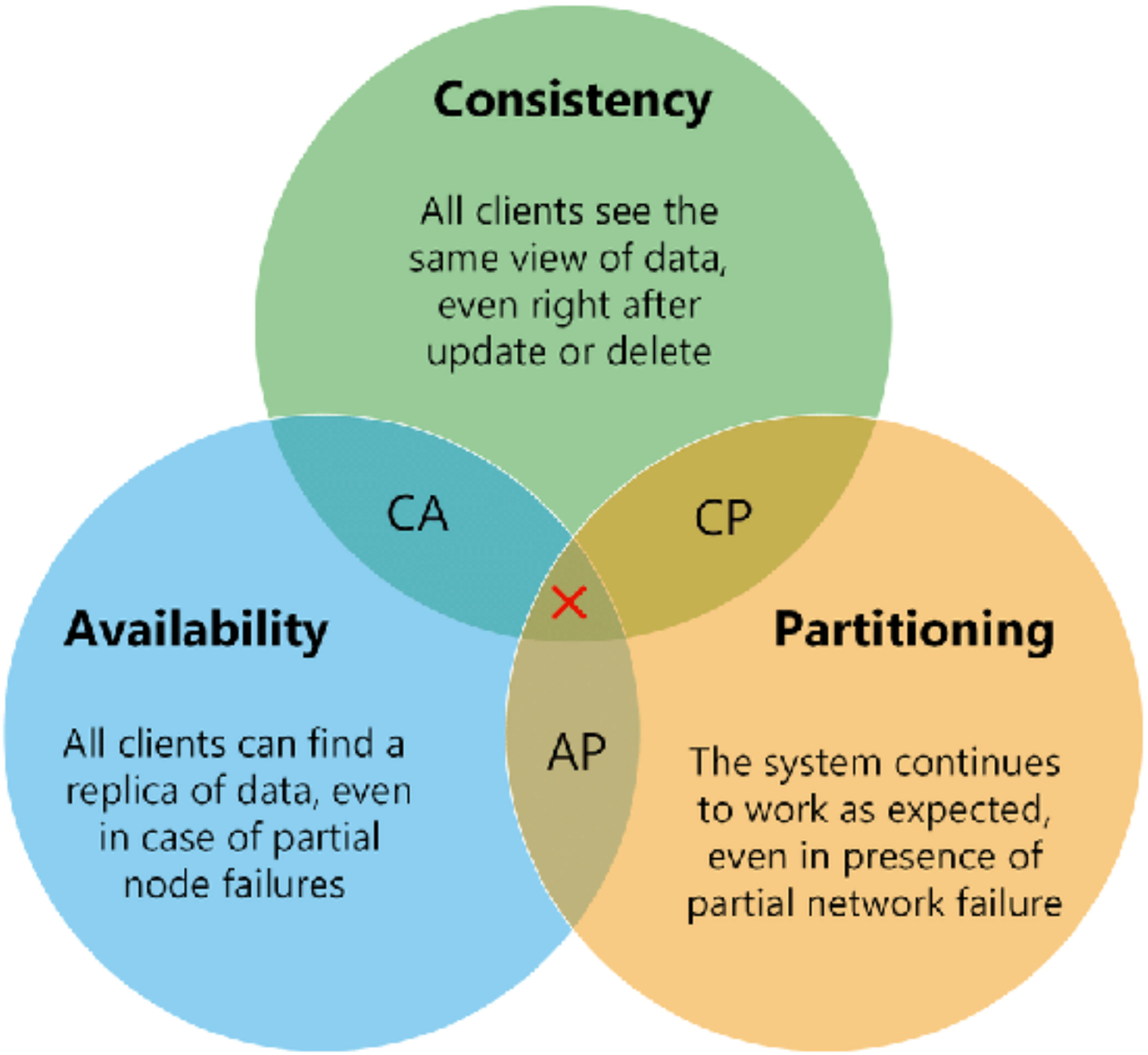
Committed Data is Never Lost

(c) http://blog.sqlauthority.com



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CAP Theorem



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BASE

Basic **A**vailable

Soft state

Eventually consistent



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ACID vs BASE

ACID	BASE
Scale up (vertical)	Scale out (horizontal)
Strong consistency	Weak/Eventual consistency
Isolation	Last write win, availability first
Transaction	Application transaction
Fix schema	Flexible schema
Expensive joins and relations	Free



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Distributed transaction

Common approach is Two-phase commit(2PC)

Use transaction manager to split operations across multiple resources in 2 phases

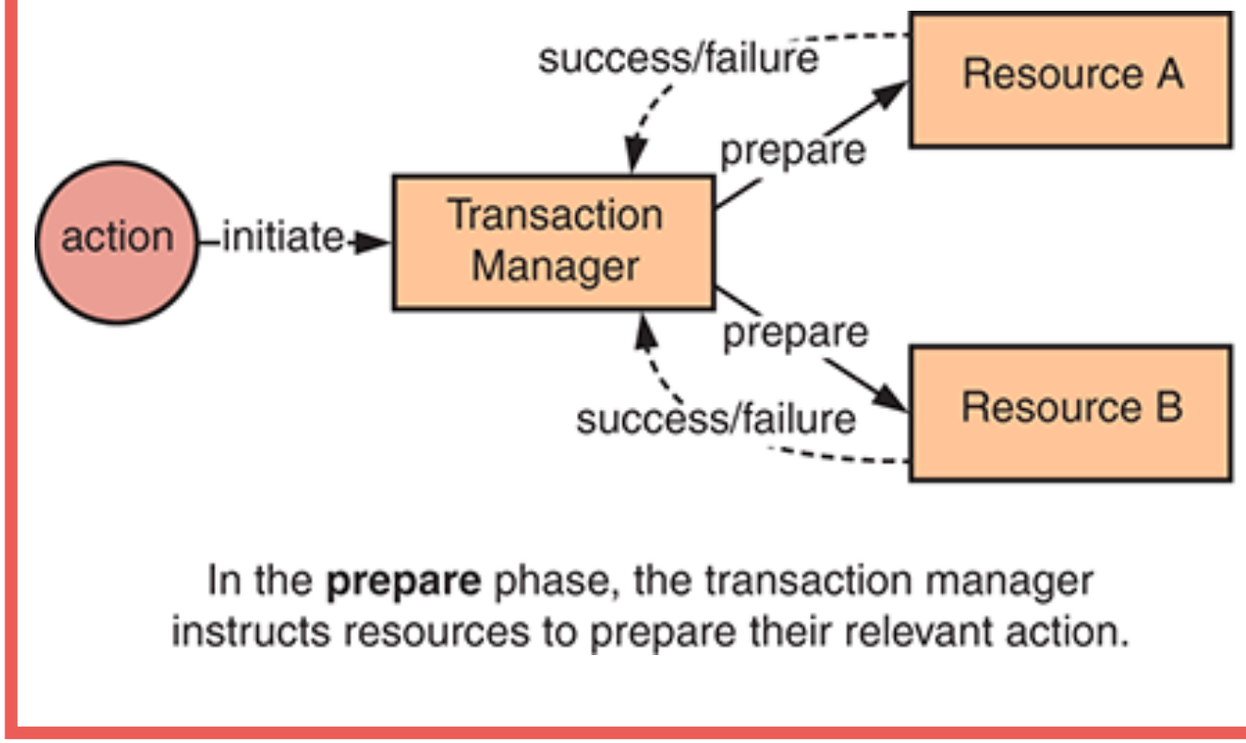
1. Prepare
2. Commit



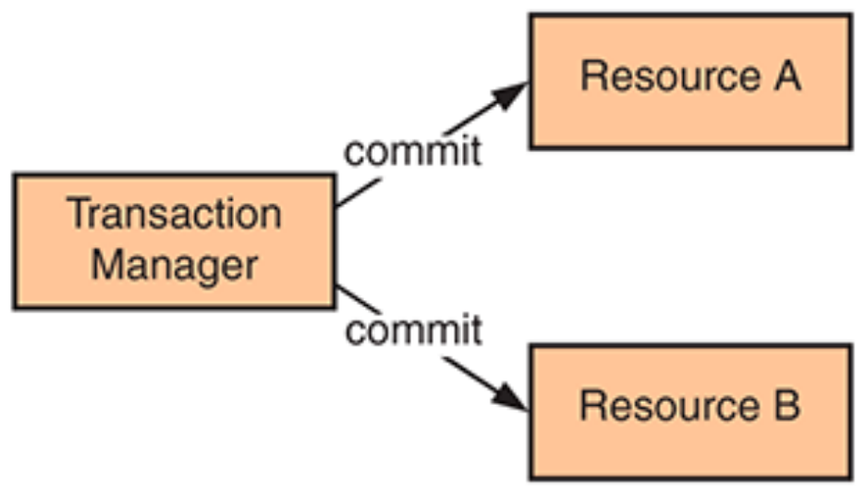
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Two-phase commit

Phase 1: prepare



In the **prepare** phase, the transaction manager instructs resources to prepare their relevant action.

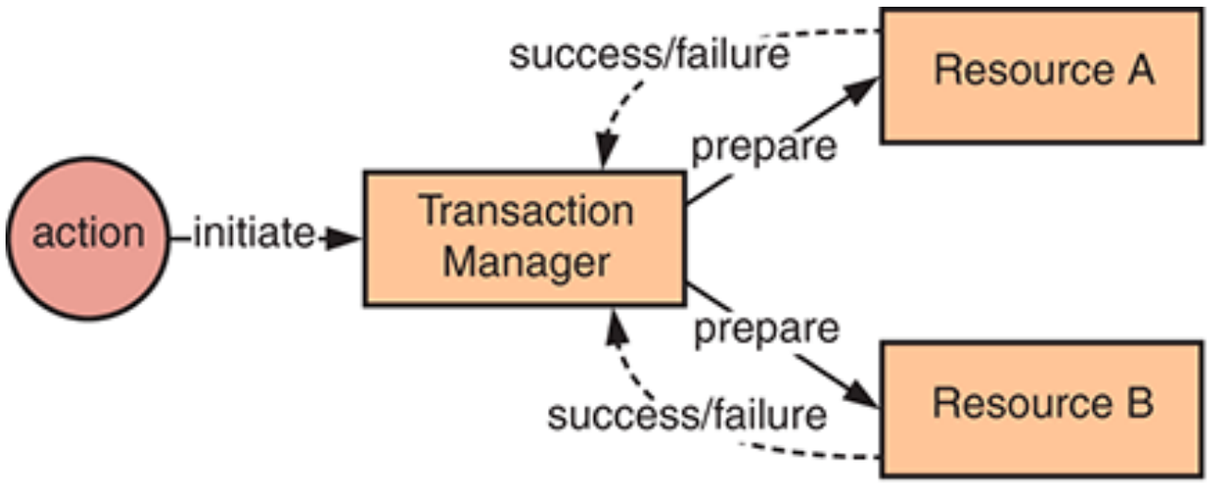


In the **commit** phase, the transaction manager instructs resources to commit or abort the prepared action.

https://en.wikipedia.org/wiki/Two-phase_commit_protocol

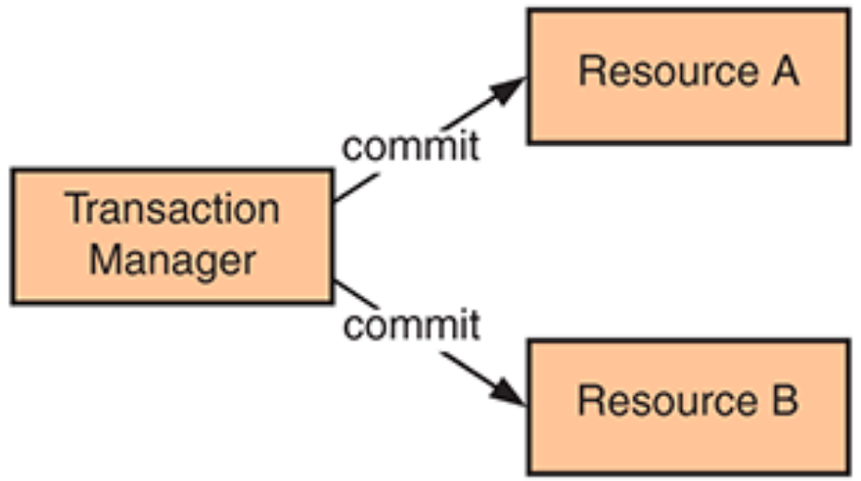
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Two-phase commit



In the **prepare** phase, the transaction manager instructs resources to prepare their relevant action.

Phase 2: commit



In the **commit** phase, the transaction manager instructs resources to commit or abort the prepared action.

https://en.wikipedia.org/wiki/Two-phase_commit_protocol

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Distributed transaction places a lock on resources under transaction to ensure isolation



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Inappropriate for long-running operations



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Increase risk of contention and deadlock



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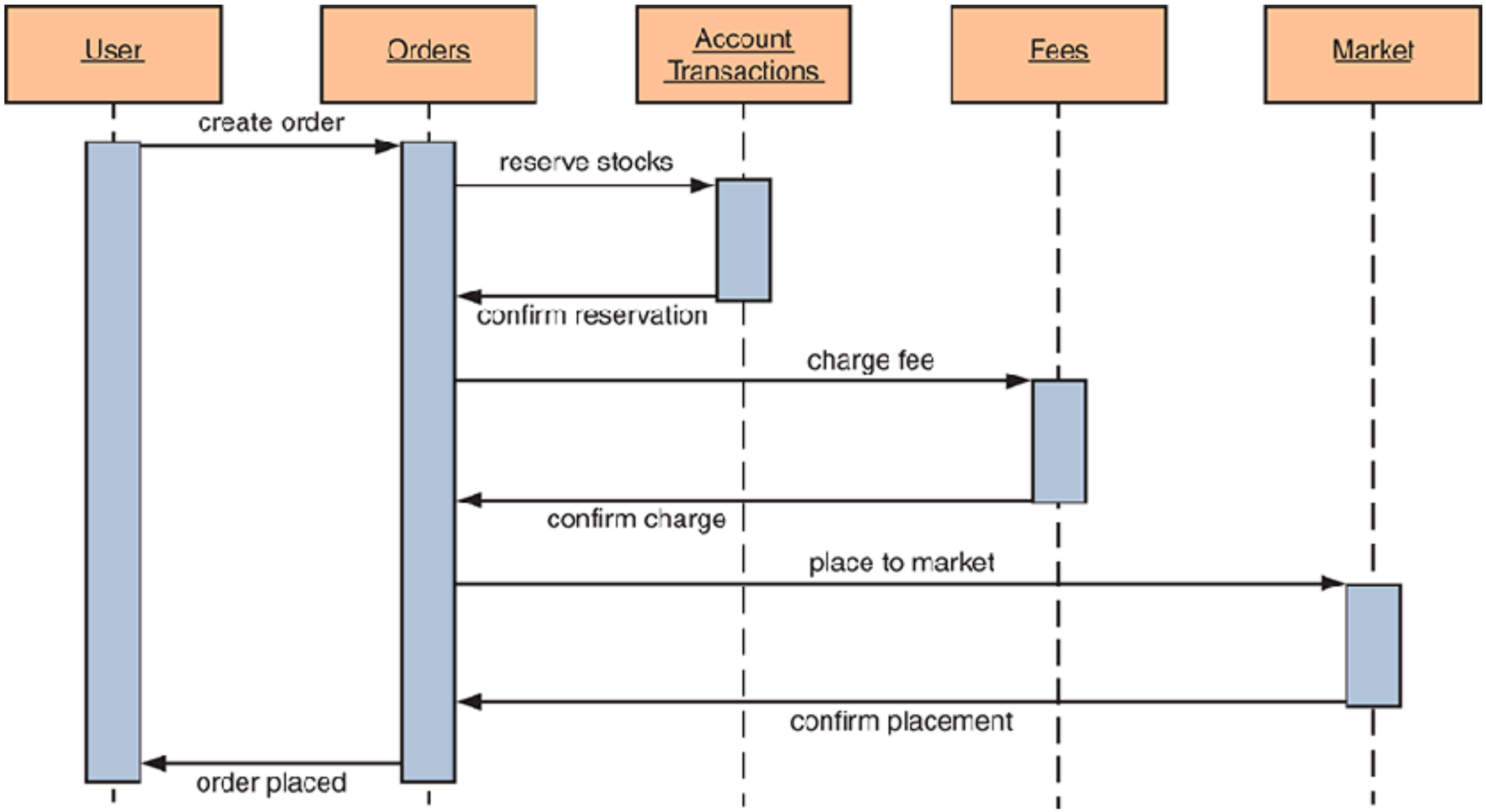
We need …

Don’t need distributed transaction
Reduce complexity of code
Reliable



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Synchronous process



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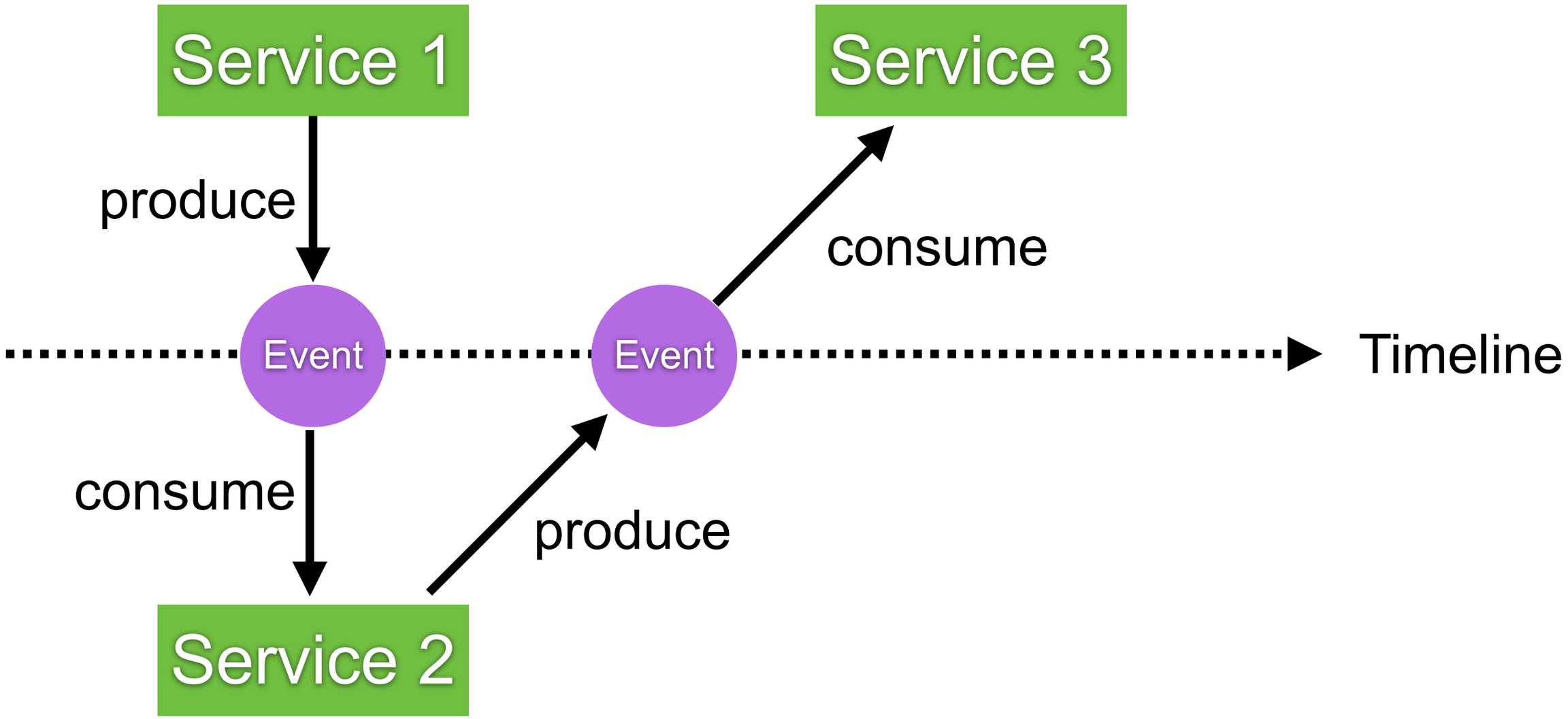
Event-Driven Architecture (EDA)



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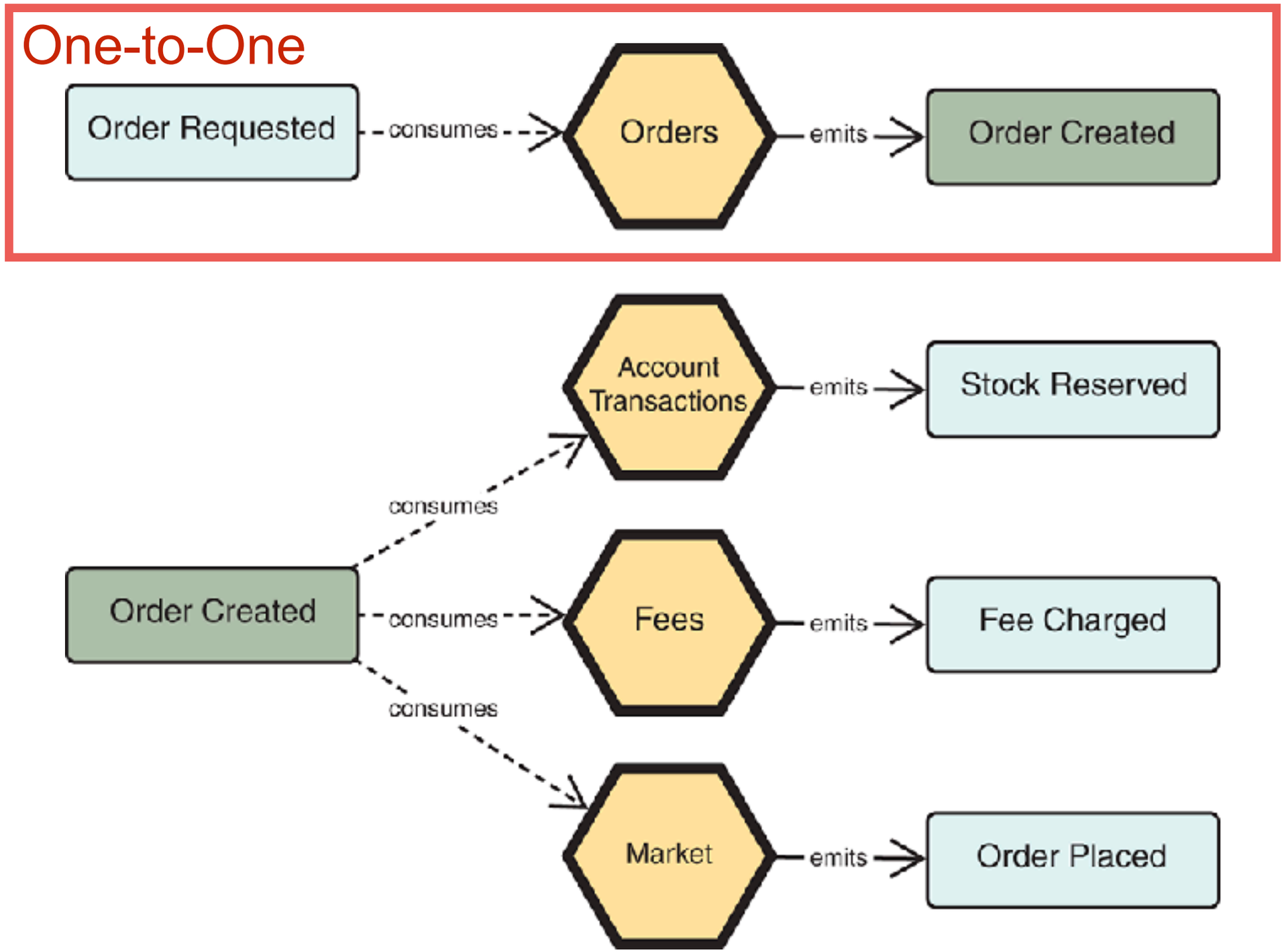
Choreography pattern

Sequence of Tx and emit **event** or **message** that trigger the next process in Tx



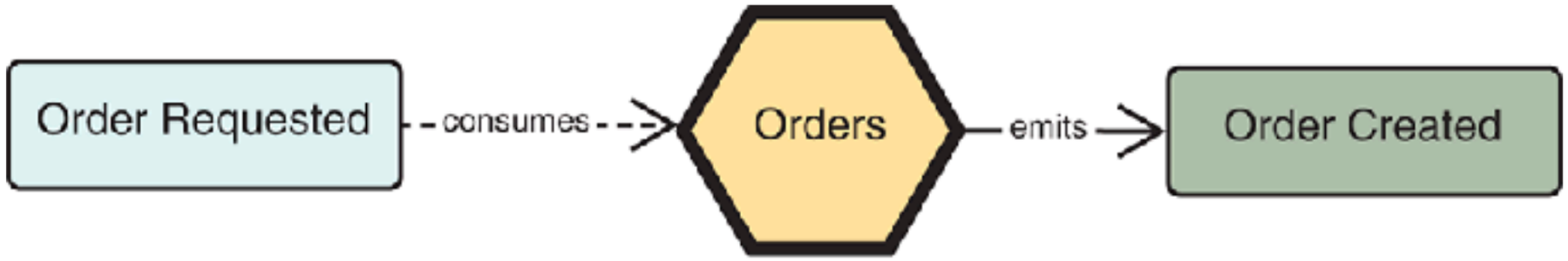
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Service consume and emit event

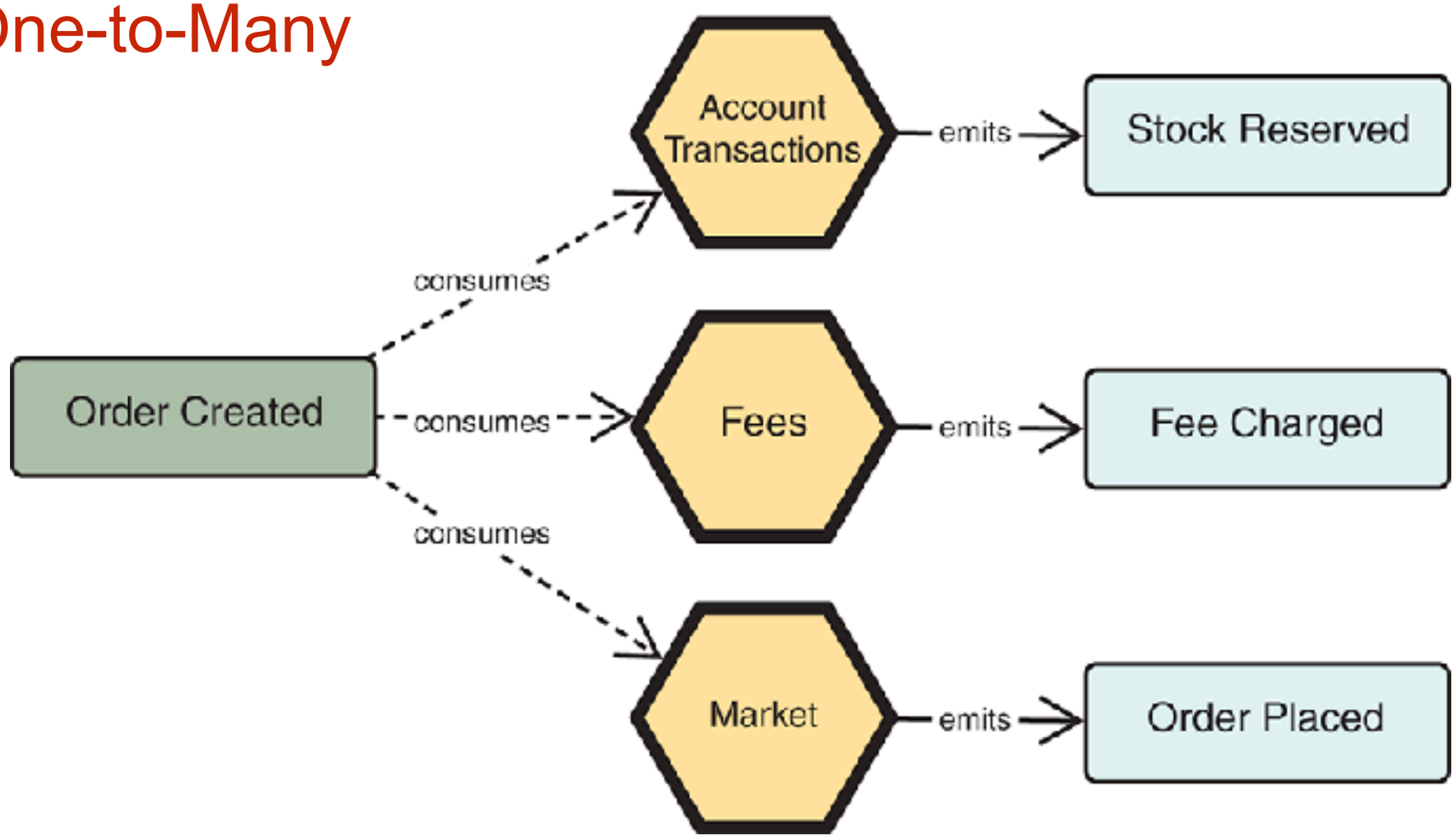


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Service consume and emit event

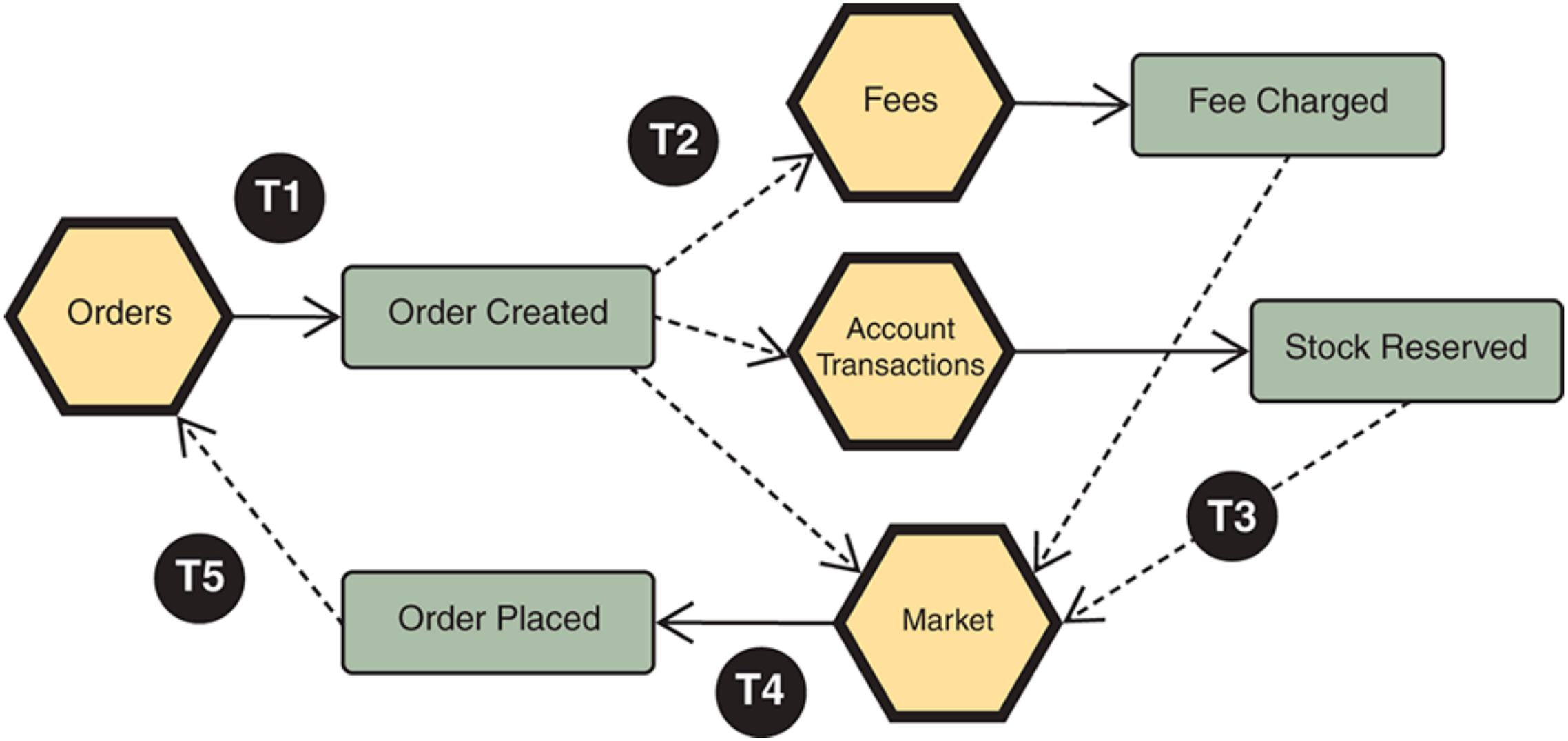


One-to-Many



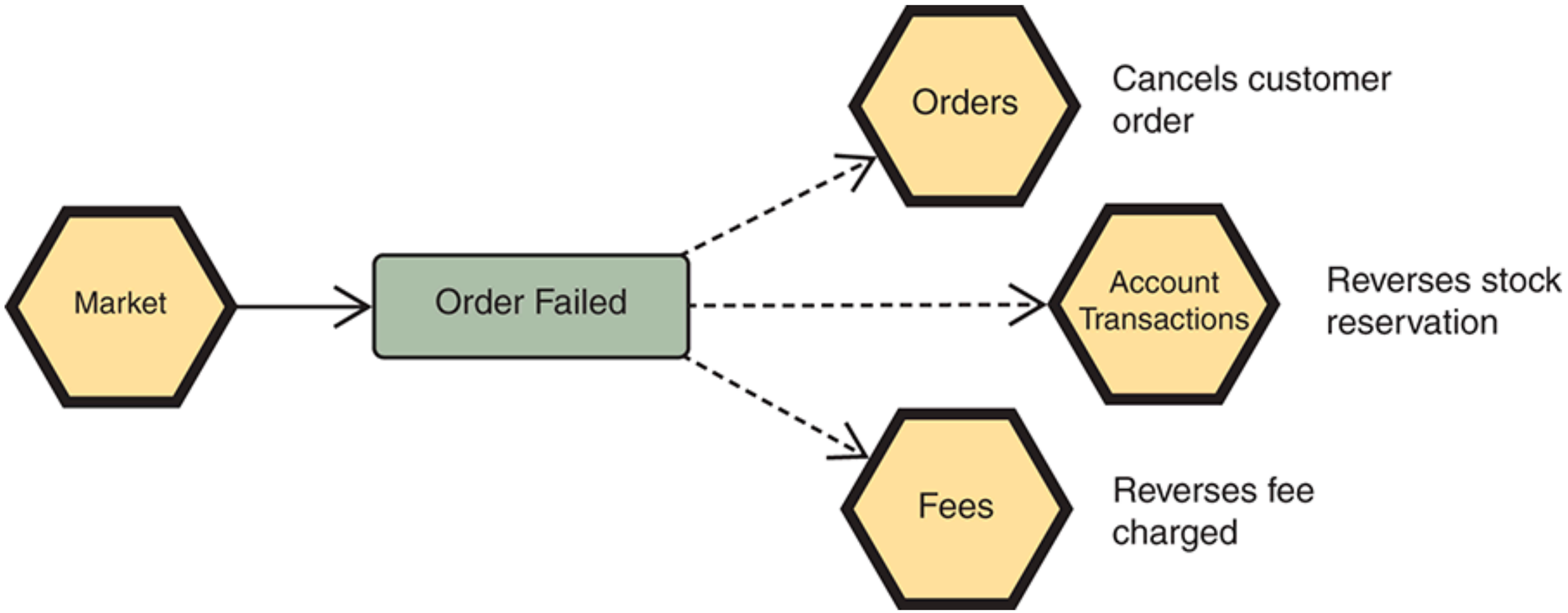
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Example (Success)



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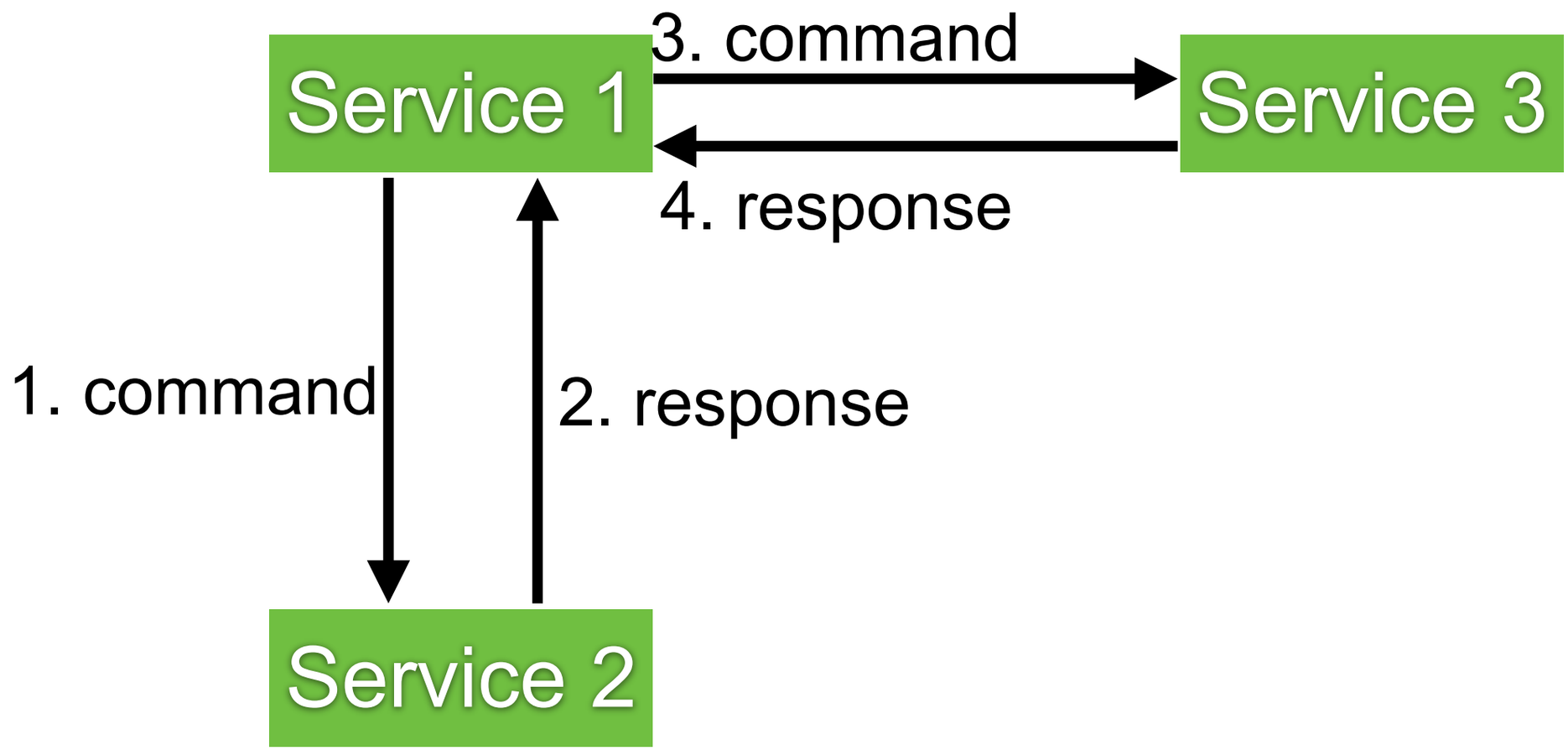
Example (Fail)



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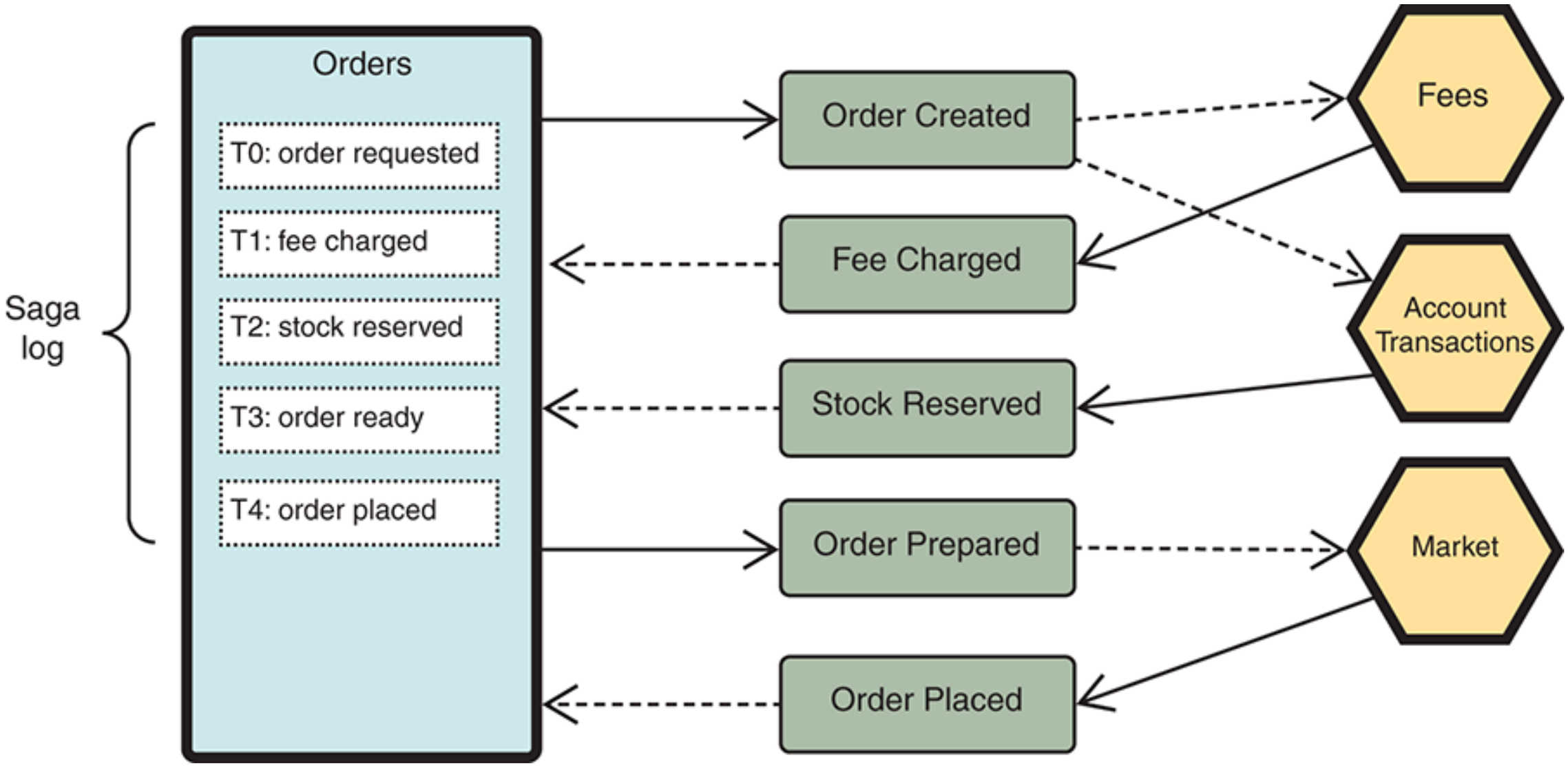
Orchestrate pattern

Sequence of Tx and emit **event** or **message** that trigger the next process in Tx



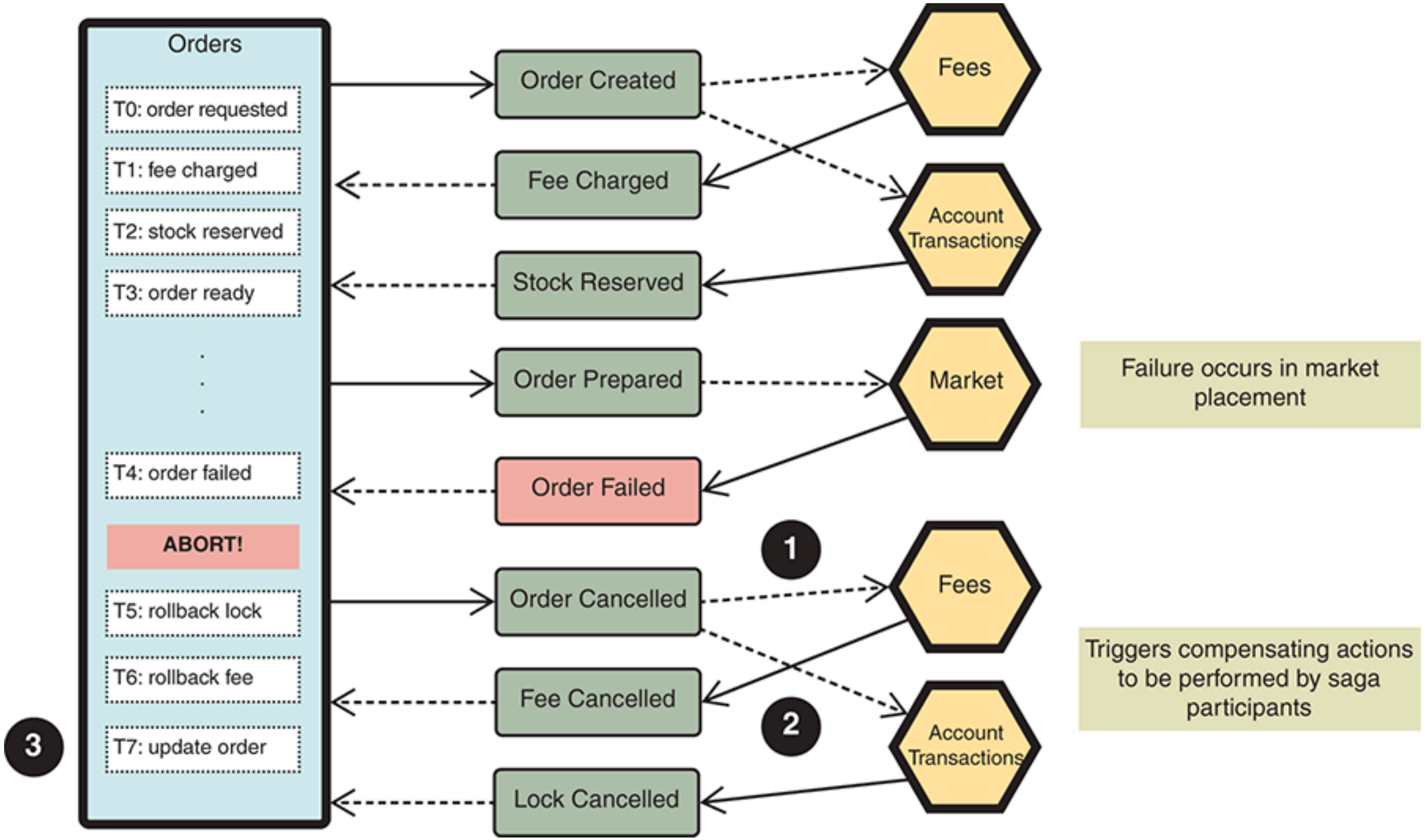
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Example (Success)



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Example (Fail)



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Consistency patterns

Name	Strategy
Compensating action	Perform action that undo prior action(s)
Retry	Retry until success of timeout
Ignore	Do nothing in the event of errors
Restart	Reset to the original state and start again
Tentative operation	Perform a tentative operation and confirm (or cancel) later



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“Saga”



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Message vs Event

Message is addressed to someone

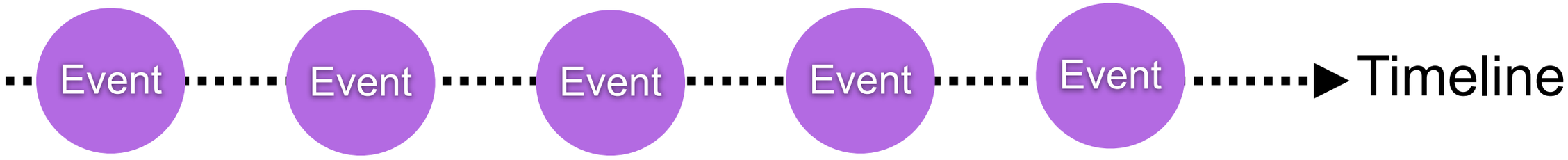
Event is something that happen and someone can react to that



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Event sourcing

Maintain source of truth of business
Immutable sequence of events



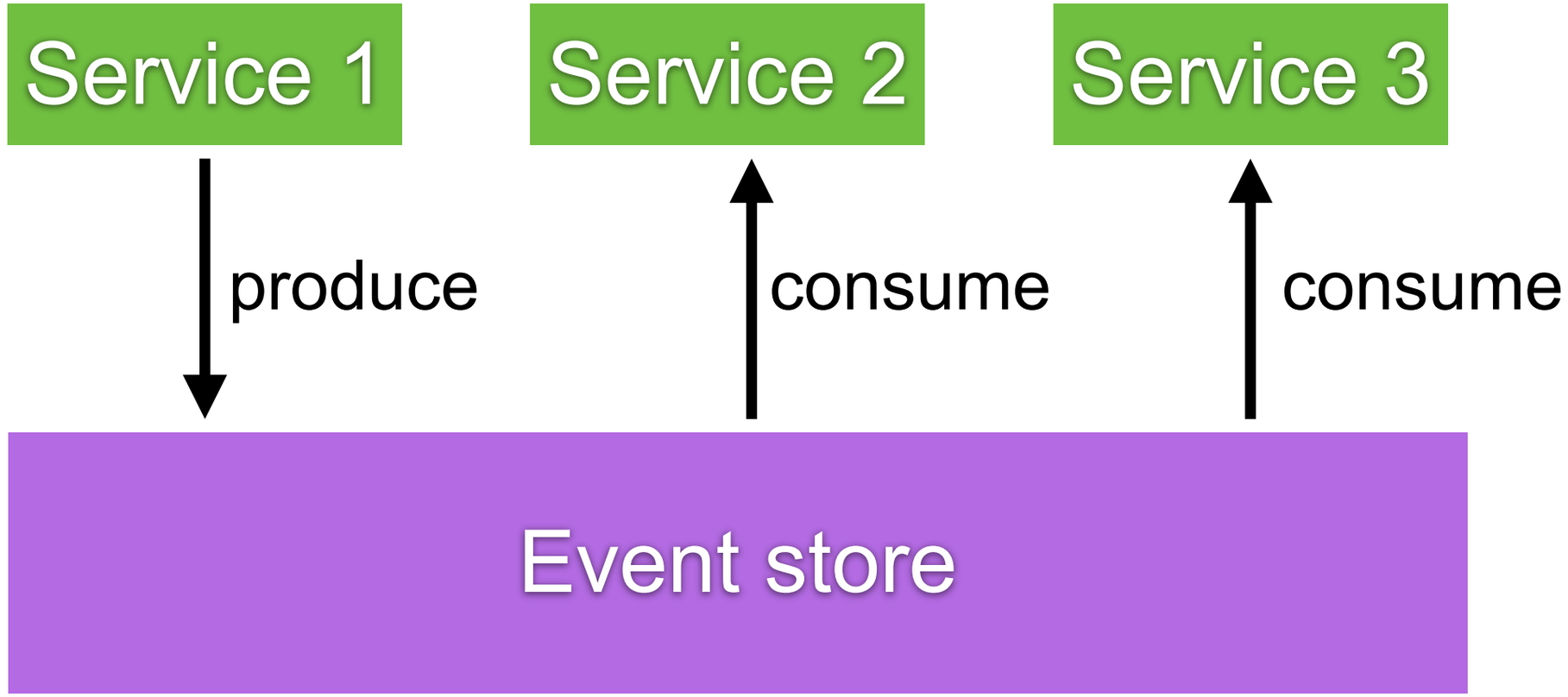
Sequence of event is keep in **Event store**



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Event store

Keep events in order
Broadcast new event



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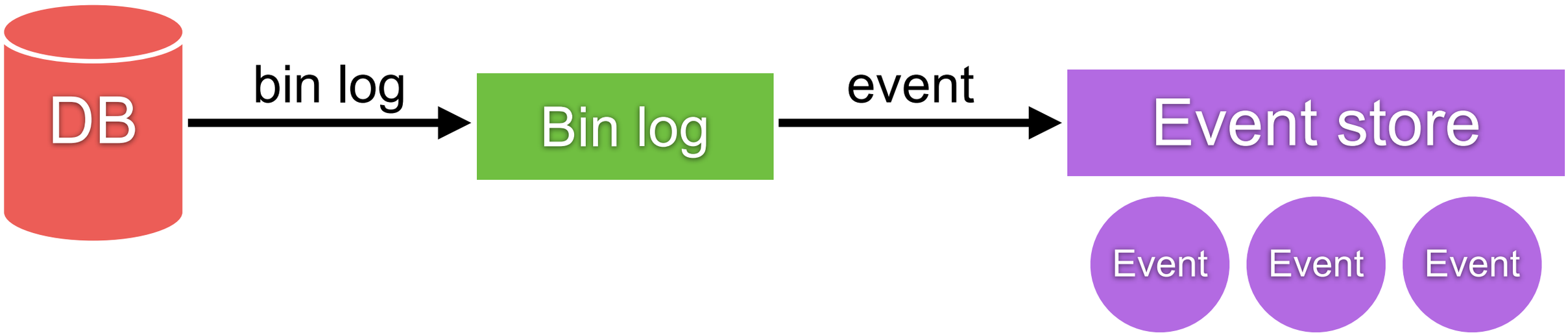
With Event sourcing, we can **decouple services** (query and command)



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Event sourcing with database

Working with database



Binlog or Binary log event is information about data modification made to database

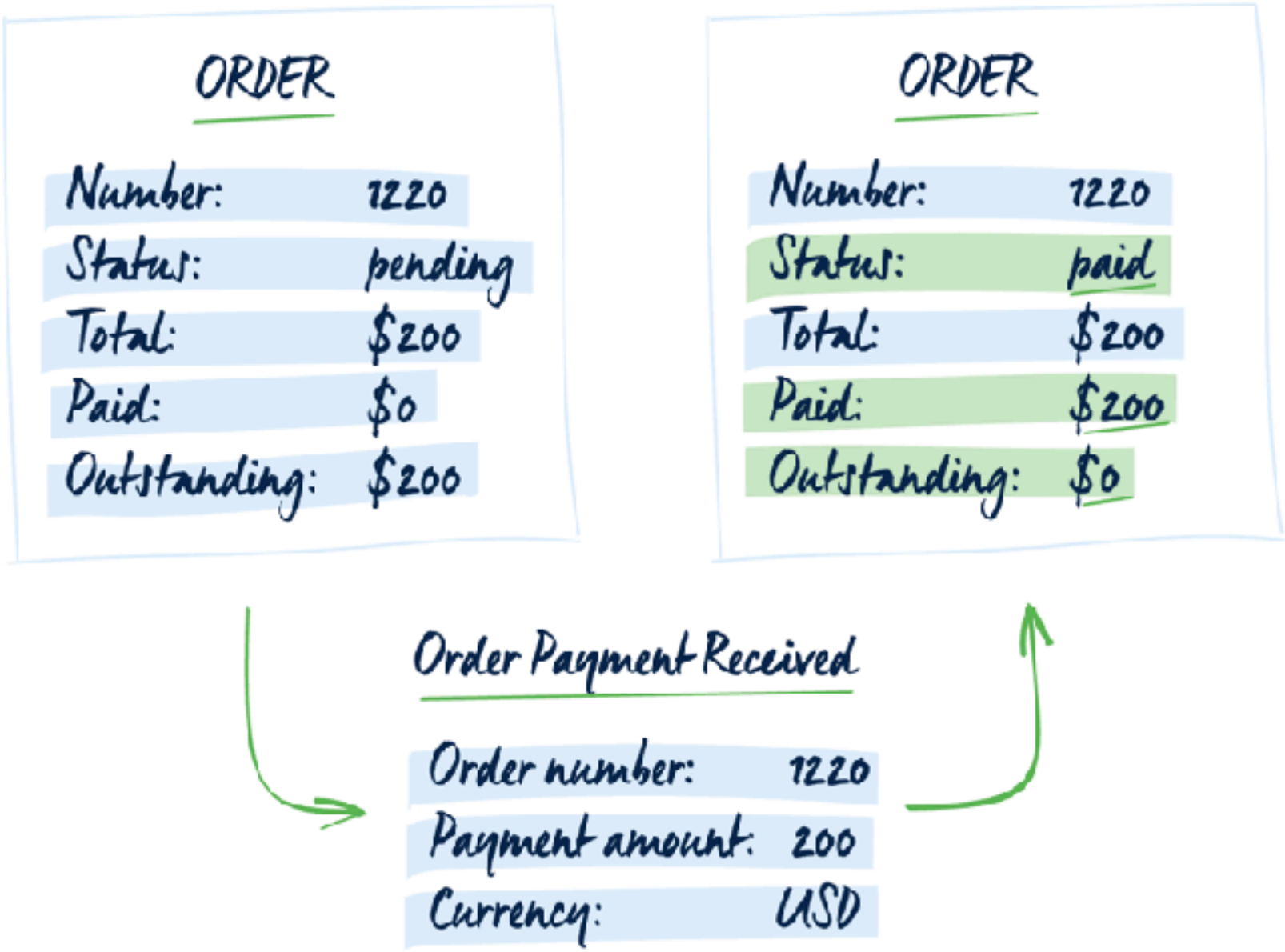
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Example of Events



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Order payment



https://www.eventstore.com/event-sourcing

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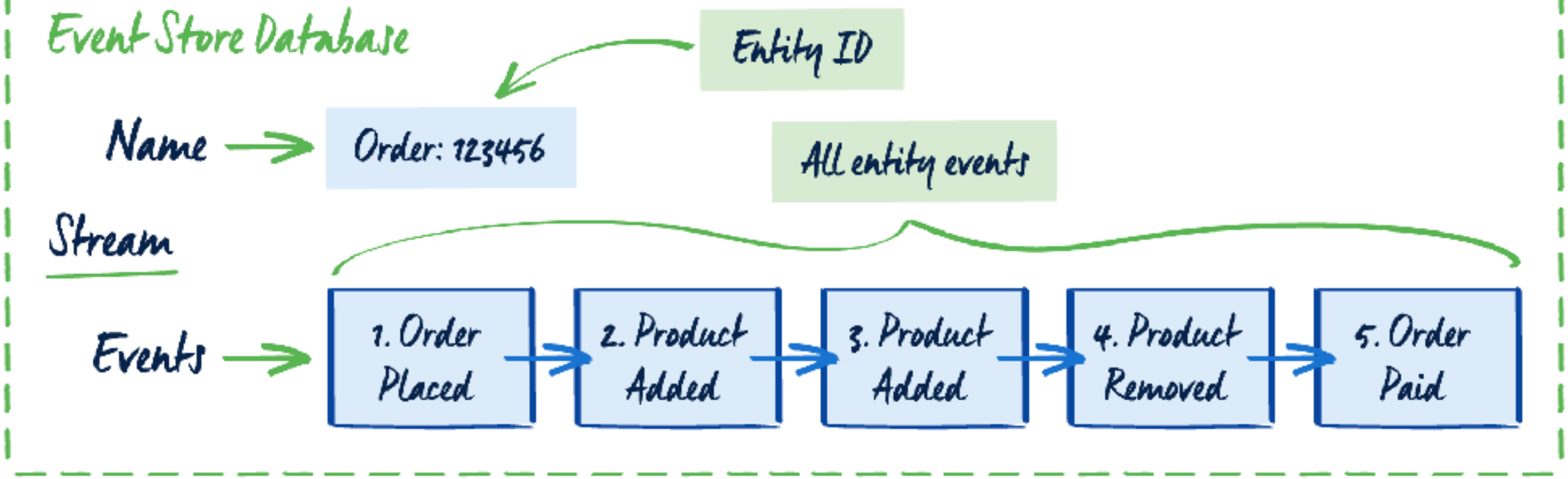
Apply discount



https://www.eventstore.com/event-sourcing

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Event sourcing



https://www.eventstore.com/event-sourcing

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How to implement queries data ?



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Query data from multiple services

API composition

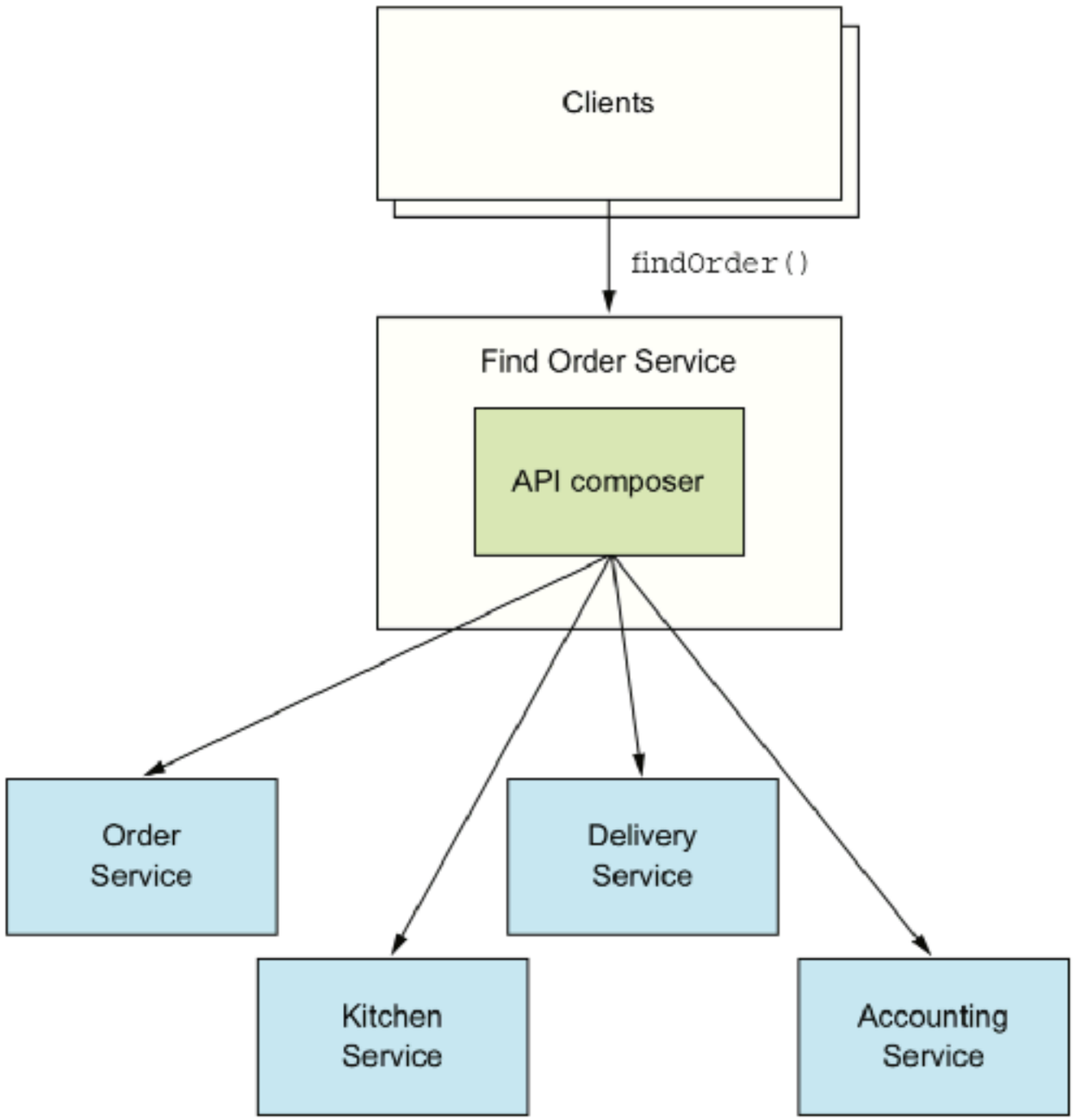
Cold data from services

CQRS (Command Query Responsibility Segregation)



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API composition pattern



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Drawbacks

Increase overhead

Lack of transactional data consistency

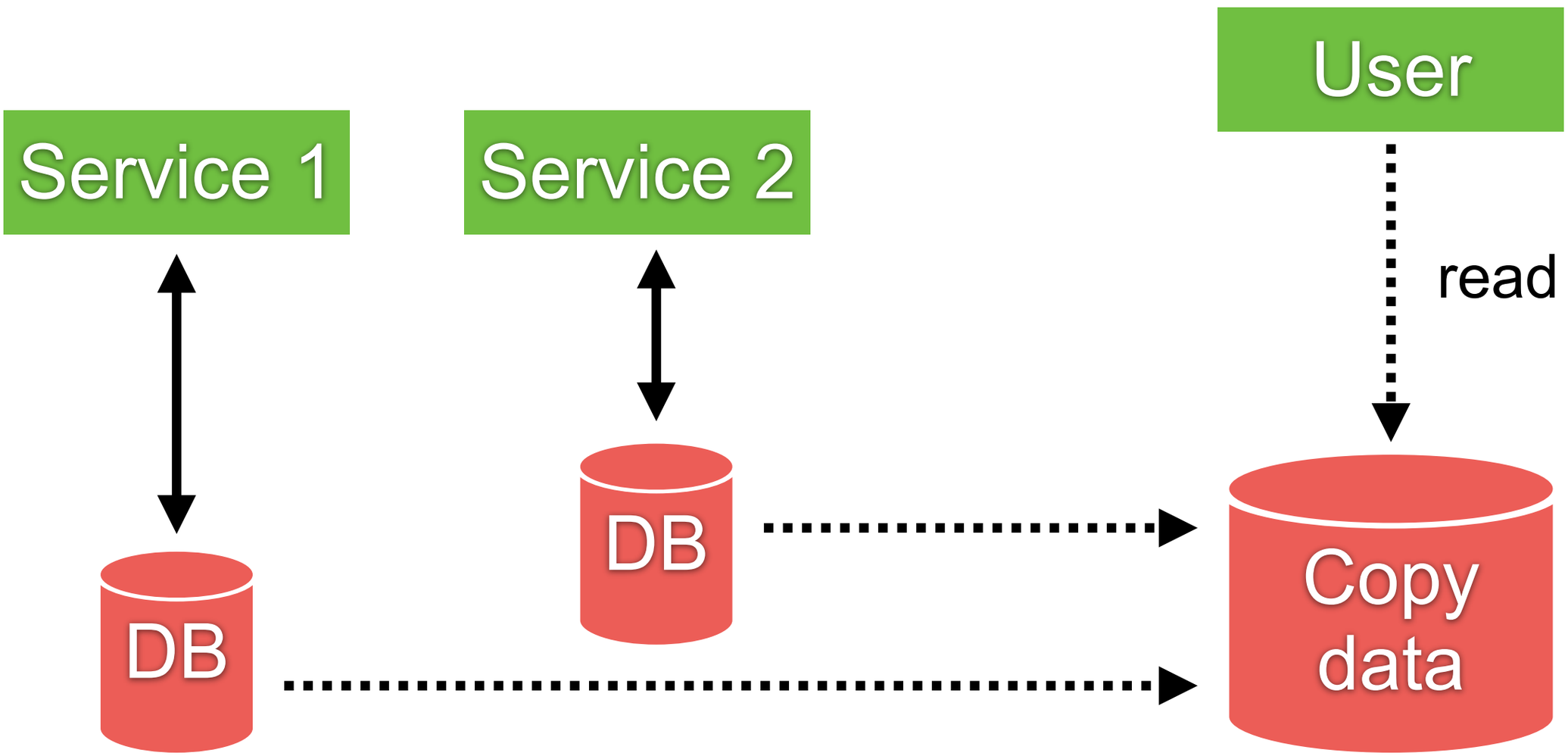
Reduce availability



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Storing copy data

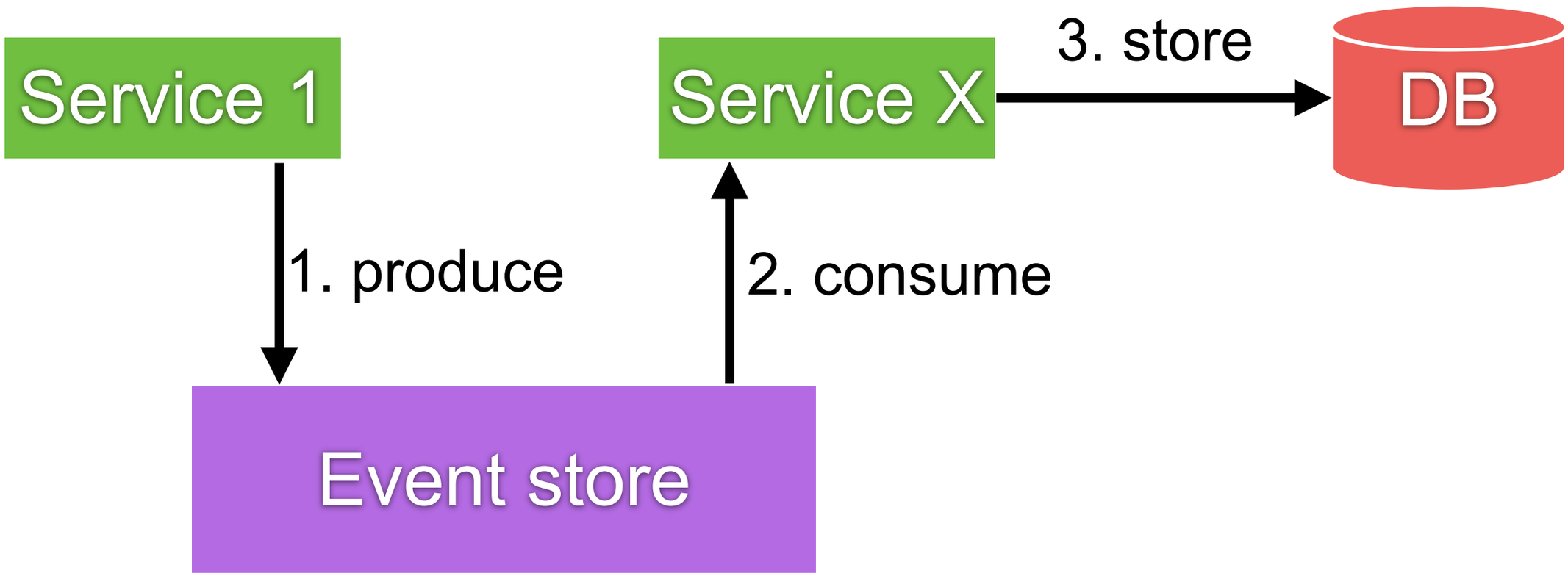
Copy or caching data to other database



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Working with Event-based

Publish event to store copy data



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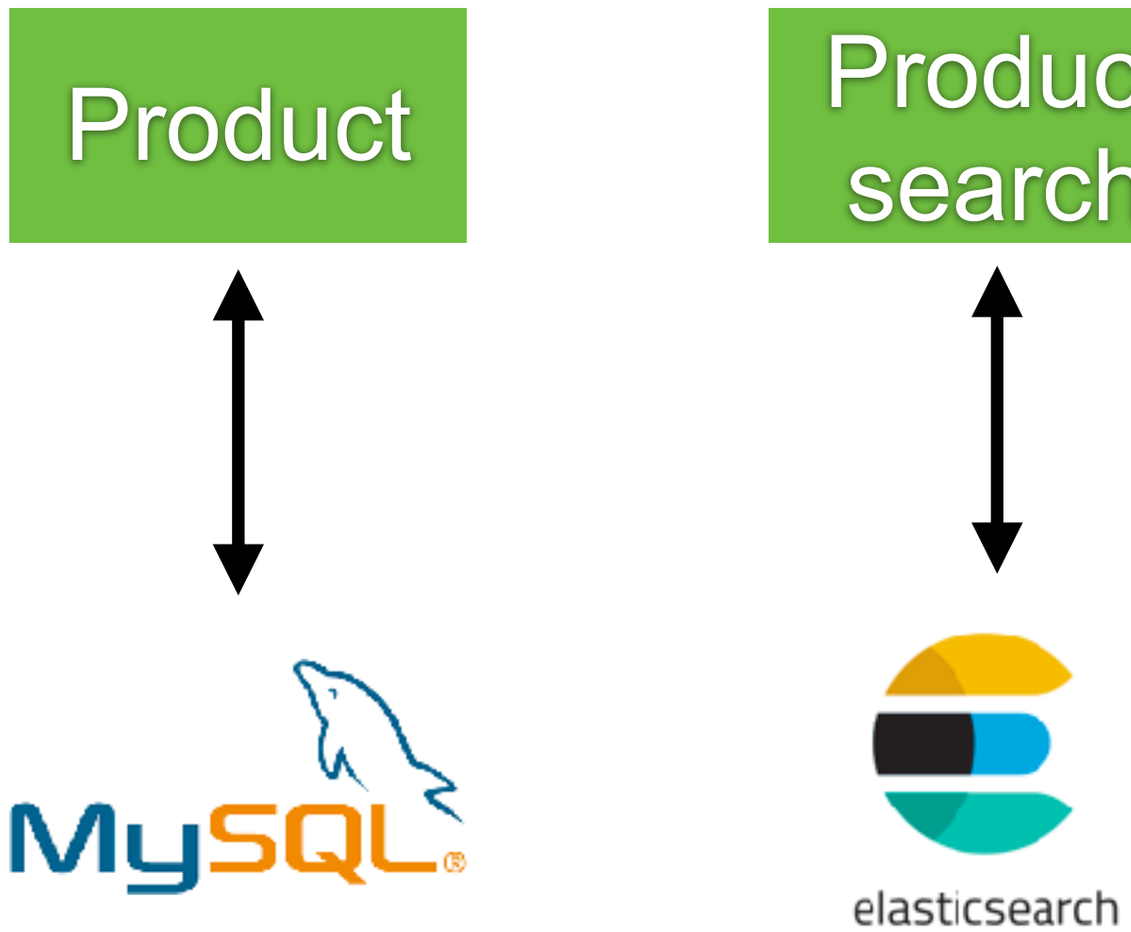
Separate query and command



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Separate data for read and write

For example MySQL to write, Elasticsearch to search



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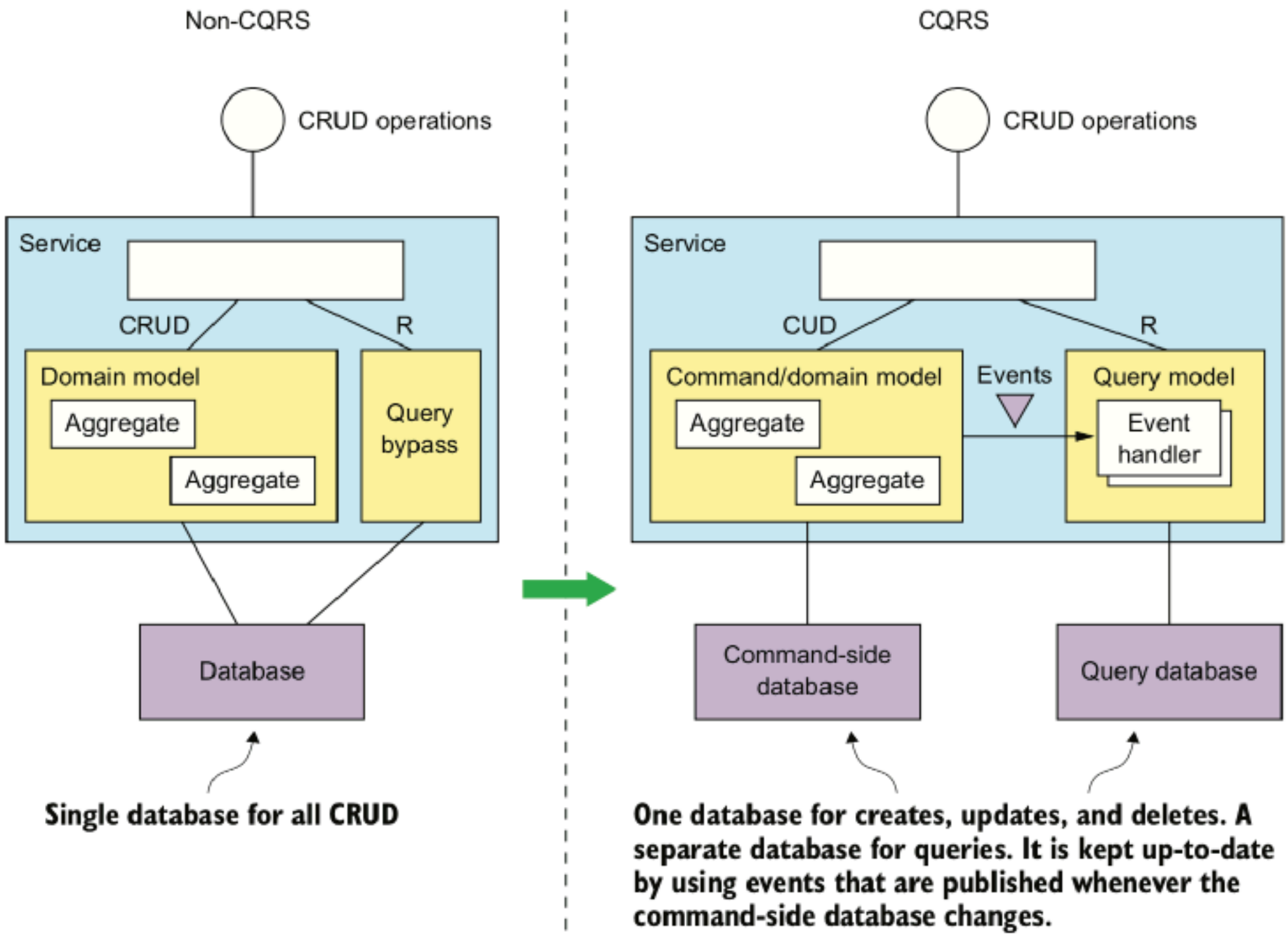
CQRS

Command **Q**uery **R**esponsibility **S**egregation



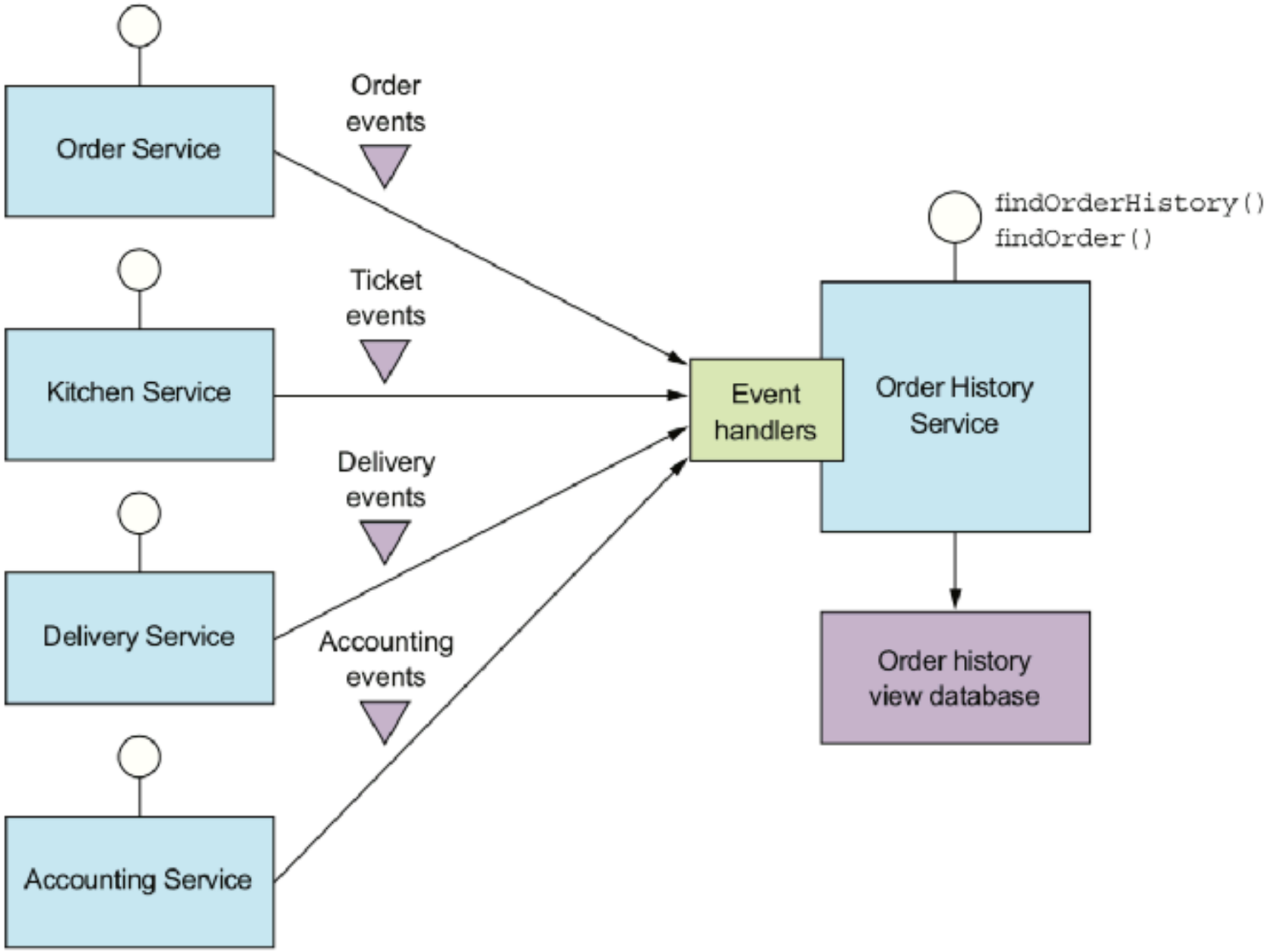
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CQRS = Separate command from query



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CQRS = Query-side service



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Benefits

Improve separation of concern
Efficiency query



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Drawbacks

More complex

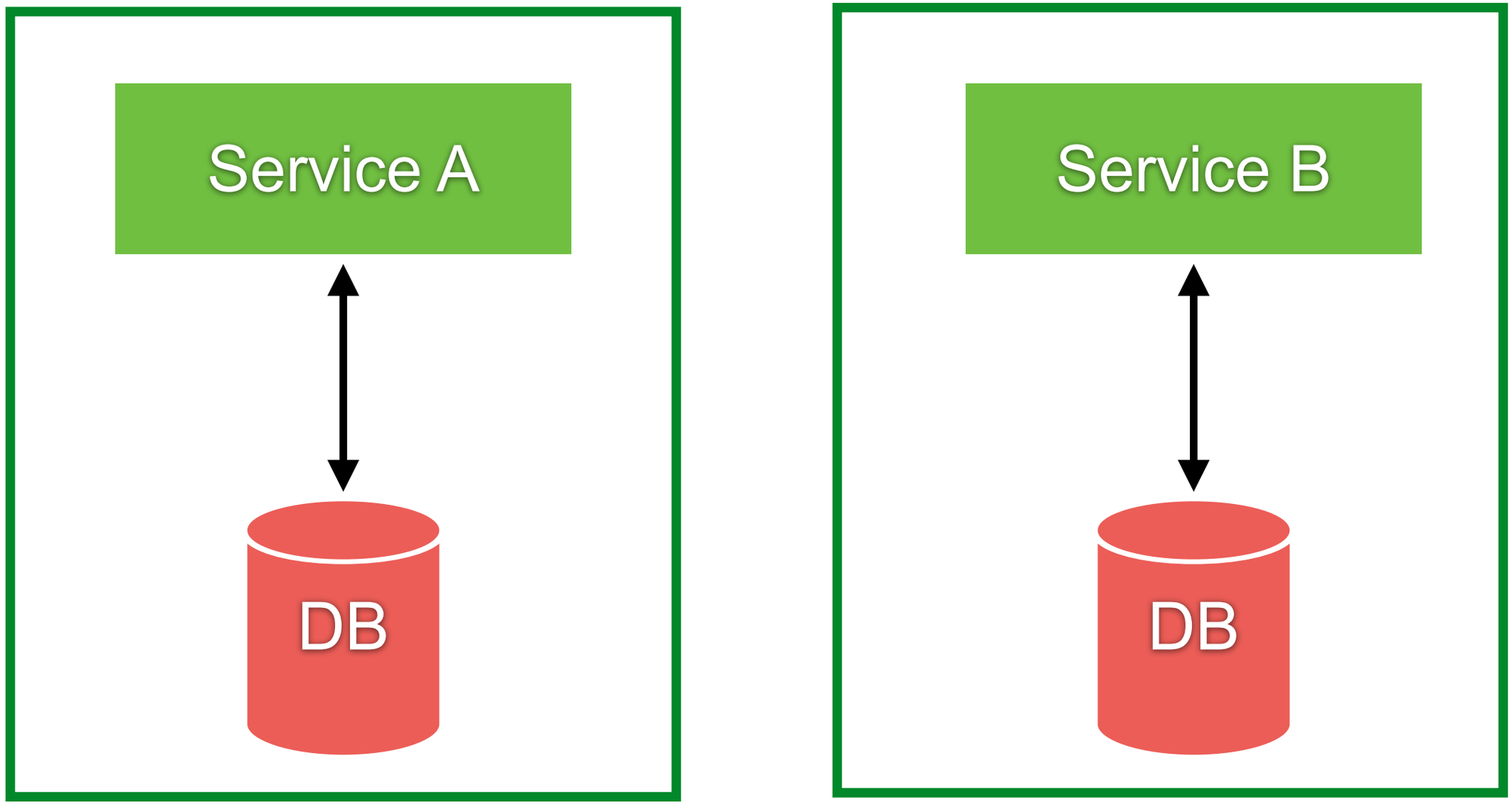
Lag between **command** and **query** side



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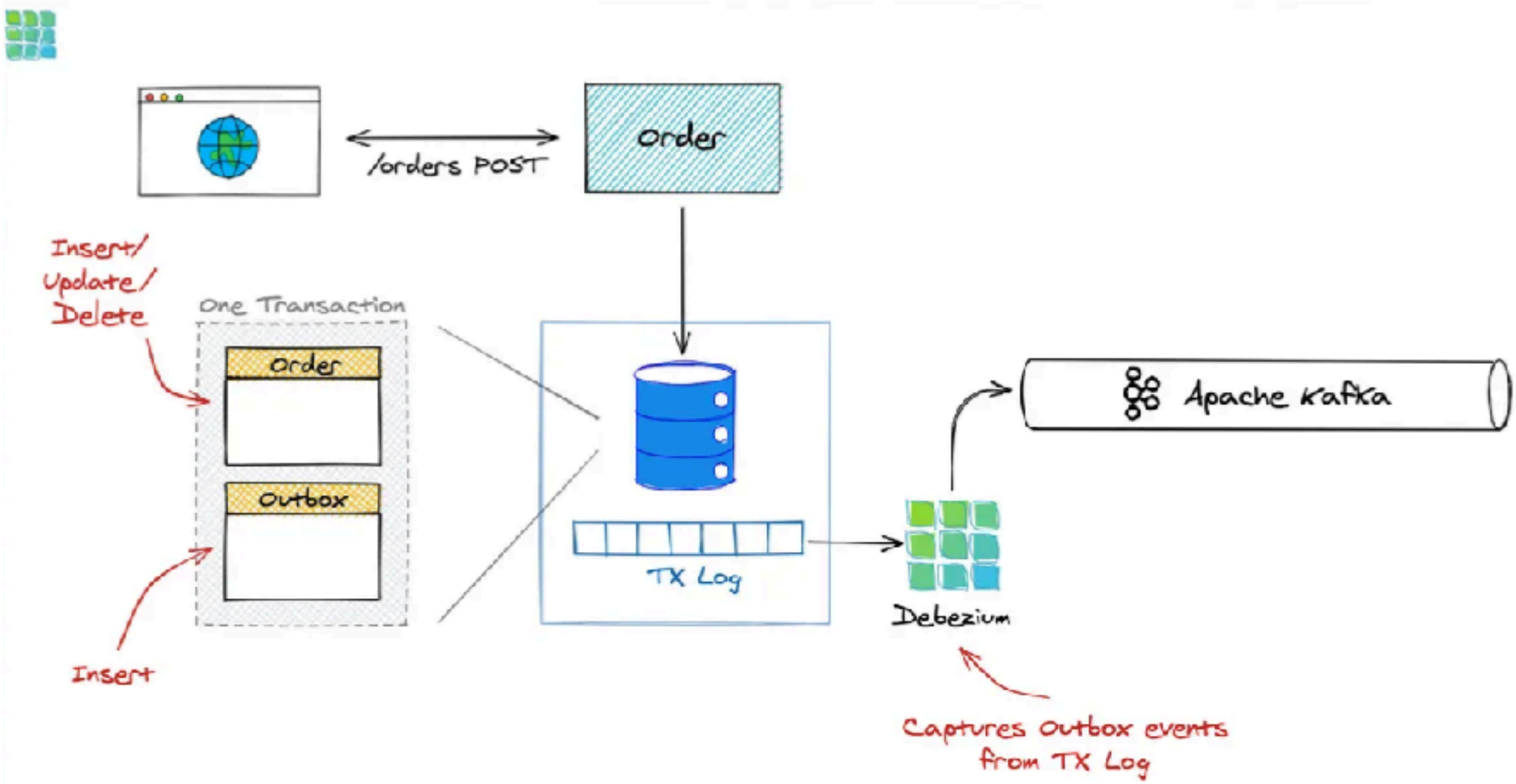
Outbox pattern

Decouple services using Pub/Sub
to exchange data between services



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Example of Outbox pattern



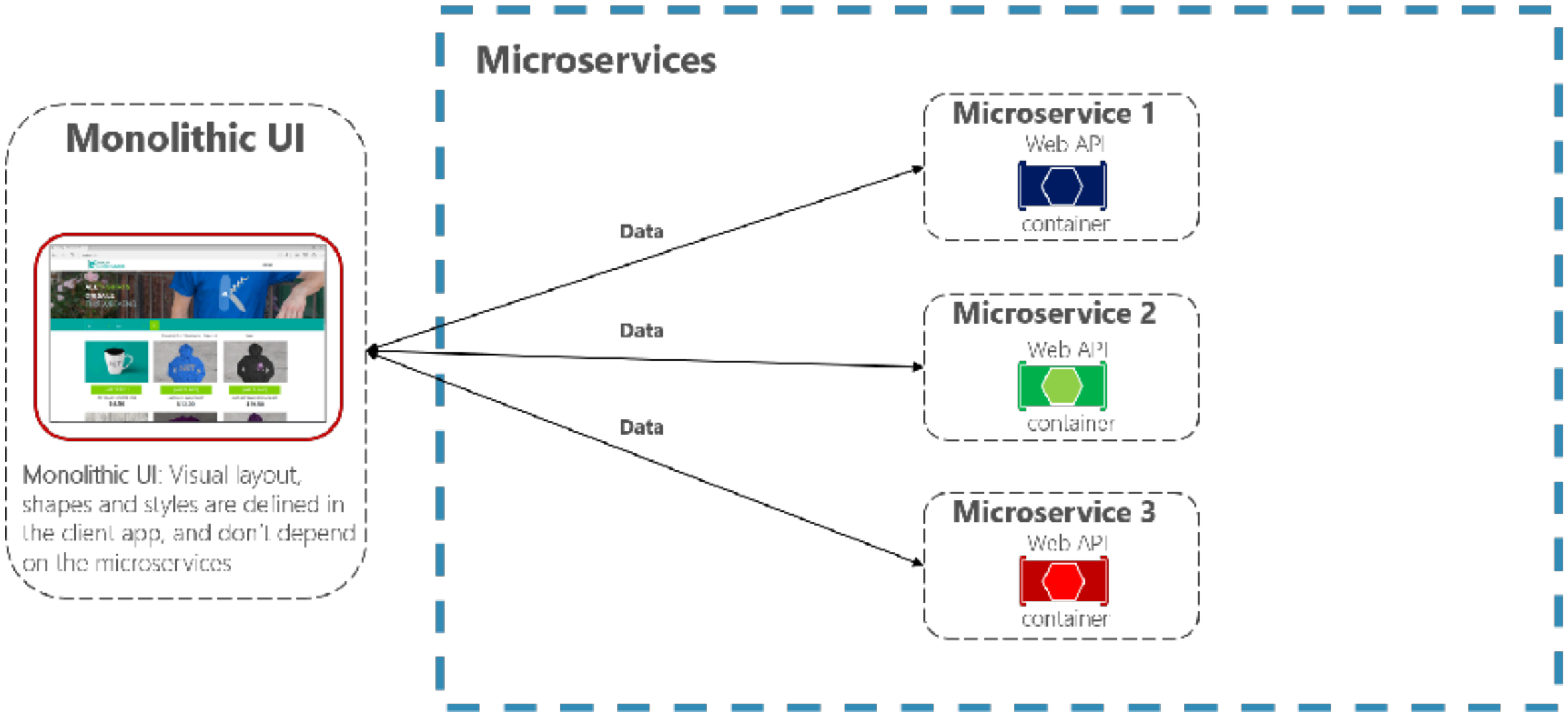
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Integrate services with User Interface



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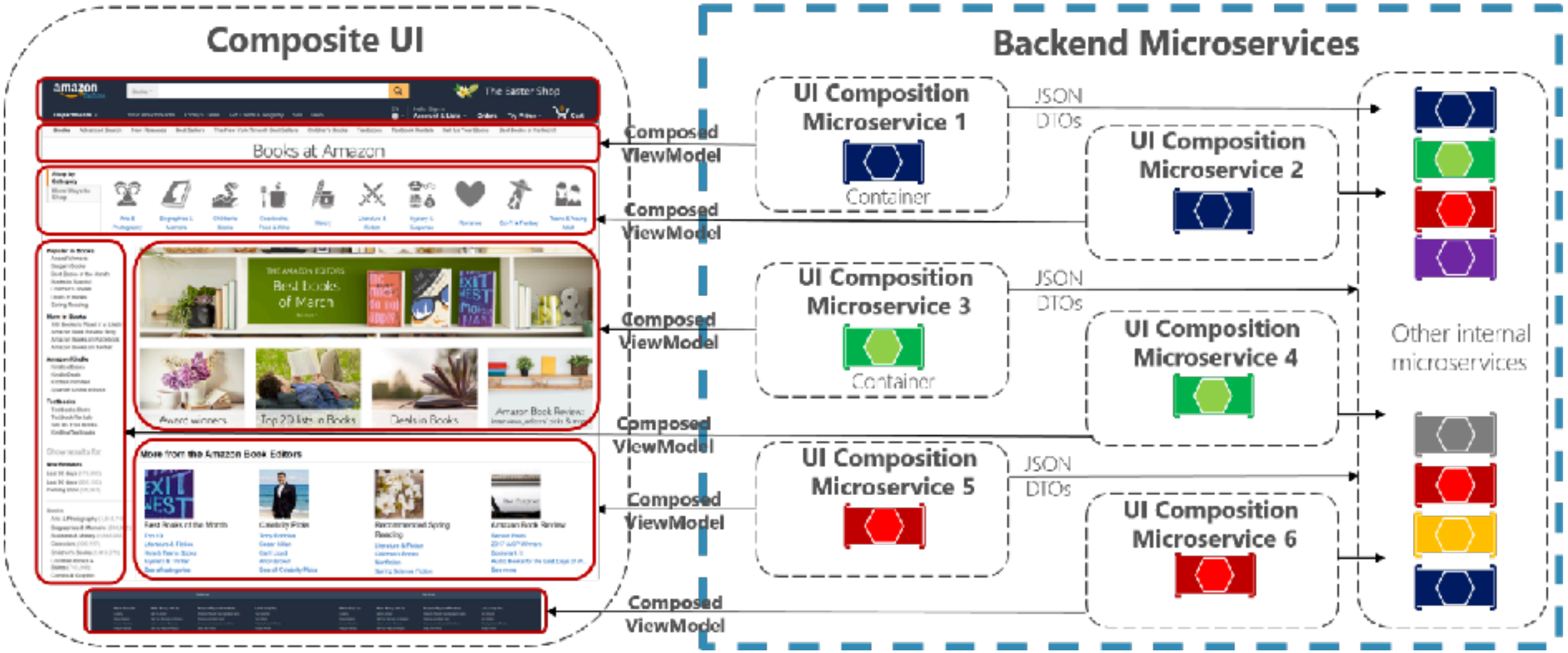
Monolithic UI consuming microservices



https://docs.microsoft.com

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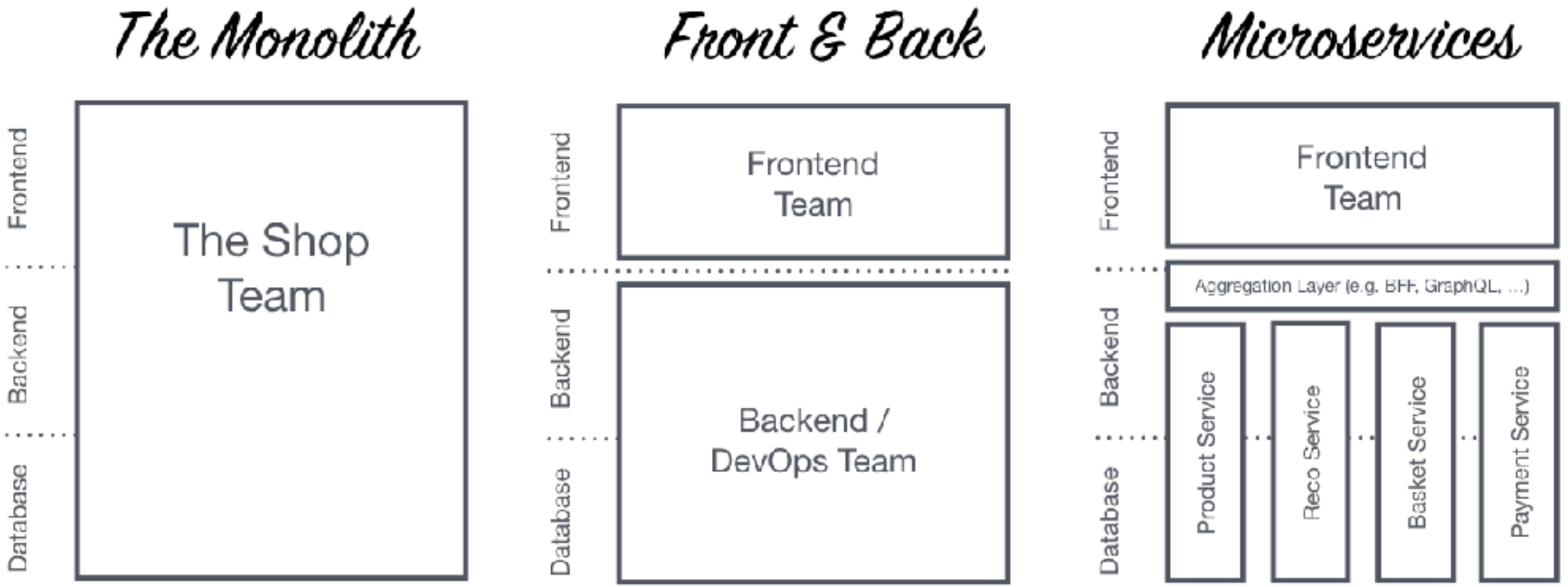
Composite UI *generated* by microservices



https://docs.microsoft.com

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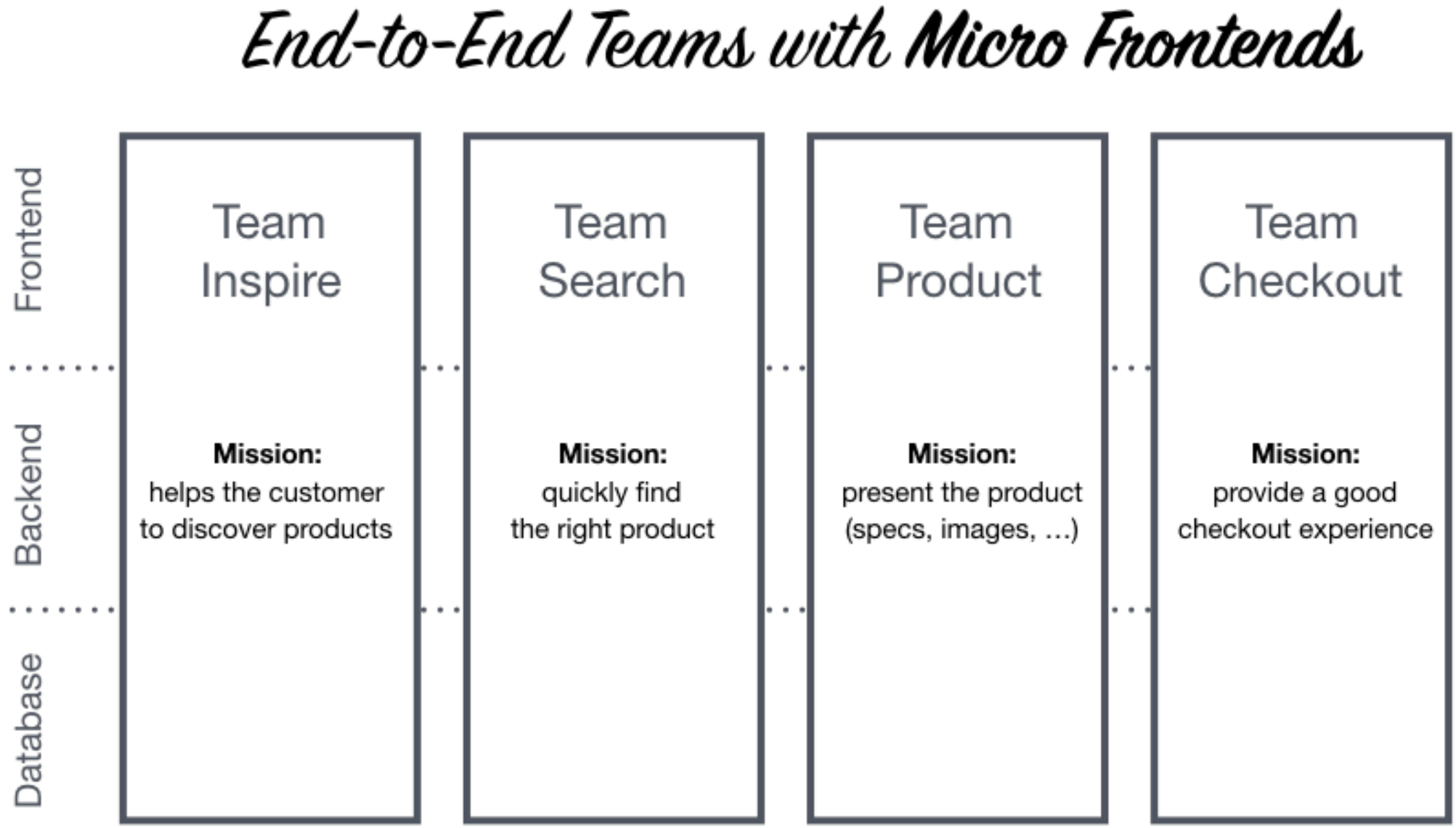
Monolith frontend



https://micro-frontends.org/

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Micro frontend



https://micro-frontends.org/



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Observability of services



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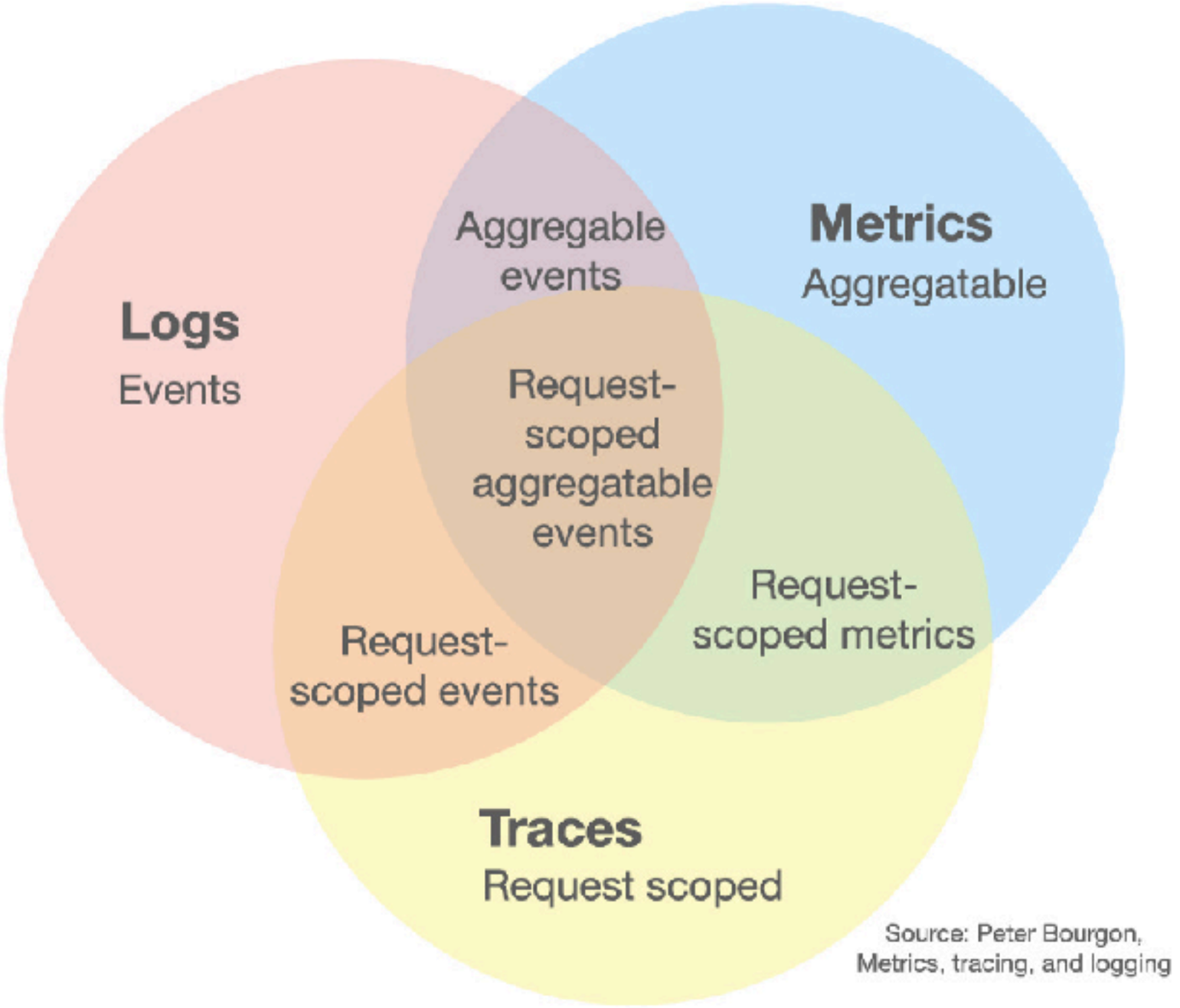
Observability of services

Application metrics
Distributed tracing
Log aggregation
Alert system
Exception tracking



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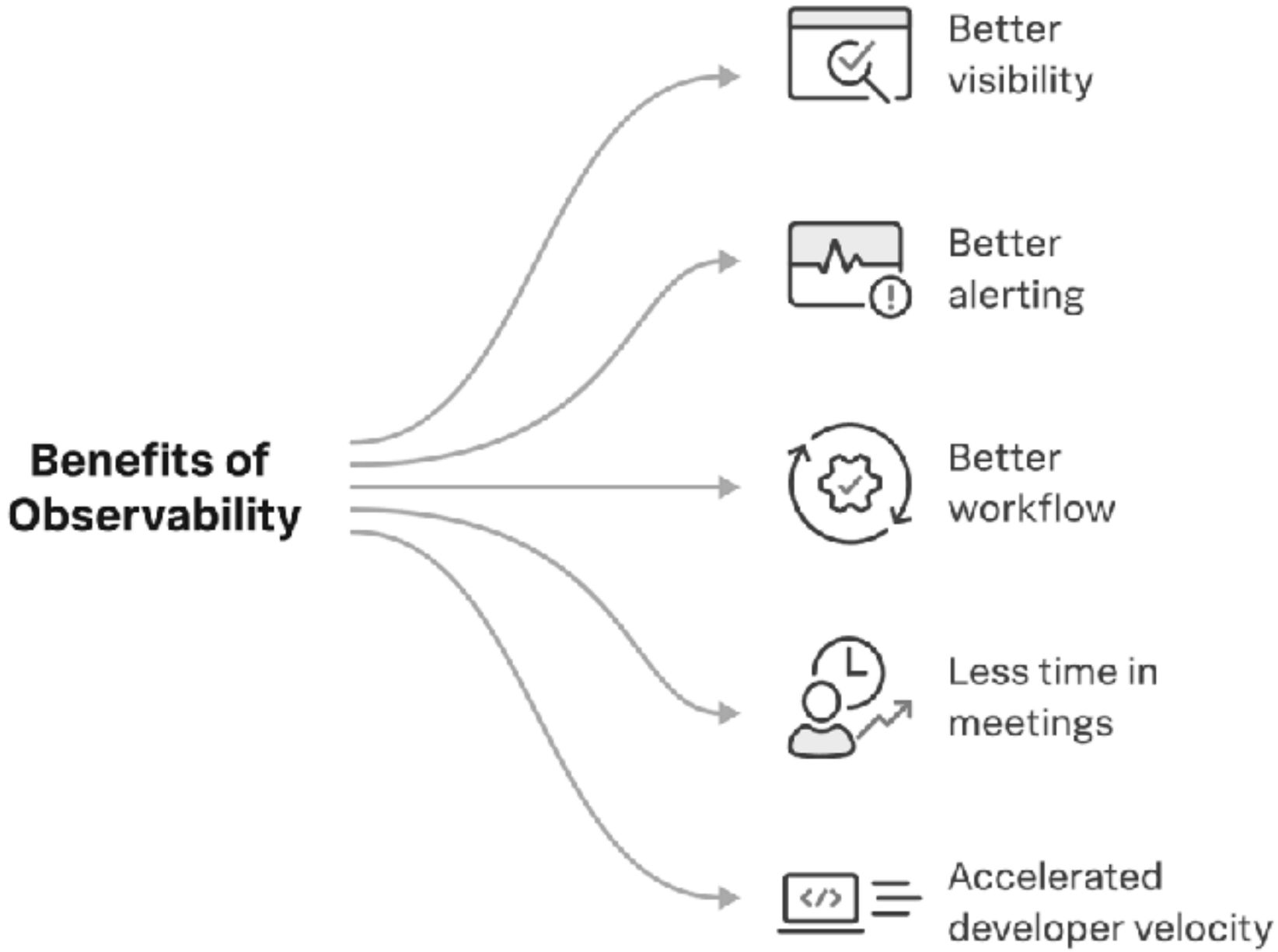
Observability of services



Source: Peter Bourgon, Metrics, tracing, and logging

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Observability of services



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Application metrics

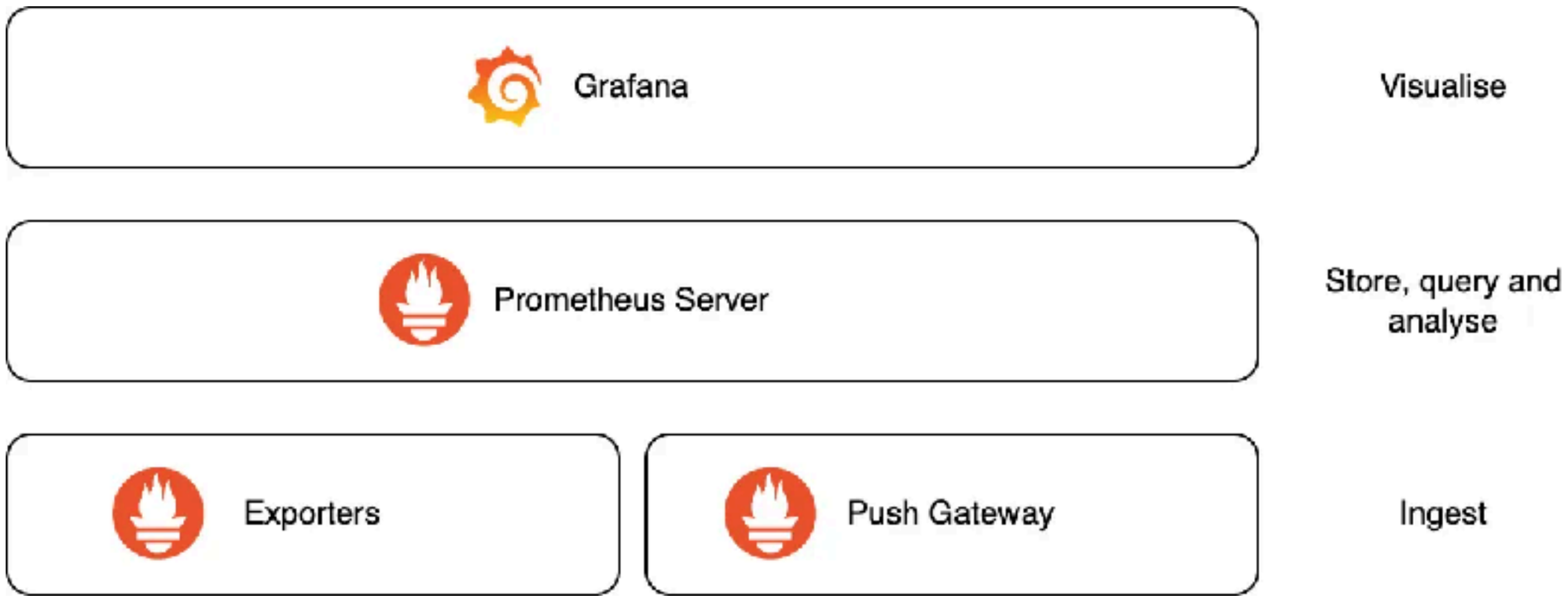
Metrics represent logs **numerically** over intervals of time

Request rate	Error rate	Health
Latency	Utilization	Availability



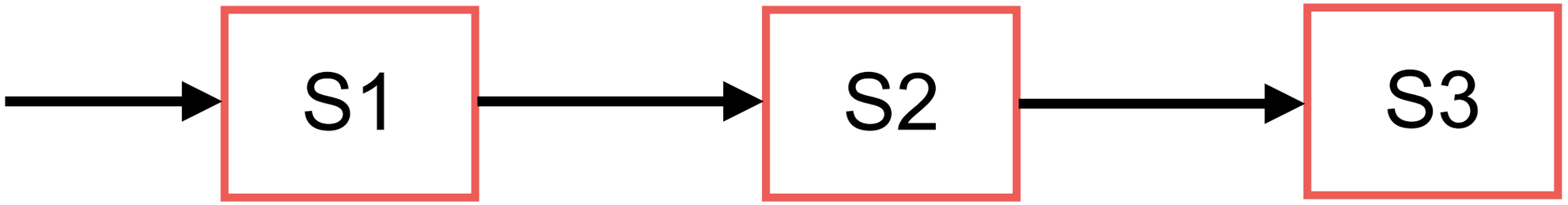
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Application metrics



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Distributed tracing



Response time

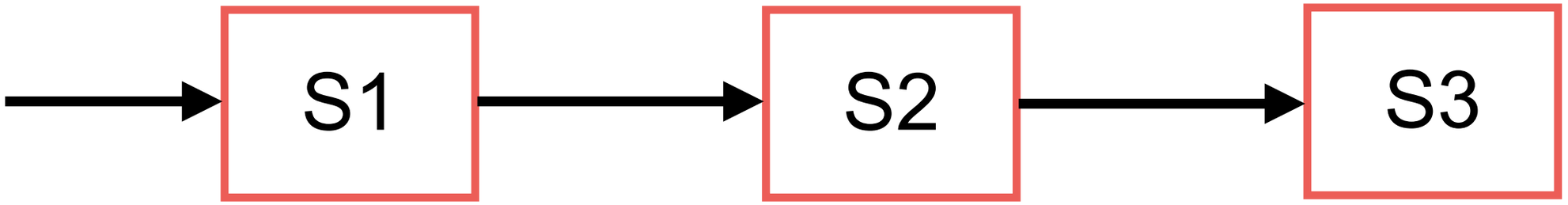
Time ?

Time ?

Time ?

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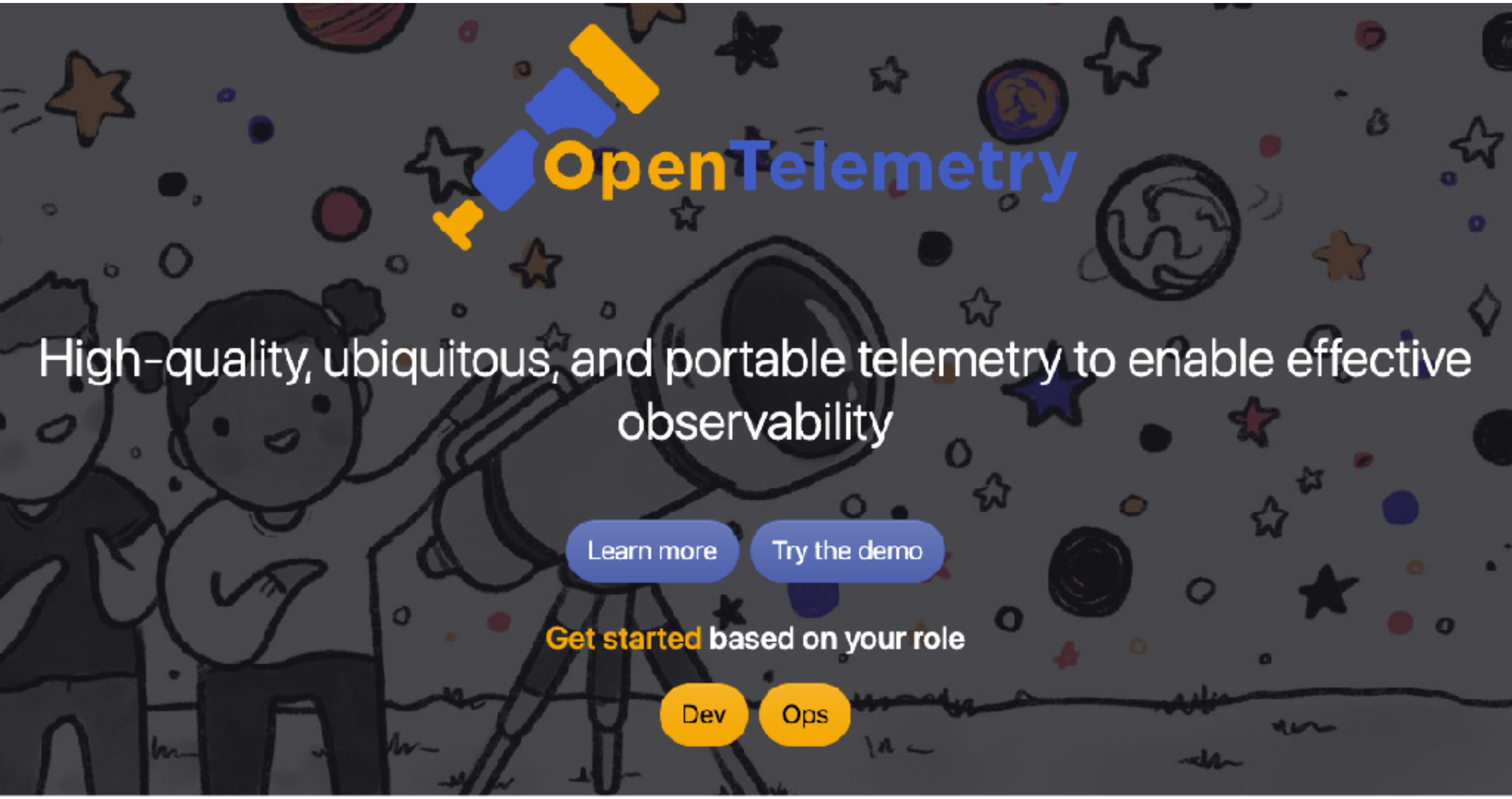
Correlation IDs



Response time **id=111**

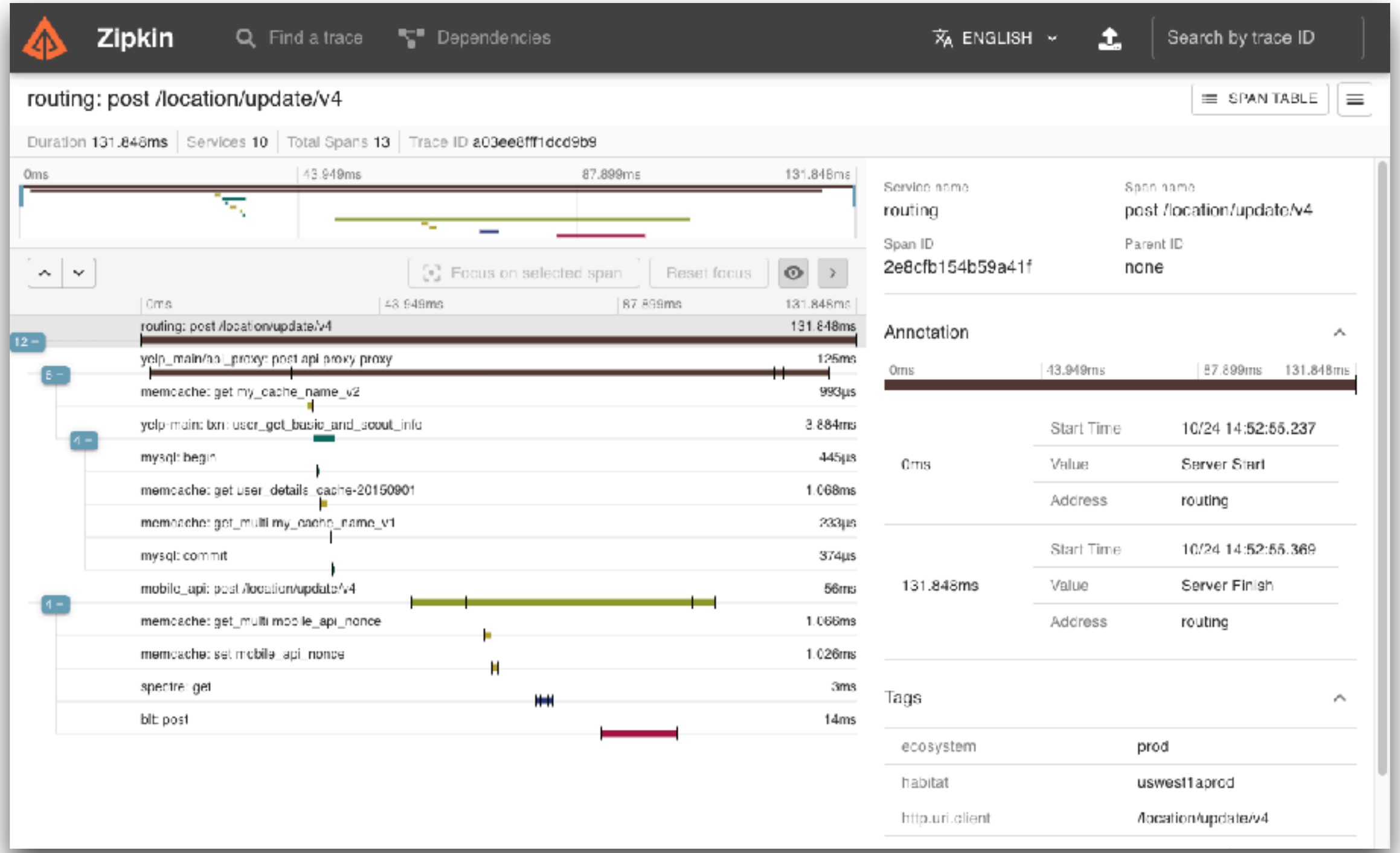
Time ?	Time ?	Time ?
id=222 **parent_id=111**	id=333 **parent_id=111**	id=444 **parent_id=111**

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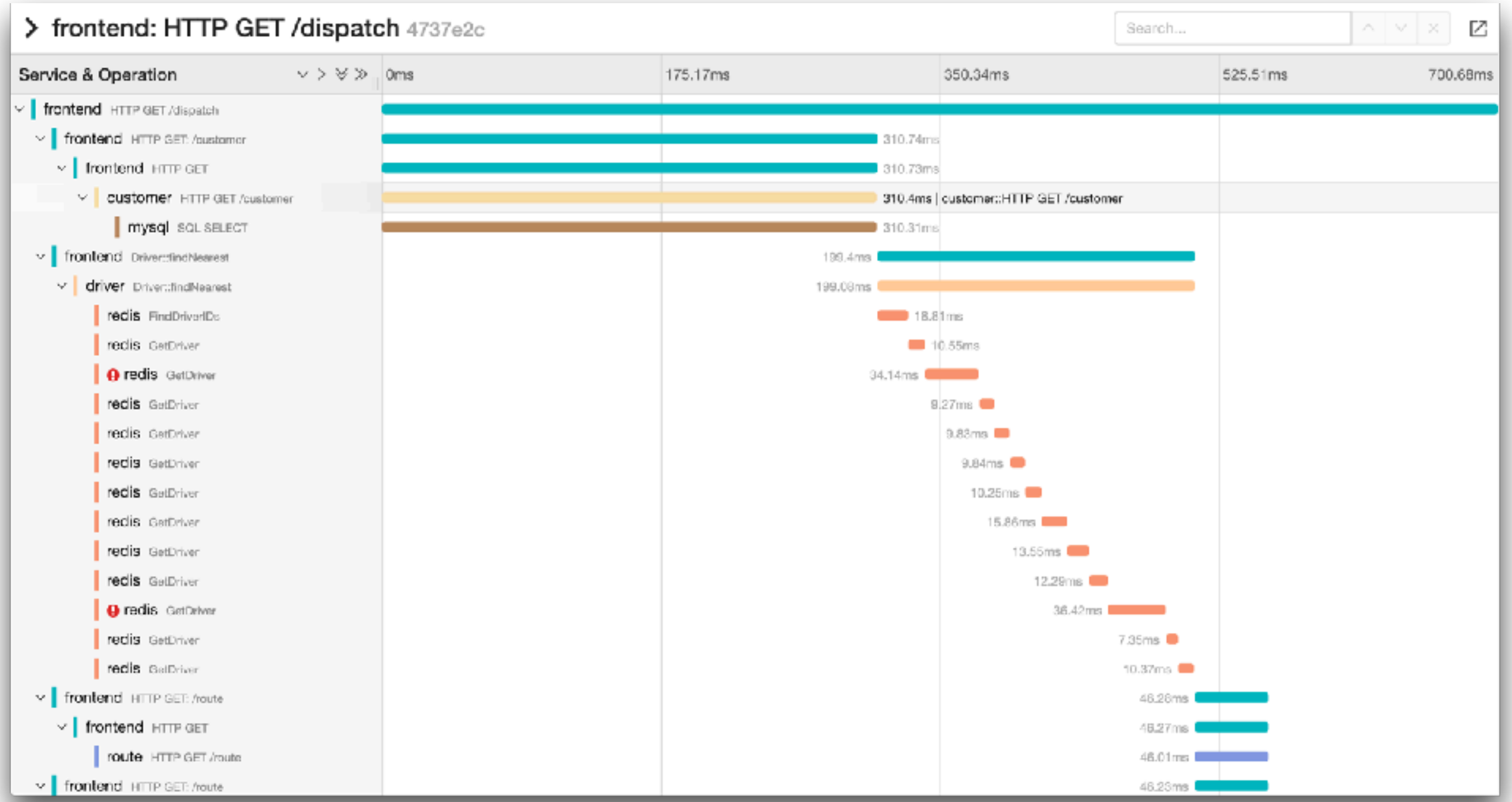
https://opentelemetry.io/

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https://zipkin.io/

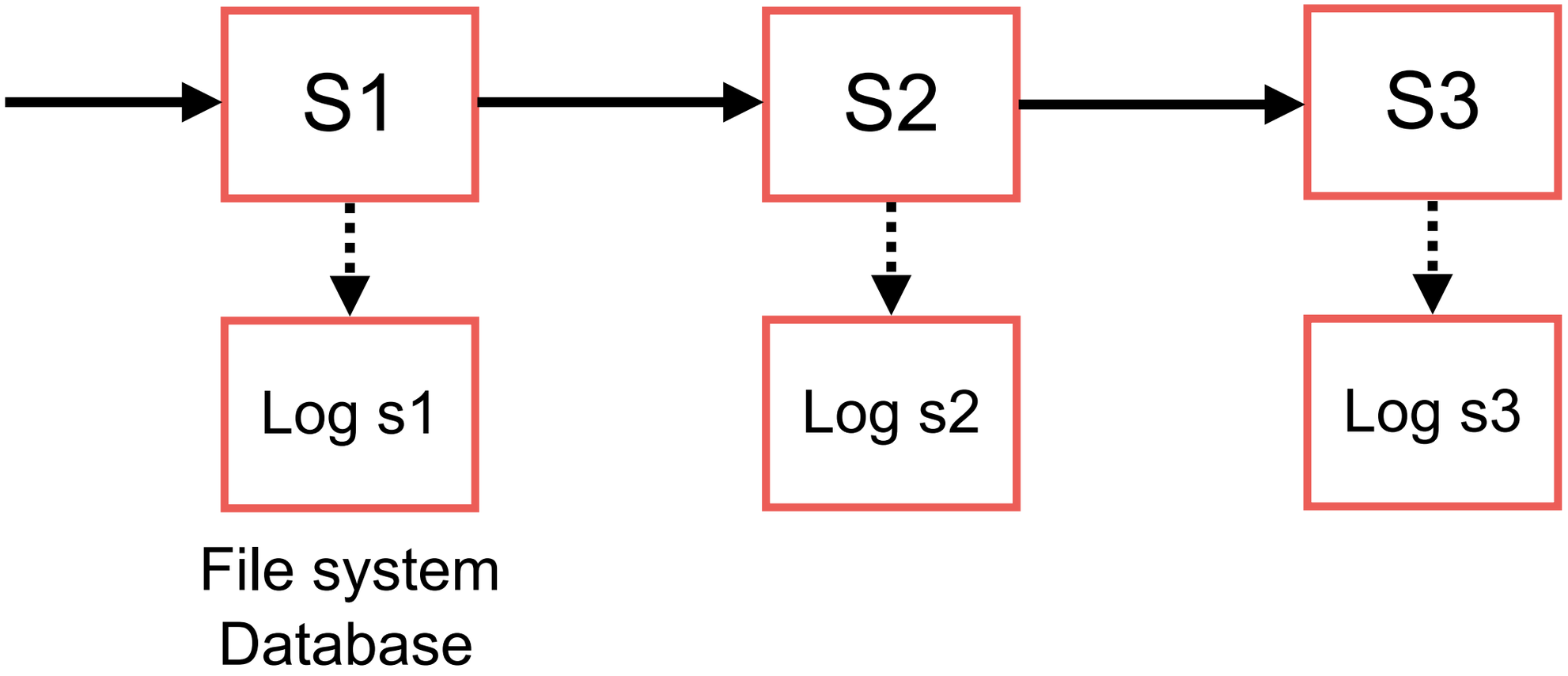
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https://www.jaegertracing.io/

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Log aggregation



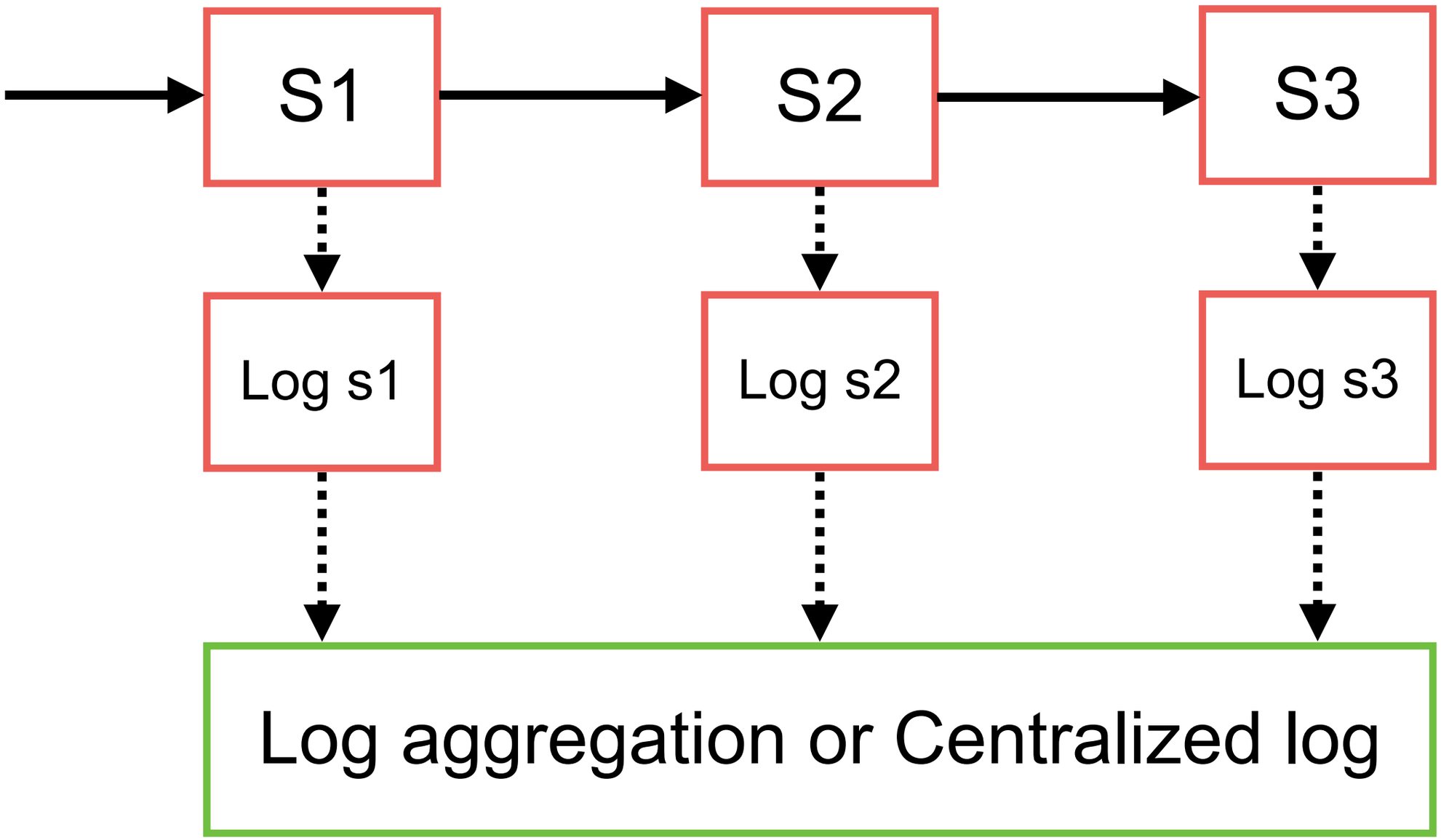
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How to find errors from log ?



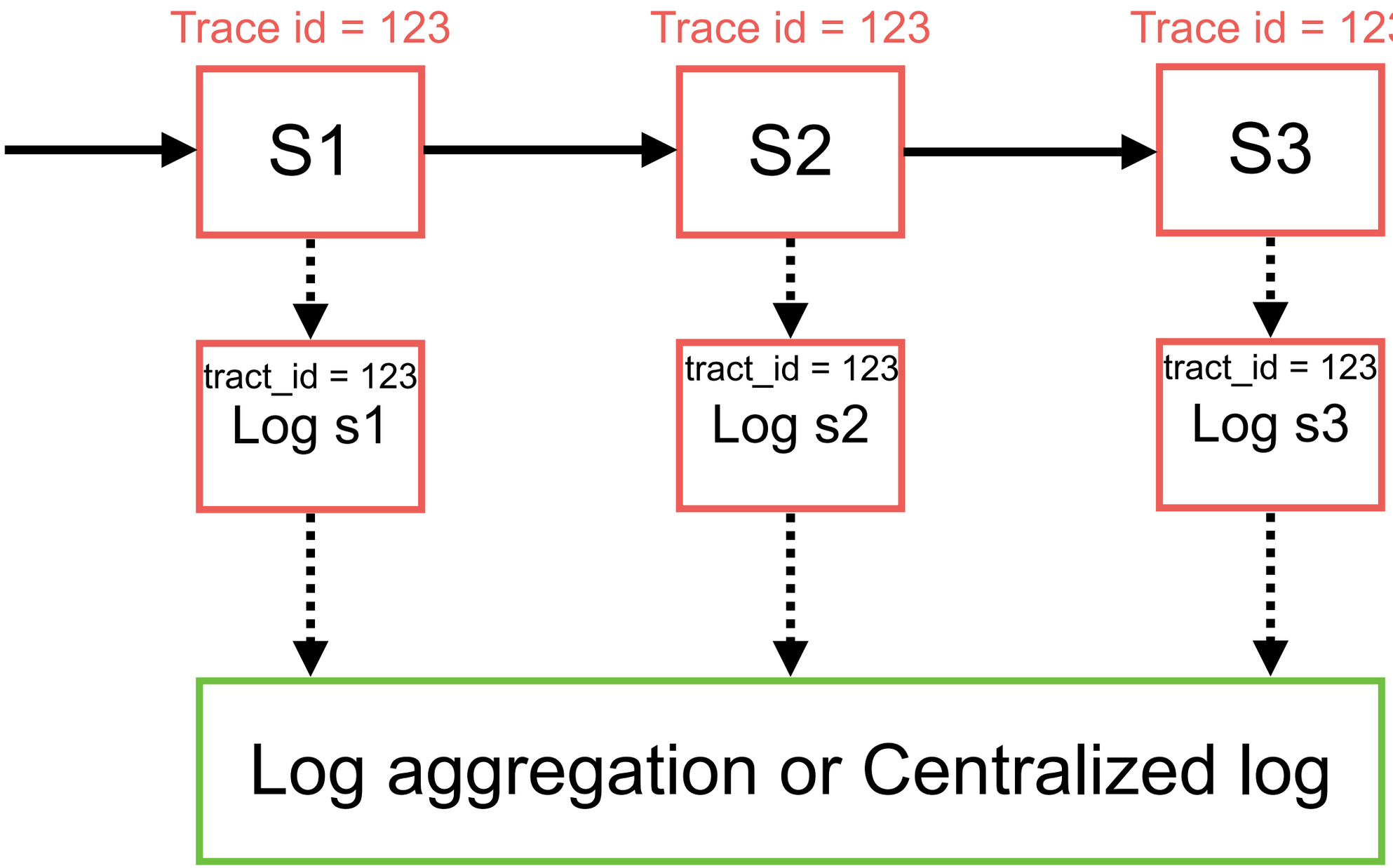
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Log aggregation



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Log aggregation + Correlation IDs



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Consistent Structure Logs

Property	Description	Example
Timestamp	Date and time of the log	2023-07-01
Log level	DEBUG, INFO, ERROR	
Trace Id or Correlation Id	Unique identifier that refer to other logs from all services	
Event/Action Name	Identify to event or action of log	Authentication fail
Service ID/ Name	Identify to service	
Request path	Path for the request	/api/products



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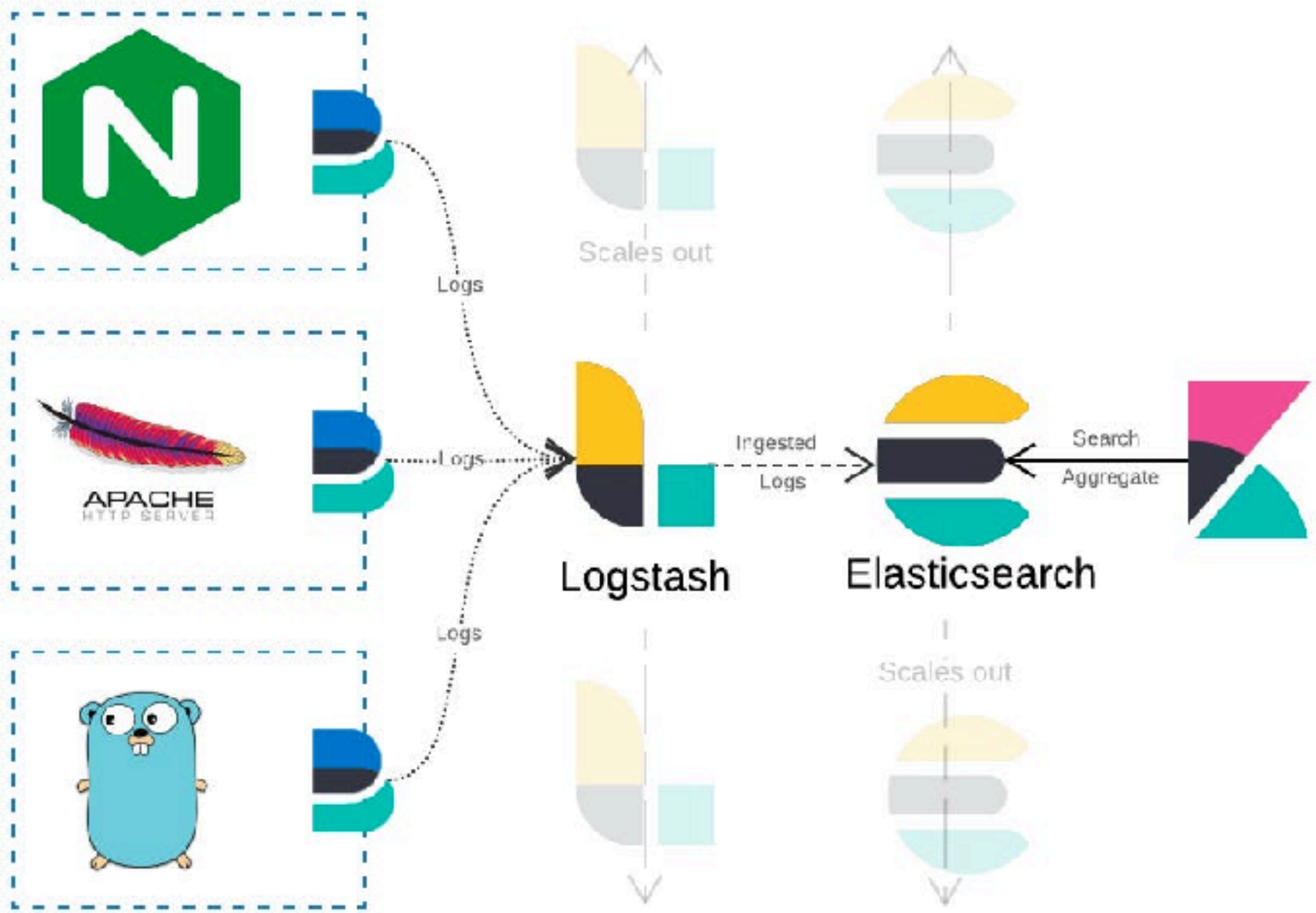
Don’t keep !!

```
com.framework.FrameworkException: Error in web request
    at com.framework.ApplicationStarter.lambda$start$0(ApplicationStarter.java:15)
    at spark.RouteImpl$1.handle(RouteImpl.java:72)
    at spark.http.matching.Routes.execute(Routes.java:61)
    at spark.http.matching.MatcherFilter.doFilter(MatcherFilter.java:134)
    at spark.embeddedserver.jetty.JettyHandler.doHandle(JettyHandler.java:50)
    at org.eclipse.jetty.server.session.SessionHandler.doScope(SessionHandler.java:1568)
    at org.eclipse.jetty.server.handler.ScopedHandler.handle(ScopedHandler.java:144)
    at org.eclipse.jetty.server.handler.HandlerWrapper.handle(HandlerWrapper.java:132)
    at org.eclipse.jetty.server.Server.handle(Server.java:503)
    at org.eclipse.jetty.server.HttpChannel.handle(HttpChannel.java:364)
    at org.eclipse.jetty.server.HttpConnection.onFillable(HttpConnection.java:260)
    at org.eclipse.jetty.io.AbstractConnection$ReadCallback.succeeded(AbstractConnection.java:305)
    at org.eclipse.jetty.io.FillInterest.fillable(FillInterest.java:103)
    at org.eclipse.jetty.io.ChannelEndPoint$2.run(ChannelEndPoint.java:118)
    at org.eclipse.jetty.util.thread.QueuedThreadPool.runJob(QueuedThreadPool.java:765)
    at org.eclipse.jetty.util.thread.QueuedThreadPool$2.run(QueuedThreadPool.java:683)
    at java.base/java.lang.Thread.run(Thread.java:834)
Caused by: com.project.module.MyProjectFooBarException: The number of FooBars cannot be zero
    at com.project.module.MyProject.anotherMethod(MyProject.java:20)
    at com.project.module.MyProject.someMethod(MyProject.java:12)
    at com.framework.ApplicationStarter.lambda$start$0(ApplicationStarter.java:13)
    ... 16 more
Caused by: java.lang.ArithmeticException: The denominator must not be zero
    at org.apache.commons.lang3.math.Fraction.getFraction(Fraction.java:143)
    at com.project.module.MyProject.anotherMethod(MyProject.java:18)
    ... 18 more
```



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ELK stack



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How to develop ?
How to test ?
How to deploy ?



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Summary



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Beyond Microservice
Process and **Organization**



Flexible organizational structure

With clear roles and accountabilities



Efficient meeting formats

Geared toward action and eliminating over-analysis



More autonomy to teams and individuals

Individuals solve issues directly without bureaucracy



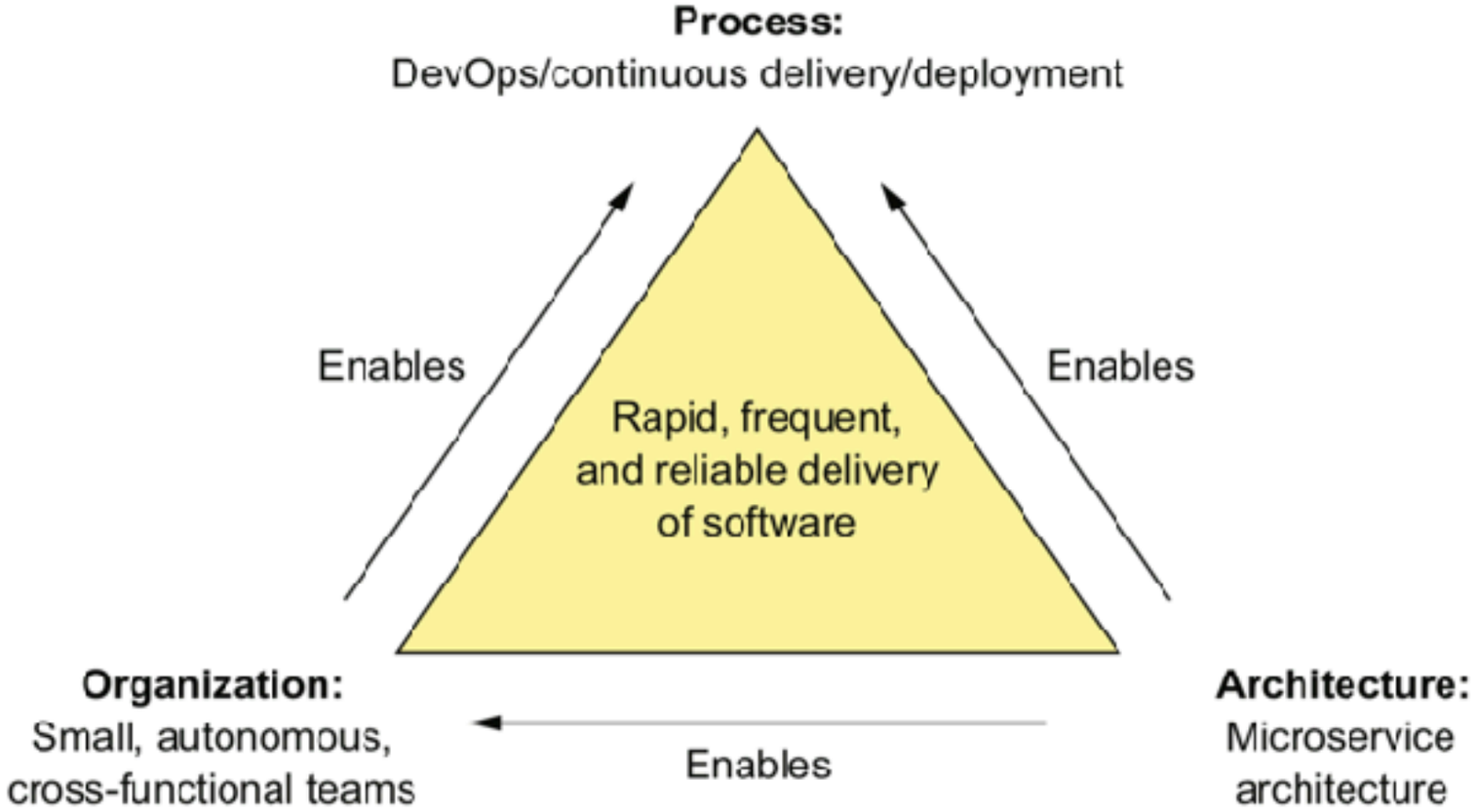
Unique decision-making process

To continuously evolve the organization's structure.

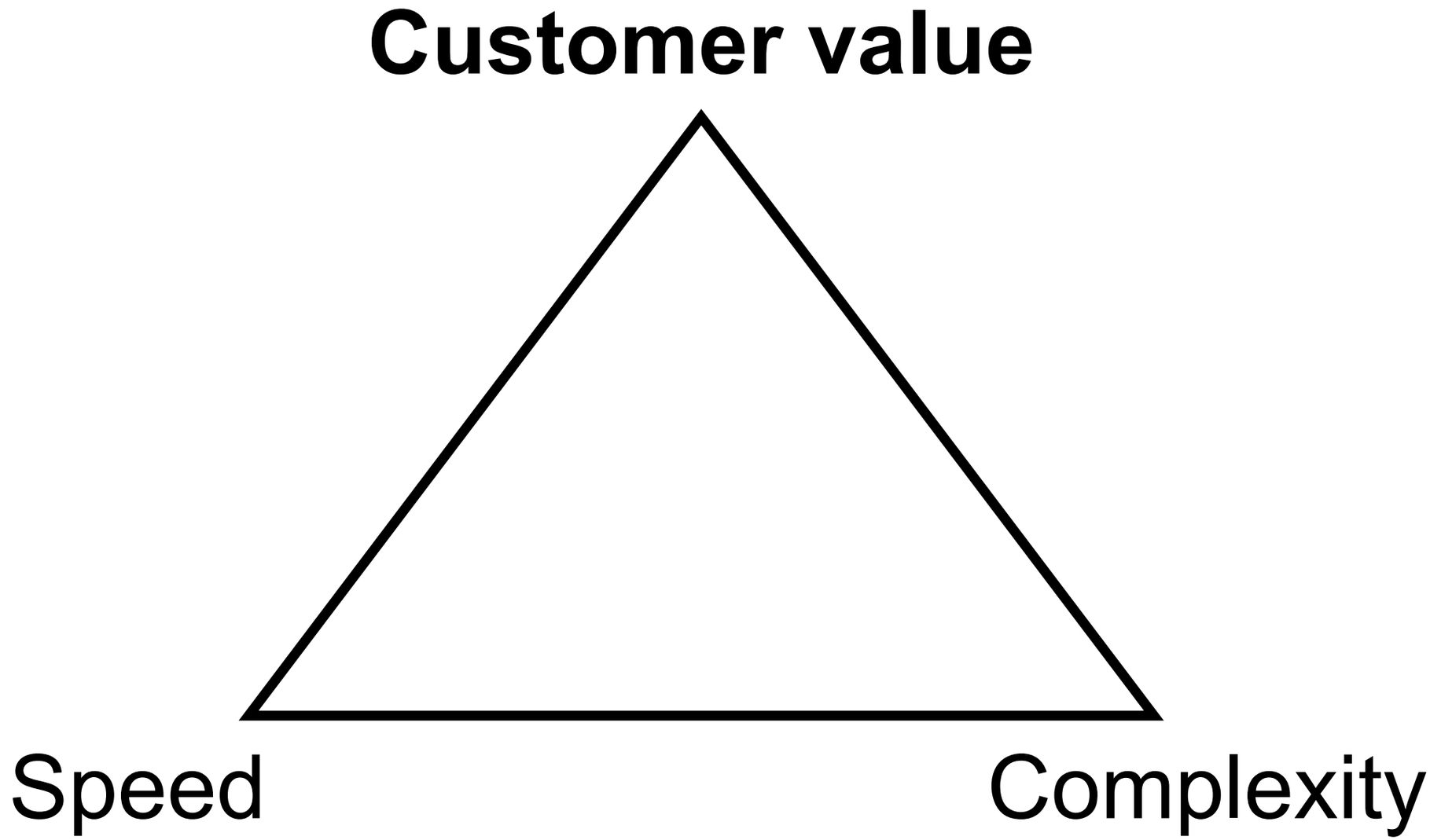
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Success project needs …

Organization process
Development process
Delivery process



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Don’t forget about the **human** side when adopt Microservice

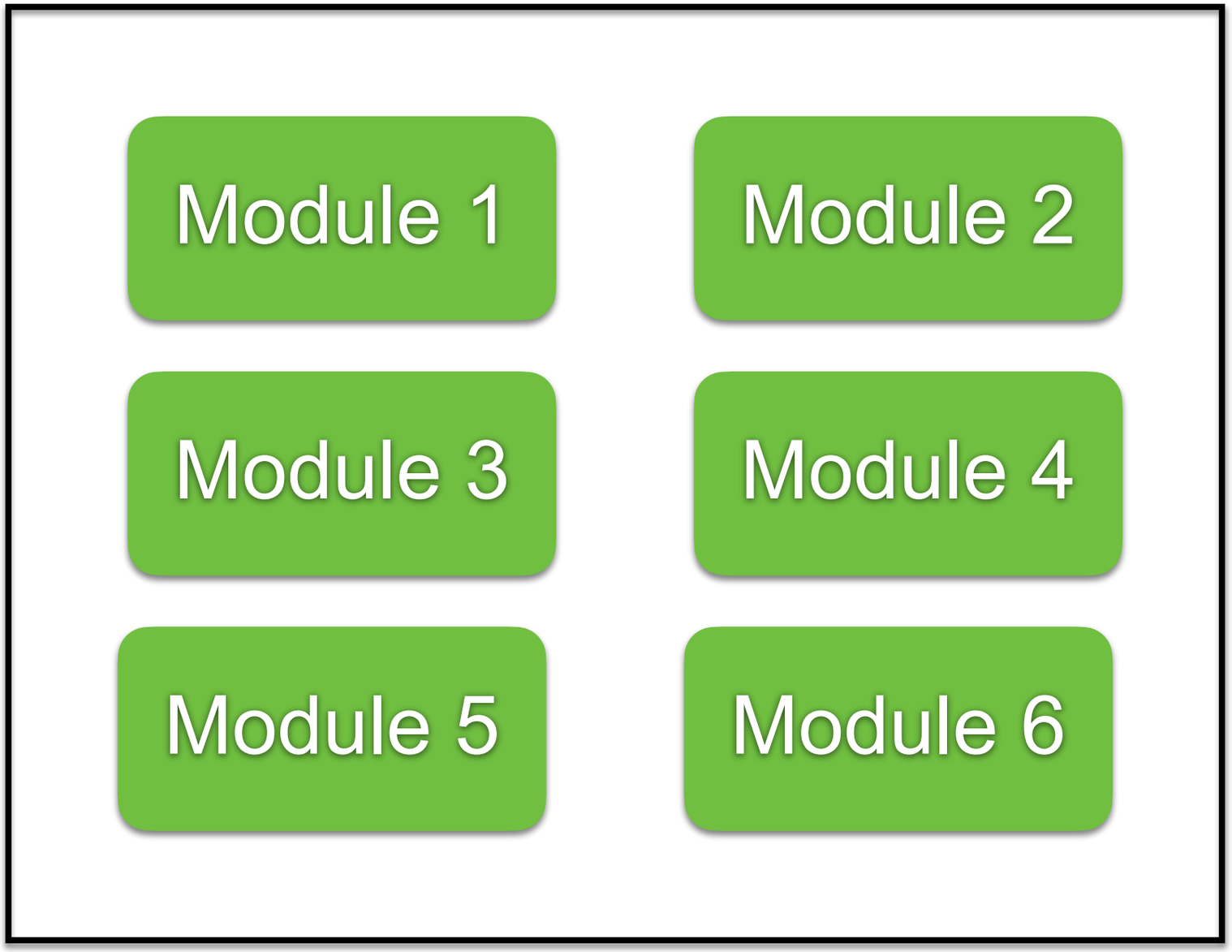


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Let’s start with good monolith



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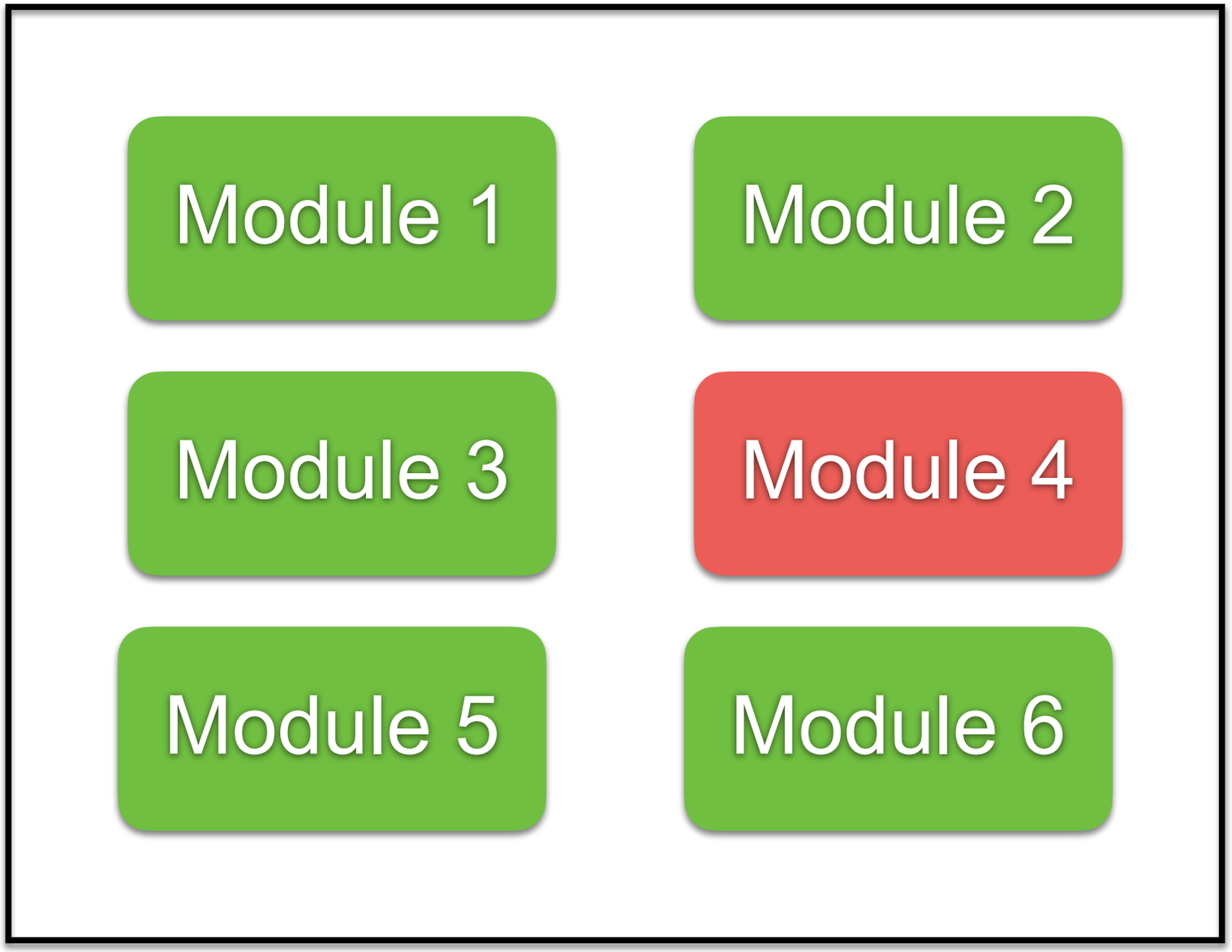


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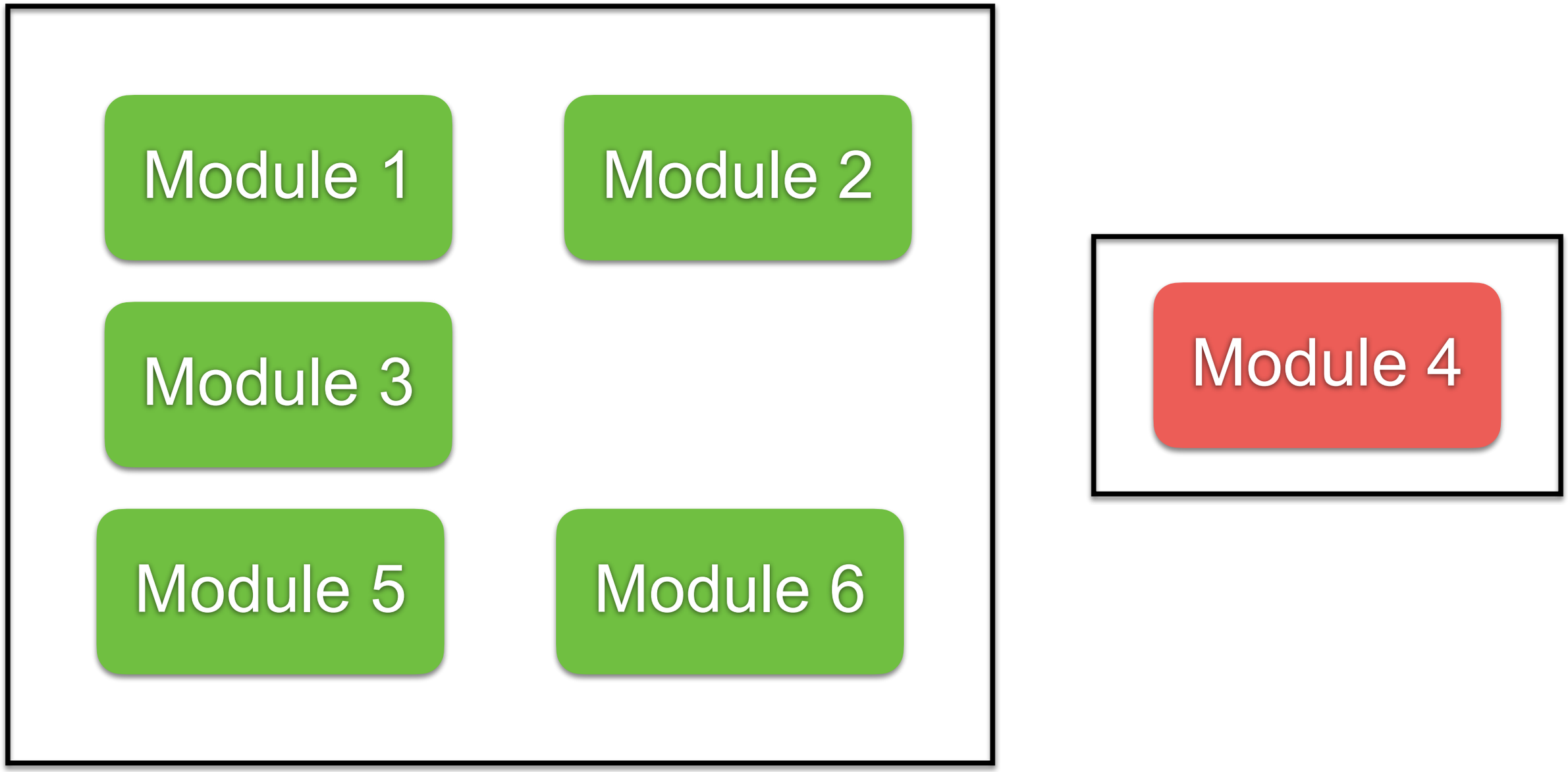
Find your problems



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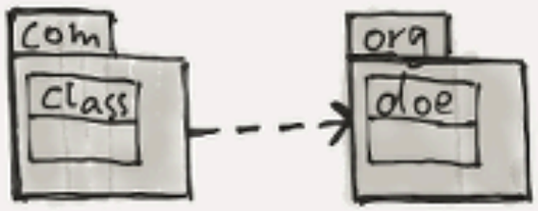


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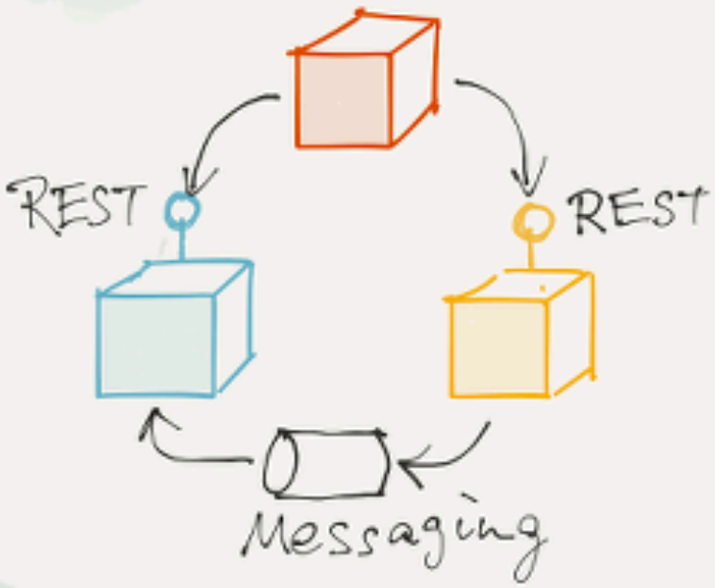


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Architecture



Microservices



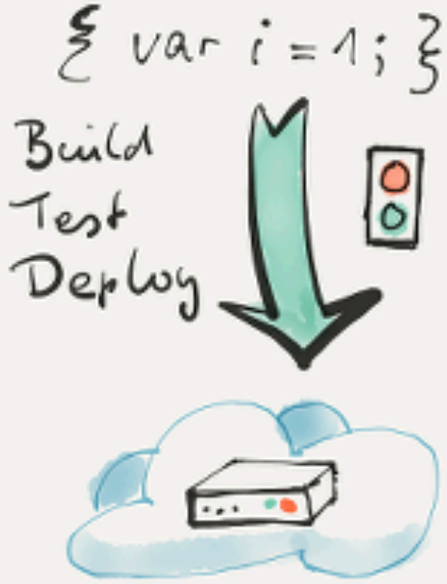
People & Teams



Communication
Collaboration

Deployment

Continous Delivery

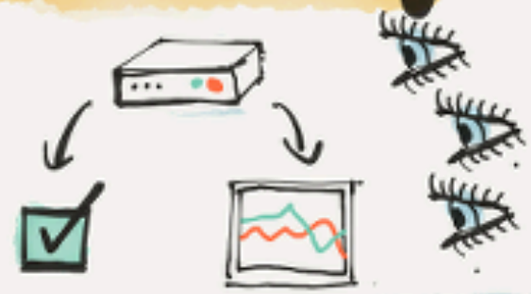


Infrastructure



Provisioning

Monitoring

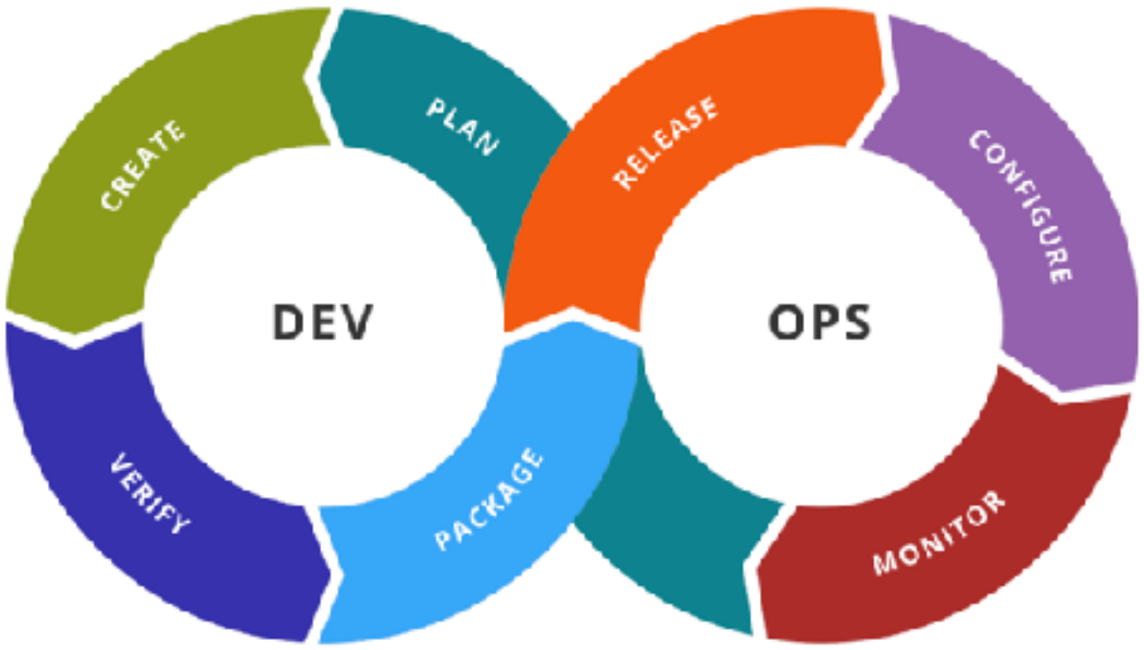


Features & Technology

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Microservice prerequisites

Rapid provisioning
Basic monitoring
Rapid Application Deployment
DevOps culture



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Let’s start your journey !!



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